

Popular Supplements: The Noise, The Science & The Verdict

An open-ended, evidence-graded field guide to 244 popular supplements, herbs, mushrooms, nootropics and beverages — each rated 1–5 on the strength of **industry-independent human-trial evidence**, with what the marketing claims, what the science actually shows, the verdict, the best form to buy, and whether you can get it on Amazon or direct.

Compiled 2026-08-04 from a multi-workflow adversarial research pass (~150 AI research agents, ~5.7M tokens). Grades measure EVIDENCE QUALITY, not popularity or personal fit. **Not medical advice.**

How to read every entry

- **THE NOISE** what the marketing, influencers and label claim it does — the hype.
- **THE SCIENCE** what independent human trials actually show (effect size, whether it clears the minimum clinically important difference, the population, RCT vs cohort).
- **THE VERDICT** the honest bottom line: buy, skip, or conditional — and for whom.
- **BEST FORM** the most bioavailable / best-tolerated form to buy (e.g. magnesium glycinate over oxide).
- **WATCH** what to be aware of — absorption, timing, upper limits, interactions, safety.
- **Amazon ✓/✗ Website ✓/✗** — where you can actually buy it. ✗ usually means prescription-only or banned from Amazon (e.g. most racetams, phenibut).

The 1–5 scale: **5** large independent RCTs, would bet on it · **4** real but modest, or narrower than marketed · **3** real signal but small/conditional · **2** mechanism outruns outcome (industry-funded/low-quality) · **1** marketing, no credible human outcome. A few compounds appear under more than one outcome because the evidence differs by use (e.g. creatine is a 5 for muscle, 4 for memory).

15

score ≥ 4

44

3 – 3.5

134

2 – 2.5

51

below 2

The short list — what actually earns a place (≥3.5)

Distinct items scoring 3.5+ across all categories. Everything else is conditional, niche, or not worth it.

- **Creatine monohydrate** (5, Muscle) — Buy. The single best-evidenced sports supplement. Large, mostly independent RCT base shows a real, clinically meaningful add-on to resistance trainin...
- **Caffeine (the honest energy answer - added)** (4.5, Energy) — This is the one 'energy' agent that actually works. Large independent literature: reliable, dose-related reduction in fatigue and improvement in aler...
- **Whey / protein powder** (4.5, Muscle) — Buy only if you are not already hitting your protein target from food. It is a convenient food, not a magic compound - the benefit vanishes once tota...
- **Iron (oral ferrous salts, for fatigue/energy)** (4, Energy) — Buy ONLY if serum ferritin is low. In iron-deficient menstruating women (ferritin <50 ug/L, no anemia) oral iron produces a real, modest reduction in...
- **Psyllium husk (ispaghula)** (4, Gut) — Buy — the strongest evidence in the category. Cheap, independent, guideline-endorsed soluble/gel-forming fiber for IBS global symptoms and for chroni...
- **Cranberry (proanthocyanidins) - UTI prevention** (4, Women's) — Conditional BUY for women with recurrent UTIs or peri-procedural risk, using a product delivering ≥36 mg standardized proanthocyanidins/day; SKIP if...
- **Modafinil (prescription - included for context)** (4, Nootropic) — Conditional and by prescription only. Clearly effective for diagnosed sleep-disorder wakefulness; for healthy adults the cognitive lift is modest, ta...
- **Red yeast rice (monacolin K)** (4, Metabolic) — Skip in favor of a prescribed statin - you get the same drug with a known dose, quality control and monitoring. Buying red yeast rice is self-prescri...
- **Folate / folic acid** (4, Vit/Min) — Buy - but it is a targeted drug, not a wellness pill. Every woman who could become pregnant should take 400 mcg/day (higher if prior NTD or on certai...
- **Beetroot / dietary nitrate** (4, Ergogenic) — Buy - conditional. Worthwhile for recreational/sub-elite endurance and short TTs; take a concentrated 'shot' delivering ≥6.4 mmol (400 mg) nitrate 2...

- **Sodium bicarbonate** (4, Ergogenic) — Buy - conditional, for trained athletes doing repeated high-intensity efforts (45 s-8 min). 0.2-0.3 g/kg (~15-20 g for a 70 kg adult) 60-180 min befo...
- **Coffee (caffeinated, paper-filtered / drip)** (4, Coffee) — Drink it if you enjoy it — 2-4 cups/day of filtered coffee is associated with the best outcomes and is safe for most adults; it is not a reason to ST...
- **Caffeine + L-theanine** (3.5, Focus) — Buy if you want a smoother acute alertness/reaction-time boost than caffeine alone, but understand the measurable focus benefit is driven by the caff...
- **Nicotinamide (niacinamide, oral vitamin B3)** (3.5, Skin) — Buy ONLY if you are a high-risk immunocompetent patient with a prior nonmelanoma skin cancer, and dose it correctly at 500 mg TWICE daily. This is a...
- **Collagen / gelatin + vitamin C, dosed 30-60 min pre-load (timing protocol)** (3.5, Tendon) — Conditional buy, and only as an adjunct. The strongest claim - that 15-30 g collagen/gelatin + vitamin C ~60 min before loading increases tendon cros...
- **Melatonin (dose & timing)** (3.5, Sleep) — Buy for the RIGHT problem: circadian misalignment (delayed sleep-wake phase disorder, jet lag, shift work, non-24, ASD/child sleep). For garden-varie...
- **Saccharomyces boulardii (CNCM I-745)** (3.5, Gut) — Conditional buy — genuinely effective for a specific job (preventing antibiotic-associated diarrhea, shortening acute infectious diarrhea), NOT for r...
- **Essential amino acids (EAA)** (3.5, Muscle) — Conditional. Genuinely stimulates muscle protein synthesis, but offers no proven hypertrophy advantage over cheaper whole protein for a healthy adult...
- **Beta-alanine** (3.5, Muscle) — Buy only for its real domain: high-intensity efforts lasting ~1-4 minutes (repeated sprints, rowing, CrossFit-style work). It is NOT a maximal-streng...
- **Ginger (Zingiber officinale)** (3.5, Herb) — Buy for nausea (pregnancy morning sickness, chemo-adjunct, motion) and period pain at 1-1.5 g/day of powdered ginger - one of the few evidence-backed...
- **Soy isoflavones (genistein/daidzein)** (3.5, Women's) — Conditional BUY for menopausal women wanting a modest, non-hormonal option - use a genistein-rich extract (>=18-20 mg genistein, ~50 mg total isoflav...
- **Plant sterols / phytosterols (and stanols)** (3.5, Metabolic) — Conditional buy - a reasonable, well-tolerated LDL-lowering adjunct at ~2 g/day (spreads/drinks or capsules with meals) for people with mild hypercho...
- **Hibiscus tea (Hibiscus sabdariffa)** (3.5, Drink) — A legitimate adjunct for prehypertension/mild hypertension: 2-3 cups/day of steeped hibiscus. Real but modest - not a replacement for prescribed anti...

Master ranked table (all 244)

Sorted by score. Amz = on US Amazon, Web = direct from a website. Full reasoning in the entries that follow.

#	SUPPLEMENT	SECTION	SCORE	CF	AMZ	WEB	TYPE OF EFFECT	BEST FORM (SHORT)
1	Creatine monohydrate	Muscle	5	H	✓	✓	muscle / strength	Plain creatine monohydrate is the gold standard - m...
2	Caffeine (the honest energy answer - ad...	Energy	4.5	H	✓	✓	energy / fatigue	Caffeine anhydrous in a measured tablet/capsule (10...
3	Whey / protein powder	Muscle	4.5	H	✓	✓	muscle / strength	Whey isolate if you're lactose-sensitive or watchin...
4	Creatine monohydrate	Mind	4	H	✓	✓	cognition / memory	Plain creatine monohydrate is the gold standard - m...
5	Caffeine (alone)	Focus	4	H	✓	✓	acute focus / attention	Caffeine anhydrous in a measured tablet/capsule (10...
6	Iron (oral ferrous salts, for fatigue/e...	Energy	4	H	✓	✓	energy / fatigue	Ferrous bisglycinate (chelated) is notably gentler...
7	Caffeine	Muscle	4	H	✓	✓	muscle / strength	n/a for efficacy - caffeine anhydrous (capsule/tabl...
8	Psyllium husk (ispaghula)	Gut	4	H	✓	✓	gut / digestion	Pure blond psyllium husk powder (ispaghula), unswee...
9	Modafinil (prescription - included for...	Nootropic	4	H	✗	✗	wakefulness (strong); modest exec...	Prescription-only; not a retail supplement. Where a...
10	Cranberry (proanthocyanidins) - UTI pre...	Women's	4	H	✓	✓	recurrent UTI prevention (women w...	Concentrated extract delivering >=36 mg soluble A-t...
11	Red yeast rice (monacolin K)	Metabolic	4	H	✓	✓	LDL-cholesterol lowering (statin...	Buy a product that discloses/standardizes monacolin...
12	Folate / folic acid	Vit/Min	4	H	✓	✓	prevention of neural tube defects...	L-5-methylfolate (5-MTHF) at ~400 mcg DFE for gener...
13	Beetroot / dietary nitrate	Ergogenic	4	H	✓	✓	endurance time-trial performance...	A concentrated beetroot juice/shot standardised to...
14	Sodium bicarbonate	Ergogenic	4	H	✓	✓	high-intensity anaerobic capacity...	Dose by body weight: ~0.2-0.3 g/kg taken 60-180 min...
15	Coffee (caffeinated, paper-filtered / d...	Coffee	4	H	✓	✓	CV & all-cause mortality (cohort...	Paper-filtered drip/pour-over is the best-tolerated...
16	Caffeine + L-theanine	Focus	3.5	H	✓	✓	acute focus / attention	The synergy is about the combo/ratio, not a special...
17	Creatine monohydrate (for 'energy'/anti...	Energy	3.5	H	✓	✓	energy / fatigue	Plain creatine monohydrate is the gold standard - m...
18	Melatonin (dose & timing)	Sleep	3.5	H	✓	✓	sleep	LOW dose - 0.5-1 mg is physiologic and as effective...
19	Beta-alanine	Muscle	3.5	H	✓	✓	muscle / strength	Sustained-release beta-alanine (e.g. CarnoSyn SR),...

#	SUPPLEMENT	SECTION	SCORE	CF	AMZ	WEB	TYPE OF EFFECT	BEST FORM (SHORT)
20	Essential amino acids (EAA)	Muscle	3.5	M	✓	✓	muscle / strength	A full-spectrum, free-form EAA blend containing all...
21	Saccharomyces boulardii (CNCM I-745)	Gut	3.5	H	✓	✓	gut / digestion	The specific documented strain S. boulardii CNCM I-...
22	Nicotinamide (niacinamide, oral vitamin...	Skin	3.5	H	✓	✓	skin	Niacinamide (nicotinamide) - the non-flushing amide...
23	Collagen / gelatin + vitamin C, dosed 3...	Tendon	3.5	M	✓	✓	tendon / ligament	~15 g hydrolyzed collagen peptides OR gelatin, plus...
24	Ginger (Zingiber officinale)	Herb	3.5	H	✓	✓	anti-nausea (pregnancy/chemo), dy...	For consistent dosing, an extract standardised to g...
25	Soy isoflavones (genistein/daidzein)	Women's	3.5	H	✓	✓	vasomotor symptoms (hot-flash fre...	Extract standardised to total isoflavones with geni...
26	Plant sterols / phytosterols (and stano...	Metabolic	3.5	H	✓	✓	LDL-cholesterol lowering (surroga...	Sterols and stanols lower LDL comparably, so isomer...
27	Hibiscus tea (Hibiscus sabdariffa)	Drink	3.5	M	✓	✓	blood pressure lowering (RCT)	n/a - the beverage is the form; brewed hibiscus (2-...
28	B-vitamins (B12/folate/B6, homocysteine...	Mind	3	M	✓	✓	cognition / memory	Use active/coenzyme forms: methylcobalamin (or hydr...
29	Bacopa monnieri	Mind	3	M	✓	✓	cognition / memory	Standardized extract to bacosides, ideally the clin...
30	Omega-3 DHA (fish/algal oil)	Mind	3	H	✓	✓	cognition / memory	Triglyceride (rTG/natural TG) form absorbs better t...
31	L-tryptophan	Sleep	3	M	✓	✓	sleep	n/a - a single amino acid; any USP/pharma-grade L-t...
32	Vitamin D (strength in deficiency)	Muscle	3	H	✓	✓	muscle / strength	Vitamin D3 (cholecalciferol) in an oil-based softge...
33	Probiotics for URTI prevention	Immune	3	M	✓	✓	immune / infection	Strain-specific, not just 'CFU count' - choose prod...
34	Zinc (acute cold lozenges vs prophylaxi...	Immune	3	M	✓	✓	immune / infection	Acute colds: zinc ACETATE (or gluconate) LOZENGES t...
35	L-glutamine (post-infectious IBS-D)	Gut	3	L	✓	✓	gut / digestion	Any pharmaceutical-grade free-form L-glutamine powd...
36	Partially hydrolyzed guar gum (PHGG) [a...	Gut	3	L	✓	✓	gut / digestion	Partially hydrolyzed guar gum specifically (e.g. Su...
37	Peppermint oil (enteric/gastro-resistan...	Gut	3	M	✓	✓	gut / digestion	Enteric-coated / gastro-resistant capsules, ideally...
38	Polypodium leucotomos extract (Fernbloc...	Skin	3	M	✓	✓	skin	A standardized Fernblock extract (the ingredient in...
39	Calcium + vitamin D (combined)	Bones	3	M	✓	✓	bone / fracture	Calcium citrate + vitamin D3. Citrate absorbs with...
40	Curcumin / Curcuma longa (turmeric) ext...	Joint	3	M	✓	✓	joint / arthritis pain	An enhanced-bioavailability form - phytosome (Meriv...
41	St John's Wort (Hypericum perforatum)	Mood	3	H	✓	✓	antidepressant effect in mild-to-...	A standardized extract used in trials - standardize...

#	SUPPLEMENT	SECTION	SCORE	CF	AMZ	WEB	TYPE OF EFFECT	BEST FORM (SHORT)
42	Beta-sitosterol	Men's	3	M	✓	✓	BPH/LUTS urinary symptom-score an...	A defined-dose beta-sitosterol (or mixed phytostero...
43	Chasteberry (Vitex agnus-castus) - PMS	Women's	3	M	✓	✓	premenstrual syndrome symptom rel...	Standardised fruit (berry) extract as used in trial...
44	Berberine	Metabolic	3	M	✓	✓	fasting glucose / HbA1c / LDL & t...	Dihydroberberine (roughly 5x the absorption, allowi...
45	Potassium / electrolyte blends	Vit/Min	3	H	✓	✓	blood-pressure lowering and strok...	For most, get potassium from food; for exercise/ele...
46	Chlorella	Ergogenic	3	M	✓	✓	modest total-cholesterol/LDL and...	Buy 'broken cell wall' or 'cracked cell wall' chlor...
47	Krill oil	Ergogenic	3	M	✓	✓	raises omega-3 index / EPA-DHA st...	Krill delivers its omega-3s bound to phospholipids,...
48	L-carnitine (L-tartrate)	Ergogenic	3	M	✓	✓	reduced muscle soreness / lowered...	For exercise/recovery buy L-carnitine L-tartrate (L...
49	Spirulina	Ergogenic	3	M	✓	✓	modest reductions in blood pressu...	Form does not matter - spirulina is a whole dried c...
50	Black tea (brewed)	Tea	3	H	✓	✓	CV/all-cause mortality associatio...	n/a - the beverage is the form. Full oxidation conv...
51	Green tea (brewed)	Tea	3	H	✓	✓	acute alertness (caffeine/L-thean...	n/a - the beverage is the form. To maximize catechi...
52	Matcha (powdered whole green tea leaf)	Tea	3	M	✓	✓	acute attention/stress-buffering...	n/a - the powder whisked into water IS the form, an...
53	Sencha / Gyokuro (premium Japanese gree...	Tea	3	M	✓	✓	acute alertness/calm (caffeine +...	n/a - premium brewed green teas; the leaf-and-water...
54	Cocoa flavanols / dark chocolate (COSMO...	Coffee	3	H	✓	✓	Blood pressure (small, RCT); CV d...	For the evidence-based cardiovascular benefit, use...
55	Decaffeinated coffee	Coffee	3	M	✓	✓	Type 2 diabetes risk & mortality...	Choose paper-filtered decaf, and if solvent exposur...
56	Espresso	Coffee	3	M	✓	✓	Small rise in total/LDL cholester...	Fine as a small serving; note it is semi-filtered (...)
57	Instant coffee	Coffee	3	M	✓	✓	Shares coffee's cohort mortality/...	n/a - the beverage is the form. Convenient and, bec...
58	Unfiltered coffee (French press, boiled...	Coffee	3	H	✓	✓	Raises LDL/total cholesterol (cau...	If you love it, keep intake modest and monitor LDL;...
59	Ginger tea (Zingiber officinale)	Drink	3	M	✓	✓	nausea reduction (RCT evidence fo...	For flavor/warmth, fresh grated ginger simmered ~10...
60	L-Tyrosine	Focus	2.5	M	✓	✓	acute focus / attention	Free-form L-tyrosine (not D/DL). N-acetyl-L-tyrosin...
61	L-theanine (alone)	Focus	2.5	M	✓	✓	acute focus / attention	L-theanine specifically (branded Suntheanine is the...
62	Rhodiola rosea	Focus	2.5	L	✓	✓	acute focus / attention	Extract standardized to 3% rosavins AND 1% salidro...
63	Ashwagandha (KSM-66, Withania somnifera)	Sleep	2.5	M	✓	✓	sleep	A standardized root extract with a named clinical g...

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64	Glycine	Sleep	2.5	L	✓	✓	sleep	Glycine is a single amino acid - the molecule is th...
65	L-theanine	Sleep	2.5	M	✓	✓	sleep	L-theanine specifically (branded Suntheanine is the...
66	Lavender oil - Silexan (Lavandula angus...	Sleep	2.5	M	✓	✓	sleep	Silexan specifically (patented 80 mg oral lavender...
67	Magnesium glycinate (bisglycinate)	Sleep	2.5	M	✓	✓	sleep	Magnesium bisglycinate - the glycine-chelated form...
68	Montmorency tart cherry (Prunus cerasus)	Sleep	2.5	L	✓	✓	sleep	n/a - the fruit IS the form; the point is getting t...
69	Betaine anhydrous	Muscle	2.5	L	✓	✓	muscle / strength	Betaine anhydrous (trimethylglycine/TMG), 2.5 g/day...
70	Citrulline malate	Muscle	2.5	M	✓	✓	muscle / strength	Either plain L-citrulline (~6 g) or citrulline mala...
71	Andrographis paniculata	Immune	2.5	M	✓	✓	immune / infection	Extract STANDARDIZED to andrographolides, ideally a...
72	Beta-glucans (yeast 1,3/1,6, Wellmune)	Immune	2.5	M	✓	✓	immune / infection	YEAST-derived (Saccharomyces cerevisiae) 1,3/1,6 be...
73	Bovine colostrum	Immune	2.5	L	✓	✓	immune / infection	First-milking, low-heat (not high-temperature) proc...
74	N-acetylcysteine (NAC)	Immune	2.5	M	✓	✓	immune / infection	n/a - plain N-acetylcysteine is a single active com...
75	Vitamin C (regular prophylaxis and cold...	Immune	2.5	H	✓	✓	immune / infection	Plain ascorbic acid in divided doses is fine and ch...
76	Vitamin D (respiratory infection preven...	Immune	2.5	H	✓	✓	immune / infection	Vitamin D3 (cholecalciferol) in an oil-based softge...
77	Lactobacillus rhamnosus GG (LGG)	Gut	2.5	M	✓	✓	gut / digestion	The defined strain L. rhamnosus GG, ATCC 53103 (Cul...
78	Multi-strain probiotics (IBS / general...	Gut	2.5	L	✓	✓	gut / digestion	Products whose exact strains were trialed for IBS -...
79	Zinc-carnosine (polaprezinc)	Gut	2.5	L	✓	✓	gut / digestion	The intact 1:1 zinc-L-carnosine chelate (polaprezin...
80	Astaxanthin (oral, Haematococcus pluvia...	Skin	2.5	L	✓	✓	skin	Natural astaxanthin from Haematococcus pluvialis al...
81	Oral hyaluronic acid (sodium hyaluronat...	Skin	2.5	L	✓	✓	skin	Sodium hyaluronate (the stable salt) is essentially...
82	Vitamin K2 (MK-7)	Bones	2.5	M	✓	✓	bone / fracture	MK-7 in the all-trans configuration - long half-lif...
83	Avocado-soybean unsaponifiables (ASU) [...	Joint	2.5	L	✓	✓	joint / arthritis pain	The studied ASU preparation - Piascledine 300 (300...
84	Boswellia serrata extract (AKBA-standar...	Joint	2.5	L	✓	✓	joint / arthritis pain	Extract STANDARDIZED to AKBA (acetyl-11-keto-beta-b...

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85	MSM (methylsulfonylmethane)	Joint	2.5	L	✓	✓	joint / arthritis pain	MSM is a single defined molecule, so the compound i...
86	UC-II (undenatured type II collagen)	Joint	2.5	M	✓	✓	joint / arthritis pain	Genuinely UNDENATURED type II collagen (UC-II brand...
87	GlyNAC (glycine + N-acetylcysteine)	Longevity	2.5	M	✓	✓	longevity / healthspan	Buy the two components separately at the studied ~1...
88	NMN (nicotinamide mononucleotide)	Longevity	2.5	M	✗	✓	longevity / healthspan	A third-party-tested (COA/HPLC-verified) NMN powder...
89	NR (nicotinamide riboside)	Longevity	2.5	M	✓	✓	longevity / healthspan	Niagen (patented nicotinamide riboside chloride, Ch...
90	Rapamycin / sirolimus (drug - noted)	Longevity	2.5	M	✗	✗	longevity / healthspan	Prescription drug, not a supplement - only legitima...
91	Urolithin A (Mitopure)	Longevity	2.5	M	✓	✓	longevity / healthspan	Direct/preformed urolithin A (Mitopure by Amazentis...
92	Turkey tail (Trametes/Coriolus versicol...	Mushroom	2.5	H	✓	✓	adjuvant overall-survival signal...	Fruiting-body extract with a stated beta-glucan per...
93	Holy basil / Tulsi (Ocimum sanctum / te...	Herb	2.5	M	✓	✓	fasting glucose reduction (in met...	A standardised leaf extract for reliable dosing (te...
94	Maca (Lepidium meyenii)	Herb	2.5	M	✓	✓	subjective sexual desire/libido;...	Gelatinised maca - the traditional preparation, bet...
95	Milk thistle (silymarin)	Herb	2.5	H	✓	✓	small transaminase (ALT/AST) redu...	A silybin phytosome (silybin-phosphatidylcholine) f...
96	Lemon balm (Melissa officinalis)	Mood	2.5	L	✓	✓	mild acute anxiolytic / calming a...	A standardized aerial-part extract such as Cyracos...
97	SAM-e (S-adenosyl-L-methionine)	Mood	2.5	M	✓	✓	possible adjunctive antidepressan...	Enteric-coated tablets of a stabilized salt (butane...
98	Tongkat ali (Eurycoma longifolia)	Men's	2.5	M	✓	✓	total testosterone increase in hy...	Standardised water extract with a stated eurycomano...
99	Yohimbine / yohimbe	Men's	2.5	M	✓	✓	modest erectile-function improvem...	Standardised yohimbine HCl so you get an exact mill...
100	Alpha-lipoic acid (ALA)	Metabolic	2.5	M	✓	✓	diabetic neuropathy symptom relie...	R-ALA is the biologically active isomer; a stabiliz...
101	Chromium picolinate	Metabolic	2.5	H	✓	✓	fasting glucose (small, low-certa...	Chromium picolinate (or nicotinate) is the well-abs...
102	Glucomannan (konjac fibre)	Weight-loss	2.5	M	✓	✓	Satiety / appetite; modest LDL re...	High-viscosity pure konjac glucomannan (the EFSA we...
103	White kidney bean extract (Phaseolus vu...	Weight-loss	2.5	M	✓	✓	Reduced starch digestion / modest...	A standardised Phase 2 (Phaseolamin) extract taken...
104	Multivitamin/multimineral (daily)	Vit/Min	2.5	H	✓	✓	small episodic-memory/global-cogn...	A reputable multi in active/chelated forms at rough...
105	Chamomile tea (Matricaria chamomilla)	Drink	2.5	M	✓	✓	sleep quality; mild GAD (mostly e...	The brewed tea IS the form, but preparation matters...

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106	Guarana (Paullinia cupana, drink/added...	Drink	2.5	M	✓	✓	acute alertness/reaction time (ca...	Prefer a standardized guarana extract with the caff...
107	Yerba mate (Ilex paraguariensis, brewed)	Drink	2.5	M	✓	✓	acute alertness (caffeine); esoph...	n/a - the beverage is the form. Brew and drink it W...
108	Alpha-GPC (choline alphoscerate)	Mind	2	M	✓	✓	cognition / memory	Alpha-GPC is one active molecule; the differentiato...
109	Citicoline (CDP-choline / Cognizin)	Mind	2	M	✓	✓	cognition / memory	Citicoline is a single active compound; Cognizin is...
110	Ginkgo biloba (EGb761)	Mind	2	H	✓	✓	cognition / memory	The EGb761 standardized extract (24% flavone glycos...
111	Lion's mane (Hericium erinaceus)	Mind	2	M	✓	✓	cognition / memory	Fruiting-body extract standardized to beta-glucans...
112	Phosphatidylserine (soy-derived)	Mind	2	M	✓	✓	cognition / memory	Soy- or sunflower-lecithin-derived phosphatidylseri...
113	Citicoline (CDP-choline)	Focus	2	M	✓	✓	acute focus / attention	Citicoline is a single active compound; Cognizin is...
114	Panax ginseng	Focus	2	L	✓	✓	acute focus / attention	Standardized to ginsenosides (commonly 4-7%), e.g....
115	Ashwagandha (Withania somnifera) for en...	Energy	2	M	✓	✓	energy / fatigue	A standardized root extract with a named clinical g...
116	CoQ10 / Ubiquinol	Energy	2	M	✓	✓	energy / fatigue	Ubiquinol (the reduced form) in an oil-based softge...
117	Cordyceps (Cs-4 / C. sinensis / C. mili...	Energy	2	L	✓	✓	energy / fatigue	Cultivated fruiting body of C. militaris (highest c...
118	Rhodiola rosea (added - common in 'ener...	Energy	2	L	✓	✓	energy / fatigue	Extract standardized to 3% rosavins AND 1% salidros...
119	5-HTP (5-hydroxytryptophan)	Sleep	2	L	✓	✓	sleep	Griffonia simplicifolia seed extract standardized t...
120	Apigenin / Chamomile (Matricaria chamom...	Sleep	2	M	✓	✓	sleep	German chamomile (Matricaria) extract standardized...
121	GABA (gamma-aminobutyric acid, oral)	Sleep	2	M	✓	✓	sleep	PharmaGABA (naturally fermented via Lactobacillus)...
122	Kava (Piper methysticum)	Sleep	2	H	✓	✓	sleep	Noble-cultivar kava, WATER-extracted from the ROOT/...
123	Magnesium L-threonate (Magtein)	Sleep	2	M	✓	✓	sleep	Magnesium L-threonate marketed as Magtein - the spe...
124	Passionflower (Passiflora incarnata) -...	Sleep	2	L	✓	✓	sleep	Standardized Passiflora incarnata aerial-part extra...
125	Valerian (Valeriana officinalis)	Sleep	2	M	✓	✓	sleep	A root extract STANDARDIZED to valerenic acids (~0....
126	Ashwagandha (Withania somnifera)	Muscle	2	M	✓	✓	muscle / strength	A standardized root extract with a named clinical g...

#	SUPPLEMENT	SECTION	SCORE	CF	AMZ	WEB	TYPE OF EFFECT	BEST FORM (SHORT)
127	HMB (beta-hydroxy-beta-methylbutyrate)	Muscle	2	M	✓	✓	muscle / strength	HMB free acid (HMB-FA, the gel-cap liquid form) rea...
128	Echinacea	Immune	2	H	✓	✓	immune / infection	Species and plant part are everything: E. purpurea...
129	Elderberry (Sambucus nigra)	Immune	2	M	✓	✓	immune / infection	A standardized, properly cooked Sambucus nigra extr...
130	Ginseng (Panax quinquefolius, COLD-fx/C...	Immune	2	L	✓	✓	immune / infection	The proprietary CVT-E002 poly-furanosyl-pyranosyl-s...
131	Pelargonium sidoides (EPs 7630, Umckalo...	Immune	2	L	✓	✓	immune / infection	The clinically validated EPs 7630 standardized root...
132	Butyrate / tributyrin (oral)	Gut	2	M	✓	✓	gut / digestion	Tributyryn (a glycerol-esterified tributyrin) is b...
133	Digestive enzymes (lactase, alpha-galac...	Gut	2	M	✓	✓	gut / digestion	Match the enzyme to the trigger: lactase for dairy/...
134	Prebiotics (inulin, GOS/FOS, inulin-typ...	Gut	2	M	✓	✓	gut / digestion	For tolerability, galacto-oligosaccharides (GOS) an...
135	Synbiotics (probiotic + prebiotic combi...	Gut	2	M	✓	✓	gut / digestion	A 'synergistic' synbiotic where the prebiotic is sp...
136	Collagen peptides (hydrolyzed collagen)	Skin	2	H	✓	✓	skin	Hydrolyzed collagen peptides (low molecular weight,...
137	Calcium alone	Bones	2	M	✓	✓	bone / fracture	Calcium citrate - absorbs with or without food, doe...
138	Collagen peptides (specific bioactive c...	Bones	2	L	✓	✓	bone / fracture	Hydrolyzed collagen peptides (low molecular weight,...
139	Magnesium	Bones	2	M	✓	✓	bone / fracture	Magnesium glycinate (bisglycinate) for best toleran...
140	Curcumin (turmeric extract)	Tendon	2	H	✓	✓	tendon / ligament	An enhanced-bioavailability form - phytosome (Meriv...
141	Omega-3 (EPA/DHA)	Tendon	2	M	✓	✓	tendon / ligament	Re-esterified triglyceride (rTG) or natural triglyc...
142	Chondroitin sulfate	Joint	2	H	✓	✓	joint / arthritis pain	High-purity pharmaceutical-grade chondroitin sulfat...
143	Glucosamine (sulfate or HCl)	Joint	2	H	✓	✓	joint / arthritis pain	Crystalline glucosamine SULFATE (the stabilized pot...
144	Glucosamine + chondroitin combination	Joint	2	H	✓	✓	joint / arthritis pain	Pair crystalline glucosamine SULFATE (1500 mg) with...
145	Fisetin (senolytic)	Longevity	2	M	✓	✓	longevity / healthspan	A liposomal or phospholipid/fat-formulated fisetin...
146	Metformin (drug - noted)	Longevity	2	M	✗	✗	longevity / healthspan	Extended-release (metformin ER/XR, e.g. Glucophage...
147	Resveratrol	Longevity	2	H	✓	✓	longevity / healthspan	Trans-resveratrol (the active isomer), micronized o...
148	Spermidine	Longevity	2	H	✓	✓	longevity / healthspan	A standardized wheat-germ extract stating spermidin...

#	SUPPLEMENT	SECTION	SCORE	CF	AMZ	WEB	TYPE OF EFFECT	BEST FORM (SHORT)
149	Sulforaphane (broccoli sprout extract)	Longevity	2	M	✓	✓	longevity / healthspan	Either a preformed/stabilized sulforaphane product...
150	Taurine	Longevity	2	M	✓	✓	longevity / healthspan	n/a - plain taurine (single amino-acid form); cheap...
151	Reishi (Ganoderma lucidum)	Mushroom	2	H	✓	✓	immune-marker modulation (surroga...	Fruiting-body dual extract - hot water pulls the im...
152	Shiitake / lentinan / AHCC (Lentinula e...	Mushroom	2	M	✓	✓	adjuvant survival in gastric canc...	Match to purpose: for general use a fruiting-body e...
153	Astragalus membranaceus (Huang Qi)	Herb	2	M	✓	✓	adjunctive renal biomarkers in di...	A root extract from a reputable supplier, ideally S...
154	Cinnamon (Cinnamomum cassia / verum)	Herb	2	H	✓	✓	modest fasting-glucose reduction;...	Ceylon (C. verum) for any regular/supplemental inta...
155	Myo-inositol	Mood	2	M	✓	✓	none demonstrated for depression/...	Plain myo-inositol powder is a single, freely water...
156	Saffron (Crocus sativus / affron)	Mood	2	M	✓	✓	small improvement in self-rated (...)	A standardized stigma extract with defined crocin/s...
157	Aniracetam	Nootropic	2	M	✗	✓	possible slowing of decline in AI...	Single defined molecule - form/salt irrelevant; buy...
158	Piracetam	Nootropic	2	H	✗	✓	small/uncertain global-rating cha...	Single well-absorbed molecule - form does not matte...
159	Sulbutiamine	Nootropic	2	M	✓	✓	possible fatigue reduction in spe...	Single defined molecule (a fat-soluble thiamine der...
160	Vinpocetine	Nootropic	2	H	✓	✓	cerebral vasodilation; no reliabl...	Single defined molecule - form/salt does not matter...
161	Acetyl-L-carnitine (ALCAR)	Nootropic	2	M	✓	✓	small slowing of decline in MCI/m...	Acetyl-L-carnitine — the acetylated form crosses th...
162	Centrophenoxine (meclofenoxate)	Nootropic	2	M	✗	✓	possible modest memory effect in...	One defined molecule, so no superior 'form' to sele...
163	PQQ (pyrroloquinoline quinone)	Nootropic	2	M	✓	✓	modest attention/working-memory a...	PQQ disodium salt is the one bioavailable, stable r...
164	DHEA (dehydroepiandrosterone)	Men's	2	M	✓	✓	inconsistent/modest ED and androg...	Micronised, USP-grade DHEA from a brand with third...
165	Fenugreek (Trigonella foenum-graecum)	Men's	2	M	✓	✓	self-reported libido/sexual funct...	Seed extract standardised to the relevant saponin/g...
166	Saw palmetto (Serenoa repens)	Men's	2	H	✓	✓	none demonstrated - no improvemen...	Liposterolic extract standardised to ~85-95% free f...
167	Shilajit (purified, PrimaVie)	Men's	2	L	✓	✓	total/free testosterone increase...	Purified, clinically-studied PrimaVie shilajit stan...
168	Tribulus terrestris	Men's	2	H	✓	✓	possible modest self-reported lib...	Fruit extract standardised to steroidal saponins as...
169	Black cohosh (Cimicifuga racemosa)	Women's	2	M	✓	✓	vasomotor symptoms - inconsistent...	Standardised rhizome/root extract as used in menopa...
170	Red clover isoflavones (Trifolium prate...	Women's	2	M	✓	✓	vasomotor symptoms - not reliably...	Extract standardised to total isoflavones with the...

#	SUPPLEMENT	SECTION	SCORE	CF	AMZ	WEB	TYPE OF EFFECT	BEST FORM (SHORT)
171	Bergamot (Citrus bergamia)	Metabolic	2	M	✓	✓	LDL / total cholesterol / triglyc...	Choose a standardized bergamot polyphenolic fractio...
172	Nattokinase	Metabolic	2	L	✓	✓	small blood-pressure reduction; f...	One enzyme, so buy on standardized activity (aim ~2...
173	Niacin (immediate-release, for lipids)	Metabolic	2	H	✓	✓	lipid biomarkers (HDL up, TG/Lp(a...	For lipids use immediate-release nicotinic acid (ni...
174	Apple cider vinegar (ACV)	Weight-loss	2	M	✓	✓	Modest post-meal glycemic bluntin...	Liquid ACV (~1-2 tablespoons = ~750-1500 mg acetic...
175	Capsaicin / Capsimax (capsaicinoids, ca...	Weight-loss	2	M	✓	✓	Acute thermogenesis / fat oxidati...	Capsimax controlled-release capsaicinoids (~2 mg ca...
176	Conjugated linoleic acid (CLA)	Weight-loss	2	H	✓	✓	Small fat-mass reduction (not bod...	A standardised ~80% CLA softgel (Tonalin or Clarino...
177	Green coffee bean extract (chlorogenic...	Weight-loss	2	H	✓	✓	Possible small body-weight change...	Pick an extract standardized to chlorogenic acid co...
178	Iodine / kelp	Vit/Min	2	H	✓	✓	corrects deficiency-related goite...	A precisely dosed potassium iodide supplement at ~1...
179	Selenium	Vit/Min	2	H	✓	✓	none demonstrated in replete adul...	L-selenomethionine (or selenium-enriched yeast), ~5...
180	CBD (cannabidiol)	Ergogenic	2	M	✗	✓	proven only for specific epilepsi...	Oral CBD has poor and variable bioavailability with...
181	Cod liver oil	Ergogenic	2	H	✓	✓	none for infection prevention; su...	Form is not the issue - it is a whole fish-liver oi...
182	Exogenous ketones (BHB salts/esters)	Ergogenic	2	H	✓	✓	none demonstrated for endurance p...	Ketone monoesters raise blood BHB highest and faste...
183	Glutathione (oral / liposomal)	Ergogenic	2	M	✓	✓	raises measured blood glutathione...	Plain oral glutathione is largely broken down in th...
184	Greens / superfood powders	Ergogenic	2	M	✓	✓	raises circulating micronutrient...	Form does not meaningfully matter - these are dried...
185	L-arginine	Ergogenic	2	M	✓	✓	inconsistent; small aerobic/anaer...	Counter-intuitively, L-citrulline (3-6 g, or citrul...
186	MCT oil	Ergogenic	2	M	✓	✓	small weight/body-composition red...	Buy C8 (caprylic acid) or a labelled C8/C10 blend...
187	Green tea EXTRACT / EGCG capsules (conc...	Tea	2	H	✓	✓	clinically trivial weight change;...	If you use it at all, choose a modest-dose, standar...
188	Oolong tea	Tea	2	M	✓	✓	acute fat-oxidation marker (small...	n/a - the beverage is the form; partial oxidation g...
189	Pu-erh tea (fermented)	Tea	2	M	✓	✓	lipid lowering in dyslipidemic su...	n/a - the beverage is the form; microbial fermentat...
190	Cacao (raw cacao powder / nibs / 'super...	Coffee	2	H	✓	✓	None demonstrated for 'cacao' as...	Choose natural (non-alkalized/non-Dutched) cocoa or...
191	Chicory 'coffee' (Cichorium intybus roo...	Drink	2	M	✓	✓	none clinically meaningful from t...	The roasted-root brew IS the form. Pick 100% roaste...

#	SUPPLEMENT	SECTION	SCORE	CF	AMZ	WEB	TYPE OF EFFECT	BEST FORM (SHORT)
192	Peppermint tea (<i>Mentha x piperita</i>)	Drink	2	H	✓	✓	IBS symptoms (peppermint OIL only...	For a beverage, loose dried leaf steeped covered 5-...
193	Rooibos (<i>Aspalathus linearis</i>)	Drink	2	L	✓	✓	lipids/oxidative markers (single...	Green (unfermented) rooibos over the common red (fe...
194	Huperzine A	Mind	1.5	M	✓	✓	cognition / memory	Huperzine A is a single purified alkaloid; ensure t...
195	NR / NMN (nicotinamide riboside / monon...	Mind	1.5	H	✓	✓	cognition / memory	For NR, branded Niagen is the stability-tested, GRA...
196	BCAAs (branched-chain amino acids)	Muscle	1.5	H	✓	✓	muscle / strength	Honestly, skip standalone BCAAs in favor of complet...
197	Garlic / allicin	Immune	1.5	M	✓	✓	immune / infection	Two valid routes: an ENTERIC-COATED garlic powder t...
198	Quercetin	Immune	1.5	M	✓	✓	immune / infection	An absorption-enhanced form: quercetin PHYTOSOME (p...
199	Boron	Bones	1.5	M	✓	✓	bone / fracture	Chelated/organic boron - calcium fructoborate, boro...
200	Vitamin D alone	Bones	1.5	H	✓	✓	bone / fracture	Vitamin D3 (cholecalciferol) - raises and sustains...
201	Vitamin C (alone, as a standalone tendo...	Tendon	1.5	M	✓	✓	tendon / ligament	Plain ascorbic acid in divided doses is fine and ch...
202	Calcium alpha-ketoglutarate (CaAKG / Re...	Longevity	1.5	H	✓	✓	longevity / healthspan	Calcium alpha-ketoglutarate (Ca-AKG, the form used...
203	Pterostilbene	Longevity	1.5	M	✓	✓	longevity / healthspan	n/a - plain pterostilbene is already the optimized...
204	Quercetin (senolytic)	Longevity	1.5	H	✓	✓	longevity / healthspan	Quercetin phytosome (e.g. Quercetin Phytosome/Querc...
205	Maitake (<i>Grifola frondosa</i> ; D/MD-fractio...	Mushroom	1.5	M	✓	✓	immune-marker (NK/cytokine) modul...	Fruiting-body extract standardised to the D-fractio...
206	<i>Poria cocos</i> (<i>Wolfiporia extensa</i> / fu li...	Mushroom	1.5	L	✓	✓	modest sleep-duration signal (unc...	A quality sclerotium extract or traditional decocti...
207	<i>Tremella fuciformis</i>	Mushroom	1.5	L	✓	✓	possible small glycaemic effect (...)	Fruiting-body (dual or hot-water) extract with a st...
208	Eleuthero / Siberian ginseng (Eleuthero...	Herb	1.5	M	✓	✓	none demonstrated in rigorous hum...	Root (not root+rhizome fillers) standardised to ele...
209	Gotu kola (<i>Centella asiatica</i>)	Herb	1.5	M	✓	✓	none demonstrated for cognition;...	An extract standardised to the total triterpenic (t...
210	<i>Moringa oleifera</i>	Herb	1.5	M	✓	✓	none robustly demonstrated (glyce...	Dried LEAF powder or leaf-only capsules (shade-drie...
211	<i>Schisandra chinensis</i> (Wu Wei Zi)	Herb	1.5	M	✓	✓	possible small muscle-strength ga...	A berry extract standardised to schisandra lignans...
212	Oxiracetam	Nootropic	1.5	M	✗	✓	cognitive score change in vascula...	Single molecule, water-soluble and well absorbed -...
213	Choline bitartrate	Nootropic	1.5	H	✓	✓	none demonstrated on memory in he...	For cognition, skip bitartrate and choose Alpha-GPC...

#	SUPPLEMENT	SECTION	SCORE	CF	AMZ	WEB	TYPE OF EFFECT	BEST FORM (SHORT)
214	Uridine monophosphate (UMP)	Nootropic	1.5	H	✓	✓	none demonstrated for standalone...	Retail UMP is effectively a single practical form:...
215	Evening primrose oil (GLA)	Women's	1.5	H	✓	✓	none reliably demonstrated in hum...	Softgels stating actual GLA content (evening primro...
216	Forskolin (Coleus forskohlii)	Weight-loss	1.5	M	✓	✓	None reliably demonstrated for we...	A root extract standardised to 10% forskolin (ForsL...
217	Garcinia cambogia (hydroxycitric acid,...	Weight-loss	1.5	H	✓	✓	Marginal, sub-clinical body-weigh...	If used at all, choose an extract standardized to ~...
218	Synephrine (bitter orange / Citrus aura...	Weight-loss	1.5	H	✓	✓	Acute thermogenesis / mild BP+HR...	A single-ingredient extract standardised to p-synep...
219	Vitamin A / beta-carotene	Vit/Min	1.5	H	✓	✓	none beneficial in replete adults...	For most people, get vitamin A largely as beta-caro...
220	Vitamin B6 (pyridoxine)	Vit/Min	1.5	H	✓	✓	none demonstrated for the markete...	Pyridoxal-5'-phosphate (P5P) at a low dose - needs...
221	Vitamin E (alpha-tocopherol, high-dose)	Vit/Min	1.5	H	✓	✓	none beneficial; net harm (increa...	Natural d-alpha (RRR-alpha-tocopherol), ideally as...
222	Energy drinks (caffeine + taurine, e.g....	Drink	1.5	H	✓	✓	acute alertness (caffeine); adver...	Choose a SUGAR-FREE product with a clearly LABELED...
223	Kombucha (fermented tea)	Drink	1.5	M	✓	✓	none clinically demonstrated; mic...	n/a - the beverage is the form. Choose commercial,...
224	Adaptogen blends (proprietary multi-her...	Energy	1	H	✓	✓	energy / fatigue	Skip proprietary multi-herb blends; buy single, sta...
225	Vitamin B12 (cyanocobalamin/methylcobal...	Energy	1	H	✓	✓	energy / fatigue	Methylcobalamin or hydroxocobalamin (active/retaine...
226	Testosterone boosters (Tribulus, D-aspa...	Muscle	1	H	✓	✓	muscle / strength	n/a - no retail form reliably raises testosterone i...
227	Slippery elm / marshmallow root / demul...	Gut	1	H	✓	✓	gut / digestion	Mucilage works by local contact, so a powder made i...
228	Biotin (vitamin B7)	Skin	1	H	✓	✓	skin	Plain d-biotin; there is one active form and no mea...
229	Strontium (retail strontium citrate)	Bones	1	H	✓	✓	bone / fracture	If used at all, retail strontium CITRATE is the OTC...
230	Agaricus blazei Murill (A. subrufescens)	Mushroom	1	M	✓	✓	immune-marker modulation only; no...	Fruiting-body extract with a stated beta-glucan % f...
231	Chaga (Inonotus obliquus)	Mushroom	1	H	✓	✓	none demonstrated in humans (anti...	Wild-harvested sclerotium, dual (water + alcohol) e...
232	DMAE (dimethylaminoethanol / deanol)	Nootropic	1	H	✓	✓	none demonstrated in humans (null...	The retail form is essentially always DMAE bitartra...
233	Noopept	Nootropic	1	H	✗	✓	possible cognition/anxiolytic eff...	Single defined peptide - form does not matter; buy...
234	Phenibut	Nootropic	1	H	✗	✓	sedation/anxiolysis (GABA-B agoni...	HCl is the standard, reliable retail form; FAA (fre...
235	Phenylpiracetam	Nootropic	1	H	✗	✓	acute psychostimulation/endurance...	Single, orally well-absorbed molecule - form irrele...

#	SUPPLEMENT	SECTION	SCORE	CF	AMZ	WEB	TYPE OF EFFECT	BEST FORM (SHORT)
236	7,8-Dihydroxyflavone (7,8-DHF)	Nootropic	1	H	✗	✓	none demonstrated in humans (anim...	Sold as a research chemical, not a regulated supple...
237	Fadogia agrestis	Men's	1	H	✓	✓	none demonstrated in humans (rode...	No reliable standardised retail form exists, there...
238	Horny goat weed (Epimedium / icariin)	Men's	1	H	✓	✓	none demonstrated in humans as mo...	Aerial-part extract with a stated icariin content (...)
239	DIM (diindolylmethane)	Women's	1	H	✓	✓	urinary estrogen-metabolite ratio...	A bioavailability-enhanced (microencapsulated/absor...
240	Dong quai (Angelica sinensis)	Women's	1	H	✓	✓	none demonstrated in humans	Form/salt is not the issue - there is no single wel...
241	Policosanol	Metabolic	1	H	✓	✓	none demonstrated in independent...	Only the original Cuban sugarcane-wax material show...
242	Raspberry ketones	Weight-loss	1	H	✓	✓	None demonstrated in humans	Form is irrelevant because there is no credible hum...
243	Sea moss (Irish moss / Chondrus crispus)	Ergogenic	1	H	✓	✓	none demonstrated in humans; deli...	Form does not matter - gel, powder and capsule are...
244	White tea	Tea	1	H	✓	✓	none demonstrated in humans (in-v...	n/a - the beverage is the form; minimally processed...

Part I — By outcome (what you're trying to fix)

Mind / memory (11)

4 Creatine monohydrate Amazon ✓ Website ✓

Evidence: good (high confidence) | Effect: cognition / memory

THE NOISE Marketed as the go-to strength supplement that increases muscle power, size, and training performance, now also pushed for brain and healthy-aging benefits.

THE SCIENCE Effect: Memory SMD ~ 0.3 (GRADE moderate) in 2024 meta; processing-speed/attention low certainty. Acute trial: improved PVT and processing speed during 21 h sleep deprivation, effect from ~ 3.5 h, peak ~ 4 h. Clears MCID: Borderline. Meaningful under metabolic stress (sleep loss, vegetarian repletion); marginal in rested omnivores. Largest independent trial: No single large independent healthy-adult cognition RCT; evidence rests on the pooled 16-trial meta (Prokopidis, Frontiers in Nutrition 2024) and the $n=15$ MRS sleep-dep trial. Population: Mixed: healthy young adults, vegetarians, older adults, and sleep-deprived young men. Largest cognitive effects in metabolically stressed or creatine-depleted subgroups.

THE VERDICT The best-evidenced item in this category, but the prior grade of 5 overstates it for the marketed use. Worth buying if you are vegetarian, older, or acutely sleep-deprived; the general healthy-rested-adult memory benefit is small. Cheap, safe, well-characterized.

BEST FORM Plain creatine monohydrate is the gold standard - micronized for easier mixing, and Creapure-grade for purity if you want third-party assurance. Fancier 'forms' are not worth paying for. (Creatine monohydrate BEST (most-studied, cheapest, proven); HCl, buffered 'Kre-Alkalyn', ethyl ester and 'nitrate' forms cost more with NO demonstrated bioavailability or performance advantage - some (ethyl ester) are actually worse.)

WATCH 3-5 g/day is sufficient; a loading phase is optional, not required. Expect ~ 1 -2 kg water-weight gain early. It raises serum creatinine harmlessly - warn your doctor so it isn't misread as kidney injury (interferes with that lab interpretation). Stay hydrated; take consistently, timing doesn't matter much.

Sources: Prokopidis K, et al. Frontiers in Nutrition 2024;11:1424972 (16-RCT meta, PMC11275561); Gordji-Nejad A, et al. Scientific Reports 2024;14:4937, doi:10.1038/s41598-024-54249-9; Xu C, et al. Nutrients 2026;18:1192 (single-dose creatine, sleep deprivation); EFSA Journal 2024;22:e9100 (creatine + cognition health-claim evaluation)

3 B-vitamins (B12/folate/B6, homocysteine-lowering) - added, in-category Amazon ✓ Website ✓

Evidence: moderate/conditional (medium confidence) | Effect: cognition / memory

THE NOISE Sold as an energy, brain and mood booster and a homocysteine/heart protector.

THE SCIENCE Effect: VITACOG: reduced rate of brain atrophy ($\sim 30\%$ overall; up to $\sim 53\%$ in the high-homocysteine subgroup); cognitive benefit concentrated where homocysteine elevated and omega-3 adequate. Clears MCID: In the targeted subgroup, plausibly; in unselected adults, no. Largest independent trial: VITACOG (Smith AD, et al. PLoS ONE 2010;5:e12244; de Jager, Int J Geriatr Psychiatry 2012). Population: Older adults with MCI and elevated homocysteine (VITACOG); benefit modified by omega-3 status. Null in unselected/replete populations.

THE VERDICT Conditional buy only. In healthy replete adults, large trials are null; benefit appears in a specific subgroup, elevated homocysteine AND adequate omega-3 status. This is a lab-value-gated intervention, not a general memory pill.

BEST FORM Use active/coenzyme forms: methylcobalamin (or hydroxocobalamin) for B12, L-methylfolate (5-MTHF) instead of folic acid, and P-5-P (pyridoxal-5-phosphate) instead of plain pyridoxine. Matters for the large share of people with MTHFR variants. (Methylfolate > folic acid (bypasses MTHFR, avoids unmetabolized folic acid); methylcobalamin/hydroxo > cyanocobalamin for some; P-5-P > pyridoxine and is safer on nerves.)

WATCH The homocysteine-lowering cognitive benefit is greatest when omega-3 status is adequate. Folate can mask B12 deficiency, dose them together. Chronic high B6 ($> \sim 50$ -100 mg long-term, especially as pyridoxine) causes peripheral neuropathy, respect the upper limit. Check B12 blood level before high-dose supplementation.

Sources: Smith AD, et al. PLoS ONE 2010;5:e12244 (VITACOG, brain atrophy); de Jager CA, et al. Int J Geriatr Psychiatry 2012;27:592-600 (cognitive outcomes); Jerneren F, et al. Am J Clin Nutr 2015;102:215-221 (omega-3 interaction)

3 Bacopa monnieri Amazon ✓ Website ✓

Evidence: moderate/conditional (medium confidence) | Effect: cognition / memory

THE NOISE Promoted as an Ayurvedic 'brain herb' that improves memory, learning and recall.

THE SCIENCE Effect: Meta: significant improvement in Trail-Making/choice-RT and speed of attention; decreased reaction time. Memory (free recall) improvements less consistent. Clears MCID: Modest; attention/RT changes small-to-moderate, no formal MCID. Largest independent trial: No single large independent trial; Stough 2001/2008 (CDRI-08, ~n=46-107) and Calabrese 2008 (n=54, 300 mg, 12 wk) are the anchors. Population: Healthy adults, medical students, and older adults with age-associated memory impairment; 300 mg standardized extract, 12 wk.

THE VERDICT **Conditional buy at best. Real but modest signal, mainly on speed of attention / reaction time rather than memory, and it takes ~12 weeks. Use a standardized extract (55% bacosides / Bacognize) at 300 mg. GI side effects common. Prior of 3 holds.**

BEST FORM Standardized extract to bacosides, ideally the clinically-tested Bacognize (~45%) or CDRI 08/BacoMind. Aim for a 300-320 mg standardized dose delivering ~50-55% bacosides equivalent, not raw unstandardized powder. (Standardized bacoside extract (Bacognize/CDRI 08) = trial-backed; generic 'Bacopa powder' with no bacoside % = unreliable potency.)

WATCH Effects are chronic, not acute, needs 8-12 weeks. GI upset/cramping/loose stools are common, so take WITH food. Can be sedating. May increase thyroid hormone and affect some meds; caution with thyroid conditions.

Sources: Kongkeaw C, et al. J Ethnopharmacol 2014;151:528-535 (9-RCT meta); Calabrese C, et al. J Altern Complement Med 2008;14:707-713 (300 mg, 12 wk, healthy elderly); Stough C, et al. Psychopharmacology 2001;156:481-484 (CDRI-08)

3 Omega-3 DHA (fish/algal oil) Amazon ✓ Website ✓

Evidence: moderate/conditional (high confidence) | Effect: cognition / memory

THE NOISE Sold as the essential 'brain-building' fat that supports memory, focus and long-term cognitive health at every age.

THE SCIENCE Effect: VITAL: no significant difference in global cognition change vs placebo (null). USC: biomarker positive (CSF DHA +17%), cognition null. Clears MCID: No, primary cognitive endpoints did not differ from placebo in the large trials. Largest independent trial: VITAL (Kang JH et al., Alz Dement TRCI 2022;8:e12288), n=4218, marine n-3 1 g/day, NO cognitive benefit over 2-3 yr. Population: Generally healthy older adults (VITAL) and older adults at dementia risk with low dietary DHA (<200 mg/day) and ≥1 risk factor (USC trial).

THE VERDICT **Do not buy for memory as a healthy, omega-3-replete adult. The large independent trials are null. A conditional signal survives only in people with low omega-3 status or genetic/early-neurodegeneration risk. Prior of 3.5 slightly generous.**

BEST FORM Triglyceride (rTG/natural TG) form absorbs better than the cheap ethyl-ester (EE) form. For DHA specifically, algal oil is a clean vegan source; choose a product listing actual EPA/DHA mg, not just 'fish oil' total. (Triglyceride/re-esterified TG = best absorbed; phospholipid (krill) = well absorbed but low dose per pill and pricey; ethyl ester = cheapest, ~lower absorption especially without fat.)

WATCH Take with a fat-containing meal for absorption. Read the label for EPA+DHA content, not fish-oil weight. Refrigerate/check for rancidity (fishy burps = oxidation); enteric-coated or freezing helps reflux. High doses (>3 g/day) can thin blood, caution with anticoagulants/before surgery.

Sources: Kang JH, et al. Alzheimers Dement (N Y) TRCI 2022;8:e12288 (VITAL cognition, null); USC high-dose DHA trial, eBioMedicine 2026, PIIS2352-3964(26)00198-2; REDUCE-IT (Bhatt DL, NEJM 2019) / STRENGTH (Nicholls SJ, JAMA 2020) atrial-fibrillation signal

2 Alpha-GPC (choline alphoscerate) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: cognition / memory

THE NOISE Sold as a high-potency choline that boosts memory, focus and 'mind-muscle' cognitive performance by raising brain acetylcholine.

THE SCIENCE Effect: ASCOMALVA: modest cognitive/behavioral benefit as add-on to donepezil in AD (small, industry-associated). No healthy-adult effect size available. Clears MCID: Not established in any non-dementia population. Largest independent trial: Largest dataset is a SAFETY study: Lee G, et al. (Korean NHIS cohort) 2021, >12M subjects, dose-dependent higher 10-yr stroke incidence in alpha-GPC users. Population: Alzheimer's and vascular-cognitive-impairment patients (efficacy trials). No healthy-adult cognitive-enhancement evidence.

THE VERDICT Do not buy. Efficacy data are confined to add-on dementia trials (mostly industry/AD, not healthy adults), and a large cohort shows a dose-dependent 10-year stroke-risk signal. Risk outruns any consumer benefit; the safety signal makes this a firm avoid.

BEST FORM Alpha-GPC is one active molecule; the differentiator is purity/moisture. It's very hygroscopic, so choose capsules or a stabilized branded grade (e.g. AlphaSize) over loose powder that cakes. (Alpha-GPC (this item) vs other choline sources: alpha-GPC = highest choline yield and crosses BBB well; citicoline = adds cytidine/uridine; choline bitartrate = poor brain delivery.)

WATCH Typical 300-600 mg. A recent observational signal linked high supplemental alpha-GPC intake to stroke risk (association, not proven causal), worth noting for those with vascular risk. Cholinergic side effects: headache, nausea; can lower mood in susceptible people. Don't double up with other high-dose choline sources.

Sources: Lee G, et al. Association of L-alpha-GPC with subsequent stroke risk (Korean NHIS cohort, >12M), 2021; Amenta F, et al. (ASCOMALVA) J Neurol Sci / J Alzheimers Dis 2012-2014 (add-on to donepezil in AD)

2 Citicoline (CDP-choline / Cognizin) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: cognition / memory · **literature compromised (funding/outlier)**

THE NOISE Branded (as Cognizin) as clinically proven to boost memory, focus and mental energy by supporting brain-cell membranes and acetylcholine.

THE SCIENCE Effect: Nakazaki 2022: significant improvement in composite/episodic memory vs placebo at 12 wk (industry-funded); not replicated at 1 g/day. Clears MCID: Uncertain, small trial, no established MCID; effect not robust across doses. Largest independent trial: No large independent healthy-adult trial; stroke trials (ICTUS, Davalos 2012, n=2298) were NEUTRAL for a different indication. Population: Healthy older adults (55-91 yr) with age-associated memory impairment (AAMI); also stroke/vascular-cognitive populations (different indication).

THE VERDICT Do not buy on current evidence. The one supportive healthy-adult RCT is industry-funded and was not replicated at higher dose, and EFSA (2024) rejected the memory health claim. Prior of 3 is slightly optimistic.

BEST FORM Citicoline is a single active compound; Cognizin is simply a branded, patented, clinically-studied citicoline used in most trials. Generic citicoline of equivalent dose is chemically the same, so pay for verified purity/dose rather than the name. (Cognizin-branded vs generic citicoline = same molecule; the real fork is citicoline vs alpha-GPC as your choline source (citicoline also yields cytidine/uridine).)

WATCH Typical 250-500 mg/day. Well tolerated; occasional headache/GI upset. Non-stimulant, timing flexible. Don't stack redundantly with high-dose alpha-GPC (both raise choline). Generally avoid combining potent cholinergics if prone to depression/low mood.

Sources: Nakazaki E, et al. J Nutr 2022;152:2415-2425, doi:10.1016/j.tjnut.2022.06.001 (Cognizin, manufacturer-funded); EFSA Journal 2024;22:e8861 (citicoline + memory health claim, negative); Davalos A, et al. (ICTUS) Lancet 2012;380:349-357 (citicoline in stroke, neutral)

2 Ginkgo biloba (EGb761) Amazon ✓ Website ✓

Evidence: weak/mostly null (high confidence) | Effect: cognition / memory

THE NOISE Marketed as a circulation-boosting herb that sharpens memory and fights age-related mental decline.

THE SCIENCE Effect: GEM: no reduction in dementia incidence (HR ~1.12, 95% CI 0.94-1.33, no benefit). Cochrane: little-to-no effect on global clinical status at 6 mo (moderate certainty). Clears MCID: No, prevention trials null; treatment effects inconsistent and below reliable clinical significance. Largest independent trial: GEM Study (DeKosky ST, JAMA 2008;300:2253-2262), n=3069, EGb761 240 mg/day, NULL for dementia prevention. Population: Older adults (GEM, GuidAge prevention); dementia/cognitive-impairment patients (treatment trials). Little healthy-young-adult data.

THE VERDICT Do not buy for memory. Large independent prevention trials are null and Cochrane calls the dementia evidence inconsistent and unreliable. Carries a real bleeding/anticoagulant interaction. Prior of 2 holds.

BEST FORM The EGb761 standardized extract (24% flavone glycosides, 6% terpene lactones, and critically <5 ppm ginkgolic acids). Buy a product that states the 24/6 standardization AND low ginkgolic-acid limit. (EGb761 / 24-6 standardized low-ginkgolic-acid extract = studied and safer; cheap crude leaf or unstandardized extract = high allergenic ginkgolic acids, inconsistent.)

WATCH Antiplatelet effect, real bleeding-risk concern with aspirin, NSAIDs, warfarin, DOACs, and before surgery (stop ~2 weeks prior). Effects build over weeks. Mild GI/headache possible.

Sources: DeKosky ST, et al. (GEM) JAMA 2008;300:2253-2262; Vellas B, et al. (GuidAge) Lancet Neurol 2012;11:851-859; Birks J / Wieland LS, Cochrane Database Syst Rev (CD003120; CD013661)

2 Lion's mane (Hericium erinaceus) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: cognition / memory

THE NOISE Promoted as a nerve-growth-factor mushroom that regenerates neurons and boosts memory and focus.

THE SCIENCE Effect: Mori: cognitive-scale (HDS-R) rise during dosing that reversed after washout, small n. Docherty: significant Stroop speed at 60 min only; memory endpoints null; earlier healthy-adult study reported reduced delayed word recall. Clears MCID: No, acute attention change is minor; no durable memory benefit clearing an MCID. Largest independent trial: Docherty S, et al. (Frontiers in Nutrition / Nutrients 2023-2025), n=41 healthy adults, 1.8 g/28 d: acute Stroop speed only, no durable memory benefit. Population: Japanese older adults with MCI (Mori 2009); healthy young adults 18-45 (Docherty 2023). Mixed and small.

THE VERDICT Do not buy for memory in a healthy adult. The one supportive cognition trial is a small industry-linked MCI study; healthy-adult trials show at most an acute processing-speed blip, and earlier healthy-adult work reported a delayed-recall decrement. Prior of 2 holds.

BEST FORM Fruiting-body extract standardized to beta-glucans (with hericenones/erinacines), NOT grain-grown mycelium-on-substrate powder which is mostly starch. Look for '100% fruiting body,' a beta-glucan % (not just 'polysaccharides'), and low starch. (Fruiting-body extract with stated beta-glucans = potent; mycelium-on-grain (myceliated biomass) = cheap, high starch filler, low active content.)

WATCH 'Polysaccharide %' on labels is inflated by grain starch, insist on beta-glucan number. Effects are gradual. Skin/GI or allergic reactions possible, especially in those with mushroom allergies; rare reports of respiratory allergy. Some anecdotal bleeding/mood effects.

Sources: Mori K, et al. Phytother Res 2009;23:367-372 (MCI, n=30); Docherty S, et al. Nutrients/Frontiers in Nutrition 2023-2025 (healthy adults, 1.8 g); La Monica MB, et al. (acute cognition/mood, healthy young adults)

2 Phosphatidylserine (soy-derived) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: cognition / memory · **literature compromised (funding/outlier)**

THE NOISE Marketed as a brain-cell-membrane nutrient clinically shown to support memory and slow cognitive aging.

THE SCIENCE Effect: Vakhapova 2010 (soy-PS 300 mg): improvement in a memory subgroup with low baseline; several endpoints not significant vs placebo. Mixed. Clears MCID: Not reliably; soy-PS effects small and inconsistent. Largest independent trial: No large independent soy-PS trial; Cenacchi 1993 (bovine, n=494) is the largest but is the wrong molecule. Population: Older adults with age-associated memory impairment or subjective memory complaints. Bovine-PS trials in the elderly; soy-PS trials smaller.

THE VERDICT **Do not buy the retail (soy) product for memory. The good early data used bovine-cortex PS (withdrawn over prion risk); the soy PS that is actually sold has weak, mixed trials, and FDA allows only a heavily-qualified claim. Prior of 2 holds.**

BEST FORM Soy- or sunflower-lecithin-derived phosphatidylserine (e.g. Sharp-PS/SerinAid) standardized to actual PS mg (~100 mg x3/day). Sunflower-sourced avoids soy allergen and GMO concerns; both are the safe modern form. (Plant (soy/sunflower) PS = current safe standard; old bovine-cortex PS = effective historically but abandoned over prion/BSE risk, avoid.)

WATCH ~300 mg/day total, effect builds over weeks. Take with food. Mild insomnia if taken late at high dose; some GI upset. Mild blood-thinning potential; caution with anticoagulants.

Sources: Cenacchi T, et al. Aging Clin Exp Res 1993;5:123-133 (bovine-cortex PS, n=494); Vakhapova V, et al. Dement Geriatr Cogn Disord 2010;29:467-474 (soy-PS 300 mg); FDA qualified health claim letter for phosphatidylserine (2003)

1.5 Huperzine A Amazon ✓ Website ✓

Evidence: none / marketing (medium confidence) | Effect: cognition / memory · **literature compromised (funding/outlier)** | Unregulated potent AChE inhibitor; sold OTC but pharmacologically drug-like.

THE NOISE Sold as a natural memory enhancer that raises acetylcholine to sharpen recall and learning.

THE SCIENCE Effect: Yang 2013: significant MMSE and ADL improvement vs placebo in AD at 8-16 wk; effect grew with duration, but pooled from high-bias trials. Clears MCID: Reported MMSE gains in AD may reach clinical relevance, but bias undermines the estimate; irrelevant to healthy adults. Largest independent trial: No rigorous Western independent trial; a US NIH-funded phase II (Rafii 2011, n=210) was essentially NEGATIVE at 200 mcg twice daily. Population: Alzheimer's disease and vascular dementia patients in China. No healthy-adult cognitive-enhancement RCTs.

THE VERDICT **Do not buy as a consumer memory supplement. It is a genuine acetylcholinesterase inhibitor (same pharmacology as donepezil) sold unregulated; the AD efficacy meta-analyses are entirely China-based with high risk of bias, and there is no healthy-adult benefit data. Prior of 1.5 holds.**

BEST FORM Huperzine A is a single purified alkaloid; ensure the label gives an absolute microgram dose (e.g. 50-200 mcg), NOT just a percentage of a Huperzia serrata extract, which hides the true amount. (Purified huperzine A with mcg dosing = controllable; 'Huperzia serrata extract 1%' with no mcg = ambiguous potency.)

WATCH Potent, long-acting acetylcholinesterase inhibitor, treat like a drug. Dose in mcg, not mg. Cholinergic overdose: nausea, sweating, cramping, slowed heart rate, vivid dreams. Cycle it (don't take daily indefinitely). Dangerous to combine with prescription AChE inhibitors (donepezil) or anticholinergic meds.

Sources: Yang G, et al. PLoS ONE 2013;8:e74916 (20-RCT meta, all China); Rafii MS, et al. Neurology 2011;76:1389-1394 (US phase II, negative); Xing SH, et al. Evid Based Complement Alternat Med 2014;2014:363985

1.5 NR / NMN (nicotinamide riboside / mononucleotide) Amazon ✓ Website ✓

Evidence: none / marketing (high confidence) | Effect: cognition / memory | NMN's US supplement status is unresolved after FDA excluded it; NR remains freely sold.

THE NOISE Marketed as an anti-aging 'cellular energy' molecule that raises NAD⁺ to restore youthful brain and body function.

THE SCIENCE Effect: NAD⁺ significantly elevated; clinically relevant metabolic/cognitive outcomes not significantly different from control. Clears MCID: No cognitive outcome measured that could clear any MCID. Largest independent trial: Wu Y, et al. *Alzheimers Dement* (N Y) TRCI 2025;11:e70023 (NR in SCD/MCI): biomarker shifts, no clear cognitive benefit. Population: Mostly metabolic-health cohorts; one 2025 NR trial in subjective cognitive decline / MCI. No healthy-adult memory outcome trials.

THE VERDICT Do not buy for memory. These reliably raise blood NAD⁺ but have no downstream cognitive outcome in humans; the biology outruns the evidence. NMN also has an unresolved US regulatory status. Prior of 1.5 holds.

BEST FORM For NR, branded Niagen is the stability-tested, GRAS-affirmed form. For NMN note the US regulatory cloud (FDA has moved to exclude NMN from supplements); NR (Niagen) is the more legally secure, better-characterized retail choice. Both raise NAD⁺ similarly. (NR (Niagen) = most human data, stable, legally clearer; NMN = popular, similar NAD⁺ raise but shakier US supplement status and more purity variability; plain niacin/niacinamide = raises NAD⁺ too but flushing (niacin) and no longevity edge.)

WATCH NAD-precursor longevity/memory benefits in humans are still unproven, manage expectations. Well tolerated at ~250-500 mg. Buy third-party-tested product (marketplace purity varies widely). Theoretical caution in active cancer (NAD fuels metabolism), discuss with oncologist.

Sources: Wu Y, et al. *Alzheimers Dement* (N Y) 2025;11:e70023 (NR, SCD/MCI); NMN meta-analysis, *Crit Rev Food Sci Nutr* 2024, doi:10.1080/10408398.2024.2387324 (NAD⁺ up, outcomes null); FDA position on NMN as excluded from supplement definition (2022)

Focus (acute) (7)

4 Caffeine (alone) Amazon ✓ Website ✓

Evidence: good (high confidence) | Effect: acute focus / attention

THE NOISE Sold as a fast-acting energy and focus booster for alertness and performance.

THE SCIENCE Effect: 2025 Psychopharmacology meta-analysis (10.1007/s00213-025-06775-1) confirms a significant acute caffeine benefit on attention; classic reviews (Einother & Giesbrecht 2013; McLellan 2016) show consistent small-to-moderate improvements in alertness and RT. Clears MCID: Yes in practical terms for alertness/vigilance, though formal MCIDs for lab attention tasks are not established. Largest independent trial: Numerous independent/military-funded trials (e.g. US Army sustained-operations studies, McLellan 2016 review) show robust vigilance benefit. Population: Healthy adults across dozens of acute crossover RCTs; effect strongest on simple/sustained attention and vigilance.

THE VERDICT The honest best-in-class acute-focus agent and the reference standard the other compounds are measured against. Well-replicated improvement in vigilance, reaction time and subjective alertness at 40-200 mg within ~30-60 min. It is a stimulant drug sold as a supplement; tolerance and rebound are real, and much of the benefit in habitual users is withdrawal reversal.

BEST FORM Caffeine anhydrous in a measured tablet/capsule (100-200 mg) for dose precision, OR simply coffee/tea. Avoid loose bulk caffeine powder, teaspoon-level dosing errors have been fatal. (Standardized anhydrous tablets or brewed coffee/tea = safe, dosable; bulk powdered caffeine = lethal-overdose risk from mis-measuring; 'extended-release' offers little acute-focus benefit.)

WATCH Cap ~400 mg/day (less if sensitive, pregnant ~200 mg). None within ~8-10 h of sleep. Tolerance builds fast; withdrawal headaches. Raises BP/anxiety; interacts with some meds (fluvoxamine, ciprofloxacin raise levels). Never dry-scoop powder.

Sources: Acute caffeine on attention meta-analysis, *Psychopharmacology* 2025, DOI 10.1007/s00213-025-06775-1; Einother SJL & Giesbrecht T 2013, *Psychopharmacology* (caffeine and attention review); McLellan TM et al. 2016, *Neurosci Biobehav Rev* (caffeine, cognition, physical/military performance)

3.5 Caffeine + L-theanine Amazon ✓ Website ✓

Evidence: moderate/conditional (high confidence) | Effect: acute focus / attention

THE NOISE Marketed as the 'jitter-free focus' stack that delivers clean, calm, sustained concentration without a crash.

THE SCIENCE Effect: Combo vs placebo (2025 Nutrition Reviews meta): simple reaction time hr2 SMD -0.71 (95% CI -0.92 to -0.50); attention-switching accuracy hr2 SMD 0.33 (0.13-0.54); digit vigilance accuracy hr2 SMD 0.20 (0.02-0.38); Bond-Lader alertness hr2 SMD 0.28 (-0.08 to 0.65, CI crosses null). Clears MCID: Partial/unclear - reaction-time SMD -0.71 is moderate-to-large, but the review explicitly notes no MCID (MID) is established for these attention tasks, so clinical meaningfulness cannot be formally confirmed. Largest independent trial: No single dominant independent trial; evidence is a pool of ~16 small crossover RCTs. Camfield 2014 systematic review (PubMed 24946991) reached the same 'caffeine drives it' conclusion. Population: Healthy adults, weighted mean age 32 (2025 Nutrition Reviews meta: 16 RCTs meta-analyzed, n=484, 229 female, median n=27). Also tested in sleep-deprived young adults and children with ADHD.

THE VERDICT Buy if you want a smoother acute alertness/reaction-time boost than caffeine alone, but understand the measurable focus benefit is driven by the caffeine. Multiple acute RCTs show a real, moderate effect on reaction time and attentional switching within the first 1-2 hours. Theanine's added value is subjective (less jitter, calmer alertness), not a distinct attention gain.

BEST FORM The synergy is about the combo/ratio, not a special chemical form. Target roughly 1:2 caffeine:theanine (e.g. 100 mg caffeine + 200 mg L-theanine). Use L-theanine (the natural enantiomer, e.g. Suntheanine), not racemic DL-theanine. (L-theanine (Suntheanine) = the studied form; DL/racemic theanine = half is the less-active isomer.)

WATCH Theanine blunts caffeine jitters/anxiety without killing alertness. Keep total caffeine under ~400 mg/day and avoid within ~8-10 h of bed. Watch hidden caffeine stacking from coffee/tea. Individual caffeine metabolism varies (CYP1A2).

Sources: 2025 Nutrition Reviews meta-analysis, Nutr Rev 83(10):1873 (tea/L-theanine/L-theanine+caffeine on cognition, sleep, mood); Camfield DA et al. 2014, Nutr Rev - PubMed 24946991 (acute tea constituents on cognition/mood); High-dose L-theanine-caffeine in sleep-deprived adults, PMC12491391 (2025 crossover, 200 mg + 160 mg); Kahathuduwa 2020 ADHD neuroimaging RCT, Sci Rep - PMC7403383

2.5 L-Tyrosine Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: acute focus / attention

THE NOISE Promoted as a dopamine-precursor amino acid that boosts focus, motivation and mental performance under stress.

THE SCIENCE Effect: Military Medicine REA (10 RCTs + 4 non-RCTs) made only a WEAK recommendation; all showed directionally positive stress-mitigation but effect sizes/CIs are inconsistent and small-sample. No robust pooled CI exists. Clears MCID: Unclear - benefits are context-specific (preventing decline under stress) rather than boosting baseline; no MCID framework applied. Largest independent trial: US military rapid evidence assessment (Military Medicine 2015, 180(7):754) synthesizing government-funded stress trials - independent but weak recommendation. Population: Healthy adults under acute stress: military trainees, sleep-deprived subjects, cold/altitude exposure, high cognitive load (Jongkees 2015 review, J Psychiatr Res; Military Medicine 2015 rapid evidence assessment).

THE VERDICT Do not buy for everyday focus. Tyrosine only helps when catecholamines are being depleted by an acute stressor - sleep deprivation, cold, noise, sustained multitasking. It does nothing measurable for a rested healthy adult, and effective doses (~100-150 mg/kg, i.e. 7-12 g) dwarf typical retail caps.

BEST FORM Free-form L-tyrosine (not D/DL). N-acetyl-L-tyrosine (NALT) is marketed as 'better absorbed' but converts to tyrosine poorly, so plain L-tyrosine is actually the more effective, cheaper choice at ~500-2000 mg. (Free-form L-tyrosine = best bioavailability for raising catecholamine precursors; NALT = poor conversion despite hype; DL-tyrosine = avoid.)

WATCH Works mainly under acute stress/sleep deprivation, not as a general 'up' pill. Take on an empty stomach away from protein meals (competes with other amino acids for transport). Contraindicated with MAOI antidepressants (hypertensive risk) and caution in hyperthyroidism (thyroid-hormone precursor). Avoid late in day if it feels stimulating.

Sources: Jongkees BJ et al. 2015, J Psychiatr Res - PubMed 26424423 (tyrosine under stress/cognitive demand review); Attipoe S et al. 2015, Military Medicine 180(7):754 - rapid evidence assessment, tyrosine for stress/performance; Behavioral/cognitive effects of tyrosine in healthy adults, ScienceDirect S0091305715000945

2.5 L-theanine (alone) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: acute focus / attention

THE NOISE Promoted as a green-tea amino acid that produces calm, relaxed focus and reduces stress without drowsiness.

THE SCIENCE Effect: Simple RT MD -0.46 ms (95% CI -15.65 to 14.73, null); Stroop congruent MD -37.38 ms (95% CI -86.39 to 11.62, null); RVIP at 400 mg MD -15.20 ms (95% CI -28.99 to -1.41, barely clears zero). Clears MCID: No - the single significant result (a ~15 ms RT change) has no established MCID and its CI nearly touches zero. Largest independent trial: Pooled evidence only; largest contributors ~50 participants each. No large independent standalone attention RCT. Population: Healthy adults 20-58 (2025 meta-analysis PMC12609247: 5 RCTs, n=148, acute single doses, measured 40-70 min post-dose; studies from UK, Japan, Turkey, Sri Lanka).

THE VERDICT Do not buy for acute focus on its own. Across 5 acute RCTs the only significant attention signal was RVIP reaction time at a 400 mg dose; simple reaction time and Stroop were null, and GRADE certainty is low. Its defensible use is subjective calm/anti-stress and taking the edge off caffeine, not standalone attention enhancement.

BEST FORM L-theanine specifically (branded Suntheanine is the pure L-isomer used in trials), 100-200 mg. Avoid cheap 'theanine' that is DL/racemic. (L-theanine (Suntheanine) = active natural form; DL-theanine = 50% less-useful D-isomer per label weight.)

WATCH Non-sedating relaxation; good daytime or to take the edge off caffeine, and can aid sleep onset at night. Very safe, minimal interactions. May mildly lower blood pressure. Effect is subtle, don't expect a stimulant kick.

Sources: L-theanine acute cognition meta-analysis 2025, PMC12609247 (5 RCTs, n=148); Camfield DA et al. 2014, Nutr Rev - PubMed 24946991

2.5 Rhodiola rosea Amazon ✓ Website ✓

Evidence: weak/mostly null (low confidence) | Effect: acute focus / attention · **literature compromised (funding/outlier)**

THE NOISE Promoted as an adaptogen that fights fatigue, beats burnout, and boosts mental and physical energy and stress resilience.

THE SCIENCE Effect: No pooled effect size available - Ishaque 2012 could not perform meta-analysis because no two trials reported the same outcome. Individual small trials report reduced fatigue scores but with high risk of bias. Clears MCID: Cannot assess - no consistent outcome metric, no MCID benchmarking possible. Largest independent trial: No large independent acute-focus RCT; recent 2025 resistance-trained athlete crossover (PMC12693935) found only modest, outcome-specific effects. Population: Fatigued students, physicians on night duty, burnout/stress-fatigue patients (Ishaque 2012 systematic review, BMC Complement Altern Med 12:70). Not rested healthy adults seeking acute focus.

THE VERDICT Do not buy for acute baseline focus. Signals exist for mental FATIGUE and burnout, not for sharpening a rested healthy adult. The evidence base could not even be meta-analysed (no two trials shared outcomes), trials are small and methodologically flawed, and positive results cluster in manufacturer SHR-5 studies.

BEST FORM Extract standardized to 3% rosavins AND 1% salidroside (the SHR-5 ratio used in trials). Unstandardized 'rhodiola' or products high in salidroside only (or adulterated with cheaper Rhodiola crenulata) underperform. (3% rosavin / 1% salidroside standardized extract (best) > extract standardized to only one marker > raw root powder / unspecified species (worst, often crenulata adulteration).)

WATCH Take in the morning on an empty stomach - it's mildly stimulating and can cause insomnia or vivid dreams if taken late. Typical dose 200-400 mg. Can be activating/irritable in some; caution with stimulants, antidepressants (theoretical serotonergic/MAO effect) and in bipolar disorder. Effects are for stress-related fatigue, not raw physical performance.

Sources: Ishaque S et al. 2012, BMC Complement Altern Med 12:70 - PubMed 22643043 (systematic review, physical/mental fatigue); Rhodiola dose-response in resistance-trained athletes 2025, PMC12693935

2 Citicoline (CDP-choline) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: acute focus / attention · **literature compromised (funding/outlier)**

THE NOISE Marketed as a fast-acting nootropic for sharper focus, attention and mental energy.

THE SCIENCE Effect: Nakazaki 2021 reported statistically significant improvements in sustained-attention and RVIP reaction time and 'mental energy' composites over 12 weeks, but published effect sizes/CIs are not clearly reported and the trial is manufacturer-run. Clears MCID: Cannot confirm - effect sizes not transparently reported; no MCID established for the attention endpoints. Largest independent trial: No sizable INDEPENDENT acute-focus trial; the dementia meta-analyses (independent) show only modest, heterogeneous benefit in impaired patients. Population: Adults 35-75 with poor sustained attention (Nakazaki 2021); adolescent males (McGlade); MCI/dementia patients (meta-analyses). Not healthy rested adults acutely.

THE VERDICT Do not buy for ACUTE focus. There is essentially no single-dose acute-attention evidence; the supporting human trials are 12-week chronic supplementation, mostly industry-funded (Cognizin/Kyowa) and run in low-attention or clinical populations. Wrong category and compromised evidence base.

BEST FORM Citicoline is a single active compound; Cognizin is simply a branded, patented, clinically-studied citicoline used in most trials. Generic citicoline of equivalent dose is chemically the same, so pay for verified purity/dose rather than the name. (Cognizin-branded vs generic citicoline = same molecule; the real fork is citicoline vs alpha-GPC as your choline source (citicoline also yields cytidine/uridine).)

WATCH Typical 250-500 mg/day. Well tolerated; occasional headache/GI upset. Non-stimulant, timing flexible. Don't stack redundantly with high-dose alpha-GPC (both raise choline). Generally avoid combining potent cholinergics if prone to depression/low mood.

Sources: Nakazaki E et al. 2021, J Nutr - citicoline 500 mg, 12 wk, attention in poor-attention adults (industry/Cognizin); Nakazaki 2021 memory in healthy older adults, PMC8349115; Citicoline and dementia systematic review/meta-analysis, Nutrients 2023, 15(2):386 (MDPI); McGlade E et al. citicoline and attention in adolescent males

2 Panax ginseng Amazon ✓ Website ✓

Evidence: weak/mostly null (low confidence) | Effect: acute focus / attention · **literature compromised (funding/outlier)**

THE NOISE Marketed as an energizing adaptogen that improves focus, mental clarity and cognitive performance.

THE SCIENCE Effect: Reay 2010: 200 mg slowed mental-arithmetic responding (worse speed) while slowing mood decline; 400 mg improved arithmetic and calmness. Small within-subject effects, no clean pooled CI; direction differs by dose and timepoint. Clears MCID: No - effects are small, timepoint-dependent and partly in the wrong direction; no MCID cleared. Largest independent trial: No large independent acute-focus RCT; Reay 2010 (n=30) is the anchor and is proprietary-extract based. Population: Healthy young adults, mean age ~23 (Reay 2010, n=30 crossover, G115 standardized extract, Human Psychopharmacol).

THE VERDICT Do not buy for acute focus. The cleanest acute trial showed the 200 mg dose actually SLOWED mental-arithmetic responding while only slowing a mood decline; benefits were small, inconsistent in direction, single-lab, and n=30. Mechanism and marketing outrun the outcome data.

BEST FORM Standardized to ginsenosides (commonly 4-7%), e.g. the G115 extract used in cognition studies. Ensure it's true Panax ginseng (Asian/Korean) standardized by ginsenoside %, distinct from American ginseng or unstandardized root. (Ginsenoside-standardized Panax ginseng (G115) = studied for alertness; unstandardized root powder or mislabeled species (Siberian 'ginseng'/Eleuthero is a different plant) = inconsistent.)

WATCH Can be stimulating, take earlier in the day; may cause insomnia, headache, or raise blood pressure. Cycle to limit tolerance. Interacts with warfarin (may reduce effect), diabetes meds (additive glucose lowering), and MAOIs. Caution with hormone-sensitive conditions.

Sources: Reay JL et al. 2010, Hum Psychopharmacol 25:462 - DOI 10.1002/hup.1138, PubMed 20737519 (G115 working memory/calmness); Safety analysis of Panax ginseng in RCTs, systematic review - PMC5533164

4.5 Caffeine (the honest energy answer - added) Amazon ✓ Website ✓

Evidence: strong (high confidence) | Effect: energy / fatigue

THE NOISE The active ingredient in every energy drink and pre-workout; sold for energy, focus and fat-burning.

THE SCIENCE Effect: Consistent dose-related improvements in alertness, vigilance, reaction time and fatigue ratings up to ~300 mg (McLellan 2016 review; 2025 attention meta-analysis shows significant gains in reaction time and accuracy). Non-linear: benefit plateaus and side effects rise above ~300-400 mg. Clears MCID: Yes - alertness/vigilance improvements are perceptible and behaviorally meaningful, especially under fatigue. Largest independent trial: McLellan 2016 (Neurosci Biobehav Rev) comprehensive independent review; 2025 acute-attention meta-analysis - both positive. Population: Healthy adults, sleep-deprived subjects, shift/military personnel, athletes - broad and independent.

THE VERDICT This is the one 'energy' agent that actually works. Large independent literature: reliable, dose-related reduction in fatigue and improvement in alertness, vigilance and reaction time, in both rested and sleep-deprived adults. Caveat: it is a stimulant with tolerance and withdrawal, not a nutrient - benefit is largely restoring, not exceeding, baseline in habituated users.

BEST FORM Caffeine anhydrous in a measured tablet/capsule (100-200 mg) for dose precision, OR simply coffee/tea. Avoid loose bulk caffeine powder, teaspoon-level dosing errors have been fatal. (Standardized anhydrous tablets or brewed coffee/tea = safe, dosable; bulk powdered caffeine = lethal-overdose risk from mis-measuring; 'extended-release' offers little acute-focus benefit.)

WATCH Cap ~400 mg/day (less if sensitive, pregnant ~200 mg). None within ~8-10 h of sleep. Tolerance builds fast; withdrawal headaches. Raises BP/anxiety; interacts with some meds (fluvoxamine, ciprofloxacin raise levels). Never dry-scoop powder.

Sources: McLellan 2016 Neurosci Biobehav Rev 71:294, doi:10.1016/j.neubiorev.2016.09.001; Acute caffeine on attention meta-analysis 2025, doi:10.1007/s00213-025-06775-1; Caffeine and cognition in sleep deprivation, NCBI NBK209050

4 Iron (oral ferrous salts, for fatigue/energy) Amazon ✓ Website ✓

Evidence: good (high confidence) | Effect: energy / fatigue

THE NOISE Sold as an energy and anti-fatigue supplement to beat tiredness and 'low energy.'

THE SCIENCE Effect: Vaucher 2012: fatigue fell 47.7% (iron) vs 28.8% (placebo), between-group difference -18.9% (95% CI -34.5 to -3.2), p=0.02. Verdon 2003: -8 points on a 1-100 fatigue scale, benefit confined to ferritin ≤50. Clears MCID: Borderline yes in the low-ferritin subgroup; CI lower bound (-3.2%) approaches null, so real but not large. na for replete adults (no effect). Largest independent trial: Vaucher et al. 2012 CMAJ, n=198 women, independent (Swiss academic) - positive but modest. IV-iron blood-donor RCT (PMC7449957) was NULL for fatigue despite raising ferritin. Population: Non-anemic menstruating women aged 18-53 with unexplained fatigue and ferritin <50 ug/L (Vaucher, Verdon). Some blood-donor and athlete cohorts.

THE VERDICT Buy ONLY if serum ferritin is low. In iron-deficient menstruating women (ferritin <50 ug/L, no anemia) oral iron produces a real, modest reduction in fatigue. In iron-replete adults it does nothing for energy and adds overdose/GI risk. Test ferritin before buying; do not take blind.

BEST FORM Ferrous bisglycinate (chelated) is notably gentler on the gut at equivalent absorbed iron than ferrous sulfate; ferrous sulfate is cheapest and effective if tolerated. Compare by ELEMENTAL iron mg, not salt weight. (Ferrous bisglycinate = gentler, good absorption; ferrous sulfate/fumarate/gluconate = effective but more GI upset/constipation; ferric (non-heme ferric) salts and 'iron oxide' = poorly absorbed.)

WATCH Only supplement iron if labs (ferritin) show you need it, excess is harmful. Alternate-day or once-daily dosing absorbs better than multiple daily doses (hepcidin). Take with vitamin C, on empty stomach if tolerated; AVOID with coffee/tea/calcium/dairy (block absorption) and separate from thyroid meds/antibiotics/PPIs. Constipation/black stools common. Keep away from children (overdose is lethal).

Sources: Vaucher 2012 CMAJ 184(11):1247, doi:10.1503/cmaj.110950 (PMID 22586171); Verdon 2003 BMJ 326:1124 (PMID 12763985); IV iron in non-anemic donors, PMC7449957; Krayenbuehl 2011 Blood 118:3222 (IV iron, ferritin<50 women)

3.5 Creatine monohydrate (for 'energy'/anti-fatigue/mental energy) Amazon ✓ Website ✓

Evidence: moderate/conditional (high confidence) | Effect: energy / fatigue

THE NOISE Marketed as a cellular-energy and anti-fatigue supplement for physical and mental stamina.

THE SCIENCE Effect: Xu 2024 (Front Nutr, 16 RCTs, n=492): significant but modest SMDs for memory, attention and processing speed; Digit test d=0.77. Gordji-Nejad 2024 (Sci Rep): single 0.2 g/kg dose attenuated - not abolished - sleep-deprivation cognitive decline, peak ~4 h. Clears MCID: For strength/power: yes. For 'mental energy'/fatigue in rested adults: marginal; several cognition endpoints are equivocal. Largest independent trial: Xu 2024 Front Nutr meta-analysis (independent, PRISMA) - positive but modest and somewhat equivocal for cognition. Population: Healthy adults, athletes, sleep-deprived subjects, vegetarians (larger effect), some clinical cohorts.

THE VERDICT Buy if the goal is strength/power/muscle (that evidence is a 5) or resilience under sleep deprivation/stress. As a general 'energy' or anti-fatigue supplement for a rested healthy adult, the effect is real but small and condition-dependent - a 3.5, not the miracle marketing implies. Cheap and safe, so low downside.

BEST FORM Plain creatine monohydrate is the gold standard - micronized for easier mixing, and Creapure-grade for purity if you want third-party assurance. Fancier 'forms' are not worth paying for. (Creatine monohydrate BEST (most-studied, cheapest, proven); HCl, buffered 'Kre-Alkalyn', ethyl ester and 'nitrate' forms cost more with NO demonstrated bioavailability or performance advantage - some (ethyl ester) are actually worse.)

WATCH 3-5 g/day is sufficient; a loading phase is optional, not required. Expect ~1-2 kg water-weight gain early. It raises serum creatinine harmlessly - warn your doctor so it isn't misread as kidney injury (interferes with that lab interpretation). Stay hydrated; take consistently, timing doesn't matter much.

Sources: Xu 2024 Front Nutr, doi:10.3389/fnut.2024.1424972 (PMID 39070254); Gordji-Nejad 2024 Sci Rep 14:4937, doi:10.1038/s41598-024-54249-9; Kreider 2017 JISSN position stand

2 Ashwagandha (Withania somnifera) for energy/fatigue Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: energy / fatigue · **literature compromised (funding/outlier)** | Emerging regulatory liver-injury signal; some markets (e.g. Denmark) restrict it, but freely sold in the US.

THE NOISE Promoted (often as KSM-66) as a stress-lowering adaptogen that reduces fatigue and boosts energy and resilience.

THE SCIENCE Effect: KSM-66 trials report ~+18% 'energy/fatigue' and +15% QoL subscale improvements, but as secondary outcomes in sponsor-linked studies. Dose-response meta-analysis: benefit on stress and anxiety with LOW certainty (GRADE); no robust independent fatigue effect size. Clears MCID: Unclear - subjective self-report changes with no established fatigue MCID and high placebo response in this literature. Largest independent trial: Genuinely independent, adequately powered fatigue RCTs are scarce; the influential energy/QoL data are KSM-66/Ixoreal-linked. Population: Mostly stressed or anxious healthy adults; fatigue/energy is typically a secondary, subjective outcome, not the primary endpoint.

THE VERDICT Do not buy for energy. The fatigue/energy signal comes almost entirely from manufacturer-linked (KSM-66/Ixoreal) trials with subjective secondary endpoints; independent replication is thin and certainty is graded LOW. There is now a regulatory liver-injury signal. Downside outweighs a weak, industry-driven benefit.

BEST FORM A standardized root extract with a named clinical grade - KSM-66 (root-only, ~5% withanolides) or Sensoril (root+leaf, higher withanolide %) - not raw non-standardized root powder (KSM-66 / Sensoril standardized extracts (dosed to withanolide content) > generic 'ashwagandha root powder' (unstandardized, unknown potency, needs much larger doses))

WATCH Rare but real reports of liver injury - stop if jaundice/RUQ pain. Can raise thyroid hormones (caution if hyperthyroid or on levothyroxine). Sedating - additive with CNS depressants; may lower blood sugar and blood pressure. Avoid in pregnancy and with autoimmune-thyroid concerns. Take with food.

Sources: TGA safety advisory: Withania somnifera and rare liver injury (2024); NIH LiverTox NBK548536 (Ashwagandha); Ashwagandha stress/anxiety dose-response meta-analysis (low-certainty GRADE); KSM-66 QoL/energy RCTs (Ixoreal-affiliated)

2 CoQ10 / Ubiquinol Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: energy / fatigue · **literature compromised (funding/outlier)**

THE NOISE Sold as a mitochondrial 'cellular energy' nutrient that fights fatigue and restores vitality.

THE SCIENCE Effect: Tsai 2022 fatigue meta (13 RCTs, n=1126): pooled SMD favors CoQ10 but across highly heterogeneous clinical populations. Statin myopathy: 2024 meta (7 RCTs, n=389) positive in only 4/7; Taylor 2015 RCT in CONFIRMED statin myopathy found NO benefit on pain, strength or exercise capacity despite raised serum CoQ10. Clears MCID: Unclear/no in most settings; the confirmed-myopathy RCT is a clean null. Largest independent trial: Taylor 2015 Atherosclerosis, RCT in confirmed statin myopathy - independent, NULL. Population: Statin users, fibromyalgia, chronic fatigue, CKD, heart failure, older adults - NOT healthy energetic adults.

THE VERDICT Do not buy for general energy. 'Fatigue' benefit is concentrated in specific patient groups (fibromyalgia, CKD, heart failure, statin users) in small, heterogeneous, often industry-adjacent trials, and even the statin-myopathy evidence is split. No credible energy benefit in a healthy adult.

BEST FORM Ubiquinol (the reduced form) in an oil-based softgel - better absorbed, especially if you're over ~40 or on a statin. Ubiquinone (oxidized) is cheaper and fine for younger people IF taken with a fatty meal; dry tablets/powders absorb poorly regardless of type. (Ubiquinol softgel (highest, most bioavailable) > ubiquinone in oil suspension with food > ubiquinone dry tablet/powder (poor).)

WATCH Fat-soluble - always take with a meal containing fat. Statins deplete CoQ10, which is the main rationale for supplementing. It is structurally vitamin-K-like and can blunt warfarin's effect, so coordinate with your anticoagulation clinic. Split doses >200 mg for better uptake. May modestly lower blood pressure.

Sources: Tsai 2022 fatigue meta, PMC9449413; CoQ10 statin myopathy 2024 meta, PMC12554813; Qu 2018 JAHA 7:e009835, doi:10.1161/JAHA.118.009835; Taylor 2015 Atherosclerosis (confirmed statin myopathy RCT)

2 Cordyceps (Cs-4 / C. sinensis / C. militaris) Amazon ✓ Website ✓

Evidence: weak/mostly null (low confidence) | Effect: energy / fatigue

THE NOISE Marketed as a natural energy and stamina booster that increases cellular ATP and oxygen utilization to raise VO2max, endurance, and athletic performance.

THE SCIENCE Effect: Chen 2010 (Cs-4, healthy older adults): modest VO2max/ventilatory-threshold gains in a small RCT. C. militaris acute crossover (2025 PLOS One): raised resting VO2 and faster reaction time - acute, small n. Fungal-supplement meta (2025): 'potential' endurance benefit, heterogeneous, low quality. No pooled fatigue effect with tight CIs. Clears MCID: Unclear - small samples, surrogate endpoints (VO2, reaction time), not a validated energy/fatigue MCID. Largest independent trial: Chen 2010 J Altern Complement Med (Cs-4 in older adults) - small, modest positive. Population: Small groups of older adults, recreational athletes; some acute crossover work.

THE VERDICT Do not buy for energy. Endurance/VO2max signals come from small, short, often industry- or TCM-linked trials with inconsistent results, and retail products under-dose the trials. No convincing energy benefit in a healthy adult.

BEST FORM Cultivated fruiting body of C. militaris (highest cordycepin) or a genuine Cs-4 fermented mycelium (the well-studied strain). Avoid 'mycelium grown on grain' products - they're mostly starch/grain filler with low beta-glucan and negligible cordycepin. (C. militaris fruiting body (best, real cordycepin) > authenticated Cs-4 fermentate (well-researched) > wild C. sinensis (expensive, often adulterated) > mycelium-on-grain (worst - grain filler).)

WATCH Look for a stated beta-glucan % and 'fruiting body' on the label; a high 'polysaccharide' number often just means grain starch. Wild sinensis is heavily counterfeited and can carry heavy metals/lead. Mild blood-sugar lowering and possible additive effect with anticoagulants and immunosuppressants. Energy/VO2 evidence is modest.

Sources: Chen 2010 J Altern Complement Med (Cs-4, older adults; PMID 20804368); C. militaris acute crossover 2025 PLOS One, doi:10.1371/journal.pone.0351725; Fungal supplementation endurance meta-analysis 2025, PMC12631420

2 Rhodiola rosea (added - common in 'energy'/adaptogen products) Amazon ✓ Website ✓

Evidence: weak/mostly null (low confidence) | Effect: energy / fatigue

THE NOISE Adaptogen sold for energy, stamina and beating fatigue/burnout.

THE SCIENCE Effect: Ishaque 2012 systematic review (11 trials): only 2/6 physical-fatigue and 3/5 mental-fatigue RCTs positive; ALL studies had high risk of bias or reporting flaws precluding a reliable pooled effect size. Clears MCID: Cannot be determined - biased, inconsistent trials with no trustworthy pooled estimate. Largest independent trial: Ishaque 2012 BMC Complement Altern Med systematic review (independent) - verdict: contradictory, needs a rigorous RCT. Population: Healthy adults, students, physicians on night duty, stressed cohorts - small, biased trials.

THE VERDICT Do not buy for energy. The fatigue literature is contradictory and uniformly high risk of bias; positive trials are a minority and methodologically weak. Thin evidence dressed up as an adaptogen.

BEST FORM Extract standardized to 3% rosavins AND 1% salidroside (the SHR-5 ratio used in trials). Unstandardized 'rhodiola' or products high in salidroside only (or adulterated with cheaper Rhodiola crenulata) underperform. (3% rosavin / 1% salidroside standardized extract (best) > extract standardized to only one marker > raw root powder / unspecified species (worst, often crenulata adulteration).)

WATCH Take in the morning on an empty stomach - it's mildly stimulating and can cause insomnia or vivid dreams if taken late. Typical dose 200-400 mg. Can be activating/irritable in some; caution with stimulants, antidepressants (theoretical serotonergic/MAO effect) and in bipolar disorder. Effects are for stress-related fatigue, not raw physical performance.

Sources: Ishaque 2012 BMC Complement Altern Med 12:70, doi:10.1186/1472-6882-12-70 (PMID 22643043)

1 Adaptogen blends (proprietary multi-herb 'energy'/'adrenal support' formulas) Amazon ✓ Website ✓

Evidence: none / marketing (high confidence) | Effect: energy / fatigue

THE NOISE Marketed as 'adrenal support' and 'energy' blends that balance stress hormones and restore natural energy.

THE SCIENCE Effect: No credible whole-product RCT with an energy outcome. Effect size unknown. Clears MCID: No - no outcome data exist to clear anything. Largest independent trial: None on the blends. Population: None for the blends as sold; evidence is borrowed from single-herb studies at doses the blends rarely deliver.

THE VERDICT Do not buy. Proprietary blends have essentially no whole-product RCTs, hide doses behind 'proprietary blend' labels that are almost always sub-therapeutic, and lean on mechanism and single-herb extrapolation. Marketing.

BEST FORM Skip proprietary multi-herb blends; buy single, standardized, individually-dosed extracts (e.g. a KSM-66 ashwagandha plus a 3%-rosavin rhodiola) so you know the actual dose of each. Proprietary blends hide per-ingredient amounts behind one total, so most herbs are present at sub-effective 'fairy dust' levels. (Individual standardized extracts at studied doses (best) > transparent-label blends listing each mg > proprietary 'blend XY mg' (worst - undosed, often underpowered).)

WATCH 'Adrenal support' is a marketing frame, not a diagnosis - true adrenal insufficiency needs a doctor. Watch for undeclared stimulants (caffeine, synephrine, yohimbine) padding the 'energy' effect. Glandular 'adrenal extract' products can contain actual animal adrenal tissue - avoid. Cross-check every herb for your meds (thyroid, sedatives, blood pressure).

Sources: No qualifying whole-product RCT located; Single-herb reviews (Rhodiola PMID 22643043; Ashwagandha low-certainty GRADE) do not validate the blends

1 Vitamin B12 (cyanocobalamin/methylcobalamin, for fatigue/energy) Amazon ✓ Website ✓

Evidence: none / marketing (high confidence) | Effect: energy / fatigue

THE NOISE Marketed as an 'energy vitamin' and B12 'energy shot' that fights fatigue and boosts metabolism.

THE SCIENCE Effect: Markun 2021 meta-analysis (16 RCTs, 6276 participants) found NO reliable fatigue benefit in non-deficient populations; only a single eligible fatigue study, insufficient to pool. No clinically meaningful energy effect established. Clears MCID: No - no measurable fatigue effect to clear an MCID in replete adults. na in deficiency (repletion is corrective). Largest independent trial: Markun 2021 Nutrients systematic review/meta-analysis (independent) - concluded evidence for fatigue in non-deficient people is essentially absent. Population: Mostly older adults and mixed clinical cohorts; almost no eligible RCTs in non-deficient people with fatigue.

THE VERDICT Do NOT buy for energy unless you are actually B12-deficient. In deficiency (pernicious anemia, vegans, metformin/PPI users, malabsorption) repletion is a 5. In replete people - which is most tired people - B12 does nothing; the 'energy shot' is theatre. Test B12/MMA first.

BEST FORM Methylcobalamin or hydroxocobalamin (active/retained forms) over cyanocobalamin, sublingual or oral high-dose works for most; injections reserved for pernicious anemia/malabsorption. Cyanocobalamin is stable and cheap but yields a trace of cyanide and needs conversion. (Methylcobalamin/adenosylcobalamin/hydroxocobalamin = bioactive, better retained; cyanocobalamin = cheapest/most stable, fine for most but less ideal in smokers/kidney issues/MTHFR.)

WATCH B12 only relieves fatigue if you're actually deficient, common in vegans, older adults, metformin and long-term PPI users. High-dose oral works even without intrinsic factor via passive absorption. Very safe (water-soluble). Check level before/after. Biotin and even B12 supplements can interfere with some lab immunoassays, tell the lab. Folate can mask B12 deficiency.

Sources: Markun 2021 Nutrients 13(3):923, doi:10.3390/nu13030923 (PMID 33809274); LiverTox/clinical reviews of B12 repletion

Sleep (15)

3.5 Melatonin (dose & timing) Amazon ✓ Website ✓

Evidence: moderate/conditional (high confidence) | Effect: sleep | OTC in the US; prescription-only in the UK, EU and Australia.

THE NOISE Sold as a natural, non-habit-forming sleep hormone that helps you fall asleep faster, sleep more deeply and reset your body clock for jet lag and shift work.

THE SCIENCE Effect: Sleep onset latency WMD -7.06 min (95% CI -9.75 to -4.37; Ferracioli-Oda 2013). Prolonged-release: subjective SOL -6.30 min, objective SOL -5.05 min (2023 meta, n=4875). Total sleep time +8-13 min. Dose-response peaks ~4 mg. Clears MCID: For insomnia: NO - a 5-7 min SOL reduction sits below the ~10-min MCID; a null in a suit. For circadian phase disorders: YES - phase-shifting is the mechanism, not the minutes. Largest independent trial: Wade 2007/2010 prolonged-release melatonin in adults ≥55 (Neurim-funded); independent pediatric/DSWPD trials (Gringras MENDS 2017 n=146) confirm circadian use. Population: Mixed: primary insomnia adults, DSWPD, jet-lag travelers, children with ASD/ADHD, elderly. Effect is population-specific.

THE VERDICT Buy for the RIGHT problem: circadian misalignment (delayed sleep-wake phase disorder, jet lag, shift work, non-24, ASD/child sleep). For garden-variety adult primary insomnia it is a weak agent the AASM recommends AGAINST. Use LOW dose (0.5-1 mg) timed hours before target sleep for phase-shifting; high-dose 5-10 mg products are chronobiologically wrong.

BEST FORM LOW dose - 0.5-1 mg is physiologic and as effective as higher doses for sleep onset with less hangover. Immediate-release for sleep-onset trouble; a low-dose extended-release can help sleep-maintenance/older adults. The dose and timing matter far more than brand. (Low-dose immediate-release 0.5-1 mg (physiologic, clean) > 3-5 mg (common but often supraphysiologic, more grogginess) > 10 mg (usually excessive). Extended-release suits middle-of-night waking. Gummies/products are frequently mislabeled (often contain far more than stated) - buy USP-verified.)

WATCH Timing is the lever: take ~30-60 min before target bedtime for insomnia, but for circadian phase-shifting (jet lag, delayed sleep phase) dose SMALL and EARLIER in the evening. Too much causes next-day grogginess and vivid dreams. Product-content variance is notorious (studies found 0.5-4.8x label) - choose third-party tested. Interacts with anticoagulants, immunosuppressants, and diabetes/BP meds; can cause daytime sedation.

Sources: Ferracioli-Oda E 2013, PLoS One, PMID 23691095 (DOI 10.1371/journal.pone.0063773); Cruz-Sanabria F 2024, J Pineal Res, PMID 38888087 (DOI 10.1111/jpi.12985); Sateia MJ 2017 AASM Clinical Practice Guideline, J Clin Sleep Med (weak recommendation AGAINST melatonin for chronic insomnia); Gringras P 2017 MENDS RCT, J Am Acad Child Adolesc Psychiatry

3 L-tryptophan Amazon ✓ Website ✓

Evidence: moderate/conditional (medium confidence) | Effect: sleep

THE NOISE Sold as an essential amino acid and serotonin/melatonin precursor that promotes calm and helps you fall and stay asleep.

THE SCIENCE Effect: Sutanto 2022: WASO shortened ~-81 min per gram (SMD -1.08; 95% CI -1.89 to -0.28; P=0.017). Dose-gated: >=1 g gave WASO 28.9 min vs 56.6 min for <1 g (P=0.001). No significant effect on other sleep components. Clears MCID: YES for WASO at >=1 g (a ~28-min reduction is clinically meaningful) - but only that endpoint. Largest independent trial: Historical Hartmann/Schneider-Helmert 1 g trials (independent, academic). Population: Mild insomnia and normal subjects with longer-than-average sleep latency; some clinical insomnia.

THE VERDICT A real but narrow effect: at >=1 g it reduces wake-after-sleep-onset, best in mild insomnia or long-latency normal sleepers. Modest and dose-gated. Carries a serious historical safety scar (1989 contaminated-batch eosinophilia-myalgia syndrome, >1500 cases, ~30+ deaths - a manufacturing contaminant, not the molecule). Buy only reputable, high-purity product at adequate dose.

BEST FORM n/a - a single amino acid; any USP/pharma-grade L-tryptophan is equivalent. What matters is HOW you take it, not the form. (Single active form - buy tested pharmaceutical-grade for purity)

WATCH Take on an empty stomach, away from other protein/amino acids (they compete for the same LAT1 brain transporter); a small carbohydrate aids uptake. Serotonin-syndrome risk with SSRIs/MAOIs. Historical EMS contamination - buy quality-tested product. Causes drowsiness; don't drive.

Sources: Sutanto CN 2022, Nutr Rev, PMID 33942088 (DOI 10.1093/nutrit/nuab027; WASO -81 min/g); Hartmann E 1974/1977 (L-tryptophan sleep latency, mild insomnia); Belongia EA 1990, NEJM (EMS / L-tryptophan contamination)

2.5 Ashwagandha (KSM-66, Withania somnifera) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: sleep · **literature compromised (funding/outlier)** | Under regulatory scrutiny for liver-injury reports; warning-labeled or restricted in some markets (e.g. Australia TGA), still OTC in the US.

THE NOISE Marketed as a clinically studied adaptogen (branded KSM-66) that lowers stress and cortisol and improves sleep quality and how rested you feel.

THE SCIENCE Effect: Pooled SMD on overall sleep -0.59 (95% CI -0.75 to -0.42; Cheah 2021). Larger in the insomnia subgroup, dose >=600 mg/day, >=8 wk. HAM-A anxiety and sleep-onset latency improved (Langade 2019/2021). Clears MCID: Plausibly yes on PSQI in the insomnia subgroup, but confounded by uniform industry funding. Largest independent trial: No large independent (non-manufacturer) RCT identified - the central weakness. Population: Adults with insomnia/stress, largely Indian trial sites; healthy stressed adults.

THE VERDICT Conditional and safety-flagged. Meta-analysis shows a moderate sleep effect, but essentially all trials are industry-funded (Ixoreal/KSM-66), single-region, and short. More important: multiple regulators (TGA 2024 and 2026, MHRA Yellow Card, Netherlands Lareb, NIH LiverTox) have posted liver-injury signals. Efficacy is real-ish; the risk-benefit for a sleep aid is unattractive.

BEST FORM A standardized root extract with a named clinical grade - KSM-66 (root-only, ~5% withanolides) or Sensoril (root+leaf, higher withanolide %) - not raw non-standardized root powder (KSM-66 / Sensoril standardized extracts (dosed to withanolide content) > generic 'ashwagandha root powder' (unstandardized, unknown potency, needs much larger doses))

WATCH Rare but real reports of liver injury - stop if jaundice/RUQ pain. Can raise thyroid hormones (caution if hyperthyroid or on levothyroxine). Sedating - additive with CNS depressants; may lower blood sugar and blood pressure. Avoid in pregnancy and with autoimmune-thyroid concerns. Take with food.

Sources: Cheah KL 2021, PLoS One, PMID 34559859 (DOI 10.1371/journal.pone.0257843; 5 RCTs, SMD -0.59); Langade D 2019/2021, Cureus / J Ethnopharmacol S0378874120331585 (KSM-66, Ixoreal-supplied); TGA 2024/2026 safety advisory 'Withania somnifera (Ashwagandha) and very rare risk of liver injury'; NIH LiverTox, NBK548536

2.5 Glycine Amazon ✓ Website ✓

Evidence: weak/mostly null (low confidence) | Effect: sleep · **literature compromised (funding/outlier)**

THE NOISE Promoted as an amino acid that, taken at 3 g before bed, helps you fall asleep faster, improves sleep quality and leaves you less tired and sharper the next day.

THE SCIENCE Effect: PSG: shortened latency to sleep onset and to slow-wave sleep; improved subjective sleep quality/efficiency and reduced daytime sleepiness. No CIs reported in the small crossover papers; direction consistent but underpowered. Clears MCID: Unclear - PSG latency shortened but small-n crossover without CI/MCID framing. Largest independent trial: None truly independent identified. Population: Adults with mild/unsatisfactory sleep (self-reported poor sleepers), small samples.

THE VERDICT **A genuine but fragile signal. Two small polysomnography crossover studies show faster sleep onset and better subjective quality at 3 g, but both trace to the amino-acid manufacturer and neither has been independently replicated at scale. Low-risk to try; not decision-grade.**

BEST FORM Glycine is a single amino acid - the molecule is the same everywhere; just buy pure pharmaceutical/USP-grade powder or capsules. Studied sleep dose is 3 g taken ~30-60 min before bed. It is naturally sweet, so powder in water is pleasant. (Free glycine powder = capsules = the glycine inside 'magnesium glycinate' (same amino acid) - no meaningful form/absorption difference. Only purity/label accuracy varies. No need to pay for proprietary 'forms'.)

WATCH 3 g pre-bed is the trial dose (improves subjective sleep quality and lowers core body temperature to aid onset); much higher doses can cause mild GI upset. Well tolerated with few interactions. Note the clozapine interaction (glycine may reduce its efficacy). It is calming, not sedating - no hangover.

Sources: Yamadera W 2007, Sleep Biol Rhythms 5:126-131, DOI 10.1111/j.1479-8425.2007.00262.x (PSG crossover); Inagawa K 2006, Sleep Biol Rhythms, DOI 10.1111/j.1479-8425.2006.00193.x; Bannai M 2012, Front Neurol (mechanism)

2.5 L-theanine Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: sleep

THE NOISE Marketed as a green-tea amino acid that promotes calm relaxation and focus without drowsiness and supports better, more restful sleep.

THE SCIENCE Effect: PSQI global and subscales (latency, disturbance, sleep-med use) improved vs placebo in stressed adults (Hidese 2019, n=30, 200 mg x4 wk). Objective-sleep effects weak; benefit skews to acute-stress anxiety reduction. Clears MCID: Borderline - PSQI subscale shifts small; unlikely to clear a clinical insomnia MCID. Largest independent trial: Hidese 2019 (n=30) is the cleanest sleep RCT; still small. Population: Healthy stressed adults, anxiety, ADHD boys, schizophrenia adjunct - not primary insomnia.

THE VERDICT **Modest and mostly about relaxation/anxiety, not a hypnotic. Improves subjective sleep-quality subscales in stressed/anxious adults; weak-to-absent effect on objective sleep latency. Reasonable for stress-linked poor sleep, low harm; not a fix for clinical insomnia.**

BEST FORM L-theanine specifically (branded Suntheanine is the pure L-isomer used in trials), 100-200 mg. Avoid cheap 'theanine' that is DL/racemic. (L-theanine (Suntheanine) = active natural form; DL-theanine = 50% less-useful D-isomer per label weight.)

WATCH Non-sedating relaxation; good daytime or to take the edge off caffeine, and can aid sleep onset at night. Very safe, minimal interactions. May mildly lower blood pressure. Effect is subtle, don't expect a stimulant kick.

Sources: Hidese S 2019, Nutrients, PMID 31623400 (200 mg RCT, PSQI); Rao TP / Williams JL L-theanine sleep systematic review; 2025 systematic review of L-theanine on sleep, PMID 41176609 (DOI 10.1080/1028415X.2025.2556925)

2.5 Lavender oil - Silexan (*Lavandula angustifolia*) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: sleep · **literature compromised (funding/outlier)** | Silexan is sold as a licensed OTC/Rx drug (Lasea) in Europe; US Amazon carries generic lavender-oil softgels rather than Silexan itself.

THE NOISE Marketed as a clinically proven oral lavender-oil capsule that relieves anxiety and restlessness and thereby improves disturbed sleep, without sedation or dependence.

THE SCIENCE Effect: Kasper 2014 GAD: HAMA reduction superior to placebo, comparable to paroxetine (HAMA -14.1 vs paroxetine -11.3 vs placebo -9.5). Sleep sub-scores and PSQI improved secondary to anxiolysis. Effect sizes moderate for anxiety; sleep effect derivative. Clears MCID: For anxiety: YES (HAMA change clinically meaningful). For sleep as a primary endpoint: not established independent of anxiety. Largest independent trial: No large sponsor-independent Silexan trial identified. Population: GAD, subsyndromal/subthreshold anxiety, 'restlessness and disturbed sleep' out-patients (n=170) - anxious populations, not primary insomnia.

THE VERDICT The best-evidenced botanical here - but it is an ANXIOLYTIC, and sleep improvement is downstream of anxiety relief, not a primary hypnotic action. Multiple RCTs (incl. vs paroxetine) show real HAMA reductions in GAD/subsyndromal anxiety; disturbed sleep improves as the anxiety lifts. Most trials are manufacturer-funded (Schwabe), which tempers confidence. Reasonable if the problem is anxiety-driven sleeplessness.

BEST FORM Silexan specifically (patented 80 mg oral lavender oil softgel; sold as Lasea/CalmAid) - the exact form used in the clinical trials. Generic lavender oil capsules are not equivalent. (Silexan 80 mg standardized softgel (trial-validated for anxiety/sleep) > generic ingested lavender oil (unstandardized) >> topical/aromatherapy lavender (weaker evidence for sleep))

WATCH Take with water, not a large fatty meal that worsens reflux; the main side effect is lavender-scented eructation (burping). Very low interaction profile. Reports linking lavender to gynecomastia are weak/contested.

Sources: Kasper S 2014, Int J Neuropsychopharmacol, DOI 10.1017/S1461145714000017 (vs paroxetine & placebo); Kasper S 2015, Eur Neuropsychopharmacol S0924977X15002424 (restlessness/disturbed sleep, n=170); Yap WS 2019, Sci Rep, DOI 10.1038/s41598-019-54529-9 (network meta-analysis)

2.5 Magnesium glycinate (bisglycinate) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: sleep

THE NOISE Marketed as a highly absorbable, gentle-on-the-stomach magnesium that calms the nervous system, relaxes muscles and promotes deep, restful sleep.

THE SCIENCE Effect: Pooled SOL -17.36 min in older adults (Mah 2021, 3 RCTs n=151, but low quality). Bisglycinate 2024: ISI reduction vs placebo at wk 4, Cohen's d ~0.2 (small). Clears MCID: Borderline/NO - the elderly SOL figure would clear MCID but rests on low-quality trials; the healthy-adult d=0.2 does not. Largest independent trial: Rawji 2024 (n=155, healthy poor sleepers) - small ISI benefit, larger in low-intake subgroup. Population: Older adults with insomnia (Abbasi 2012); healthy adults reporting poor sleep (Rawji 2024). Benefit skews to low-baseline-intake subgroups.

THE VERDICT Weak, condition-dependent. A reasonable low-risk trial IF dietary magnesium is low; do not expect a hypnotic effect in a replete adult. Effect sizes are small and the older-adult meta-analysis is explicitly rated substandard by its own authors.

BEST FORM Magnesium bisglycinate - the glycine-chelated form is well absorbed, gentle on the gut, and the glycine itself is mildly calming, making it the go-to for sleep. Dose by ELEMENTAL magnesium (~200-400 mg elemental at night). (Bisglycinate / glycinate (well-absorbed, non-laxative, calming) & citrate/malate (good absorption, citrate mildly laxative) >> magnesium OXIDE (~4x lower absorption, mostly a laxative). Threonate is separate (brain-targeted). Oxide is the cheap filler to avoid for sleep.)

WATCH Read ELEMENTAL magnesium, not the chelate weight - '1000 mg magnesium glycinate' may be only ~140 mg elemental. Overshooting causes loose stools (glycinate least likely). Separate from thyroid meds, bisphosphonates, and certain antibiotics (tetracyclines/fluoroquinolones) by 2+ hours. Caution in renal impairment. UL for supplemental magnesium is 350 mg elemental (GI-based, not toxicity).

Sources: Abbasi B 2012, J Res Med Sci, PMID 23264875 (elderly primary insomnia RCT); Mah J & Pitre T 2021, BMC Complement Med Ther, DOI 10.1186/s12906-021-03297-z (authors call evidence 'substandard'); Rawji A 2024/2025, Nat Sci Sleep, DOI 10.2147/NSS.S524348 (bisglycinate RCT n=155)

2.5 Montmorency tart cherry (*Prunus cerasus*) Amazon ✓ Website ✓

Evidence: weak/mostly null (low confidence) | Effect: sleep · **literature compromised (funding/outlier)**

THE NOISE Promoted as a natural source of melatonin and anthocyanins that increases sleep time and efficiency while also aiding exercise recovery.

THE SCIENCE Effect: Howatson 2012: +total sleep time and +sleep efficiency, raised urinary 6-sulfatoxymelatonin (n=20 healthy crossover). Losso 2018: +84 min sleep time on PSG but n=8 pilot - hypothesis-generating only. Pigeon 2010: modest ISI improvement. Clears MCID: Possibly in the tiny insomnia pilots, but n=8-15 with wide uncertainty - not decision-grade. Largest independent trial: 2024 MDPI Nutrients RCT (tart cherry powder, overweight/obese) - NO improvement in sleep or inflammation. Population: Healthy adults (Howatson n=20), older adults with insomnia (Losso n=8; Pigeon n=15). Null in overweight/obese (2024).

THE VERDICT Thin and small-study-driven. A few small trials show more sleep time/efficiency and raised urinary melatonin, but samples are tiny (n=8-20), some industry-linked, and a 2024 RCT in overweight/obese adults was null. High-anthocyanin juice concentrate was tested; retail gummies/capsules are nowhere near that dose.

BEST FORM n/a - the fruit IS the form; the point is getting the Montmorency variety and enough of it. Best consumed as unsweetened concentrate (~1 oz twice daily) or a standardized capsule/powder standardized to anthocyanin content; skip sugary cocktail 'cherry juice.' Whole-food matrix (melatonin + anthocyanins + tryptophan) is the active principle. (Montmorency concentrate or standardized extract (concentrated anthocyanins/melatonin) ≈ standardized capsule > diluted/sweetened cherry juice drinks (mostly sugar, little active) > sweet Bing cherries (different variety, lower actives). Use Montmorency specifically.)

WATCH Juice/concentrate carries significant sugar - use unsweetened and mind the calories/glycemic load; capsules avoid this. Provides natural melatonin plus tryptophan and anti-inflammatory anthocyanins (also studied for exercise recovery/gout). Effects are modest and build over ~1-2 weeks. High anthocyanin/polyphenol load - theoretical additive effect with anticoagulants at large doses.

Sources: Howatson G 2012, Eur J Nutr, DOI 10.1007/s00394-011-0263-7; Losso JN 2018, Am J Ther, PMID 28901974 (n=8 PSG pilot); Pigeon WR 2010, J Med Food, PMID 20438325; 2024 Nutrients tart cherry powder RCT, MDPI 2072-6643/16/23/4125 (null)

2 5-HTP (5-hydroxytryptophan) Amazon ✓ Website ✓

Evidence: weak/mostly null (low confidence) | Effect: sleep | OTC in the US; prescription-restricted or banned in some countries (e.g. Australia, Canada limits).

THE NOISE Promoted as a serotonin precursor that lifts mood, curbs appetite and improves sleep by boosting the body's own melatonin.

THE SCIENCE Effect: Singapore older-adult RCT: favorable change in some PSQI components and reduced sleep latency at 4-8 wk, more evident in poor sleepers; raised serum serotonin. Modest, subjective, not PSG. No pooled CI for stand-alone insomnia. Clears MCID: Unclear - subjective subscale changes; no PSG MCID demonstrated. Largest independent trial: Singapore older-adult 5-HTP RCT (Clinical Nutrition 2024, PMID 38309227). Population: Older adults (Singapore RCT), often poor sleepers; some combination-product trials (GABA+5-HTP).

THE VERDICT Thin. No adequately powered stand-alone insomnia RCT; recent evidence is a modest older-adult trial helping poor sleepers on some PSQI components, plus combination products that confound the ingredient. Serotonin-load and historical Peak-X contamination concerns argue for caution. Not decision-grade for sleep.

BEST FORM Griffonia simplicifolia seed extract standardized to 95-99% 5-HTP; an enteric-coated or sustained-release version reduces nausea and smooths delivery. (Standardized Griffonia 5-HTP (enteric/SR to reduce GI upset) - essentially one active form; quality/purity is the real variable)

WATCH Do NOT combine with SSRIs/SNRIs/MAOIs/triptans - serotonin syndrome risk. Start low (nausea is common); take at night. Take away from protein. Historical EMS-contamination concern - buy tested pharma-grade. Avoid with carbidopa unless medically supervised.

Sources: Sutanto CN 2024, Clin Nutr, PMID 38309227 (5-HTP older adults, sleep & gut microbiota); Byun JI 2018, J Clin Neurol (GABA/5-HTP combination - confounded); NIH ODS / LiverTox 5-HTP entries; Peak-X/EMS historical literature

2 Apigenin / Chamomile (*Matricaria chamomilla*) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: sleep · **literature compromised (funding/outlier)**

THE NOISE Marketed as the calming flavonoid in chamomile (popularized as a nightly 'apigenin' sleep aid) that binds relaxation receptors to ease you into sleep.

THE SCIENCE Effect: Zick 2011 (270 mg chamomile BID x28 d, n=34): NO significant difference vs placebo in total sleep time, sleep efficiency, latency, WASO, or quality. 2024 meta: significant PSQI reduction but across heterogeneous populations and low study quality. Clears MCID: NO in primary insomnia (null); questionable in the positive elderly subgroups. Largest independent trial: Zick SM 2011 (rigorous, placebo-controlled, primary insomnia) - NULL. Population: DSM-IV primary insomnia (Zick, null); elderly nursing-home residents and postpartum women (positive).

THE VERDICT Do not buy for insomnia. The best-designed primary-insomnia RCT (Zick 2011) was flatly NULL on every sleep-diary measure. Positive results are in special populations (elderly nursing-home, postpartum) and heterogeneous low-quality meta-analyses. Isolated apigenin has essentially no direct human sleep outcome data - it is a mechanism (GABA-A ligand), not a proven product.

BEST FORM German chamomile (*Matricaria*) extract standardized to ~1.2% apigenin, or a phospholipid/absorption-enhanced apigenin - because pure apigenin is poorly water-soluble with low oral bioavailability. (Standardized apigenin-rich German chamomile extract > chamomile tea (low, variable apigenin) > plain unformulated apigenin powder (poorly absorbed))

WATCH Cross-reactive allergy in ragweed/*Asteraceae*-sensitive people. Contains coumarins - additive with anticoagulants/antiplatelets. Take ~30-60 min before bed. Mildly sedating; additive with other sedatives.

Sources: Zick SM 2011, BMC Complement Altern Med, PMID 21939549 (DOI 10.1186/1472-6882-11-78; NULL primary-insomnia RCT); Abdullahzadeh M 2017 (elderly nursing-home, positive); 2024 chamomile-on-sleep meta-analysis, PMID 39106912

2 GABA (gamma-aminobutyric acid, oral) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: sleep · **literature compromised (funding/outlier)**

THE NOISE Sold as the brain's main calming neurotransmitter in a pill, promising reduced stress and anxiety and faster, more relaxed sleep.

THE SCIENCE Effect: Modest reductions in sleep latency and small non-REM/EEG changes in small trials (100-300 mg); one Western RCT reported sleep-efficiency and HRV improvement. No robust PSG replication; CIs not meaningful given size. Clears MCID: NO / unclear - endpoints soft and small. Largest independent trial: No convincing independent PSG trial; no direct evidence of human BBB crossing. Population: Small Japanese RCTs; sedentary overweight women (one Western RCT); healthy adults.

THE VERDICT Do not buy. Oral GABA poorly crosses the blood-brain barrier; the small positive sleep RCTs are largely industry-funded (PharmaGABA/Pharma Foods), soft-endpoint, and any effect likely runs through peripheral gut-vagal signaling, not central inhibition. Mechanism outruns outcome.

BEST FORM PharmaGABA (naturally fermented via *Lactobacillus*) is the better-studied retail form vs synthetic GABA, though the difference is modest. Take sublingually or on an empty stomach. (PharmaGABA (fermented, some human sleep data) > synthetic GABA (cheaper, thinner evidence) - both share the same core limitation)

WATCH Oral GABA crosses the blood-brain barrier poorly; benefits likely act via the enteric/vagal nervous system, so effects are inconsistent. Take on an empty stomach. Can cause transient flushing/tingling and may lower blood pressure at higher doses. Additive with other sedatives.

Sources: Yamatsu A 2016, J Nutr Sci Vitaminol (PharmaGABA sleep latency); Byun JI 2018, J Clin Neurol (GABA/5-HTP combination - confounded by co-ingredient); Boonstra E 2015, Front Psychol (review: oral GABA BBB uncertainty)

2 Kava (Piper methysticum) Amazon ✓ Website ✓

Evidence: weak/mostly null (high confidence) | Effect: sleep | Carries an FDA hepatotoxicity advisory and was banned or restricted in parts of Europe; sold OTC in the US.

THE NOISE Marketed as a South Pacific root that promotes relaxation, relieves stress and anxiety and eases you toward sleep.

THE SCIENCE Effect: Anxiety: significant HAMA reduction vs placebo (Cochrane weighted mean difference favoring kava). Sleep-specific outcome data are thin and secondary. Clears MCID: For anxiety possibly; for sleep as a primary endpoint - not established. Largest independent trial: Kava-anxiety Cochrane pooled analysis (Pittler & Ernst 2003). Population: Generalized/situational anxiety adults (efficacy); sleep is not the primary studied endpoint.

THE VERDICT Recommend AGAINST for sleep. Efficacy evidence is for ANXIETY (Cochrane-level, kavain), not sleep per se, and it is overshadowed by a serious hepatotoxicity signal: FDA 2002 consumer advisory, temporary bans in Germany/UK/Switzerland, >100 suspected severe liver-injury cases including transplants and deaths. The risk-benefit for an OTC sleep aid is unacceptable.

BEST FORM Noble-cultivar kava, WATER-extracted from the ROOT/rhizome, standardized to kavalactones - NOT acetone/ethanol extracts and NOT aerial parts (stems/leaves), which drive hepatotoxicity. (Noble cultivar + water extraction of root (traditional, safest) > standardized noble root extract > acetonic/ethanolic extracts and non-noble/aerial material (higher liver risk - avoid))

WATCH Hepatotoxicity risk: avoid alcohol and other hepatotoxic drugs, limit duration, avoid with liver disease. Strong CYP450 inhibition (many drug interactions). Additive sedation - don't drive. Chronic heavy use causes reversible scaly 'kava dermatopathy.'

Sources: Pittler MH & Ernst E 2003, Cochrane Database Syst Rev (kava for anxiety); FDA 2002 Consumer Advisory on kava-containing supplements; FDA 2023 scientific memorandum (fda.gov/media/169556); NIH LiverTox Kava, NBK548637

2 Magnesium L-threonate (Magtein) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: sleep · **literature compromised (funding/outlier)**

THE NOISE Sold as the only form of magnesium that crosses the blood-brain barrier, raising 'brain magnesium' to sharpen memory and cognition and improve sleep.

THE SCIENCE Effect: Subjective sleep and daytime-functioning improvements; wearable (Oura) deltas. No polysomnography. Effect sizes small and endpoint-soft; headline claims (e.g. '7.5-year brain-age reduction') are cognitive, not sleep, and industry-reported. Clears MCID: NA - no validated objective sleep MCID reported; subjective-only. Largest independent trial: None identified - this is the core problem. Population: Adults 18-45 with self-reported dissatisfied sleep (Lopresti 2024/2026); prior work in older adults.

THE VERDICT Do not pay the premium. Every supportive sleep/cognition trial is tied to the ingredient owner (Magtein/Threotech/Neurocentria) and relies on subjective or wearable endpoints. No independent replication. The 'brain-bioavailable magnesium' story is marketing ahead of independent outcome data.

BEST FORM Magnesium L-threonate marketed as Magtein - the specific form shown to raise brain magnesium (crosses the blood-brain barrier) for cognition/sleep-quality claims. Full studied dose is ~1.5-2 g of the compound (delivering only ~144 mg elemental), often split or taken before bed. (L-threonate/Magtein (uniquely brain-penetrant, targeted at CNS/cognition/sleep - but LOW elemental magnesium per gram and expensive) vs glycinate/citrate (far more elemental magnesium per dollar for whole-body repletion). Threonate is a niche cognitive tool, not a general magnesium source.)

WATCH Very low elemental magnesium (~8%), so it is a poor choice for correcting a systemic deficiency - use it for the CNS effect only. Costly and requires the full ~2 g/day. Some feel it activating - trial timing. Same drug-spacing and renal cautions as other magnesium. Don't double-count it as your magnesium RDA.

Sources: Lopresti AL & Smith SJ 2026, Front Nutr, PMID 41601871 (DOI 10.3389/fnut.2025.1729164; Magtein supplied by Threotech); Hausenblas HA / Lopresti 2024, J Affect Disord Rep, ScienceDirect S2590142724000193 (MgT self-reported sleep RCT)

2 Passionflower (*Passiflora incarnata*) - added, in-category Amazon ✓ Website ✓

Evidence: weak/mostly null (low confidence) | Effect: sleep

THE NOISE Herbal calm/sleep/anti-anxiety aid.

THE SCIENCE Effect: Ngan 2011: small but significant improvement in subjective sleep-quality diary rating vs placebo tea; no objective PSG effect. Underpowered. Clears MCID: NO - subjective-only, small. Largest independent trial: Ngan & Conduit 2011, *Phytother Res*. Population: Healthy adults with mild sleep complaints (Ngan & Conduit 2011, n=41 tea crossover).

THE VERDICT Thin. One small positive tea RCT and a few low-quality trials; frequently bundled with valerian, confounding attribution. Not decision-grade as a stand-alone sleep aid.

BEST FORM Standardized *Passiflora incarnata* aerial-part extract (or a well-made tincture/tea) - species and plant part matter; use *P. incarnata*, not ornamental *Passiflora* species. (Standardized *P. incarnata* extract > tincture > tea (mild) - unstandardized products vary widely in flavonoid content)

WATCH Additive sedation with alcohol, benzodiazepines, and other sleep aids. Avoid in pregnancy (theoretical uterine stimulation). Take ~30-60 min before bed. Generally well tolerated; occasional drowsiness/dizziness.

Sources: Ngan A & Conduit R 2011, *Phytother Res*, PMID 21294203 (DOI 10.1002/ptr.3400)

2 Valerian (*Valeriana officinalis*) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: sleep · literature compromised (funding/outlier)

THE NOISE Sold as a natural herbal sedative that helps you fall asleep faster and sleep more soundly without next-day grogginess or dependence.

THE SCIENCE Effect: Bent 2006: RR 1.8 (95% CI 1.2-2.9) for dichotomous 'improved sleep' - an outcome prone to response bias; objective PSG endpoints largely null. 2024 umbrella review: no consistent efficacy. Clears MCID: NO - objective sleep parameters show no reliable clinically meaningful change. Largest independent trial: Independent objective-endpoint trials (Diaper & Hindmarch) are null on PSG. Population: Mixed insomnia/poor-sleep adults; highly heterogeneous extracts and doses.

THE VERDICT Do not buy for efficacy. The 2024 umbrella review found NO evidence of efficacy for insomnia. Older positive meta-analyses rest on a subjective dichotomous 'improved sleep?' outcome, heterogeneous and poorly standardized extracts, and industry involvement. Good safety is not the same as working.

BEST FORM A root extract STANDARDIZED to valerenic acids (~0.8%), 300-600 mg of extract ~30-60 min before bed. Aqueous/hydroalcoholic extracts of the root are the studied preparations; standardization is what makes dosing meaningful. (Standardized valerenic-acid root extract (consistent, studied) > generic valerian root powder (variable potency, unreliable) > loose teas (dose uncertain). Potency varies enormously batch to batch, so standardization matters more than brand.)

WATCH Notorious sulfur/sweaty-sock smell - capsules help. Effects often build over ~2 weeks rather than working the first night. Additive sedation with alcohol, benzodiazepines, and other CNS depressants. Rare reports of hepatotoxicity - avoid with liver disease and don't combine with other hepatotoxic herbs (kava). Can cause morning grogginess or, paradoxically, stimulation in some.

Sources: Bent S 2006, *Am J Med*, PMID 17145239 (meta-analysis, dichotomous outcome); Baglioni C et al. umbrella review 2024, *Eur Neuropsychopharmacol*, DOI 10.1016/j.euroneuro.2024.01.008 ('no evidence of efficacy'); Taibi DM 2007, *Sleep Med Rev* (critical review)

Muscle / strength (12)

5 Creatine monohydrate [Amazon ✓](#) [Website ✓](#)

Evidence: strong (high confidence) | Effect: muscle / strength

THE NOISE Marketed as the go-to strength supplement that increases muscle power, size, and training performance, now also pushed for brain and healthy-aging benefits.

THE SCIENCE Effect: Added to RT vs RT+placebo: lean mass +1.1 kg; upper-body strength ~+4.4 kg, lower-body ~+11 kg (1RM). Branch 2003 pooled ES ~0.24 (95% CI 0.16-0.31) for strength/performance. Clears MCID: Yes - a ~1 kg LBM and multi-kg 1RM increment over placebo across a training block is a meaningful add-on. Largest independent trial: Numerous independent university RCTs (e.g. Candow/Chilibeck aging trials, Volek 1999). No single trial carries the grade. Population: Healthy young and older adults doing resistance training; both sexes (larger LBM effect in males). Also studied in sarcopenia and clinical wasting.

THE VERDICT **Buy. The single best-evidenced sports supplement. Large, mostly independent RCT base shows a real, clinically meaningful add-on to resistance training. Cheap, off-patent, well-tolerated.**

BEST FORM Plain creatine monohydrate is the gold standard - micronized for easier mixing, and Creapure-grade for purity if you want third-party assurance. Fancier 'forms' are not worth paying for. (Creatine monohydrate BEST (most-studied, cheapest, proven); HCl, buffered 'Kre-Alkalyn', ethyl ester and 'nitrate' forms cost more with NO demonstrated bioavailability or performance advantage - some (ethyl ester) are actually worse.)

WATCH 3-5 g/day is sufficient; a loading phase is optional, not required. Expect ~1-2 kg water-weight gain early. It raises serum creatinine harmlessly - warn your doctor so it isn't misread as kidney injury (interferes with that lab interpretation). Stay hydrated; take consistently, timing doesn't matter much.

Sources: Kreider 2017 ISSN Position Stand, J Int Soc Sports Nutr 14:18 (PMID 28615996); Branch 2003 meta-analysis, Int J Sport Nutr Exerc Metab 13:198 (PMID 12945830); Chilibeck 2017 older-adults meta, Open Access J Sports Med 8:213 (PMID 29138605); Delpino 2022 Clin Nutr 41:551 (PMID 35081505)

4.5 Whey / protein powder [Amazon ✓](#) [Website ✓](#)

Evidence: strong (high confidence) | Effect: muscle / strength

THE NOISE Marketed as essential fuel that builds muscle, speeds recovery and 'unlocks gains' with every scoop, implying more protein always means more muscle.

THE SCIENCE Effect: Morton 2018: +2.49 kg 1RM strength (95% CI 0.64-4.33), +0.30 kg fat-free mass (95% CI 0.09-0.51), +310 μ m² fiber CSA. Benefit plateaus above 1.62 g/kg/day. Clears MCID: Strength effect yes; the +0.30 kg FFM increment is small and only meaningful because it compounds over months. Largest independent trial: Morton 2018 pools many independent university RCTs; no single-study dependence. Population: Healthy adults on prolonged resistance training; effect attenuates with age, larger in already-trained lifters.

THE VERDICT **Buy only if you are not already hitting your protein target from food. It is a convenient food, not a magic compound - the benefit vanishes once total intake reaches ~1.6 g/kg/day.**

BEST FORM Whey isolate if you're lactose-sensitive or watching fat/carbs (>90% protein, minimal lactose); whey concentrate is perfectly fine and cheaper if you tolerate dairy. Prioritize leucine/EAA content and a third-party purity seal over 'form' hype. (Isolate (low-lactose, leaner) and concentrate (cheaper, some lactose) both GOOD; hydrolysate is fastest-absorbing but pricey with marginal real-world benefit; casein is slower (good pre-sleep). Collagen is NOT a muscle-building protein (low leucine, incomplete).)

WATCH Concentrate contains lactose (GI upset in intolerant users - choose isolate). Watch added sugars and proprietary 'amino spiking'. Choose NSF Certified for Sport / Informed Sport tested products to avoid heavy-metal and banned-substance contamination. High intake is fine for healthy kidneys but discuss if you have kidney disease.

Sources: Morton 2018 meta-analysis, Br J Sports Med 52:376 (PMID 28698222)

4 Caffeine Amazon ✓ Website ✓

Evidence: good (high confidence) | Effect: muscle / strength

THE NOISE Sold as a pre-workout ergogenic that boosts energy, focus, strength, power and endurance and delays fatigue.

THE SCIENCE Effect: Grgic 2018 meta: strength Hedges g 0.20, power g 0.17 (both significant). Upper-body strength significant; lower-body less clear. Small effect, large evidence base. Clears MCID: Modestly - a small acute uplift that is genuine and repeatable, meaningful for competition/hard sessions. Largest independent trial: Many independent university crossover RCTs; effect is robust to any single study's removal. Population: Trained and untrained adults, acute pre-exercise dosing; strength and power tasks.

THE VERDICT Buy as an acute ergogenic. One of the few genuinely proven performance aids - small but real, independently replicated strength/power effect. Cheap, generic, well-characterized.

BEST FORM n/a for efficacy - caffeine anhydrous (capsule/tablet) is preferred only because it lets you dose precisely; coffee works too but its content varies widely (Caffeine anhydrous = precise, cheap; coffee = same active but variable dose; 'dicafeine malate/Infinergy' and extended-release give smoother curves but no extra ergogenic effect)

WATCH Ergogenic at ~3-6 mg/kg ~45-60 min pre-exercise. Half-life ~5 h (longer in slow CYP1A2 metabolizers, pregnancy, some meds) - avoid within ~8-10 h of sleep. Tolerance builds; cycle if needed. Raises BP/anxiety; caution stacking stimulants. Keep total daily intake under ~400 mg for most adults; bulk powdered caffeine is dangerous to eyeball-dose.

Sources: Grgic 2018 meta, J Int Soc Sports Nutr 15:11 (PMID 29527137); Grgic 2020 meta-of-metas, Sports Med (PMID 32488520)

3.5 Beta-alanine Amazon ✓ Website ✓

Evidence: moderate/conditional (high confidence) | Effect: muscle / strength

THE NOISE Marketed (via the signature skin-tingle) as a buffer that delays fatigue and boosts strength, power, endurance and muscle growth across the board.

THE SCIENCE Effect: Saunders 2017 (Br J Sports Med, 40 studies): median ~2.85% performance improvement; pooled effect concentrated in 60-240 s efforts (ES ~0.18, 95% CI ~0.08-0.31). Minimal/absent effect on maximal strength. Clears MCID: Yes for high-intensity endurance/repeated-effort tasks; no for maximal strength. Largest independent trial: Multiple independent carnosine-loading trials (Hill 2007, Hoffman); no single-study dependence. Population: Trained and untrained adults in high-intensity/anaerobic tasks; buffering-limited efforts.

THE VERDICT Buy only for its real domain: high-intensity efforts lasting ~1-4 minutes (repeated sprints, rowing, CrossFit-style work). It is NOT a maximal-strength or hypertrophy aid - marketing overreaches on that.

BEST FORM Sustained-release beta-alanine (e.g. CarnoSyn SR), or plain beta-alanine split into smaller doses; the molecule is identical, but SR/split dosing is the meaningful choice (Sustained-release CarnoSyn > standard beta-alanine dumped in one dose (same efficacy, far worse tingling) >> 'buffered/instant' marketing variants (no advantage))

WATCH Works by raising muscle carnosine over WEEKS, not acutely - needs chronic ~3.2-6.4 g/day, timing irrelevant. Single large doses cause harmless paresthesia (skin tingling); split into ≤1.6 g doses or use SR to avoid it. Take with a meal. No acute pre-workout benefit.

Sources: Saunders 2017 meta, Br J Sports Med 51:658 (PMID 27797728); Hill 2007 Amino Acids 32:225 (PMID 16868650)

3.5 Essential amino acids (EAA) Amazon ✓ Website ✓

Evidence: moderate/conditional (medium confidence) | Effect: muscle / strength

THE NOISE Sold as a superior, faster-absorbing way to 'maximize muscle protein synthesis' that beats regular protein for building and preserving muscle.

THE SCIENCE Effect: Robust acute MPS elevation; chronic hypertrophy RCTs are few and largely redundant with adequate protein. No pooled long-term effect size that beats whole protein. Clears MCID: na - acute signalling endpoint, not a clinical outcome; long-term outcome data insufficient to judge MCID. Largest independent trial: Acute crossover MPS trials (e.g. leucine-enriched EAA, n~20-30); no large chronic independent hypertrophy RCT. Population: Acute MPS studies in young and older adults; chronic hypertrophy data thinner and often in older/low-protein populations.

THE VERDICT Conditional. Genuinely stimulates muscle protein synthesis, but offers no proven hypertrophy advantage over cheaper whole protein for a healthy adult eating enough. Useful mainly in older adults or when protein intake is low.

BEST FORM A full-spectrum, free-form EAA blend containing all nine essential amino acids with a high leucine share (~2.5-3 g leucine per serving). Free-form EAAs are rapidly absorbed and drive muscle protein synthesis better than BCAAs. (Complete 9-EAA blends GOOD (contain everything needed to actually build muscle protein); BCAA-only products INFERIOR (missing the other 6 EAAs, can't sustain synthesis); whole-protein/whey is the cheaper 'form' of EAAs for most people.)

WATCH Largely redundant if you already hit ~1.6 g/kg/day total protein - they mainly help when whole protein is impractical (fasted training, elderly, very low appetite). Free-form aminos taste bitter/sour. Check that all nine EAAs are present and leucine is adequate, not just a token amount.

Sources: Wolfe 2017 review, J Int Soc Sports Nutr 14:30 (PMID 28852372); Jackman 2017 Front Physiol 8:390 (PMID 28638350)

3 Vitamin D (strength in deficiency) Amazon ✓ Website ✓

Evidence: moderate/conditional (high confidence) | Effect: muscle / strength

THE NOISE Pitched as a universal strength, testosterone, immunity and athletic-performance booster that everyone should megadose year-round.

THE SCIENCE Effect: Deficient (<25 nmol/L): large effect on hip/proximal strength; athletes <75 nmol/L: small strength effect (2023 Frontiers meta, 11 RCTs, n=436). Replete: no grip-strength effect. Falls reduced by active analogues in elderly. Clears MCID: Yes in deficiency (proximal weakness reversal, fall reduction); no in repletion. Largest independent trial: Burt 2019 (Calgary, JAMA 322:736, n=311) - independent: 4000 and 10000 IU/day LOWERED radial/tibial bone density vs 400 IU, demonstrating the wrong-direction dose-response and repletion harm. Population: Deficient/older adults (25(OH)D <25-50 nmol/L) and athletes with low baseline; distinct from replete healthy adults.

THE VERDICT **Conditional buy - correct a measured deficiency, then stop. Real strength/fall benefit only in deficient people; null in the already-replete, and high doses actively harm. This is a repletion story, not an ergogenic one.**

BEST FORM Vitamin D3 (cholecalciferol) in an oil-based softgel, taken with a fat-containing meal; ideally paired with vitamin K2 and adequate magnesium (D3 (cholecalciferol) raises and sustains 25-OH-D better than D2 (ergocalciferol); oil softgel/liquid > dry tablet for absorption)

WATCH Fat-soluble - take with the day's fattiest meal. Benefit for strength is mainly correcting deficiency, not boosting the replete. Test a 25-OH-D level rather than guessing; routine UL ~4000 IU/day. Magnesium is a cofactor; K2 helps direct calcium. Excess causes hypercalcemia. High-dose biotin can distort some vitamin D immunoassays.

Sources: Burt 2019 JAMA 322:736 (PMID 31454046); 2023 athletes meta, Front Nutr 10:1163313 (PMID 37841405); Beaudart 2014 muscle-strength meta, J Clin Endocrinol Metab (PMID 25162434)

2.5 Betaine anhydrous Amazon ✓ Website ✓

Evidence: weak/mostly null (low confidence) | Effect: muscle / strength · **literature compromised (funding/outlier)**

THE NOISE Advertised as boosting power, strength, muscle size and body composition by supporting cellular hydration and creatine synthesis.

THE SCIENCE Effect: Pooled maximal strength SMD 0.47 (95% CI 0.04-0.89); lower-body SMD 0.49 (95% CI 0.01-0.98) - both lower bounds essentially at zero. No effect on upper-body strength, power, or endurance. Ashtary-Larky 2022: no body-composition effect. Clears MCID: Borderline - point estimate is moderate but the near-zero CI bound means the true effect may be trivial. Largest independent trial: No decisive large independent RCT; base is small early-phase studies. Population: Young resistance-trained men, short (1-6 week) trials.

THE VERDICT **Skip pending better data. A borderline-significant strength signal exists but rests on small trials with a confidence interval nearly touching zero, and there is no body-composition benefit.**

BEST FORM Betaine anhydrous (trimethylglycine/TMG), 2.5 g/day - NOT betaine HCl, which is a stomach-acid product with far less TMG per gram (Betaine anhydrous (TMG) = the studied performance form > betaine HCl (sold for digestion/low stomach acid, much less usable betaine, acidic) - do not substitute)

WATCH Take with food, daily; workout timing doesn't matter. Lowers homocysteine but may nudge LDL/total cholesterol up in some people. Mild GI or 'fishy' body odor at high doses. Plausible synergy with creatine but not required.

Sources: Betaine strength meta-analysis (Gao/Cholewa lineage); Ashtary-Larky 2022 body-composition meta, Br J Nutr (PMID via Cambridge Core, failed body-comp)

2.5 Citrulline malate Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: muscle / strength

THE NOISE Sold as a nitric-oxide 'pump' booster that increases blood flow, power output, endurance and reduces muscle soreness.

THE SCIENCE Effect: Vialpando/Varvik 2021 meta (8 studies, n=137): +6.4% reps to failure, SMD 0.196 (small); subgroups non-significant and authors concluded evidence does not support a recommendation. Trujillo-Colmena/three-level metas confirm SMD ~0.2 (95% CI ~0.01-0.39), fragile. Clears MCID: No - a fraction of a rep advantage that does not reliably exceed noise. Largest independent trial: Varkik 2021 pooled analysis is the best synthesis; no single large independent confirmatory trial exists. Population: Mostly young resistance-trained men, acute pre-exercise dosing.

THE VERDICT **Borderline / skip. The vasodilator pump is real, but the performance benefit is small, inconsistent, and the meta-analysis authors themselves declined to recommend it. Effect sits near the null.**

BEST FORM Either plain L-citrulline (~6 g) or citrulline malate 2:1 (~8 g to deliver ~6 g citrulline) - just confirm the label states the ratio so you know the actual citrulline dose (L-citrulline / citrulline malate 2:1 (label states ratio) > citrulline malate 1:1 (more malate filler, less citrulline) >> L-arginine (destroyed first-pass; citrulline raises plasma arginine more effectively))

WATCH 'Malate' is filler weight - an 8 g 2:1 scoop is only ~5.3 g citrulline, so dose to the citrulline fraction. Take ~60 min pre-workout on an empty-ish stomach. Well tolerated; high doses can cause mild GI upset. Theoretical additive hypotension with nitrates/PDE5 inhibitors.

Sources: Varkik 2021 meta, Int J Sport Nutr Exerc Metab 31:350 (PMID 34010809); Trujillo-Colmena 2025 three-level meta, Nutrients 18:1881

2 Ashwagandha (Withania somnifera) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: muscle / strength · **literature compromised (funding/outlier)** | Widely sold; some regulators flag liver/thyroid/reproductive safety concerns.

THE NOISE Marketed as an adaptogen that dramatically raises strength, muscle mass and testosterone while slashing cortisol and stress.

THE SCIENCE Effect: Wankhede 2015: bench-press 1RM +46 kg vs +26 kg placebo and large testosterone/muscle-size gains - effect sizes implausibly large for 8 weeks and unreplicated independently. Clears MCID: Nominally yes in the sponsor trial, but the result is not trustworthy enough to credit. Largest independent trial: No adequately powered independent strength RCT exists; the evidence base is sponsor-dependent. Population: Young men with little resistance-training experience (Wankhede 2015, n=57); small single-site trials.

THE VERDICT **Do not buy for muscle/strength. The headline strength gains come from extract-manufacturer-funded trials with implausibly large effects and no independent replication - and multiple regulators have now flagged liver, thyroid and reproductive safety.**

BEST FORM A standardized root extract with a named clinical grade - KSM-66 (root-only, ~5% withanolides) or Sensoril (root+leaf, higher withanolide %) - not raw non-standardized root powder (KSM-66 / Sensoril standardized extracts (dosed to withanolide content) > generic 'ashwagandha root powder' (unstandardized, unknown potency, needs much larger doses))

WATCH Rare but real reports of liver injury - stop if jaundice/RUQ pain. Can raise thyroid hormones (caution if hyperthyroid or on levothyroxine). Sedating - additive with CNS depressants; may lower blood sugar and blood pressure. Avoid in pregnancy and with autoimmune-thyroid concerns. Take with food.

Sources: Wankhede 2015, J Int Soc Sports Nutr 12:43 (PMID 26609282) - industry-funded; Bjornsson 2020 liver-injury case series, Liver Int (PMID 31879990); NIH LiverTox NBK548536; EU HoA Article 8 prioritization 2024; Danish/Icelandic advisories 2023

2 HMB (beta-hydroxy-beta-methylbutyrate) [Amazon](#) ✓ [Website](#) ✓

Evidence: weak/mostly null (medium confidence) | Effect: muscle / strength · **literature compromised (funding/outlier)**

THE NOISE Promoted as an anti-catabolic agent that preserves lean mass, accelerates strength gains and boosts recovery in hard-training lifters.

THE SCIENCE Effect: Sanchez-Martinez 2018 and Jakubowski 2019: no significant body-composition or strength effect in young/trained. The oversized Wilson/Lowery 2014 HMB-FA result (large FFM/strength gains) is industry-linked and widely regarded as implausible. Older-adult metas show small, heterogeneous effects. Clears MCID: No in healthy young adults; marginal at best in sarcopenia. Largest independent trial: Jakubowski 2019 (J Appl Physiol / MSSE-adjacent, resistance-trained men, no benefit over placebo) and Teixeira/Sanchez-Martinez independent work - null. Population: Young trained/untrained (null in independent work) vs older adults and sarcopenia/clinical wasting (small positive signal).

THE VERDICT **Do not buy for hypertrophy/strength in healthy trained adults. The large effects come from manufacturer-linked studies; independent replications in young trainees are null. A modest signal survives only in older/sarcopenic and clinical-wasting populations.**

BEST FORM HMB free acid (HMB-FA, the gel-cap liquid form) reaches a higher, faster plasma peak and better retention than the calcium salt. If only the salt is available, calcium-HMB works but is absorbed more slowly. (HMB free acid (HMB-FA) BETTER (superior pharmacokinetics, quicker uptake); calcium-HMB (Ca-HMB) salt is the older, cheaper, slower-absorbed form used in most tubs.)

WATCH Standard dose is 3 g/day, split. Benefits are clearest in untrained beginners, older adults, and catabolic/muscle-wasting states - effects in well-trained lifters are modest at best. Take with meals; time a dose near training. Ca-HMB adds a little calcium to your daily total. Generally very safe.

Sources: Sanchez-Martinez 2018 meta, J Sci Med Sport 21:727 (PMID 29249685); Jakubowski 2019 (independent RCT, resistance-trained); Wilson/Lowery 2014 Eur J Appl Physiol 114:1217 (industry-linked outlier, PMID 24599749)

1.5 BCAAs (branched-chain amino acids) [Amazon](#) ✓ [Website](#) ✓

Evidence: none / marketing (high confidence) | Effect: muscle / strength

THE NOISE Hyped as intra-workout fuel that prevents muscle breakdown, slashes soreness and triggers growth via leucine.

THE SCIENCE Effect: Jackman 2017: BCAA-alone raised post-exercise MPS ~22% but that is ~50% lower than whole-protein/EAA. IV BCAA trials showed DECREASED muscle protein turnover. No hypertrophy outcome advantage. Clears MCID: No - fails to produce a sustained anabolic outcome; the acute bump is a null for real hypertrophy endpoints. Largest independent trial: Jackman 2017 (n=10 crossover) - the definitive mechanistic study, independent, showing sub-maximal response. Population: Acute MPS studies in young men; a few soreness/recovery trials.

THE VERDICT **Do not buy. Mechanism outruns outcome, and it is actively inferior to whole protein or EAA. Isolated BCAAs cannot sustain muscle protein synthesis because the other essential amino acids are missing.**

BEST FORM Honestly, skip standalone BCAAs in favor of complete EAAs or whey - BCAAs alone cannot sustain muscle protein synthesis without the other essential aminos. If you still use them, the standard 2:1:1 leucine:isoleucine:valine ratio is fine; higher-leucine ratios add little. (2:1:1 ratio is the researched standard; 4:1:1 / 8:1:1 'high-leucine' versions are marketing with no clear added benefit. All BCAA-only products are INFERIOR to EAAs/whole protein for muscle building.)

WATCH The key caveat is efficacy, not chemistry: with adequate total protein, BCAAs add essentially nothing and are an overhyped category. They can compete with tryptophan uptake. Many products are underdosed or flavored with lots of sweeteners. Spend the money on whey or EAAs instead.

Sources: Wolfe 2017, J Int Soc Sports Nutr 14:30 (PMID 28852372); Jackman 2017, Front Physiol 8:390 (PMID 28638350)

1 Testosterone boosters (Tribulus, D-aspartic acid, fenugreek blends) Amazon ✓ Website ✓

Evidence: none / marketing (high confidence) | Effect: muscle / strength

THE NOISE Marketed as natural test boosters that spike testosterone, libido, strength and muscle growth in men.

THE SCIENCE Effect: D-aspartic acid: Melville 2015 RCT (n=24) - no rise in testosterone, 6 g/day trended DOWN. Tribulus: Rogerson 2007 and others - no testosterone or performance effect. Pooled: null. Clears MCID: No - no effect to clear it. Largest independent trial: Melville 2015 (independent, D-aspartic acid, n=24) - no benefit, higher dose counterproductive. Population: Mostly young men; a few small RCTs, largely null.

THE VERDICT **Do not buy. Marketing. No credible human data showing raised testosterone or strength/hypertrophy in eugonadal men; the best-controlled trials are null. Included because it dominates this category's shelf space.**

BEST FORM n/a - no retail form reliably raises testosterone in men with normal levels; save your money and fix sleep, training, body fat, and get bloodwork if truly hypogonadal (Tribulus - does NOT raise testosterone (libido at best); D-aspartic acid - inconsistent, some trials show no rise or a drop; fenugreek - may modestly help libido, T rise unconfirmed; proprietary blends - unquantified, often underdosed)

WATCH Proprietary blends hide per-ingredient doses. Some 'boosters' have been adulterated with actual anabolic/SARM compounds - a real risk for drug tests and liver. Tribulus has case reports of hepato/nephrotoxicity. If genuinely low-T, see a clinician for TRT. Insist on third-party testing (NSF/Informed Sport) if used at all.

Sources: Melville 2015, J Int Soc Sports Nutr 12:15 (PMID 25844073); Rogerson 2007 Tribulus RCT, J Strength Cond Res 21:348 (PMID 17530942)

Immunity (14)

3 Probiotics for URTI prevention Amazon ✓ Website ✓

Evidence: moderate/conditional (medium confidence) | Effect: immune / infection

THE NOISE Sold as 'immune-support' probiotics that keep your gut-immune axis strong so you catch fewer colds and respiratory infections.

THE SCIENCE Effect: At least one URTI episode: RR ~0.76 (0.65-0.89). Episode rate ratio 0.79 (0.71-0.89), p<0.001. Also reduced antibiotic use and school absence. GRADE low-to-moderate. Clears MCID: Borderline/modest — meaningful at population level (fewer episodes, less antibiotic use) but individual effect is small. Largest independent trial: Mixed; several independent daycare/school trials contribute alongside sponsor trials. Population: General adults and children (community, daycare, some athletes)

THE VERDICT **Conditional and modest. Cochrane finds probiotics reduce the number of people getting at least one URTI and cut episode rates, but certainty is low-to-moderate, effects are strain-specific, heterogeneity is high, and many trials are industry-funded. A reasonable but not compelling buy, and only specific strains.**

BEST FORM Strain-specific, not just 'CFU count' - choose products naming trial-supported strains (e.g., Lactobacillus rhamnosus GG, L. paracasei/casei, Bifidobacterium animalis lactis, or the L. acidophilus NCFM + B. lactis combo). Multi-strain with guaranteed CFU through expiry. (Documented URTI strains at labeled CFU-through-end-of-shelf-life > high-CFU products with undisclosed/unstudied strains > generic 'probiotic blends'; strain identity matters more than raw CFU)

WATCH Take consistently for weeks - it's preventive, not acute. Separate from antibiotics by ~2 h. Check viability/expiry and storage (refrigerated vs shelf-stable). Rare bloodstream infection risk in severely immunocompromised or critically ill patients - caution there.

Sources: Zhao Y et al. Cochrane Database Syst Rev 2022;CD006895.pub4, PMID 36001877

3 Zinc (acute cold lozenges vs prophylaxis) Amazon ✓ Website ✓

Evidence: moderate/conditional (medium confidence) | Effect: immune / infection

THE NOISE Marketed as an immune mineral, with lozenges promising to stop a cold in its tracks or cut its length if taken at the first sign of symptoms.

THE SCIENCE Effect: Cold duration -33% (95% CI 21-45%); acetate -40%, gluconate -28% (difference NS). No added benefit above ~100 mg/day. 2024 Cochrane (Nault) rated certainty low and was more equivocal; Hemilä published a formal rebuttal. Clears MCID: Yes if formulation is right — roughly 2 fewer days of a 7-day cold clears the threshold Largest independent trial: Hemilä 2017 pooled (independent reanalysis of Prasad, Petrus, Mossad et al.) Population: Adults with naturally acquired common colds, treatment started within 24-48h

THE VERDICT **Conditional buy for TREATMENT only, and only a correctly formulated lozenge. High-dose ionic zinc acetate/gluconate lozenges started within 24h can meaningfully shorten a cold. Most retail products fail on formulation, not the headline mg. Prophylactic daily zinc does little or nothing for colds.**

BEST FORM Acute colds: zinc ACETATE (or gluconate) LOZENGES that dissolve in the mouth, taken within 24 h of onset every ~2-3 h - crucially WITHOUT citric acid, sorbitol, or mannitol (they chelate the ionic zinc that does the work). Daily prophylaxis: zinc picolinate, citrate, or bisglycinate. Avoid zinc oxide. (Lozenges: acetate ~ gluconate (release ionic zinc) - avoid citrate/tartrate-buffered candies. Oral: picolinate ~ citrate ~ bisglycinate (well absorbed) > sulfate > oxide (poor))

WATCH Lozenges: let dissolve, don't chew; expect metallic taste/nausea, don't exceed the labeled short-course. Daily zinc on an empty stomach causes nausea - take with food. Chronic >40 mg/day causes copper deficiency (upper limit 40 mg). Separate from calcium, iron, and quinolone/tetracycline antibiotics by 2 h.

Sources: Hemilä H. JRSM Open 2017;8(5), PMID 28515951; Hemilä H et al. Open Forum Infect Dis 2017 IPD, PMID 28480298; Nault D et al. Cochrane 2024;CD014914, PMID 38719213

2.5 Andrographis paniculata Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: immune / infection | Active regulatory safety signal for anaphylaxis/allergic reactions; sold OTC in the US but flagged by some agencies abroad.

THE NOISE Sold as the 'Indian echinacea' or 'king of bitters' immune herb that relieves cold and flu symptoms and speeds recovery.

THE SCIENCE Effect: Hu 2017: improved cough and sore throat and shortened symptom course vs placebo/usual care; effect real but heterogeneity high and several trials low quality / combination products confound the single-agent claim Clears MCID: Plausibly for symptom relief, but confounded by combination formulas and trial quality Largest independent trial: Hu 2017 pooled base; Coon & Ernst 2004 earlier systematic review Population: Adults and children with acute respiratory tract infections; many trials use fixed combos (Kan Jang = andrographis + eleuthero)

THE VERDICT **Efficacy is the strongest of the herbals here — meta-analysis shows genuine improvement in cough and sore throat versus placebo — but a serious, active regulatory safety signal caps this. Not a recommended buy: the anaphylaxis risk outweighs a symptom-relief benefit you can get more safely.**

BEST FORM Extract STANDARDIZED to andrographolides, ideally a branded/clinically-tested form (e.g., KalmCold, ParActin) delivering a defined andrographolide dose - crude herb is too variable. (Standardized branded andrographolide extract (defined actives, cold-duration data) > generic standardized extract > raw powder/tea (bitter, inconsistent))

WATCH Very bitter. Use short-term at cold onset. Can lower blood pressure and blood sugar; antiplatelet effect (bleeding risk with anticoagulants). Avoid in pregnancy/trying to conceive (anti-fertility/abortifacient signals in animals) and in autoimmune disease. Occasional allergic/rash reactions.

Sources: Hu XY et al. PLoS One 2017;12:e0181780, PMID 28783758 (+2018 correction); Coon JT, Ernst E. Planta Med 2004;70:293, PMID 15095142; TGA safety advisory, Andrographis and anaphylaxis, updated 2024-2025

2.5 Beta-glucans (yeast 1,3/1,6, Wellmune) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: immune / infection · **literature compromised (funding/outlier)**

THE NOISE Branded (Wellmune) as clinically proven to 'prime' your innate immune cells so you get sick less often and have fewer sick days.

THE SCIENCE Effect: Reductions in URTI symptom-days and symptom severity (e.g., fewer symptomatic days post-marathon; fewer symptom days in older adults). Effect sizes modest with wide CIs; endpoints often self-reported symptom diaries. Clears MCID: Uncertain — symptom-day reductions are modest and rest on soft, self-reported endpoints Largest independent trial: No adequately powered independent trial; Talbott/Fuller/McFarlin studies are manufacturer-linked Population: Marathon runners, stressed adults, older adults (50-70), some general adults

THE VERDICT **Do not buy on current evidence. Trials show modest reductions in cold/flu symptom-days, but the positive literature is overwhelmingly funded by the ingredient manufacturer, with no adequately powered independent RCT. Mechanism (innate immune priming) outruns the outcome data.**

BEST FORM YEAST-derived (*Saccharomyces cerevisiae*) 1,3/1,6 beta-glucan - the branded Wellmune WGP (~250 mg) is the immune-studied particulate form. This is NOT the same as cereal beta-glucan. (Yeast 1,3/1,6 beta-glucan / Wellmune (immune-active) and mushroom 1,3/1,6 > oat/barley 1,3/1,4 beta-glucan (good for cholesterol/glucose, NOT the immune form))

WATCH Don't confuse yeast/mushroom 1,3/1,6 (immune) with oat/barley 1,3/1,4 (metabolic) - the label must specify. Generally very well tolerated. Theoretical caution in autoimmune conditions (immune activation). Consistency over weeks matters for the immune benefit.

Sources: Fuller R et al. Nutrition 2017 (older adults, yeast beta-glucan); Talbott SM, Talbott JA. J Sports Sci Med 2009 (marathoners); McFarlin BK et al. 2013

2.5 Bovine colostrum Amazon ✓ Website ✓

Evidence: weak/mostly null (low confidence) | Effect: immune / infection

THE NOISE Hyped as 'nature's first food' loaded with antibodies and immunoglobulins that boost immunity, gut health, and athletic recovery.

THE SCIENCE Effect: 2016 meta (Jones): ~44% fewer URTI symptom-days, ~38% fewer episodes. 2022 meta: RR 0.64 for URTI, duration-dependent. Individual estimates imprecise; authors call for an adequately powered trial. Clears MCID: Uncertain — nominal reductions are meaningful for athletes but precision is low Largest independent trial: Jones AW et al. meta 2016; no single large definitive trial exists Population: Endurance/exercise-trained adults under training load (not sedentary healthy adults)

THE VERDICT **Do not buy at retail doses. Meta-analyses in exercising adults show reduced URTI incidence, but trials are small and imprecise, the population is athletes under training load, and — critically — retail servings are a fraction of the 20-60 g/day used in trials.**

BEST FORM First-milking, low-heat (not high-temperature) processed colostrum standardized to a stated IgG content (~20-40% IgG); reputable, tested sourcing. Some use liposomal/enteric delivery to protect immunoglobulins. (Standardized high-IgG, first-milking, low-temp-processed colostrum > generic colostrum (undisclosed IgG, heat-damaged) > skim/defatted low-Ig products)

WATCH Contains milk proteins and lactose - avoid with milk allergy; may bother the lactose-intolerant. Take on an empty stomach so the immunoglobulins survive digestion. Choose BSE-free, reputably sourced product. Naturally contains IGF-1 (debated relevance). Store per label (some need refrigeration).

Sources: Jones AW et al. Br J Nutr 2016;116:1226, PMID 27462401; Systematic review/meta-regression, J Funct Foods 2022 (ScienceDirect S1756464622003863)

2.5 N-acetylcysteine (NAC) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: immune / infection | Delisted by Amazon in 2021 after FDA warning letters; restored following FDA's 2022 enforcement-discretion stance, so listings can be inconsistent.

THE NOISE Marketed as a master-antioxidant and glutathione precursor that detoxifies, thins mucus, and shields the lungs and immune system from respiratory infection.

THE SCIENCE Effect: De Flora 1997: seroconversion to A/H1N1 unchanged (NAC did not prevent infection), but only 25% of infected NAC subjects became symptomatic vs 79% on placebo; fewer/less severe influenza-like episodes. Single-trial, not replicated for this endpoint. Clears MCID: Plausibly in the studied elderly/comorbid group; unproven and untransferred to healthy adults Largest independent trial: De Flora S et al. 1997, n=262 (multicenter Italian, independent) Population: Elderly (78% ≥65y), 62% with chronic disease (De Flora 1997) — NOT healthy adults

THE VERDICT Do not buy as a general immune supplement. Its real, drug-grade evidence is as a mucolytic that reduces COPD exacerbations and as the paracetamol-overdose antidote. The single positive flu-symptom trial is old, in elderly/comorbid patients, and unreplicated; it did not prevent infection, only symptomatic expression.

BEST FORM n/a - plain N-acetylcysteine is a single active compound; standard capsules, effervescent, and sustained-release perform similarly. Buy a reputable brand; timing matters more than form. (Single active form - effervescent/SR are convenience variants, not meaningfully more effective; quality/purity is the variable)

WATCH Take on an empty stomach for better absorption; has a sulfur smell and can cause GI upset/nausea. Additive hypotension with nitroglycerin. Rare bronchospasm concern is specific to INHALED NAC, not oral. It's a glutathione precursor. Note fluctuating US regulatory/FDA status affects availability.

Sources: De Flora S et al. Eur Respir J 1997;10:1535, PMID 9230243; FDA Final Guidance on NAC enforcement discretion, Aug 2022

2.5 Vitamin C (regular prophylaxis and cold treatment) Amazon ✓ Website ✓

Evidence: weak/mostly null (high confidence) | Effect: immune / infection

THE NOISE Sold as the classic immune booster that helps prevent colds and shortens them when you feel one coming on.

THE SCIENCE Effect: Incidence general pop: RR ~0.97, no effect. Duration: -8% adults (95% CI 3-13%), -14% children (5-22%). Therapeutic dosing at onset: no consistent benefit. Extreme-stress subgroup incidence: RR 0.48 (0.35-0.64). Clears MCID: No — 8% of a ~7-day cold is roughly half a day, below any meaningful threshold for a healthy adult Largest independent trial: Hemilä Cochrane pooled base (independent); no single sponsor drives it Population: General population (incidence/duration) and a distinct subgroup under extreme physical stress (marathon runners, skiers, soldiers)

THE VERDICT Do not buy to avoid catching colds — regular supplementation does not lower incidence in the general population. Daily supplementation shaves a modest, sub-clinical fraction off cold duration; taking it only after symptoms start does nothing. Genuine benefit exists ONLY for people under extreme acute physical stress.

BEST FORM Plain ascorbic acid in divided doses is fine and cheapest; use buffered (sodium/calcium ascorbate) or liposomal if you get GI upset or want higher single doses. Ester-C/rose-hip premiums add little. (Ascorbic acid (divided doses) ~ buffered ascorbate (gentler on gut) ~ liposomal (better high-dose tolerance/uptake) > 'Ester-C' marketing premium; fractional absorption falls above ~200-400 mg)

WATCH Doses >2 g/day cause diarrhea/GI upset (upper limit 2000 mg); split doses. Raises urinary oxalate - caution with kidney-stone history. Interferes with glucose meters and fecal occult-blood tests. Enhances non-heme iron absorption (helpful, or a caution in iron overload). Regular prophylaxis modestly shortens colds; starting at onset does little.

Sources: Hemilä H, Chalker E. Cochrane Database Syst Rev 2013;CD000980, PMID 23440782; Douglas RM et al. earlier Cochrane iterations

2.5 Vitamin D (respiratory infection prevention) Amazon ✓ Website ✓

Evidence: weak/mostly null (high confidence) | Effect: immune / infection

THE NOISE Marketed as an 'immune-support' vitamin that strengthens your defenses and protects against colds, flu and respiratory infections.

THE SCIENCE Effect: 2021 aggregate: OR 0.94 (0.88-1.00), $p=0.057$ — NULL. 2017 IPD overall OR 0.88 (0.81-0.96) but diluted to null once VIDARIS/VIDA/VITAL added. Deficiency subgroup (baseline <25 nmol/L) 2017 OR 0.30 (0.17-0.53). Dose-response trap: on bone, 4000-10000 IU LOWERED BMD vs 400 IU (Burt JAMA 2019) — more is not better. Clears MCID: No in replete adults (null); yes only when correcting frank deficiency Largest independent trial: VITAL (Manson/Martineau, NEJM 2019; respiratory analysis Thorax 2021), $n=25,871$, 2000 IU/day — no reduction in ARI Population: Mixed: general adults, children, and deficient subgroups across ~40 RCTs. Benefit concentrated in profoundly deficient participants.

THE VERDICT Do not buy for immunity if you are vitamin D replete. The one honest benefit is correcting genuine deficiency, which is cheap and does not require an 'immune' product. In replete adults the effect is null.

BEST FORM Vitamin D3 (cholecalciferol) in an oil-based softgel, taken with a fat-containing meal. D3 raises and sustains 25-OH-D better than D2. (D3 cholecalciferol (superior, longer-acting) > D2 ergocalciferol (weaker at raising blood levels); oil/softgel > dry tablet for absorption)

WATCH Fat-soluble - take with dietary fat. For infection prevention, regular daily/weekly dosing outperforms large infrequent bolus doses. General adult upper limit ~4000 IU/day; get levels checked. Needs adequate magnesium as a cofactor. Interacts with thiazides (hypercalcemia risk); some pair with K2.

Sources: Jolliffe DA et al. Lancet Diabetes Endocrinol 2021; PMID 33798465; Martineau AR et al. BMJ 2017;356:i6583, PMID 28202713; VITAL respiratory analysis (Thorax 2021); Burt LA et al. JAMA 2019;322:736 (bone dose-response), PMID 31454046

2 Echinacea Amazon ✓ Website ✓

Evidence: weak/mostly null (high confidence) | Effect: immune / infection

THE NOISE Marketed as an immune 'booster' that helps you fight off colds and shortens how long they last if you take it at the first sniffle.

THE SCIENCE Effect: Prevention: post-hoc pooled RRR ~10-20%, not statistically significant, deemed of questionable clinical relevance. Treatment: only 2 of 6 duration trials significant; no consistent effect. Clears MCID: No — even the weak prevention trend does not reach clinical relevance Largest independent trial: Cochrane pooled base (Karsch-Volk 2014) — no single positive independent trial stands out Population: Adults and children, prevention and treatment of the common cold

THE VERDICT Do not buy. Across 24 blinded trials and 4,600+ participants, echinacea shows no reliable treatment benefit and at best a weak, non-significant, clinically irrelevant prevention trend. Preparations are heterogeneous and non-comparable.

BEST FORM Species and plant part are everything: *E. purpurea* expressed juice of the aerial parts, or an alcoholic extract of *E. purpurea*/*E. angustifolia* root standardized to alkylamides (the tongue-tingling actives). Products vary enormously. (*E. purpurea* pressed aerial juice or alkylamide-standardized root tincture (best evidence) > generic capsules (often under-dosed/wrong part) > *E. pallida* (weaker data))

WATCH Cross-reactive allergy in ragweed/Asteraceae-sensitive people (rare anaphylaxis). Best started at symptom onset and not used continuously long-term. Caution in autoimmune disease. Mild theoretical CYP interactions. Product quality/labeling is notoriously inconsistent - buy standardized brands.

Sources: Karsch-Volk M et al. Cochrane Database Syst Rev 2014;CD000530, PMID 24554461; Shah SA et al. Lancet Infect Dis 2007 meta (DARE)

2 Elderberry (*Sambucus nigra*) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: immune / infection · **literature compromised (funding/outlier)**

THE NOISE Promoted as an antioxidant-rich berry that boosts immunity and shortens the duration and severity of colds and flu.

THE SCIENCE Effect: Hawkins 2019 reported a large pooled symptom-reduction SMD, but the estimate is dominated by Tiralongo 2016 (industry-funded) and small n; wide, fragile CIs Clears MCID: Cannot be trusted — nominal reductions exist but rest on compromised inputs Largest independent trial: No adequately powered independent trial exists; the largest data are sponsor-linked (Tiralongo n=312) Population: Small trials in air travelers and self-limited colds/flu; tiny samples

THE VERDICT **Do not buy. The favorable meta-analysis is built almost entirely on small, manufacturer-funded trials, and the pooled effect leans on the industry-sponsored air-traveler study. Independent replication is essentially absent, and a subsequent trial in influenza failed to show benefit.**

BEST FORM A standardized, properly cooked *Sambucus nigra* extract or syrup (e.g., anthocyanin-standardized types like Sambucol). The berries **MUST** be heat-processed. (Standardized cooked extract/syrup (consistent anthocyanins) > generic syrup/gummies (variable, sugary) >> raw/homemade preparations (safety risk))

WATCH Raw/unripe berries, leaves, stems, and bark contain cyanogenic glycosides and are toxic - only use cooked/processed products. Take at first sign of a cold. Theoretical cytokine-stimulation concern in autoimmune disease (weak evidence). Watch sugar content in syrups/gummies for diabetics.

Sources: Tiralongo E et al. *Nutrients* 2016;8(4):182, PMID 27023596 (Iprona-funded); Hawkins J et al. *Complement Ther Med* 2019;42:361, PMID 30670267

2 Ginseng (*Panax quinquefolius*, COLD-fX/CVT-E002) Amazon ✓ Website ✓

Evidence: weak/mostly null (low confidence) | Effect: immune / infection · **literature compromised (funding/outlier)**

THE NOISE Branded (COLD-fX) as a proprietary American ginseng extract that reduces the frequency, severity, and duration of colds and flu.

THE SCIENCE Effect: Predy 2005: fewer recurrent colds and reduced total symptom-days over 4 months; effect modest, trials small and sponsor-linked, later independent reviews (Seida 2011) judged evidence insufficient/low-quality Clears MCID: Uncertain — modest and not independently confirmed Largest independent trial: Predy 2005 (CMAJ) is sponsor-linked; no strong independent trial Population: Community adults and older adults, cold prevention over a winter season

THE VERDICT **Do not buy. A few trials report modest reductions in cold frequency/duration, but the evidence is concentrated in manufacturer-funded studies of a single proprietary extract, with no convincing independent confirmation and low overall quality.**

BEST FORM The proprietary CVT-E002 poly-furanosyl-pyranosyl-saccharide extract of North American ginseng (COLD-fX) - a standardized polysaccharide fraction, not generic ginseng root or ginsenoside-standardized products. (CVT-E002 / COLD-fX standardized polysaccharide extract (the form used in cold-prevention RCTs) GOOD; ordinary *Panax* root, energy 'ginseng' blends, and ginsenoside-standardized extracts POOR proxies - different active fraction.)

WATCH Used preventively across cold season (daily dosing), not as a single acute dose. American ginseng lowers INR - genuine interaction with warfarin (avoid). Additive hypoglycemia with diabetes drugs; theoretical conflict with immunosuppressants/autoimmune disease; avoid with MAOIs. Can cause insomnia if taken late.

Sources: Predy GN et al. *CMAJ* 2005;173:1043, PMID 16247099; Seida JK et al. *Evid Based Complement Alternat Med* 2011 (review)

2 Pelargonium sidoides (EPs 7630, Umckaloabo) Amazon ✓ Website ✓

Evidence: weak/mostly null (low confidence) | Effect: immune / infection · **literature compromised (funding/outlier)**

THE NOISE Branded (Umcka) as an African geranium root remedy that shortens colds and bronchitis and eases symptoms when taken at onset.

THE SCIENCE Effect: Cochrane: may improve symptom resolution in acute bronchitis (reduced symptom scores by day 7), but evidence graded low quality with high risk of bias and reliance on one manufacturer's trials Clears MCID: Uncertain given quality limitations Largest independent trial: Few/none independent; base is manufacturer-driven Population: Adults and children with acute bronchitis, sinusitis, or common cold (treatment, not prevention)

THE VERDICT **Do not buy for general immunity. Trials suggest faster symptom resolution in acute bronchitis/cold, but nearly all come from a single manufacturer, methodological/quality concerns are noted by Cochrane, and it is a treatment for an active infection, not prevention.**

BEST FORM The clinically validated EPs 7630 standardized root extract (Umckaloabo / Umcka ColdCare) as liquid drops or film-coated tablets - not generic pelargonium tinctures, which are not therapeutically equivalent to the trialed extract. (EPs 7630 standardized extract (proven in acute bronchitis/URTI RCTs) GOOD; generic/unstandardized pelargonium tinctures and 'root powder' capsules unproven and not interchangeable.)

WATCH Start at first sign of a cold; typical course ~7 days (do not use long-term). EMA flagged rare hepatotoxicity - stop and seek care with dark urine/jaundice/RUQ pain, and avoid in existing liver disease. Contains coumarins with mild antiplatelet effect: caution with warfarin/anticoagulants and bleeding disorders. Not for pregnancy/children under 6 without medical advice.

Sources: Timmer A et al. Cochrane Database Syst Rev 2013;CD006323, PMID 24146345

1.5 Garlic / allicin Amazon ✓ Website ✓

Evidence: none / marketing (medium confidence) | Effect: immune / infection · **literature compromised (funding/outlier)**

THE NOISE Aged garlic and high-allyl extracts are sold as natural immune boosters that cut how often you catch colds.

THE SCIENCE Effect: Josling 2001: 24 colds (garlic) vs 65 (placebo) over 12 weeks; fewer days ill. Single trial, high risk of bias, no CI-robust replication. Clears MCID: Cannot be relied upon — one biased trial Largest independent trial: None exists Population: Healthy adults, 12-week winter prevention (single trial)

THE VERDICT **Do not buy. The entire prevention claim rests on a single low-quality, manufacturer-funded trial that has never been replicated. Cochrane could include only that one study. This is marketing propped up by one weak result.**

BEST FORM Two valid routes: an ENTERIC-COATED garlic powder tablet standardized to 'allicin-releasing potential' (protects the alliin + alliinase enzyme so allicin forms in the gut), OR aged garlic extract (Kyolic), standardized to S-allylcysteine. Freshly crushed raw garlic works but is odorous/unstable. (Enteric-coated standardized allicin-potential tablet ~ aged garlic extract (stable, odor-reduced, S-allylcysteine) > non-enteric garlic powder (stomach acid destroys allicin) > garlic oil (little allicin))

WATCH Antiplatelet effect - bleeding risk with warfarin/aspirin/NSAIDs and before surgery (stop ~7-10 days prior). Can lower levels of some drugs (e.g., saquinavir). GI upset, reflux, body/breath odor. Enteric coating is needed for non-aged forms to survive stomach acid.

Sources: Lissiman E et al. Cochrane Database Syst Rev 2014;CD006206, PMID 25386977; Josling P. Adv Ther 2001;18:189, PMID 11697022

1.5 Quercetin Amazon ✓ Website ✓

Evidence: none / marketing (medium confidence) | Effect: immune / infection

THE NOISE Promoted as a plant antioxidant that strengthens immunity and wards off colds, often stacked with zinc and vitamin C as a 'zinc ionophore' cold-defense combo.

THE SCIENCE Effect: Overall: no significant effect on URTI incidence or severity in any prespecified analysis. Post-hoc subgroup (fit, >=40, 1000 mg): -36% severity, -31% sick days — hypothesis-generating only, wide uncertainty. Clears MCID: No — overall null; subgroup effect is exploratory Largest independent trial: Nieman/Heinz 2010, n=1002 — null overall Population: Community adults 18-85; positive signal only in post-hoc subgroup (age >=40, self-rated fit)

THE VERDICT **Do not buy. The large, well-powered RCT was flatly null for URTI overall. The only positive signal is a post-hoc subgroup (fit, over-40, at the higher dose). Poor oral bioavailability compounds the weakness. Mechanism-driven marketing.**

BEST FORM An absorption-enhanced form: quercetin PHYTOSOME (phospholipid-bound, e.g., Quercefit), enzymatically-modified isoquercitrin (EMIQ), or isoquercetin - plain quercetin aglycone is very poorly absorbed. Taking with vitamin C/bromelain is a cheaper boost. (Quercetin phytosome (up to ~20x uptake) > EMIQ/isoquercetin (glycoside, better absorbed) > plain quercetin dihydrate aglycone (poorly bioavailable))

WATCH Take with food/fat to aid absorption. Inhibits CYP3A4 and P-glycoprotein - can raise levels of various drugs (e.g., cyclosporine, some statins). Very high chronic doses raise theoretical kidney concerns. Often paired with vitamin C and bromelain.

Sources: Heinz SA, Henson DA, Nieman DC et al. Pharmacol Res 2010;62:237, PMID 20478383

4 Psyllium husk (ispaghula) Amazon ✓ Website ✓

Evidence: good (high confidence) | Effect: gut / digestion

THE NOISE Marketed as a daily soluble-fiber that promotes regularity, relieves constipation, and supports gut and heart health.

THE SCIENCE Effect: IBS: symptoms not improving RR 0.83 (95% CI 0.73-0.94), NNT≈7 (Ford/Moayyedi 2014, BMJ). Constipation: significant increase in stool frequency and improved consistency (AJCN 2024 meta-analysis; ~66% responders vs ~41% control). Clears MCID: Yes — modest but clinically usable; effect largest in IBS-C and constipation. Largest independent trial: Bijkerk et al. 2009, BMJ (n=275, primary care): psyllium 10 g/day significantly improved IBS symptom severity vs placebo and bran at 3 months. Population: Adults with IBS (all subtypes, best in IBS-C) and adults/elderly with chronic idiopathic constipation.

THE VERDICT **Buy — the strongest evidence in the category. Cheap, independent, guideline-endorsed soluble/gel-forming fiber for IBS global symptoms and for chronic constipation. Titrate to an effective dose and give it 4 weeks.**

BEST FORM Pure blond psyllium husk powder (ispaghula), unsweetened/sugar-free; whole or coarsely ground husk. Capsules work but need many to reach an effective 5-10 g/day dose. Avoid products loaded with added sugar or sweeteners. (Pure psyllium husk powder GOOD (gel-forming soluble fiber, best IBS/cholesterol/regularity data); sugary flavored blends and low-dose capsules INFERIOR (dose burden, additives); psyllium is better evidenced for IBS than wheat bran (insoluble, can worsen symptoms).)

WATCH Always take with a full glass of water and adequate daily fluids - inadequate liquid risks choking/esophageal or bowel obstruction. Separate from medications by 2-4 h: it blunts absorption of levothyroxine, lithium, carbamazepine, digoxin and some others. Start low (1 tsp) and titrate to limit gas/bloating.

Sources: Moayyedi P, Ford AC, et al. Am J Gastroenterol 2014 / BMJ soluble fiber meta-analysis (RR 0.83); Bijkerk CJ, et al. BMJ 2009;339:b3154 (independent primary-care RCT) PMID 19713235; van der Schoot A, et al. Am J Clin Nutr 2024 (fiber for chronic constipation meta-analysis); Foods 2025;14:2315 (fiber for constipation in elderly meta-analysis)

3.5 Saccharomyces boulardii (CNCM I-745) Amazon ✓ Website ✓

Evidence: moderate/conditional (high confidence) | Effect: gut / digestion

THE NOISE Sold as a hardy 'good yeast' probiotic that restores gut balance, prevents antibiotic-associated and traveler's diarrhea, and supports overall digestive health.

THE SCIENCE Effect: AAD prevention: RR 0.47 (95% CI 0.35-0.63); absolute risk ~18.7% → 8.5%, NNT≈10 (95% CI 9-13). Pediatric acute diarrhea: reduces duration by ~1 day. Clears MCID: Yes for AAD and acute pediatric diarrhea (NNT≈10; ~1 day shorter illness). Largest independent trial: Pooled Szajewska & Kołodziej 2015 (Aliment Pharmacol Ther) across ~21 RCTs; consistent 50%+ relative risk reduction. Population: Adults and children on antibiotics (AAD prevention) and children with acute gastroenteritis; NOT healthy adults seeking general gut benefits.

THE VERDICT **Conditional buy — genuinely effective for a specific job (preventing antibiotic-associated diarrhea, shortening acute infectious diarrhea), NOT for routine 'gut health.' Independent evidence, meaningful NNT. Avoid in the critically ill.**

BEST FORM The specific documented strain *S. boulardii* CNCM I-745 (Florastor), with a stated CFU count. Being a yeast, it is unaffected by antibacterial antibiotics and can be co-administered during antibiotic courses. (CNCM I-745 (most-studied strain, antibiotic-associated and travelers'/C. difficile diarrhea data) GOOD; unnamed '*S. boulardii*' or generic-strain products with no CFU/strain disclosure less reliable.)

WATCH Do NOT use in immunocompromised, critically ill, or patients with central venous catheters - documented risk of fungemia (open capsules away from IV lines). Mix into food/liquid that is warm, not hot (heat kills yeast). Not cleared by antifungals. Generally well tolerated otherwise (mild gas).

Sources: Szajewska H, Kołodziej M. Aliment Pharmacol Ther 2015 (AAD meta-analysis, RR 0.47); McFarland LV. World J Gastroenterol 2010 (*S. boulardii* meta-analysis); EMA 2018 referral/PRAC restriction on *S. boulardii* medicinal products (ICU/CVC contraindication); Rannikko J, et al. Emerg Infect Dis 2021;27(8) (fungemia case series)

3 L-glutamine (post-infectious IBS-D) Amazon ✓ Website ✓

Evidence: moderate/conditional (low confidence) | Effect: gut / digestion · **literature compromised (funding/outlier)**

THE NOISE Sold as a gut-lining amino acid that repairs 'leaky gut,' restores the intestinal barrier, and calms IBS and digestive distress.

THE SCIENCE Effect: ≥ 50 -point IBS-SS reduction in 79.6% (glutamine) vs 5.8% (placebo) — very large; also reduced daily stool frequency and normalized lactulose/mannitol ratio (Zhou Q, et al. Gut 2019;68:996-1002). Clears MCID: Yes in the studied population (huge responder gap) — but plausibility strained by effect size and lack of replication. Largest independent trial: Zhou Q, et al. Gut 2019;68:996-1002 (n=115) — the only trial; not yet independently reproduced. Population: Adults with post-infectious IBS-D AND confirmed increased intestinal permeability (urinary lactulose/mannitol ratio ≥ 0.07), documented enteric infection ≥ 1 year prior.

THE VERDICT **Conditional and narrow. One striking single-center trial in post-infectious IBS-D with measured increased gut permeability — huge effect, but unreplicated, single research group, and needs 15 g/day (retail sells ~5 g). Not for healthy adults or general 'leaky gut.'**

BEST FORM Any pharmaceutical-grade free-form L-glutamine powder (not the D- or DL- isomer). It's a single active molecule - no premium 'form' outperforms plain L-glutamine; buy on purity/price, not marketing. (Free-form L-glutamine powder is the only meaningful form; 'peptide-bound' or exotic glutamine products offer no proven advantage for gut use.)

WATCH Trial dose in the IBS-D study was high (5 g three times daily). Glutamine is unstable in solution over time - mix and consume promptly, don't pre-make. Caution in cirrhosis/hepatic encephalopathy (converts to ammonia) and severe renal disease. High doses can cause GI upset; may affect some cancer regimens - check with oncologist.

Sources: Zhou Q, et al. Gut 2019;68:996-1002 PMID 29122852 (single RCT); NCT01414244 (registry); NCT06291038 (ongoing replication in IBS with impaired permeability)

3 Partially hydrolyzed guar gum (PHGG) [added, in-category] Amazon ✓ Website ✓

Evidence: moderate/conditional (low confidence) | Effect: gut / digestion

THE NOISE Prebiotic gut-health and regularity fibre.

THE SCIENCE Effect: Parisi 2002: PHGG improved IBS symptom scores comparably to wheat bran with better tolerability and greater patient preference; effect sizes modest, trials small. Clears MCID: Borderline — modest symptom/bowel-habit improvement, better tolerability profile than inulin. Largest independent trial: Parisi GC, et al. Dig Dis Sci 2002 (n=188): PHGG comparable to wheat bran for IBS with better tolerability. Population: Adults with IBS (mixed subtypes) and functional constipation.

THE VERDICT **Conditional. A better-tolerated soluble fiber than inulin, with small RCTs suggesting improved IBS symptoms/bowel habit and less bloating than wheat bran, but trials are small and some industry-linked. Reasonable low-gas fiber alternative to psyllium; not first-line.**

BEST FORM Partially hydrolyzed guar gum specifically (e.g. Sunfiber) - a low-viscosity, largely tasteless, well-tolerated soluble fiber. This hydrolyzed form is the point: it is NOT the same as raw/native guar gum. (PHGG GOOD (low viscosity, minimal gas, good regularity/IBS tolerability, mixes clear); native/unhydrolyzed guar gum POOR and unsafe as a supplement - extremely viscous, historically caused esophageal and bowel obstruction.)

WATCH Very well tolerated but still fermentable - start ~3-5 g and build to avoid initial mild gas. Take with adequate fluid. Like other soluble fibers, separate from medications by a couple of hours. Confirm the label says 'partially hydrolyzed' guar gum, not plain guar gum.

Sources: Parisi GC, et al. Dig Dis Sci 2002;47:1697-1704 (PHGG vs bran in IBS); Yasukawa Z, et al. Nutrients 2019 (PHGG bowel habit)

3 Peppermint oil (enteric/gastro-resistant, IBS) Amazon ✓ Website ✓

Evidence: moderate/conditional (medium confidence) | Effect: gut / digestion · **literature compromised (funding/outlier)**

THE NOISE Sold as enteric-coated capsules that calm IBS by relaxing gut muscle spasms, easing abdominal pain, bloating, and cramping.

THE SCIENCE Effect: Global symptoms not improving RR 0.65 (95% CI 0.43-0.98), NNT=4; abdominal pain not improving RR 0.76 (95% CI 0.62-0.93), NNT=7 (Black 2022 meta-analysis, Aliment Pharmacol Ther). Clears MCID: Yes on pooled dichotomous responder outcomes (NNT 4-7 is clinically meaningful) — but the largest trial failed the FDA-standard primary endpoint. Largest independent trial: Weerts et al. 2020, Gastroenterology (n=190/analyzed ~178): small-intestinal- and ileocolonic-release peppermint oil vs placebo, NO significant benefit on co-primary abdominal pain response or global relief at 8 weeks; only secondary pain-severity outcomes improved. Population: Adults with IBS (Rome II-IV), mixed subtypes; most trials 2-4 weeks, few high-quality.

THE VERDICT **Conditional buy for IBS abdominal pain/global symptoms — the best OTC option in this category, but the effect is fragile: it holds in pooled older trials yet the largest independent trial missed its primary endpoint. Reasonable to trial for 4 weeks; stop if no benefit.**

BEST FORM Enteric-coated / gastro-resistant capsules, ideally sustained-release microsphere formulations (e.g. IBgard-type) that survive the stomach and release oil in the small intestine/colon where it relaxes gut smooth muscle. (Enteric-coated / SR microsphere capsules GOOD (target site, less reflux); plain (non-coated) peppermint oil softgels and liquid oil POOR - dumped in the stomach they relax the lower esophageal sphincter and cause heartburn/reflux.)

WATCH Take 30-60 min before meals. Do NOT co-take with antacids, PPIs, H2 blockers or milk - raising gastric pH dissolves the enteric coating early, causing reflux and blunting effect. Common: heartburn and perianal/anal burning. Avoid in significant GERD or hiatal hernia; caution in pregnancy.

Sources: Black CJ, et al. Aliment Pharmacol Ther 2022 (10-RCT meta-analysis, NNT 4/7); Weerts ZZRM, et al. Gastroenterology 2020;158:123-136 (independent RCT, null primary endpoint) PMID 32289372; Alamar N, et al. BMC Complement Altern Med 2019;19:21 (pooled RR 2.39 global improvement) PMID 30654773; Ford AC, et al. ACG monograph / IBS guideline conditional recommendation

2.5 Lactobacillus rhamnosus GG (LGG) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: gut / digestion · **literature compromised (funding/outlier)**

THE NOISE Marketed as the world's most-studied probiotic strain that supports immunity, digestive balance, and relief from diarrhea in kids and adults.

THE SCIENCE Effect: Schnadower 2018 NEJM (n=943): NO benefit — moderate/severe disease in 11.8% LGG vs 12.6% placebo (RR 0.96, 95% CI 0.68-1.35). Freedman 2018 (PROBIOTIC, n=886) also null. Older meta-analyses claimed ~1 day duration but on low-quality evidence. Clears MCID: No — the high-quality independent trials show no clinically meaningful effect on acute gastroenteritis. Largest independent trial: Schnadower D, et al. NEJM 2018;379:2002-2014 (n=943, 10 US pediatric EDs, NIH-funded): LGG 10¹⁰ CFU BID = NO benefit over placebo for acute gastroenteritis. Population: Children with acute gastroenteritis (trials); some AAD-prevention data in children and adults. Adult IBS/general use largely untested.

THE VERDICT **Weak buy at best. Its flagship claim (acute pediatric diarrhea) was overturned by two large independent trials; residual benefit for antibiotic-associated diarrhea is modest. Do not buy for adult IBS or general gut health.**

BEST FORM The defined strain L. rhamnosus GG, ATCC 53103 (Culturelle), with a labeled CFU count and end-of-shelf-life viability guarantee. Buy by strain designation, not just 'L. rhamnosus'. (LGG / ATCC 53103 (best-studied Lactobacillus strain - antibiotic-associated and pediatric diarrhea) GOOD; unspecified 'L. rhamnosus' or species-only labels can be unrelated strains with different effects.)

WATCH Effects are strain-specific - species-only claims don't transfer. Check refrigeration requirement and CFU at expiry, not at manufacture; take consistently. Separate from antibiotic dose by ~2 h. Caution (rare bacteremia) in immunocompromised, critically ill, or central-line patients. Modest evidence for IBS specifically.

Sources: Schnadower D, et al. NEJM 2018;379:2002-14 PMID 30462939 (large independent null); Freedman SB, et al. NEJM 2018;379:2015-26 (PROBIOTIC trial, null); Szajewska H, et al. Aliment Pharmacol Ther 2019 (WJG meta-analysis; low-quality evidence, effect attenuated with 2018 trials); Szajewska H, Kolodziej M. Aliment Pharmacol Ther 2015 (LGG for AAD)

2.5 Multi-strain probiotics (IBS / general gut) Amazon ✓ Website ✓

Evidence: weak/mostly null (low confidence) | Effect: gut / digestion · **literature compromised (funding/outlier)**

THE NOISE Sold as high-CFU multi-strain 'gut health' formulas that rebalance the microbiome to relieve IBS, bloating, and support immunity and mood.

THE SCIENCE Effect: Pooled global symptom benefit exists (e.g. RR of symptoms persisting ~0.7-0.8) but GRADE certainty low/very low; effect not reproducible across strains. No robust, MCID-clearing, replicated class effect. Clears MCID: Uncertain — statistical signal, but low certainty and heterogeneity make clinical significance unreliable. Largest independent trial: No single large independent trial establishes a class effect; guideline bodies (ACG 2020, AGA 2020) explicitly decline to recommend for global IBS. Population: Adults with IBS (Rome II-IV), highly heterogeneous strains/doses/durations.

THE VERDICT **Do not buy as a class for IBS or 'gut health.'** Meta-analyses show a signal but at low/very-low GRADE certainty with heavy heterogeneity and industry funding; ACG and AGA recommend against routine use outside a trial. Any benefit is strain-and-product-specific, not a class effect.

BEST FORM Products whose exact strains were trialed for IBS - e.g. Bifidobacterium longum subsp. infantis 35624 (Alflorex) for general IBS, or a high-CFU multi-strain like the Visbiome/former VSL#3 formula for pouchitis/UC-adjacent use. Match strain blend to your indication. (Indication-matched, strain-disclosed formulas (35624; high-dose 8-strain blends) GOOD; generic 'billions of CFU, 10 strains' tubs with undisclosed strains POOR - CFU count alone does not predict benefit.)

WATCH Give a full 4-week trial before judging; transient bloating/gas early is common. Verify CFU at end of shelf life and any refrigeration need. Caution in suspected SIBO (may worsen) and in immunocompromised/critically ill. Don't assume more strains or higher CFU is better.

Sources: Goodoory VC, Ford AC, et al. Gastroenterology 2023 (probiotics in IBS meta-analysis); AGA Clinical Practice Guideline on Probiotics, Gastroenterology 2020 (recommend against outside trials); ACG Monograph / IBS Guideline 2020 (Ford et al.); Network meta-analysis, Nutrients 2024;16:2114 (strain-specific, prebiotics/synbiotics not significant)

2.5 Zinc-carnosine (polaprezinc) Amazon ✓ Website ✓

Evidence: weak/mostly null (low confidence) | Effect: gut / digestion

THE NOISE Sold as a stomach-lining supplement that soothes and heals the gut, protects against ulcers, and repairs 'leaky gut.'

THE SCIENCE Effect: Gastric ulcer: endoscopic improvement ~60% and higher symptom response vs comparator in an ~258-patient RCT. Permeability: prevented the ~3-fold indomethacin-induced rise in gut permeability in healthy volunteers (Mahmood 2007, n=10 crossover). Clears MCID: Plausibly for ulcer healing as adjunct; unproven for functional gut symptoms/IBS. Largest independent trial: Mahmood A, et al. Gut 2007 (n=10 crossover): zinc-carnosine 37.5 mg BID prevented indomethacin-induced increase in small-intestinal permeability and stimulated repair. Ulcer efficacy from larger Japanese RCTs. Population: Gastric ulcer patients (adjunct to standard therapy); healthy volunteers given indomethacin; small UC enema pilots — largely Japanese cohorts.

THE VERDICT **Weak/conditional. Real but small-trial evidence for gastric mucosal protection and ulcer healing (mostly Japanese, adjunct to PPI) and one tiny crossover showing it blunts NSAID-induced gut permeability. Not established for 'leaky gut' or IBS. Mechanism ahead of outcomes.**

BEST FORM The intact 1:1 zinc-L-carnosine chelate (polaprezinc), typically 75 mg twice daily. The point is to buy this specific chelate for mucosal/gastric use - it acts locally and is not interchangeable with ordinary zinc salts. (Polaprezinc (zinc-L-carnosine chelate) is the researched gut/mucosal form (adheres to ulcerated mucosa); plain zinc oxide/sulfate/gluconate do NOT reproduce the local mucosal-protective effect.)

WATCH Each 75 mg polaprezinc delivers ~16 mg elemental zinc - long-term high zinc intake causes copper deficiency (anemia, neuropathy); watch total zinc vs the 40 mg/day adult upper limit. Take between meals for mucosal contact. Separate from antibiotics (tetracyclines/quinolones) and iron. Mild GI upset possible.

Sources: Mahmood A, et al. Gut 2007;56:168-175 PMID 16777920 (permeability crossover); Polaprezinc gastric ulcer multicenter RCT (2022, positive-controlled); Polaprezinc enema in ulcerative colitis, pilot PMID 24286534

2 Butyrate / tributyrin (oral) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: gut / digestion

THE NOISE Marketed as an oral short-chain fatty acid that feeds colon cells, heals the gut lining, and reduces inflammation and IBS symptoms.

THE SCIENCE Effect: Pediatric IBD: 12-week sodium butyrate add-on = NO difference in remission vs placebo (multicenter RCT). Some small adult UC trials show improvement in secondary markers (ESR, Mayo subscores, mood) but not robust clinical remission endpoints. Clears MCID: No — no consistent clinically meaningful outcome; the cleanest add-on remission trial was null. Largest independent trial: Pieczarkowski/pediatric IBD multicenter RCT (PMC9414716): 12-week oral sodium butyrate add-on did NOT improve remission over placebo. Population: Ulcerative colitis and IBD patients; pediatric IBD and pediatric IBS — small heterogeneous trials.

THE VERDICT Do not buy. Compelling colonocyte biology, but oral butyrate is largely absorbed before reaching the colon and human outcome trials are small, inconsistent, and include a clear null in pediatric IBD. Mechanism outruns outcome.

BEST FORM Tributyrin (a glycerol-esterified tributyrate) is better absorbed and largely odorless, releasing butyrate more distally than the mineral salts. If using salts, calcium/magnesium butyrate is less malodorous than sodium butyrate. (Tributyrin GOOD (higher bioavailability, minimal smell, more colon delivery); sodium/calcium-magnesium butyrate salts INFERIOR (strong rancid-butter odor, much absorbed proximally before reaching the colon).)

WATCH Overall clinical evidence is still limited/emerging - manage expectations. Main issues are the notorious butyric odor (worst with sodium salt), belching and GI upset. Salt forms add sodium or calcium/magnesium to your daily load. Take with food to reduce reflux/aftertaste.

Sources: Pediatric IBD sodium butyrate multicenter RCT, PMC9414716 (null on remission); UC butyrate RCT, PMC11930386 (secondary-marker improvement); Calcium butyrate pediatric IBS RCT, PMC12408980

2 Digestive enzymes (lactase, alpha-galactosidase, blends) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: gut / digestion

THE NOISE Marketed as broad-spectrum enzyme blends that help you break down food, so you get less bloating, gas, and discomfort after meals.

THE SCIENCE Effect: Alpha-galactosidase: significant reduction in flatulence/bloating after high-GOS meals in sensitive subjects (small crossover RCTs). Generic blends: no credible IBS outcome data. Clears MCID: Only for the matched deficiency (lactose intolerance, GOS sensitivity); not demonstrated for undifferentiated IBS. Largest independent trial: Di Stefano M, et al. 2007 and Hillilä M, et al. 2016 — small alpha-galactosidase RCTs showing reduced gas symptoms in GOS-sensitive subjects; no large trial for generic blends. Population: Lactose-intolerant individuals (lactase); GOS-sensitive individuals (alpha-galactosidase); EPI patients (PERT — prescription).

THE VERDICT Do not buy generic 'digestive enzyme' blends for IBS/gut health. The only justified uses are enzyme REPLACEMENT for a specific deficiency — lactase for lactose intolerance, and prescription PERT for pancreatic insufficiency, which is not an OTC supplement.

BEST FORM Match the enzyme to the trigger: lactase for dairy/lactose; alpha-galactosidase (Beano) for beans/cruciferous/FODMAP oligosaccharide gas. For diagnosed pancreatic insufficiency use prescription enteric-coated pancrelipase, not OTC blends. Broad-spectrum 'digestive blends' have the weakest evidence. (Targeted single enzymes (lactase, alpha-galactosidase) GOOD for their specific substrate; generic multi-enzyme blends WEAK/unproven for otherwise-healthy people; OTC enzymes are NOT a substitute for prescription pancreatic enzyme replacement.)

WATCH Timing is critical - take with the first bite of the offending food (alpha-galactosidase before eating; lactase with the dairy). Alpha-galactosidase does nothing for lactose and vice versa. Enteric coating matters for acid-labile pancreatic enzymes. Check units (FCC/ALU/GalU), not just 'mg'.

Sources: Di Stefano M, et al. Dig Liver Dis 2007 (alpha-galactosidase, gas); Hillilä M, et al. 2016 (alpha-galactosidase RCT); AGA/ACG guidance on PERT for exocrine pancreatic insufficiency

2 Prebiotics (inulin, GOS/FOS, inulin-type fructans) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: gut / digestion

THE NOISE Marketed as prebiotic 'fuel for good bacteria' that feeds the microbiome, improves digestion, and reduces bloating.

THE SCIENCE Effect: No significant difference vs placebo in abdominal pain, bloating, or flatulence overall; flatulence improved only at ≤ 6 g/day non-inulin types, while inulin-type fructans **WORSENE**d flatulence (So D, et al. AJCN 2022). Clears MCID: No — null for the symptoms patients care about. Largest independent trial: So D, Gibson PR, et al. Am J Clin Nutr 2022 meta-analysis (prebiotics in IBS/functional bowel disorders): no symptom benefit; inulin-type worsens flatulence. Population: Adults with IBS and functional bowel disorders.

THE VERDICT **Do not buy for IBS symptoms. The best meta-analysis shows no effect on abdominal pain or bloating overall, and inulin-type fructans can WORSENE flatulence. Rising bifidobacteria is a biomarker, not a symptom outcome. Wrong dose-response direction.**

BEST FORM For tolerability, galacto-oligosaccharides (GOS) and partially hydrolyzed guar gum are gentler than inulin/FOS; start at a low dose (2-3 g) and titrate. Inulin/FOS give more bifidogenic effect but far more gas. (GOS and PHGG **BETTER TOLERATED** (slower, distal fermentation); short-chain FOS and native inulin ferment fast and proximally - most bloating/flatulence, and they are high-FODMAP.)

WATCH All are fermentable and cause dose-dependent gas/bloating/cramping; inulin and FOS are FODMAPs that commonly flare IBS - not ideal for symptomatic IBS. Start very low and build up over weeks. Avoid abrupt high doses; not for known fructan intolerance.

Sources: So D, et al. Am J Clin Nutr 2022 (prebiotics in IBS meta-analysis); Wilson B, et al. Am J Gastroenterol 2019 (prebiotic systematic review); Huaman JW, et al. Gastroenterology 2018 (low-FODMAP vs prebiotic)

2 Synbiotics (probiotic + prebiotic combinations) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: gut / digestion

THE NOISE Sold as synergistic probiotic-plus-prebiotic combos where the fiber 'feeds' the bacteria for superior gut-health and IBS results.

THE SCIENCE Effect: No significant improvement in IBS symptom scores or stool output vs control (2024 network meta-analysis, Nutrients 2024;16:2114; and 2022 chronic-constipation meta-analysis PMID 36372047 — insufficient evidence, high heterogeneity). Clears MCID: No — pooled effect not significant. Largest independent trial: No large independent trial; network meta-analysis (Nutrients 2024) finds prebiotics/synbiotics non-significant while some single-strain probiotics show a signal. Population: Adults with IBS and chronic constipation; a few RCTs, heterogeneous formulations.

THE VERDICT **Do not buy. Combining a probiotic with a prebiotic does not rescue either: network meta-analyses find no significant benefit for IBS symptoms or chronic constipation, with high heterogeneity, few RCTs, and substantial bias.**

BEST FORM A 'synergistic' synbiotic where the prebiotic is specifically chosen to feed the co-formulated strain (complementary design), with both strain identity and prebiotic type/dose disclosed - not a token pinch of inulin added to a generic probiotic. (Rationally designed synergistic synbiotics (matched strain + substrate, disclosed) **GOOD**; 'complementary' marketing blends that just co-pack an undisclosed probiotic with a little inulin offer little proven advantage over the components alone.)

WATCH The prebiotic component carries the same gas/bloating and FODMAP-flare risk in IBS - start low. Verify CFU at shelf-life end and refrigeration needs. Don't pay a premium for vague 'synbiotic' labeling without strain and prebiotic-dose transparency. Same immunocompromised/SIBO cautions as probiotics.

Sources: Nutrients 2024;16:2114 (network meta-analysis; synbiotics not significant); van der Schoot A, et al. 2022, PMID 36372047 (probiotics/synbiotics in chronic constipation; insufficient evidence); Ford AC, et al. Am J Gastroenterol 2014 (prebiotics/probiotics/synbiotics in IBS/CIC)

1 Slippery elm / marshmallow root / demulcent 'gut-soothing' herbs [added, in-category] Amazon ✓

Website ✓

Evidence: none / marketing (high confidence) | Effect: gut / digestion

THE NOISE Gut-soothing demulcent for reflux/IBS.

THE SCIENCE Effect: No controlled human outcome data isolating slippery elm/marshmallow. Clears MCID: No — no outcome data to assess. Largest independent trial: None. Population: None of adequate quality; occasional multi-herb combination products, not the single agent.

THE VERDICT Do not buy. Traditional demulcents with essentially no credible human outcome trials for IBS, reflux, or 'gut lining.' Pure marketing in this context.

BEST FORM Mucilage works by local contact, so a powder made into a fresh slurry/tea (or lozenges for throat/esophagus) beats dry capsules for coating mucosa. Choose plain, high-mucilage powder without fillers. (Powder slurry / lozenge GOOD (delivers hydrated mucilage to the mucosal surface); dry capsules WEAKER (less surface contact, must rehydrate in gut); cut herb teas are milder than concentrated powders.)

WATCH Evidence is largely traditional/soothing rather than proven disease-modifying. The mucilage physically coats the gut and can reduce absorption of other drugs/supplements - take 1-2 h apart. Use plenty of water. Marshmallow may modestly lower blood glucose; slippery elm is best avoided in pregnancy due to sourcing/quality concerns.

Sources: No credible primary RCTs identified for isolated slippery elm/marshmallow in gut conditions

Skin (6)

3.5 Nicotinamide (niacinamide, oral vitamin B3) Amazon ✓ Website ✓

Evidence: moderate/conditional (high confidence) | Effect: skin

THE NOISE Marketed as an anti-aging skin vitamin that repairs UV damage, evens tone, and keeps skin youthful and healthy from within.

THE SCIENCE Effect: ONTRAC: 23% relative reduction in new NMSC, rate ratio 0.77 (95% CI 0.63-0.95, P=0.02); SCC -30%, BCC -20% (NS); AK -11% to -20%. VA cohort: HR 0.86 any skin cancer, HR 0.78 for SCC (0.44 if started after first cancer), BCC HR 1.00 (no effect). Clears MCID: Yes for the chemoprevention endpoint — a ~1 in 4 reduction in new skin cancers is clinically meaningful in a high-risk population. NA for cosmetic outcomes (never measured). Largest independent trial: 2025 Veterans Affairs retrospective cohort (VA Corporate Data Warehouse), 33,822 patients / 12,287 nicotinamide-exposed: HR 0.86 any new skin cancer, HR 0.78 SCC — independent, supports ONTRAC in high-risk patients. Population: Immunocompetent adults with ≥2 histologically confirmed nonmelanoma skin cancers in the prior 5 years (ONTRAC); large VA cohort of veterans with prior keratinocyte cancer. NOT tested for cosmetic/anti-aging endpoints.

THE VERDICT Buy ONLY if you are a high-risk immunocompetent patient with a prior nonmelanoma skin cancer, and dose it correctly at 500 mg TWICE daily. This is a chemoprevention drug-like use, not a cosmetic/anti-aging supplement — there is essentially no credible evidence it improves the appearance of healthy skin.

BEST FORM Niacinamide (nicotinamide) - the non-flushing amide form. This is the one with skin-cancer-prevention evidence (500 mg twice daily reduced new non-melanoma skin cancers/actinic keratoses in the ONTRAC trial). Do NOT substitute niacin/nicotinic acid (causes flushing) or 'no-flush inositol hexanicotinate'. (Niacinamide (best for skin, no flush) vs. niacin/nicotinic acid (flushes, lowers lipids - different use) vs. inositol hexanicotinate ('no-flush niacin' - poorly converted, ineffective).)

WATCH Don't confuse with NR/NMN (marketed for 'NAD/anti-aging' - far pricier, weaker human evidence). Niacinamide does NOT lower cholesterol and does NOT flush. Keep below ~3 g/day; very high doses can stress the liver. The skin-cancer benefit is for higher-risk patients and reverses when stopped. Generally very well tolerated at 500 mg BID.

Sources: Chen AC et al. A Phase 3 Randomized Trial of Nicotinamide for Skin-Cancer Chemoprevention (ONTRAC). NEJM 2015;373:1618-1626. PMID 26488693; Allen NC et al. Nicotinamide for Skin-Cancer Chemoprevention in Transplant Recipients (ONTRANS). NEJM 2023;388:804-812. PMID 36856616. DOI 10.1056/NEJMoa2203086; 2025 Veterans Affairs retrospective cohort, nicotinamide and keratinocyte carcinoma (n=33,822); reported AJMC/News-Medical Sep 2025; JAAD Reviews 2025: Examining the safety of nicotinamide for skin cancer prevention. S2950-1989(25)00079-0

3 Polypodium leucotomos extract (Fernblock / Heliocare) Amazon ✓ Website ✓

Evidence: moderate/conditional (medium confidence) | Effect: skin

THE NOISE Sold as an 'oral sunscreen' fern extract that protects skin from UV damage, sunburn, and photoaging from the inside.

THE SCIENCE Effect: Surrogate endpoints only: RCTs report subjects markedly more likely to increase MED and reduce erythema (one healthy-adult RCT reported ~22x higher odds of MED increase vs placebo); a 2025 red-orange+PLE RCT reported +23.8% MED and -46.2% erythema intensity. No wrinkle/skin-cancer hard-endpoint RCT. Clears MCID: Uncertain/likely no in isolation — a fractional MED increase is far below the protection of an SPF 30-50 sunscreen; no defined MCID and no hard-outcome data. Largest independent trial: Thin — the most robust data (healthy-adult MED RCTs; El Mansouri 2023 xeroderma pigmentosum RCT) are small and frequently manufacturer-linked; no large independent hard-endpoint trial. Population: Healthy adults (MED/erythema studies) and patients with photodermatoses (polymorphic light eruption, solar urticaria), melasma, and xeroderma pigmentosum. Mostly small, short trials.

THE VERDICT Conditional / low priority. Real but modest photoprotective signal on surrogate endpoints (raises MED, blunts UV erythema). It is an ADJUNCT, never a substitute for sunscreen, and most of the literature comes from the manufacturer. Reasonable add-on for photodermatoses/melasma under a dermatologist; not worth buying as stand-alone sun defense.

BEST FORM A standardized Fernblock extract (the ingredient in Heliocare), since almost all the human photoprotection data uses it. Generic 'fern extract' of unknown standardization is a gamble. (Fernblock-standardized PLE (best, studied) > generic standardized PLE > unstandardized fern powder (unknown potency).)

WATCH It is an ADJUNCT, not a sunscreen replacement - it raises the threshold for sunburn modestly (helpful for photodermatoses, melasma, post-procedure) but you still need topical SPF and shade. Take ~240 mg before sun exposure, and re-dose for long exposure. Very safe; occasional mild GI upset. Don't use it as license to skip sunscreen.

Sources: Middelkamp-Hup/Nestor healthy-adult PLE MED/erythema RCTs (240 mg dosing); El Mansouri et al. Combined oral+topical Fernblock in xeroderma pigmentosum, Photodermatol Photoimmunol Photomed 2023. DOI 10.1111/phpp.12904; 2025 RCT: Red orange + Polypodium leucotomos on UVB-induced skin responses. PMC11990338; Clinical Efficacy & Safety of Oral PLE for Photoprotection: A Systematic Review, SKIN J Cutan Med (dermsquared #413)

2.5 Astaxanthin (oral, Haematococcus pluvialis) Amazon ✓ Website ✓

Evidence: weak/mostly null (low confidence) | Effect: skin

THE NOISE Promoted as a powerful antioxidant carotenoid that smooths wrinkles, improves skin elasticity and moisture, and protects against UV aging.

THE SCIENCE Effect: Meta-analysis: elasticity SMD 0.77 (95% CI 0.19-1.35, I²=75%, p=0.009); moisture SMD 0.53 (95% CI 0.05-1.01, I²=52%, p=0.03); wrinkle depth NOT significantly reduced. Clears MCID: Unclear — moderate instrument effects from short trials with high heterogeneity; no established MCID and wrinkle depth (the outcome consumers care about) is null. Largest independent trial: Thin — most robust trials are small and frequently industry-affiliated; no large independent RCT. Population: Healthy adults (often middle-aged, several Japanese-population trials); 6-12 week durations.

THE VERDICT Weak buy at best. A meta-analysis shows statistically significant but small improvements in skin elasticity and moisture and NO effect on wrinkle depth, from small, short (6-12 week) trials with heavy involvement from the ingredient maker (AstaReal/Fuji) and high heterogeneity. Signal is real but fragile.

BEST FORM Natural astaxanthin from Haematococcus pluvialis algae, in an oil-based softgel. Synthetic astaxanthin (petrochemical-derived, different isomer mix, sold mainly for aquaculture) has weaker antioxidant data and shouldn't be used as a human supplement. (Natural algal astaxanthin in oil softgel (best) > algal in dry capsule > synthetic astaxanthin (inferior isomers, avoid for humans).)

WATCH Strongly fat-soluble - take with a fatty meal. Typical 4-12 mg/day; skin-UV-resilience effects take 6-12 weeks. Can slightly lower blood pressure and blood sugar (additive with those meds) and has mild antiplatelet potential. Weak 5-alpha-reductase inhibition noted in vitro. Very well tolerated; harmless orange tint to skin at high doses is theoretical, not typical.

Sources: Ng QX et al. Systematic Review and Meta-Analysis on the Effects of Astaxanthin on Human Skin Ageing. Nutrients 2021;13(9):2917. PMC8472736. DOI 10.3390/nu13092917; Ito N et al. Protective Role of Astaxanthin for UV-Induced Skin Deterioration in Healthy People: RCT. Nutrients 2018. PMC6073124; Frontiers Med 2025 meta-analysis of dietary supplements for skin photoaging. PMC12318760

2.5 Oral hyaluronic acid (sodium hyaluronate) Amazon ✓ Website ✓

Evidence: weak/mostly null (low confidence) | Effect: skin · **literature compromised (funding/outlier)**

THE NOISE Sold as a beauty-from-within supplement that plumps and hydrates skin and reduces wrinkle depth by boosting internal hyaluronic acid.

THE SCIENCE Effect: 120 mg/day x12 wk RCTs report significant improvement in wrinkle assessment, stratum corneum hydration, TEWL, and elasticity; 2025 n=150 RCT reports significant gains in dermal density, hydration, elasticity, epidermal thickness, and wrinkle depth. Effect sizes/CIs inconsistently reported across the small trials. Clears MCID: Uncertain — instrument-measured improvements are statistically significant but small; no established MCID, and visible clinical relevance is modest. Largest independent trial: Thin — the largest recent trials still involve the manufacturer; no large fully independent RCT. Population: Healthy adults with dry skin/wrinkles, predominantly Japanese cohorts aged ~22-60; 8-12 week trials.

THE VERDICT **Weak/conditional. Multiple RCTs at 120 mg/day for 12 weeks report improved hydration, elasticity, and reduced wrinkle depth, but the positive evidence is concentrated in trials authored or funded by the ingredient maker (Kewpie), in small, ethnically narrow (Japanese) samples, and whether oral HA reaches the skin in humans is unresolved. Real signal, shaky foundation.**

BEST FORM Sodium hyaluronate (the stable salt) is essentially the only oral retail form - the label term varies but the molecule is the same. Molecular-weight marketing (low- vs high-MW) is debated and not a clear buying criterion; oral HA appears to work partly via gut-microbiome signaling and amino-sugar substrate rather than intact absorption. (n/a for meaningful absorption differences - low-MW is claimed to absorb better but evidence is thin; any reputable sodium hyaluronate at a studied dose is fine.)

WATCH Dose ~120-240 mg/day; skin-hydration benefit builds over ~6-12 weeks. Very safe with essentially no notable drug interactions. Don't expect the dramatic effect of injectable/topical HA - oral works modestly and systemically. Some products are fermentation-derived (vegan-friendly); rooster-comb-derived versions exist for those avoiding poultry allergens.

Sources: Oe M et al. Oral hyaluronan relieves wrinkles: 12-week double-blind placebo-controlled study. Clin Cosmet Investig Dermatol 2017;10:267-273. PMC8308347 (Kewpie-funded); Oral sodium hyaluronate improves skin hydration, barrier function and signs of aging: RCT in 150 healthy adults. Sci Rep 2025. DOI 10.1038/s41598-025-32758-5. PMC12827323; JDDonline 2025: Oral HA supplement efficacy in hydration, elasticity, wrinkle depth. PMID 40911749

2 Collagen peptides (hydrolyzed collagen) Amazon ✓ Website ✓

Evidence: weak/mostly null (high confidence) | Effect: skin · **literature compromised (funding/outlier)**

THE NOISE Marketed to reduce wrinkles and boost skin firmness, elasticity, and hydration by replenishing the body's collagen.

THE SCIENCE Effect: Batool 2026 primary pooled: hydration SMD 0.44 (95% CI 0.15-0.73), elasticity SMD 0.62 (95% CI 0.15-1.10), TEWL -0.39 (95% CI -0.62 to -0.16). After removing high-leverage outliers: wrinkle effect non-significant; hydration and elasticity diminished. Week-8 heterogeneity fell from I²=79.2% to 40.4% on outlier removal. Clears MCID: No for wrinkles (effect disappears in sensitivity/independent analysis). Borderline for hydration/elasticity — statistically present but small and attenuated once outliers/funding are controlled. Largest independent trial: Independent, high-quality studies in the AmJMed 2025 pool showed no significant effect on hydration, elasticity, or wrinkles — the benefit is confined to industry-funded and low-quality trials. Population: Predominantly women 35-65 with visible photoaging; short (8-12 week) trials.

THE VERDICT **Do not buy for wrinkles. Two 2025/2026 meta-analyses show the anti-wrinkle claim is a funding-source-and-outlier artifact: it survives pooled analysis but vanishes in independent and high-quality studies, and collapses when one outlier study is removed. A modest hydration/elasticity signal is the only thing that partly survives, and even that shrinks.**

BEST FORM Hydrolyzed collagen peptides (low molecular weight, ~2-10 kDa) - ideally a studied bioactive peptide like Verisol for skin. Plain gelatin (unhydrolyzed) and undenatured type-II collagen (a joint product, different purpose) are not the same thing. (Studied bioactive hydrolysates (Verisol, etc.) > generic hydrolyzed peptides > gelatin (poorly dissolved, larger peptides) > 'marine vs bovine' (both work; marine slightly smaller peptides).)

WATCH Dose 2.5-10 g/day; benefits (skin elasticity/hydration) show over 8-12 weeks and reverse on stopping. Take with vitamin C, a cofactor for collagen synthesis. Not vegan (bovine/marine/porcine source - check if it matters for diet/allergy). Marine collagen is a fish allergen. The protein is fully digested to amino acids, so 'topical-like' targeting claims are overstated - it works as signaling peptides + amino acid substrate.

Sources: Effects of Collagen Supplements on Skin Aging: Systematic Review and Meta-Analysis of RCTs. Am J Med 2025. S0002-9343(25)00283-9; 23 RCTs, n=1474; Batool et al. Oral Collagen Peptides and Skin Rejuvenation: Updated Meta-Analysis. J Cosmet Dermatol 2026. DOI 10.1111/jocd.71041; Pu SY et al. Effects of hydrolyzed collagen on skin aging: systematic review/meta-analysis. PMID 33742704

1 Biotin (vitamin B7) Amazon ✓ Website ✓

Evidence: none / marketing (high confidence) | Effect: skin

THE NOISE Marketed as the hair, skin, and nails vitamin that makes hair grow thicker and faster, strengthens nails, and clears skin.

THE SCIENCE Effect: No reliable placebo-controlled effect size exists for skin/hair/nails in non-deficient adults. Clears MCID: No — there is no credible controlled effect to clear any MCID. Largest independent trial: None of adequate quality; dermatology reviews (JAAD 2022) confirm the lack of RCTs. Population: Efficacy data limited to case reports and small uncontrolled series, mostly in genuine deficiency or specific genetic/nail-fragility conditions. No adequate placebo-controlled RCT in healthy, non-deficient adults.

THE VERDICT **Do not buy for skin/hair/nails unless you have a documented, genuine biotin deficiency (rare). No credible RCT shows benefit in non-deficient adults, and it actively causes dangerous laboratory-test interference — including falsely low troponin (a reported death) and false thyroid results. Actively worse than inert.**

BEST FORM Plain d-biotin; there is one active form and no meaningful absorption differences between brands or salt forms. The real issue is that megadoses are pointless - hair/nail benefit only occurs if you're genuinely biotin-deficient, which is rare. (n/a - d-biotin is the single active form; 10,000 mcg 'beauty' doses aren't better absorbed or more effective than the ~30 mcg daily requirement in most people.)

WATCH CRITICAL: high-dose biotin interferes with many lab immunoassays - it can falsely LOWER troponin (masking a heart attack) and falsely alter thyroid (TSH/T4) and hormone results, mimicking Graves' disease. Stop biotin at least 2-3 days before blood tests and tell your lab. Otherwise extremely safe (water-soluble). Skip the 5,000-10,000 mcg 'hair, skin & nails' megadoses unless a doctor confirms deficiency.

Sources: NIH ODS Biotin Health Professional Fact Sheet (efficacy and deficiency); FDA Safety Communication: Biotin (Vitamin B7) Interference with Troponin and Other Laboratory Tests (2017, updated 2019); AACC/ADLM Academy Guidance Document on Biotin Interference in Laboratory Tests; JAAD 2022: Biotin beware - perils of biotin supplementation. S0190-9622(22)00352-8

Bones (8)

3 Calcium + vitamin D (combined) Amazon ✓ Website ✓

Evidence: moderate/conditional (medium confidence) | Effect: bone / fracture

THE NOISE Sold as the foundational bone-health duo that builds strong bones and prevents osteoporosis and fractures as you age.

THE SCIENCE Effect: Yao 2019 (6 RCTs, n=49,282): hip fracture RR ~0.84 (favouring). WHI (Jackson 2006, n=36,282): overall hip fracture HR 0.88 (0.72-1.08, NS); 21% lower in age 60+ and ~29% in adherent. Zhao 2017 (33 RCTs, n=51,145 community-dwelling): hip fracture RR 1.09 (0.85-1.39), NS. Clears MCID: Yes in frail/institutionalized elderly (absolute hip-fracture reduction meaningful); No/near-null in community-dwelling adults. Largest independent trial: WHI (Jackson 2006, NEJM, PMID 16481635), NIH-funded, n=36,282: no significant overall hip-fracture reduction, +kidney stones. Population: Mostly women 65+, many institutionalized or vitamin-D-deficient/low calcium intake; some healthy postmenopausal (WHI).

THE VERDICT **Conditional buy. Real hip-fracture reduction exists but is concentrated in institutionalized, low-intake, or vitamin-D-deficient elderly (Chapuy/Decalys, French nursing homes). In healthy community-dwelling adults the effect is near-null, so for the typical retail buyer it is weak. Get calcium from diet first; supplement adds kidney-stone risk.**

BEST FORM Calcium citrate + vitamin D3. Citrate absorbs with or without food and is the right choice for anyone on a PPI/acid reducer or with low stomach acid; carbonate needs stomach acid and must be taken with food. Always pair with D3 (cholecalciferol), not D2. (Calcium citrate (absorbs any time, PPI-friendly, ~21% elemental) > calcium carbonate with food (cheaper, 40% elemental, needs acid) > coral/oyster-shell calcium (carbonate + heavy-metal risk).)

WATCH Absorption caps at ~500 mg elemental per dose - split larger amounts. Read ELEMENTAL calcium, not the compound weight. Prioritize diet first; total (diet+supplement) >2,000 mg/day and isolated calcium boluses have been linked to constipation, kidney stones and possibly cardiovascular risk. Space it 2+ hours from iron, zinc, thyroid meds (levothyroxine), and tetracycline/fluoroquinolone antibiotics - calcium blocks their absorption.

Sources: Yao P et al. JAMA Netw Open 2019;2(12):e1917789 (PMID 31860103); Zhao JG et al. JAMA 2017;318(24):2466-2482 (DOI 10.1001/jama.2017.19344); Jackson RD et al. (WHI) NEJM 2006;354:669-683 (PMID 16481635); Chapuy MC et al. (Decalys) NEJM 1992;327:1637-1642 (PMID 1331788)

2.5 Vitamin K2 (MK-7) [Amazon](#) ✓ [Website](#) ✓

Evidence: weak/mostly null (medium confidence) | Effect: bone / fracture

THE NOISE Promoted as the vitamin that directs calcium into bones and out of arteries to strengthen bone density and prevent fractures.

THE SCIENCE Effect: Knapen 2013 (n=244, 180 mcg MK-7, 3 yr): significant but small attenuation of lumbar-spine and femoral-neck BMD loss vs placebo ($P < 0.05$), no effect at total hip. 2022 meta (16 RCTs, n=6425): lumbar BMD improved ($P = 0.006$); fracture reduction driven by low-quality MK-4 Japanese trials. Clears MCID: Marginal. Lumbar BMD gains ($\sim +0.02$ g/cm²) are below what would reliably translate to fracture prevention. Largest independent trial: Knapen MHJ et al. 2013, n=244 healthy postmenopausal Dutch women - the largest well-conducted MK-7 BMD RCT; modest, not fracture-powered. Population: Postmenopausal women (osteopenia/osteoporosis); much of the fracture data is Japanese osteoporotic women on MK-4 45 mg.

THE VERDICT Do not rely on it for fractures. Biomarkers (osteocalcin carboxylation) move robustly and lumbar BMD attenuation is real but small; hard fracture evidence is weak and concentrated in older Japanese MK-4 45 mg/day pharmacological trials, a dose $\sim 250\times$ the retail MK-7 pill. As a bone insurance policy it is thin.

BEST FORM MK-7 in the all-trans configuration - long half-life (~ 3 days) so once-daily dosing keeps blood levels steady. MK-4 works but has a very short half-life needing multiple daily high doses. Ensure the label says 'all-trans' or a branded MK-7 (e.g. MenaQ7); cis-isomer content is biologically inactive filler. (All-trans MK-7 (best - potent, once-daily) > MK-4 (short half-life, needs 3x/day mega-doses) > cis-contaminated cheap MK-7 (inactive).)

WATCH MAJOR interaction: any vitamin K antagonizes warfarin - do not start K2 if you're on warfarin without physician oversight (it can destabilize INR). Fat-soluble, so take with a meal containing fat. Often paired with vitamin D (D drives calcium into blood; K2 is thought to help direct it to bone rather than arteries). Typical 90-180 mcg/day. Very safe otherwise, no known toxicity ceiling for non-anticoagulated people.

Sources: Knapen MHJ et al. Osteoporos Int 2013;24:2499-2507 (PMID 23525894); Ma ML et al. Front Public Health 2022;10:979649 (PMC9403798); Mott A et al. (K/BMD & fracture) Osteoporos Int 2019 (systematic review); Rønn SH et al. MK-7 osteopenia 3-yr RCT 2021 (PMID 33030563)

2 Calcium alone [Amazon](#) ✓ [Website](#) ✓

Evidence: weak/mostly null (medium confidence) | Effect: bone / fracture

THE NOISE Sold as the classic bone mineral that builds and maintains strong bones and prevents osteoporosis.

THE SCIENCE Effect: Bolland/Reid meta-analyses: minimal fracture benefit; calcium supplements associated with $\sim +24$ -27% MI risk in some pooled analyses (contested). WHI: +17% kidney stones. Clears MCID: No - fracture benefit small/inconsistent, does not reliably clear MCID. Largest independent trial: WHI (Jackson 2006, PMID 16481635). Population: Postmenopausal women / older adults (WHI, Bolland meta-analyses).

THE VERDICT Do not buy standalone for bones. Calcium-only supplements give small BMD changes, weak/inconsistent fracture reduction, raise kidney-stone risk, and carry a contested cardiovascular/MI signal. Dietary calcium is the better route; if supplementing, pair with vitamin D in the relevant elderly population rather than taking calcium alone.

BEST FORM Calcium citrate - absorbs with or without food, doesn't need stomach acid (best for PPI/H2-blocker users and older adults) and is gentler on the gut. Calcium carbonate is cheaper and more concentrated but must be taken WITH food and adequate stomach acid. (Citrate (any time, PPI-friendly, $\sim 21\%$ elemental, less constipating) > carbonate with food (40% elemental, cheap) > coral/oyster/bone-meal (carbonate with heavy-metal/lead risk - avoid).)

WATCH Absorption caps ~ 500 mg elemental per dose - split it. Go by ELEMENTAL calcium on the label. Food-first is best; large isolated supplemental doses are linked to constipation, kidney stones and possible cardiovascular signal - and calcium alone (without D/K2/exercise) is a weak bone strategy. Space 2+ hours from iron, zinc, thyroid meds and tetracycline/quinolone antibiotics.

Sources: Bolland MJ et al. BMJ 2010;341:c3691 (PMID 20671013); Bolland MJ et al. BMJ 2011;342:d2040; Jackson RD et al. (WHI) NEJM 2006;354:669-683 (PMID 16481635)

2 Collagen peptides (specific bioactive collagen peptides, for bone) Amazon ✓ Website ✓

Evidence: weak/mostly null (low confidence) | Effect: bone / fracture · **literature compromised (funding/outlier)**

THE NOISE Marketed as bioactive collagen peptides clinically shown to increase spine and hip bone density and strengthen the bone matrix.

THE SCIENCE Effect: König 2018 (n=131): spine vBMD +4.2%, femoral neck +7.7% vs placebo over 12 mo - effect size implausibly large for a nutritional agent. Clears MCID: Nominally yes, but the effect size is large enough to raise validity concerns rather than confidence. Largest independent trial: None independent; only König DA et al. Nutrients 2018;10(1):97 (PMID 29337906), industry-funded. Population: Postmenopausal women with age-related low BMD (n=131), single trial + open-label follow-up.

THE VERDICT **Do not buy on current evidence. The single supporting BMD trial is small, GELITA/FORTIBONE-funded, and reports implausibly large gains (spine +4.2%, femoral neck +7.7% in 12 months - exceeding bisphosphonates). No independent replication, no fracture data. Mechanism/marketing outruns outcome.**

BEST FORM Hydrolyzed collagen peptides (low molecular weight, ~2-10 kDa) - ideally a studied bioactive peptide like Verisol for skin. Plain gelatin (unhydrolyzed) and undenatured type-II collagen (a joint product, different purpose) are not the same thing. (Studied bioactive hydrolysates (Verisol, etc.) > generic hydrolyzed peptides > gelatin (poorly dissolved, larger peptides) > 'marine vs bovine' (both work; marine slightly smaller peptides).)

WATCH Dose 2.5-10 g/day; benefits (skin elasticity/hydration) show over 8-12 weeks and reverse on stopping. Take with vitamin C, a cofactor for collagen synthesis. Not vegan (bovine/marine/porcine source - check if it matters for diet/allergy). Marine collagen is a fish allergen. The protein is fully digested to amino acids, so 'topical-like' targeting claims are overstated - it works as signaling peptides + amino acid substrate.

Sources: König D et al. Nutrients 2018;10(1):97 (PMID 29337906; PMC5793325); Zdzieblik D et al. J Bone Metab 2021;28:207-215 (PMID 34520654, follow-up)

2 Magnesium Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: bone / fracture

THE NOISE Marketed as an essential bone mineral that supports bone density and structure alongside calcium.

THE SCIENCE Effect: Groenendijk 2022 meta (RCT arm): fracture RR 0.72 (0.51-1.02), NS; significant reductions in serum ALP and osteocalcin (turnover markers). Dietary meta (Farsinejad-Marj 2016): + association with hip/femoral-neck BMD. Clears MCID: No - no significant clinical fracture endpoint; turnover-marker changes are surrogate. Largest independent trial: Groenendijk I et al. systematic review/meta-analysis 2022 (PMID 34666201) - no significant fracture reduction. Population: Older adults, osteoporotic/osteopenic; much evidence is dietary-intake observational, not supplement RCT.

THE VERDICT **Do not buy for bones unless deficient. No RCT shows fracture reduction (pooled RR 0.72, CI crosses 1). Bone-turnover markers shift and observational BMD associations are positive, but that is mechanism plus epidemiology, not outcome. Any real benefit is a correction-of-deficiency effect.**

BEST FORM Magnesium glycinate (bisglycinate) for best tolerance, or citrate/malate for good absorption. Avoid magnesium oxide - roughly 4x lower absorption (~4% bioavailable) and mostly acts as a laxative. (Glycinate (gentlest, well-absorbed, best if constipation-prone) ≈ citrate/malate (well-absorbed, citrate mildly laxative) > glycerophosphate/taurate > OXIDE (poor ~4x less absorbed, laxative) ≈ sulfate.)

WATCH Check ELEMENTAL magnesium on the label (oxide lists a big compound number but little usable Mg). The classic sign of too much is loose stools - back off the dose. Aim to stay near the ~350 mg/day supplemental upper limit unless directed higher. Space 2+ hours from tetracycline/fluoroquinolone antibiotics, bisphosphonates and thyroid meds. Caution in significant kidney disease (risk of accumulation).

Sources: Groenendijk I et al. Bone 2022;154:116233 (PMID 34666201); Farsinejad-Marj M et al. Osteoporos Int 2016;27:1389-1399 (PMID 26556742); Veronese N et al. magnesium & bone reviews 2021

1.5 Boron [Amazon](#) ✓ [Website](#) ✓

Evidence: none / marketing (medium confidence) | Effect: bone / fracture

THE NOISE Promoted as a trace mineral that supports bone strength and density by aiding calcium, magnesium, and hormone metabolism.

THE SCIENCE Effect: No fracture or BMD RCT effect size exists; only shifts in urinary calcium/magnesium and serum steroid markers. Clears MCID: N/A - no clinical endpoint studied. Largest independent trial: None with a bone outcome; Nielsen WL et al. metabolic-balance work only. Population: Small metabolic-balance studies in postmenopausal women (Nielsen 1987 and similar); no outcome trials.

THE VERDICT **Do not buy for bones. Boron modestly influences calcium/magnesium and steroid-hormone metabolism in small balance studies, but there are no human fracture or BMD RCTs. This is mechanism and micronutrient physiology, not outcome evidence.**

BEST FORM Chelated/organic boron - calcium fructoborate, boron glycinate, or boron citrate at ~3 mg elemental; these are better absorbed and gentler than cheap sodium borate/borax. Boron aspartate blends are also fine. (Calcium fructoborate / boron glycinate / citrate (organic, well-absorbed, food-form) > boron aspartate > sodium tetraborate (borax, industrial-tasting, harsher). Absorption is actually high (>85%) across most forms, so tolerability and dose accuracy drive the choice more than uptake.)

WATCH Effective dose is only ~3 mg/day - don't confuse with the 20 mg/day adult Tolerable Upper Limit. Boron is hormonally active (raises free testosterone and estradiol, lowers SHBG), so it stacks with hormone therapy - be cautious. Supports bone via magnesium/calcium/vitamin D metabolism, so it is an adjunct, not a standalone bone builder. High doses cause nausea.

Sources: Nielsen FH et al. FASEB J 1987;1:394-397 (boron/calcium balance)

1.5 Vitamin D alone [Amazon](#) ✓ [Website](#) ✓

Evidence: none / marketing (high confidence) | Effect: bone / fracture

THE NOISE Sold as the sunshine vitamin essential for strong bones, calcium absorption, and preventing fractures and falls.

THE SCIENCE Effect: VITAL (LeBoff 2022, n=25,871): total fracture HR 0.98 (0.89-1.08), hip HR 1.01 (0.70-1.47) - null; no effect on falls or BMD/structure. D-Health (n=21,315): fractures null. Burt 2019 (n=311): radial vBMD change -1.2% (400 IU) vs -2.4% (4000) vs -3.5% (10000), dose-dependent LOSS. Clears MCID: No - null for fractures/falls in replete adults; where it moves BMD at high dose it moves it the WRONG way. Largest independent trial: VITAL - LeBoff MS et al. NEJM 2022;387:299-309 (PMID 35939577), NIH-funded, n=25,871: null for fractures. Population: Generally healthy, largely vitamin-D-replete midlife/older adults not selected for deficiency or low bone mass.

THE VERDICT **Do not buy for bones if you are vitamin-D-replete. Three mega-trials (VITAL, D-Health, DO-HEALTH; >65,000 people) are null for fractures and falls, and high doses actively LOWERED bone density in a dose-dependent way. Value is limited to correcting genuine deficiency/osteomalacia, ideally paired with calcium in frail elderly.**

BEST FORM Vitamin D3 (cholecalciferol) - raises and sustains blood 25(OH)D more effectively than D2 (ergocalciferol). Softgel/oil-based or a fatty-meal-taken tablet beats a dry tablet on an empty stomach. (D3 in oil/with fat (best) > D3 dry tablet with food > D2/ergocalciferol (prescription, less potent, shorter-lived) - use D2 only if you need a vegan mega-dose.)

WATCH Fat-soluble - take with your largest/fattiest meal for markedly better absorption. Get a 25(OH)D blood level rather than guessing; most people do well on 1,000-2,000 IU/day. It's fat-soluble and CAN build to toxicity (hypercalcemia) - avoid chronic mega-doses (e.g. 10,000+ IU/day) without monitoring. Magnesium and vitamin K2 are functional cofactors. Interacts with thiazides (raises calcium) and certain heart meds.

Sources: LeBoff MS et al. (VITAL) NEJM 2022;387:299-309 (PMID 35939577); Waterhouse M et al. (D-Health) Lancet Diabetes Endocrinol 2023;11:324-332; Burt LA et al. JAMA 2019;322:736-745 (PMID 31454046); Bischoff-Ferrari HA et al. (DO-HEALTH) JAMA 2020;324:1855-1868

1 Strontium (retail strontium citrate) Amazon ✓ Website ✓

Evidence: none / marketing (high confidence) | Effect: bone / fracture

THE NOISE Marketed as a bone-building mineral that increases bone density and strengthens bones to prevent osteoporosis and fractures.

THE SCIENCE Effect: Ranelate SOTI (Meunier 2004, n=1649): vertebral fracture RR 0.59 (0.48-0.73), ~41% reduction. TROPOS: nonvertebral reduction. Strontium citrate: NO RCT effect size exists. Clears MCID: N/A for citrate (no data); ranelate cleared it but is unsafe/withdrawn. Largest independent trial: None independent for citrate. Ranelate: Meunier PJ et al. (SOTI) NEJM 2004;350:459-468 (PMID 14749454) - industry-funded drug trial. Population: Ranelate: postmenopausal women with osteoporosis (SOTI/TROPOS). Citrate: essentially unstudied in humans for fracture or BMD.

THE VERDICT **Do not buy. The fracture evidence belongs to a DIFFERENT compound - prescription strontium ranelate (SOTI/TROPOS) - which the EMA restricted then withdrew for cardiovascular and clot risk. Over-the-counter strontium CITRATE has zero fracture RCTs, zero BMD RCTs, and any 'BMD gain' on a DEXA is largely a measurement artifact (strontium's high atomic number inflates the reading).**

BEST FORM If used at all, retail strontium CITRATE is the OTC form - but understand it is NOT strontium ranelate, the prescription drug that carried the bone-fracture evidence (and its cardiovascular/clot warnings). The citrate salt is assumed similar but lacks that trial evidence. Honestly, this is a supplement to approach with skepticism. (Strontium citrate (OTC, weak/assumed evidence) vs. strontium ranelate (prescription, real trial data but withdrawn in many markets over heart-attack/VTE risk). Retail chloride/gluconate are lesser variants.)

WATCH It DISTORTS DEXA bone-density scans - strontium's heavier atoms artificially inflate the BMD reading, so your scan can look better while bone isn't - tell any doctor ordering a DEXA that you take it. It competes with calcium for absorption, so take them several hours apart. Ranelate's class carried serious cardiovascular and blood-clot risk. Space from calcium and from antibiotics/thyroid meds. Many clinicians advise against routine use.

Sources: Meunier PJ et al. (SOTI) NEJM 2004;350:459-468 (PMID 14749454); Reginster JY et al. (TROPOS) J Clin Endocrinol Metab 2005;90:2816-2822; EMA PRAC press release, suspension of Protelos/Osseor, Jan 2014; EMA withdrawal of Protelos marketing authorisation, 2020

3.5 Collagen / gelatin + vitamin C, dosed 30-60 min pre-load (timing protocol) Amazon ✓ Website ✓

Evidence: moderate/conditional (medium confidence) | Effect: tendon / ligament

THE NOISE Sold for skin, joints, tendons and anti-aging.

THE SCIENCE Effect: Shaw 2017 (AJCN, PMID 27852613): 15 g gelatin + 48 mg vit C ~doubled the serum collagen-synthesis marker PINP vs placebo (crossover, n=8). Systematic review (JFMA 2026;11:130): tendon CSA +6-11% vs +1-4% placebo in 3/4 RCTs; stiffness +15.4% to +56.4%, Young's modulus +14.2% to +46.2% in the 15-30 g arms (narrative synthesis, no pooled SMD). Praet 2019 (Nutrients 11:76): VISA-A rose +12.6 (9.7-15.5) on collagen vs +5.3 (2.3-8.3) on placebo over 3 mo, all on top of a calf-loading program. Clears MCID: Partial. VISA-A gain of ~12.6 exceeds the ~7-point MCID, but that trial is small and industry-funded. Structural CSA/stiffness changes are real but no trial has shown they translate to a hard clinical endpoint (re-injury rate, return-to-sport, tendon rupture). Largest independent trial: No large independent clinical trial exists. The independent mechanistic/structural work (Shaw 2017 n=8; Lis & Baar dose-finding; various n=8-24 adaptation studies) is small; positive but underpowered for clinical endpoints. Population: 8 RCTs, n=257 (246 M : 11 F), ages 18-52, mostly trained/active young men; plus small mid-portion Achilles tendinopathy cohorts (Praet 2019, n=20) and patellar tendon studies. Very little data in women, older adults, or untrained people.

THE VERDICT Conditional buy, and only as an adjunct. The strongest claim - that 15-30 g collagen/gelatin + vitamin C ~60 min before loading increases tendon cross-sectional area and stiffness - is GRADE A but exists ONLY layered on structured high-intensity resistance loading, in mostly young healthy men, at doses above what most retail tubs deliver. The consumer-facing endpoints (pain, function, injury reduction) rest on small, largely Gelita-funded trials. Without training it does close to nothing.

BEST FORM ~15 g hydrolyzed collagen peptides OR gelatin, plus ~50 mg vitamin C, taken 30-60 min BEFORE loading exercise (Shaw et al. 2017 protocol). Peptides dissolve cold and taste neutral; gelatin needs warm liquid but performed identically in the trials. The timing window is the active ingredient here. (Hydrolyzed peptides (cold-soluble, convenient) = gelatin (equally effective, must be warm) - both deliver the glycine/proline/hydroxyproline and the amino-acid blood spike that peaks ~1 hr post-dose, coinciding with the loading bout. Undenatured/UC-II is a DIFFERENT mechanism and does not substitute here.)

WATCH Vitamin C is a mandatory cofactor for prolyl/lysyl hydroxylase - collagen synthesis fails without it, so never skip it. The pre-exercise timing is essential: dosing after training misses the amino-acid delivery window. Effect is on collagen synthesis, not a quick painkiller - expect weeks. Take on a relatively empty stomach so the amino-acid spike isn't blunted.

Sources: Shaw G, Lee-Barthel A, Ross ML, Wang B, Baar K. Vitamin C-enriched gelatin supplementation before intermittent activity augments collagen synthesis. Am J Clin Nutr 2017;105(1):136-143. PMID 27852613; Praet SFE, Purdam CR, Welvaert M, et al. Oral supplementation of specific collagen peptides combined with calf-strengthening exercises enhances function and reduces pain in Achilles tendinopathy. Nutrients 2019;11(1):76. PMID 30609761; Collagen supplementation on tendon-related structural and performance outcomes: a systematic review. J Funct Morphol Kinesiol 2026;11(1):130 (MDPI); Khatri M, et al. Does additional dietary supplementation improve physiotherapeutic treatment outcome in tendinopathy? A systematic review and meta-analysis. J Clin Med 2022;11(6):1666

2 Curcumin (turmeric extract) Amazon ✓ Website ✓

Evidence: weak/mostly null (high confidence) | Effect: tendon / ligament

THE NOISE Sold as an anti-inflammatory that soothes tendon and joint inflammation and supports connective-tissue recovery.

THE SCIENCE Effect: No standalone human tendon RCT with an effect size to report. Combination curcumin+Boswellia open-label study reported symptom reduction with no control group. Any pain benefit in the physiotherapy+supplement meta (SMD -0.74) is confounded by co-ingredients and pharma funding. Clears MCID: na - no controlled human tendon endpoint exists to compare against an MCID. Largest independent trial: None. Largest rigorous work is preclinical (rat Achilles, n=30, 20/40/200 mg/kg) showing histologic/biomechanical healing - does not transfer to a grading of human outcomes. Population: Predominantly animal models (rat Achilles collagenase injury, 20-200 mg/kg). Human data limited to open-label / post-observational tendinopathy cohorts using combination products.

THE VERDICT Do not buy for tendon health. There is no credible standalone human tendon-outcome RCT. Every positive human data point is either an open-label observational study or a curcumin+Boswellia combination product; the rest is rat collagenase and cell-culture work. Mechanism (NF-kB, anti-inflammatory) massively outruns outcome. There is also an active regulatory liver-injury signal.

BEST FORM An enhanced-bioavailability form - phytosome (Meriva/curcumin-phospholipid), micellar/nano (Theracurmin), or a piperine-standardized extract (Curcumin C3 Complex + BioPerine). Plain 95% turmeric extract is nearly worthless orally (curcumin is poorly soluble and rapidly glucuronidated). (Phytosome/Meriva (~29x) and micellar/Theracurmin (very high uptake) > piperine-boosted (~20x via UGT inhibition) >> plain curcuminoid powder (minimal systemic absorption) >> raw turmeric spice.)

WATCH Take with a fatty meal (curcumin is lipophilic). Piperine works by INHIBITING drug-metabolizing enzymes (CYP3A4, UGT) - this raises blood levels of many medications, so avoid piperine forms if on prescription drugs. Mild anticoagulant/antiplatelet effect - caution with warfarin/before surgery. Can aggravate reflux and gallbladder issues.

Sources: Fusar-Poli L, et al. Is curcumin useful in the treatment and prevention of tendinopathy and myotendinous junction injury? A scoping review. *Nutrients* 2023. PMID 36678255; Curcumin and tendon healing in a rat model of Achilles tendon injury. PMID 29922838; TGA Safety Advisory: Medicines containing turmeric or curcumin - risk of liver injury (15 Aug 2023), tga.gov.au; Curcuminoids and Boswellia serrata extract combination decreases tendinopathy symptoms - open-label post-observational study (2023)

2 Omega-3 (EPA/DHA) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: tendon / ligament

THE NOISE Marketed as an anti-inflammatory fish oil that reduces tendon and joint inflammation and supports connective-tissue healing.

THE SCIENCE Effect: Sandford 2018 (BMJ Open Sport Exerc Med): SPADI -8.3 (95% CI -15.6 to -0.94, p=0.03) favouring omega-3 at 3 months; NO between-group difference at 12 months. CI upper bound near zero. Clears MCID: Borderline/no. SPADI MCID is ~8-13 points; the -8.3 point estimate sits at the lower edge, the CI crosses toward null, and the effect is not durable. Largest independent trial: Sandford et al. (BMJ Open Sport Exerc Med 2018), rotator cuff related shoulder pain: transient SPADI improvement at 3 mo, null at 12 mo. Population: Rotator cuff related shoulder pain / rotator cuff tendinopathy patients (Sandford RCT, plus small chronic tendon-disorder athlete trials). Not healthy adults and not a tendon-structure or rupture-healing population.

THE VERDICT Do not buy for tendons. The single decent RCT (rotator cuff related shoulder pain) showed a modest, statistically borderline pain benefit at 3 months that had vanished by 12 months and sits around or below the SPADI MCID. It is a transient anti-inflammatory effect in a pain population, not evidence of tendon healing, and typical retail softgels underdose the trial.

BEST FORM Re-esterified triglyceride (rTG) or natural triglyceride form; phospholipid (krill) is also well-absorbed. Choose by combined EPA+DHA content on the back label, not total fish-oil weight. (rTG / natural triglyceride (best absorbed, ~higher than EE) ≈ phospholipid/krill > ethyl ester (EE) - EE is the cheap concentrate form, absorbed noticeably worse without a fatty meal. 'Fish oil 1000 mg' often hides only 300 mg EPA+DHA - read the actives.)

WATCH Take with a fat-containing meal (especially EE). Check for third-party oxidation testing (TOTOX) - rancid oil is common and pro-inflammatory. Enteric coating or splitting the dose reduces fishy burp. Antiplatelet effect at high doses - caution with anticoagulants and before surgery. Store cool/dark.

Sources: Sandford FM, et al. A randomised controlled trial of long-chain omega-3 PUFA in the management of rotator cuff related shoulder pain. *BMJ Open Sport Exerc Med* 2018 (KCL); Nutritional supplements in the clinical management of tendinopathy: a scoping review. *J Sport Rehabil* 2023;32(5):493; Khatri M, et al. *J Clin Med* 2022;11(6):1666 (supplement + physiotherapy meta-analysis, pain SMD -0.74); STRENGTH / OMEMI trials - high-dose omega-3 and atrial fibrillation signal (context for dose safety)

1.5 Vitamin C (alone, as a standalone tendon supplement) Amazon ✓ Website ✓

Evidence: none / marketing (medium confidence) | Effect: tendon / ligament

THE NOISE Sold as immune support and collagen/healing aid.

THE SCIENCE Effect: Preliminary rotator cuff repair study: a non-significant trend toward improved healing with vitamin C; no reported effect size clearing significance. No standalone tendinopathy pain/function RCT. Clears MCID: No - the one human tendon signal did not reach statistical significance, let alone an MCID. Largest independent trial: Preliminary rotator cuff repair vitamin C study (2021): trend only, authors called for a larger adequately powered trial. Population: Post-rotator-cuff-repair surgical patients (preliminary study); otherwise mechanistic/cell and deficiency-repletion contexts. No healthy-adult tendon outcome data.

THE VERDICT Do not buy as a standalone tendon product. Vitamin C is a genuine cofactor for prolyl hydroxylase in collagen synthesis, but the only human tendon trial (rotator cuff repair) showed a non-significant trend, and there is no evidence that high-dose vitamin C helps tendons in already-replete adults. Its real role is a ~50 mg cofactor inside the collagen+loading protocol, not a 500-1000 mg standalone pill.

BEST FORM Plain ascorbic acid in divided doses is fine and cheapest; use buffered (sodium/calcium ascorbate) or liposomal if you get GI upset or want higher single doses. Ester-C/rose-hip premiums add little. (Ascorbic acid (divided doses) ~ buffered ascorbate (gentler on gut) ~ liposomal (better high-dose tolerance/uptake) > 'Ester-C' marketing premium; fractional absorption falls above ~200-400 mg)

WATCH Doses >2 g/day cause diarrhea/GI upset (upper limit 2000 mg); split doses. Raises urinary oxalate - caution with kidney-stone history. Interferes with glucose meters and fecal occult-blood tests. Enhances non-heme iron absorption (helpful, or a caution in iron overload). Regular prophylaxis modestly shortens colds; starting at onset does little.

Sources: Does vitamin C supplementation improve rotator cuff healing? A preliminary study (2021); Shaw G, et al. Am J Clin Nutr 2017;105(1):136-143 (vitamin C as the prolyl-hydroxylase cofactor in the gelatin protocol). PMID 27852613; Effect of vitamin C on flexor tendon healing zone II RCT (PMC12925623) - injection model, adjunct context

Joint / cartilage (8)

3 Curcumin / Curcuma longa (turmeric) extract Amazon ✓ Website ✓

Evidence: moderate/conditional (medium confidence) | Effect: joint / arthritis pain

THE NOISE Promoted as a powerful natural anti-inflammatory that relieves arthritis and joint pain as effectively as NSAIDs but without the stomach or liver side effects, with branded extracts (Meriva, BCM-95, Theracurmin) touted for superior absorption.

THE SCIENCE Effect: 2025 NMA: curcumin WOMAC pain WMD -3.31 (95% CI -5.25 to -1.34), total WOMAC -15.6 (-23.5 to -7.61), function -11.7 (-17.8 to -5.52). 2022 meta-analysis (PMC9353077): significant VAS and WOMAC pain reduction, non-inferior to NSAIDs. Non-inferiority vs paracetamol in bioavailable-turmeric RCT (PMC7847013). Clears MCID: Plausibly yes for pain in enhanced-extract trials (VAS reductions ~1.5-2 cm in several studies exceed MCID), but heterogeneous and formulation-dependent. Largest independent trial: Mix of independent and sponsor trials; several academic trials (e.g., Panahi 2014, n=53, Iran) positive but small and often product-supplied. No single large fully-independent pivotal trial; effect is consistent across many small ones. Population: Adults with symptomatic knee OA; trials typically 4-16 weeks.

THE VERDICT Conditional. Best-supported item in the category: multiple RCTs show real pain reduction, roughly non-inferior to NSAIDs/paracetamol. But effect sizes come from bioavailability-enhanced extracts (Meriva, BCM-95, Theracurmin), many trials are industry-funded and short, and there is a genuine regulator liver-injury signal. Plain retail turmeric powder is not the studied product.

BEST FORM An enhanced-bioavailability form - phytosome (Meriva/curcumin-phospholipid), micellar/nano (Theracurmin), or a piperine-standardized extract (Curcumin C3 Complex + BioPerine). Plain 95% turmeric extract is nearly worthless orally (curcumin is poorly soluble and rapidly glucuronidated). (Phytosome/Meriva (~29x) and micellar/Theracurmin (very high uptake) > piperine-boosted (~20x via UGT inhibition) >> plain curcuminoid powder (minimal systemic absorption) >> raw turmeric spice.)

WATCH Take with a fatty meal (curcumin is lipophilic). Piperine works by INHIBITING drug-metabolizing enzymes (CYP3A4, UGT) - this raises blood levels of many medications, so avoid piperine forms if on prescription drugs. Mild anticoagulant/antiplatelet effect - caution with warfarin/before surgery. Can aggravate reflux and gallbladder issues.

Sources: Zeng L et al. Front Immunol 2022;13:891822 (PMC9353077); 2025 turmeric-products NMA, BMC Complement Med Ther (PMID 40731001); Bioavailable turmeric vs paracetamol RCT (PMC7847013); TGA safety alert: turmeric/curcumin liver injury, 2023

2.5 Avocado-soybean unsaponifiables (ASU) [added in-category compound] Amazon ✓ Website ✓

Evidence: weak/mostly null (low confidence) | Effect: joint / arthritis pain · **literature compromised (funding/outlier)**

THE NOISE Sold for joint/osteoarthritis cartilage support.

THE SCIENCE Effect: Meta-analyses show small symptomatic improvement (Lequesne/pain) vs placebo; ERADIAS (Maheu 2014, n=399, 3 yr) missed primary joint-space-narrowing endpoint but showed reduced proportion of progressors in a secondary analysis. Clears MCID: Symptomatic effects modest and near MCID at best; structure-modification unproven. Largest independent trial: ERADIAS (Maheu 2014, n=399) -- sponsor-linked (Expanscience); primary endpoint negative. Population: Adults with symptomatic hip or knee OA; trials up to 3 years (ERADIAS).

THE VERDICT **Borderline / do not prioritize. Some symptomatic benefit in hip/knee OA and a possible structure-modifying signal in hip OA, but pivotal trials are sponsor-funded (Expanscience/Piascledine), results are inconsistent, and it is not a US mainstream retail product at trial grade.**

BEST FORM The studied ASU preparation - Piascledine 300 (300 mg/day) with the ~1:2 avocado:soybean unsaponifiable ratio. Buy a product matching that standardized composition rather than a generic 'avocado/soy oil' blend. (Piascledine-type standardized ASU (300 mg, the ratio and processing used in European OA trials, possible structure-modifying effect on joint-space) >> generic unstandardized avocado-soy oil capsules (unproven, wrong composition). The specific unsaponifiable fraction and ratio are the active material.)

WATCH Soy-derived - soy-allergic users beware. It is the non-saponifiable lipid fraction, not just avocado/soy oil, so composition matters. Well tolerated; occasional GI upset or reflux. Slow onset (2+ months) and used as a background disease-modifier, not acute pain relief. Evidence strongest for knee and hip OA.

Sources: Maheu E et al. ERADIAS. Ann Rheum Dis 2014;73:376 (PMID 23345601); Christensen R et al. ASU meta-analysis, Osteoarthritis Cartilage 2008;16:399 (PMID 18337124)

2.5 Boswellia serrata extract (AKBA-standardized: 5-Loxin, Aflapin) Amazon ✓ Website ✓

Evidence: weak/mostly null (low confidence) | Effect: joint / arthritis pain · **literature compromised (funding/outlier)**

THE NOISE Marketed as clinically standardized 'Indian frankincense' that blocks the 5-LOX inflammation pathway to relieve knee osteoarthritis pain and improve mobility in as little as five to seven days.

THE SCIENCE Effect: 2025 NMA: Aflapin WOMAC pain WMD -22.4 (95% CI -28.4 to -16.4), function -15.8 (-21.9 to -9.68) vs placebo - from a single n=19 vs 19 trial (Sengupta 2010). 2020 meta-analysis (7 RCTs, 545 pts): WOMAC pain WMD -14.22 (95% CI -22.34 to -6.09), stiffness -10.04 (-15.86 to -4.22). Clears MCID: Point estimates exceed WOMAC MCID, but confidence undermined by tiny per-arm sample sizes and sponsor origin. Largest independent trial: No robust independent trial. The AKBA extract literature (Sengupta 2008 n=75; 2010 n=60; Vishal 2011 n=60) is dominated by Laila Nutraceuticals / PLThomas (manufacturer) sponsorship. Population: Adults with knee OA; trials 30-90 days, small samples.

THE VERDICT **Do not buy on current evidence. Reported effect sizes are large, but they come almost entirely from very small trials run or funded by the ingredient manufacturers (Laila/PLThomas), with essentially no independent replication. The signal is real-looking but compromised.**

BEST FORM Extract STANDARDIZED to AKBA (acetyl-11-keto-beta-boswellic acid) - 5-Loxin (30% AKBA) or Aflapin (AKBA + enhanced absorption). AKBA is the potent 5-LOX inhibitor; ordinary 65% 'boswellic acids' extracts often contain very little AKBA. (Aflapin / 5-Loxin (AKBA-standardized, low effective dose ~100-250 mg, fastest onset in trials) > generic Boswellia standardized only to 'total boswellic acids' (may be high in the pro-inflammatory KBA and low in AKBA) >> crude resin. Standardize to AKBA specifically, not total acids.)

WATCH Check the AKBA percentage, not just 'boswellic acids.' Take with food/fat for absorption. Generally well tolerated; can cause mild GI upset, and rarely acid reflux/nausea. May interact with drugs metabolized by CYP enzymes. Onset can be relatively fast (1-4 weeks) versus other joint agents.

Sources: Yu G et al. Boswellia meta-analysis. BMC Complement Med Ther 2020;20:225 (PMID 32723302); Sengupta K et al. Arthritis Res Ther 2008;10:R85 (PMID 18667054); Vishal AA et al. Int J Med Sci 2011;8:615 (PMID 22022214); 2025 dietary supplements NMA (PMC11925762)

2.5 MSM (methylsulfonylmethane) Amazon ✓ Website ✓

Evidence: weak/mostly null (low confidence) | Effect: joint / arthritis pain

THE NOISE Marketed as an organic-sulfur 'building block' that reduces joint pain, stiffness and inflammation while supporting healthy cartilage and flexibility.

THE SCIENCE Effect: 2025 NMA (Kim 2006 + Debbi 2011): WOMAC pain WMD -10.1 (95% CI -19.1 to -1.07), function -14.1 (-22.9 to -5.35), stiffness -16.2 (-26.8 to -5.60) on 0-100 WOMAC. Kim 2006 (n=50): significant on function/total, NOT on pain or stiffness subscales. Clears MCID: Borderline/uncertain. Wide CIs from tiny trials; point estimates approach WOMAC MCID (~15-20/100) only for stiffness, lower bound near zero for pain. Largest independent trial: Debbi EM et al. 2011 (n=50, 6 g/day): significantly greater pain and function improvement vs placebo at 12 weeks, but small and short. Population: Adults with mild-to-moderate knee OA; trials 12 weeks only.

THE VERDICT Weak buy-case at best. Nominally positive on pain and function, but the entire evidence base is a handful of small (n<50/arm), short (12-week) trials with industry ties and no large independent replication. Signal without confirmation.

BEST FORM MSM is a single defined molecule, so the compound is identical everywhere - the only real quality variable is purity. Choose a distilled, tested grade (OptiMSM is the reference) over crystallized/imported bulk that can carry contaminants. Typical dose 1.5-3 g/day. (Distilled MSM (OptiMSM - low heavy-metal/contaminant, verified) > cheap crystallized bulk MSM (same molecule, but purity/contaminant risk). No absorption difference between 'forms' - it is one compound; purity is the point.)

WATCH Generally very well tolerated; high doses can cause mild GI upset, headache, or insomnia (some find it activating - take earlier). Minimal known drug interactions. Onset over weeks. Because it is one molecule, ignore marketing about proprietary 'superior forms' - just verify purity testing.

Sources: Kim LS et al. Osteoarthritis Cartilage 2006;14:286 (PMID 16309928); Debbi EM et al. BMC Complement Altern Med 2011;11:50 (PMID 21708034); Xu G et al. Nutrients 2023;15:2995 (PMC10346176); Brien S et al. meta-analysis, Osteoarthritis Cartilage 2008 (PMC3135791)

2.5 UC-II (undenatured type II collagen) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: joint / arthritis pain · **literature compromised (funding/outlier)**

THE NOISE Sold as a single tiny 40 mg daily capsule that 'retrains' the immune system and is clinically proven to beat glucosamine plus chondroitin for joint comfort, mobility and flexibility.

THE SCIENCE Effect: Lugo 2016 (n=191, 3-arm): UC-II 40 mg/day reduced total WOMAC ~40% from baseline vs less for placebo. BUT 2025 NMA placed UC-II (Lugo 2016) LAST across pain/function/stiffness (pain SUCRA 0.03), i.e. no advantage over placebo in the pooled network. Clears MCID: Unclear/likely no. Within-arm percentage drops look large but the between-arm placebo-controlled ranking is bottom-tier. Largest independent trial: None truly independent. Lugo 2016 (largest, n=191) funded by InterHealth Nutraceuticals (now Lonza), the ingredient manufacturer; Crowley 2009 (UC-II 33% vs G/C 14% WOMAC drop) also industry-sponsored, n=52. Population: Adults with symptomatic knee OA (Lugo 2016) or exercise-related knee discomfort in healthy subjects (Lugo 2013); 90-180 days.

THE VERDICT Do not buy on current evidence. The positive trials are all manufacturer-funded, and a 2025 independent network meta-analysis ranked UC-II LAST of all supplements for pain (SUCRA 0.03) and function. The marketing (beats glucosamine+chondroitin) rests on tiny industry studies.

BEST FORM Genuinely UNDENATURED type II collagen (UC-II brand), 40 mg/day delivering ~10 mg bioactive undenatured collagen, taken on an EMPTY stomach. The undenatured 3D epitope is the active principle - heat/hydrolysis destroys it. (Undenatured UC-II (40 mg, oral-tolerance/immune mechanism - outperformed glucosamine+chondroitin in a head-to-head knee-OA trial) is completely different from hydrolyzed collagen peptides (10-15 g, structural amino-acid mechanism). More is NOT better - the low 40 mg dose is correct; do not substitute bulk collagen powder.)

WATCH Must stay undenatured - buy the standardized branded ingredient, not generic 'type II collagen.' Take on an empty stomach (bedtime or between meals) so stomach acid/enzymes don't degrade the intact epitope before it reaches gut-associated immune tissue. Onset ~4-12 weeks. The mechanism is oral tolerance, so the tiny dose is intentional.

Sources: Lugo JP et al. Nutr J 2016;15:14 (PMID 26822714); Crowley DC et al. Int J Med Sci 2009;6:312 (PMID 19847319); Efficacy of dietary supplements for knee OA: NMA 2025 (PMC11925762); Tjandra O et al. Indones Biomed J 2023

2 Chondroitin sulfate Amazon ✓ Website ✓

Evidence: weak/mostly null (high confidence) | Effect: joint / arthritis pain · **literature compromised (funding/outlier)**

THE NOISE Marketed as a cartilage-building joint supplement that eases arthritis pain and protects joints from wear.

THE SCIENCE Effect: Wandel 2010 NMA: pain -0.3 cm/10 cm vs placebo, not reaching MCID. Fransen 2015: chondroitin-alone arm not significantly better than placebo on WOMAC pain. Clears MCID: No. -0.3 cm vs ~0.9 cm MCID. Largest independent trial: Fransen M et al. LEGS 2015 (n=605 total): chondroitin alone not significantly superior to placebo for pain; combination arm showed small joint-space benefit only. Population: Adults with symptomatic knee/hip OA. Positive trials used pharmaceutical-grade chondroitin (Bioiberica/IBSA); US OTC products often low-purity or under-labeled.

THE VERDICT **Do not buy. Same picture as glucosamine: pooled pain effect -0.3 cm on 10 cm scale, below MCID, with pharmaceutical-grade positive trials industry-sponsored and independent trials weaker or null.**

BEST FORM High-purity pharmaceutical-grade chondroitin sulfate (bovine/porcine cartilage source), 800-1200 mg/day. Purity and molecular weight are the real variables - buy a brand with third-party identity/purity testing. (Verified pharmaceutical-grade chondroitin (the form used in structure-modifying trials) >> cheap bulk chondroitin (frequently adulterated or under-labeled - independent testing has found many products contain far less than claimed). Source and assay purity matter more than the salt.)

WATCH Notorious for label inaccuracy and adulteration (e.g., cut with cheaper polysaccharides) - insist on tested product. Additive bleeding risk with warfarin/antiplatelets. Large molecule, slow onset (weeks-months). Best evidence is in COMBINATION with glucosamine sulfate for moderate-severe knee OA.

Sources: Wandel S et al. BMJ 2010;341:c4675; Singh JA et al. Cochrane Database Syst Rev 2015;1:CD005614 (PMID 25629804); Fransen M et al. Ann Rheum Dis 2015;74:851

2 Glucosamine (sulfate or HCl) Amazon ✓ Website ✓

Evidence: weak/mostly null (high confidence) | Effect: joint / arthritis pain · **literature compromised (funding/outlier)**

THE NOISE Sold as a joint supplement that rebuilds cartilage, relieves joint pain, and improves mobility in arthritis and aging joints.

THE SCIENCE Effect: Wandel 2010 BMJ NMA (10 trials, 3803 pts): pain difference vs placebo -0.4 cm on 10 cm VAS, does not reach MCID. Joint-space width change minute, CrI crossing zero. Clears MCID: No. -0.4 cm vs an MCID commonly set at ~0.9 cm on a 10 cm scale. Largest independent trial: GAIT (Clegg 2006, NIH-funded, n=1583): glucosamine not superior to placebo on primary endpoint; benefit only in exploratory moderate-to-severe subgroup. LEGS (Fransen 2015) glucosamine-alone arm null. Population: Adults with symptomatic knee or hip OA (Kellgren-Lawrence 2-3); benefit signal, where present, concentrated in moderate-to-severe pain subgroups.

THE VERDICT **Do not buy for joint pain. The best network meta-analysis puts the effect at -0.4 cm on a 10 cm pain scale, below the minimal clinically important difference, and the effect shrinks further in industry-independent trials. Large public-funded trials (GAIT, LEGS) are null.**

BEST FORM Crystalline glucosamine SULFATE (the stabilized potassium/sodium-chloride co-crystal, 1500 mg once daily - the 'Rotta/pharmaceutical-grade' form used in the positive trials). Glucosamine HCl has more evidence gaps despite delivering more glucosamine per gram. (Crystalline glucosamine sulfate (the form with symptom/structure trial support) > generic glucosamine sulfate > glucosamine HCl (higher glucosamine content but weaker/inconsistent trial results). Once-daily 1500 mg dosing outperformed split dosing in trials.)

WATCH Most is shellfish-derived - shellfish-allergic users need corn/fungal-fermented versions. Can mildly raise blood glucose/insulin resistance - monitor if diabetic. Documented warfarin interaction (raises INR/bleeding). Slow onset - judge after 4-8 weeks, not days. Sulfate salt matters, so sulfate is not interchangeable with HCl.

Sources: Wandel S et al. BMJ 2010;341:c4675 (PMID 20847017); Clegg DO et al. GAIT. N Engl J Med 2006;354:795 (PMID 16495392); Fransen M et al. LEGS. Ann Rheum Dis 2015;74:851 (PMID 24395557); Runhaar J et al. OA Trial Bank IPD, Semin Arthritis Rheum 2016

2 Glucosamine + chondroitin combination Amazon ✓ Website ✓

Evidence: weak/mostly null (high confidence) | Effect: joint / arthritis pain · **literature compromised (funding/outlier)**

THE NOISE Sold as the two-in-one joint-health combo that rebuilds cartilage, relieves arthritis pain, and improves joint mobility.

THE SCIENCE Effect: Wandel 2010 NMA: combination pain -0.5 cm/10 cm vs placebo, not reaching MCID. GAIT: combination not superior to placebo overall; moderate-to-severe subgroup responder 79.2% vs 54.3% placebo (exploratory, underpowered). Clears MCID: No overall. Subgroup finding not confirmed in a prospective adequately-powered trial. Largest independent trial: GAIT (NIH, n=1583): null overall for combination; subgroup only. Population: Symptomatic knee OA; positive subgroup limited to moderate-to-severe baseline pain.

THE VERDICT **Do not buy. The combination does not beat placebo on pain overall (-0.5 cm/10 cm, below MCID); the only positive signals are an exploratory moderate-to-severe subgroup in GAIT and an industry non-inferiority-vs-celecoxib trial with no placebo arm.**

BEST FORM Pair crystalline glucosamine SULFATE (1500 mg) with tested pharmaceutical-grade chondroitin sulfate (800-1200 mg). Use the exact forms from the trials - a sulfate-glucosamine + verified-chondroitin combo, not glucosamine HCl with bargain chondroitin. (Glucosamine sulfate + high-purity chondroitin (GAIT trial subgroup benefit in moderate-severe knee OA) > glucosamine HCl + generic chondroitin (weaker data). The combination's evidence hinges on using the validated individual forms.)

WATCH Combined shellfish-allergy and warfarin-INR cautions apply. Watch total daily pill burden and added sodium/potassium from the salts. Mild glucose effects in diabetics. Slow onset (8-12 weeks) - not an acute analgesic. Buy tested products; combo products are a common spot for chondroitin under-dosing.

Sources: Clegg DO et al. GAIT. NEJM 2006;354:795; Hochberg MC et al. MOVES. Ann Rheum Dis 2016;75:37 (PMID 25589511); Wandel S et al. BMJ 2010;341:c4675

Longevity (NAD & senolytics) (14)

2.5 GlyNAC (glycine + N-acetylcysteine) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: longevity / healthspan · **literature compromised (funding/outlier)** | NAC was briefly delisted by Amazon (2021) then restored; branded GlyNAC combos are limited, often bought as separate glycine + NAC.

THE NOISE Marketed as restoring glutathione to reverse aging hallmarks, boosting strength, cognition and mitochondrial health.

THE SCIENCE Effect: Kumar et al. 2022, J Gerontol A / Kumar 2021 pilot: GlyNAC corrected glutathione deficiency, lowered oxidative stress, improved mitochondrial function, gait speed, muscle strength, cognition and insulin resistance vs alanine placebo over 16 wks. Effect sizes reported as large, but n is very small and CIs wide. Clears MCID: plausibly yes for glutathione/oxidative markers; functional gains promising but underpowered and unreplicated. Largest independent trial: None independent - all from Sekhar's Baylor group, which holds related IP/conflicts. Population: Older adults (61-80y) with glutathione deficiency/oxidative stress; a HIV-cohort variant.

THE VERDICT **Conditional, do not buy off-the-shelf. Real, sizeable improvements in glutathione, oxidative stress, strength and cognition - but from small (n=24) trials by a single conflicted group (Sekhar/Baylor), never independently replicated, in a glutathione-deficient older population. Promising within its niche, not proven for healthy adults.**

BEST FORM Buy the two components separately at the studied ~1:1 molar ratio - plain glycine powder plus N-acetylcysteine (NAC) - so you can hit the high, weight-based doses cheaply (Separate glycine + NAC (dose-flexible, cheap) > pre-blended 'GlyNAC' capsules (usually far underdosed vs trials); NAC > cystine/whey as a cysteine source for glutathione)

WATCH Trial doses are large and weight-based (~100 mg/kg/day EACH of glycine and NAC) - most capsule blends deliver a fraction. NAC causes nausea/GI and has a sulfur smell; start low. NAC has US regulatory ambiguity (FDA has questioned its supplement status). NAC may potentiate nitroglycerin (hypotension) and is mildly antiplatelet. Glycine is sedating for some (often taken at night).

Sources: Kumar et al. 2022, J Gerontol A Biol Sci Med Sci, PMID 35975308 (RCT, n=24 older adults); Kumar et al. 2021 pilot, PMID 33783984; FDA final guidance on NAC enforcement discretion (2022)

2.5 NMN (nicotinamide mononucleotide) Amazon ✗ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: longevity / healthspan | FDA no longer permits NMN as a dietary supplement; delisted from Amazon US, still sold gray-market on vendor sites.

THE NOISE Sold as an anti-aging NAD⁺ booster that restores youthful energy, metabolism and cellular function and 'reverses aging'.

THE SCIENCE Effect: NAD⁺ rises robustly (SMD large, $p < 0.001$). Physical performance: pooled effect non-significant. Igarashi 2024 (250 mg, older adults) shorter 4 m gait; Yi 2022 improved 6-min walk. Glucose/lipid meta-analysis (12 RCTs, $n=513$): NAD up, but HbA1c, fasting glucose, lipids not significantly changed vs placebo. Clears MCID: no - performance changes are small and inconsistent; pooled physical-performance effect does not reach significance, let alone MCID. Largest independent trial: Igarashi et al. 2024 (250 mg, older adults, $\sim n80$) among larger; independent adequately powered outcome trials remain scarce. Population: Healthy middle-aged (avg $\sim 58y$) and older adults; several trials in amateur athletes and Asian cohorts.

THE VERDICT Do not buy for healthspan/longevity. NMN reliably raises blood NAD⁺, but that is a biomarker, not an outcome. The largest, most rigorous syntheses show physical-function and metabolic benefits are non-significant; the scattered positive trials are small, short, and largely industry- or single-region (China/Japan) sourced.

BEST FORM A third-party-tested (COA/HPLC-verified) NMN powder or capsule stored cool and dry; ignore 'liposomal' and 'sublingual' premium claims, which lack good human absorption data (Reputable COA-verified crystalline NMN > cheap unverified NMN (purity/degradation issues) ~ 'sublingual/liposomal' NMN (marketing, unproven); NMN vs NR is an unsettled debate - both raise NAD⁺)

WATCH US regulatory limbo - FDA has taken the position NMN is excluded from the supplement definition, so availability/labeling is inconsistent; buy verified product. NMN is hygroscopic and heat-sensitive - degrades if stored warm/humid. Human longevity outcomes unproven. Typical research doses 250-1000 mg/day, taken in the morning.

Sources: PMID 36482258 (efficacy/safety RCT, healthy middle-aged, Sci Rep 2023); Cureus 2024 systematic review PMC11365583 (10 RCTs, physical performance non-significant); Crit Rev Food Sci Nutr 2024, PMID 39116016 (glucose/lipid meta-analysis, 12 RCTs $n=513$); PMC11557618 (glucose/lipid meta-analysis)

2.5 NR (nicotinamide riboside) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: longevity / healthspan

THE NOISE Marketed as the leading NAD⁺ precursor that recharges cellular energy, supports healthy aging and boosts vitality.

THE SCIENCE Effect: Martens 2018 (Nat Commun, independent, U. Colorado): 1000 mg/d x6 wk raised NAD⁺ $\sim 60\%$; systolic BP fell ~ 9 mmHg in the elevated-BP subgroup but NOT statistically significant in the full cohort. Follow-up NCT03821623 (Frontiers Cardiovasc Med 2022) found NR did not significantly reduce systolic BP or aortic stiffness (primary endpoints null). Clears MCID: no - the ~ 9 mmHg BP signal would be clinically meaningful IF real, but it did not reach significance; confirmatory trial was null. Largest independent trial: Martens et al. 2018, Nat Commun ($n=24$ crossover, healthy 55-79y): NAD⁺ up, BP/stiffness endpoints exploratory and non-significant. Population: Healthy midlife/older adults (Martens, Boulder); Parkinson's (NR-SAFE); overweight adults; hypertensive older adults.

THE VERDICT Do not buy for longevity. Best-characterised NAD precursor with genuine independent RCTs and a clean safety record - but the flagship blood-pressure/vascular endpoints were non-significant in the primary analyses. NAD⁺ goes up; hard outcomes do not.

BEST FORM Niagen (patented nicotinamide riboside chloride, ChromaDex) - the form used in essentially all human NR trials and independently assayed for purity (Niagen NR chloride (clinically studied, GRAS) > generic/unbranded NR (often degraded to nicotinamide, poor stability); vs NMN - both reliably raise blood NAD⁺, no proven winner for outcomes)

WATCH Reliably raises NAD⁺ in humans but longevity/healthspan benefits remain unproven. Generally very well tolerated; occasional nausea/flushing. Poorly made NR degrades to niacinamide - brand/COA matters. Doses 250-1000 mg/day, morning. Long-term safety data still limited.

Sources: Martens et al. 2018, Nat Commun (independent, NAD⁺ up, BP exploratory); Frontiers Cardiovasc Med 2022, PMID 35620522 / NCT03821623 (BP & stiffness null); NR-SAFE, Nat Commun 2023 (3000 mg safety in Parkinson's); Conze/ChromaDex NIAGEN safety RCT PMC6611812

2.5 Rapamycin / sirolimus (drug - noted) Amazon ✗ Website ✗

Evidence: weak/mostly null (medium confidence) | Effect: longevity / healthspan | Prescription immunosuppressant; longevity use is off-label and unproven.

THE NOISE Hyped in longevity circles as an anti-aging mTOR drug.

THE SCIENCE Effect: PEARL (Zalzala et al. 2024, medRxiv, n=114, 48 wk): safety comparable to placebo; primary healthspan endpoints largely null; women on 10 mg showed improved lean tissue mass and self-reported pain; 5 mg group improved self-reported wellbeing/general health. Visceral adiposity unchanged. Mouse ITP: intermittent/late-life rapamycin reproducibly extends median lifespan ~10-25%. Clears MCID: no in humans yet - subgroup, self-reported gains; no hard clinical endpoint moved. Largest independent trial: PEARL (AgelessRx/Lifespan.io/academic) - decentralised, largely null primary; independent of a pharma sponsor. Population: Mice (ITP lifespan); healthy adults 50-85 (PEARL); small immune/skin-aging pilots.

THE VERDICT Not a supplement. The strongest PRECLINICAL geroprotector (reproducible mouse lifespan extension), but the first dedicated healthy-human trial (PEARL) was largely null on primary endpoints, with only modest subgroup improvements and low bioavailability of the compounded drug. Mechanistically the best bet in this list, but human outcome evidence is still thin.

BEST FORM Prescription drug, not a supplement - only legitimate form is pharmaceutical sirolimus (Rapamune or verified compounded/generic) obtained via a physician; avoid grey-market 'rapamycin' powders and unverified research-chemical sources.

WATCH Off-label longevity use is experimental and requires medical supervision. It is an mTOR-inhibiting immunosuppressant: risks include mouth ulcers, raised LDL/triglycerides, glucose intolerance, delayed wound healing, infection risk, and cytopenias. Metabolized by CYP3A4/P-gp - grapefruit juice, ketoconazole, and many drugs sharply raise levels; take consistently relative to food (absorption varies with fat). Intermittent (weekly) dosing is what most longevity protocols study, not daily. Needs baseline and periodic bloodwork.

Sources: Zalzala et al. 2024, PEARL Trial, medRxiv 2024.08.21.24312372 (n=114, largely null primary); Harrison et al. 2009, Nature (ITP mouse lifespan extension); Mannick et al. 2014/2018 (rapalog RTB101, immune function in elderly)

2.5 Urolithin A (Mitopure) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: longevity / healthspan · literature compromised (funding/outlier)

THE NOISE Marketed as a mitophagy activator that revitalizes mitochondria to boost muscle strength, endurance and cellular energy.

THE SCIENCE Effect: Middle-aged (Singh/Andreux 2022, Cell Rep Med): muscle strength ~+12% and improved 6-min walk vs placebo. Older adults (Liu et al. 2022, JAMA Netw Open, 65-90y): 6-min walk distance and hand-muscle max ATP were NOT significantly different vs placebo; only muscle endurance and plasma biomarkers (acylcarnitines, CRP) improved. Clears MCID: borderline - the middle-aged strength/6MWT gains approach clinical relevance, but the older-adult functional primaries missed significance. Largest independent trial: None fully independent - Amazentis sponsored/employed authors on all pivotal trials. Population: Overweight middle-aged adults (Cell Rep Med 2022); older adults 65-90 (JAMA Netw Open 2022).

THE VERDICT Conditional / mostly do not buy. The only category member where dosed retail matches trial dose, but every RCT is Amazentis-sponsored, and in the population that matters most (older adults 65-90) the headline 6-minute-walk and muscle-ATP endpoints were NOT significant. The '12% strength' result comes from a middle-aged industry trial.

BEST FORM Direct/preformed urolithin A (Mitopure by Amazentis) - because only ~40% of people harbor the gut bacteria to convert dietary ellagitannins into urolithin A themselves (Preformed urolithin A (Mitopure) = reliable dose for everyone > pomegranate/ellagitannin extracts (depend on your microbiome; most people convert little or none))

WATCH The whole point of buying it preformed is to bypass poor/absent endogenous conversion - a pomegranate supplement won't reliably give you urolithin A. Clinical dose ~500 mg/day, best with food. Well tolerated in trials. Expensive. Muscle/mitochondrial benefits are promising but modest in size.

Sources: Singh/Andreux et al. 2022, Cell Reports Medicine, PMID 35584623 (middle-aged, +12% strength); Liu et al. 2022, JAMA Netw Open, PMID 35050355 (older adults; 6MWT & ATP non-significant); Andreux et al. 2019, Nat Metab (first-in-human safety/mitophagy biomarkers)

2 Fisetin (senolytic) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: longevity / healthspan

THE NOISE Hyped as a senolytic that clears 'zombie' senescent cells to reduce inflammation and extend healthspan.

THE SCIENCE Effect: No completed human RCT with a positive clinical endpoint. Mouse: Yousefzadeh et al. 2018 (EBioMedicine) intermittent fisetin extended median/max lifespan and reduced senescence. 2025 preclinical (PMC12341784) improved physical function and reduced skeletal-muscle senescence in aged mice. Clears MCID: na - no human outcome effect size to evaluate. Largest independent trial: AFFIRM (Mayo Clinic, FDA-approved, frail elderly women, 20 mg/kg pulsed) - enrolled >50, results not yet reported positive. Population: Mice (lifespan); humans: frail elderly women (AFFIRM), childhood-cancer survivors, COVID nursing-home residents - mostly ongoing/pilot.

THE VERDICT Do not buy yet. Compelling mouse healthspan/lifespan data, but human senolytic outcome data are still pending - the pivotal Mayo AFFIRM frailty RCT has not reported positive outcomes, and no completed human trial shows a clinical benefit. Mechanism far outruns outcome.

BEST FORM A liposomal or phospholipid/fat-formulated fisetin (or plain fisetin taken with a high-fat meal) - raw fisetin is very poorly absorbed (Liposomal / phytosome fisetin > plain fisetin powder with fat >> plain fisetin on an empty stomach (minimal absorption))

WATCH Senolytic use rests on the unproven 'hit-and-run' protocol (high dose a few consecutive days, intermittently) extrapolated largely from animal data - human evidence is preliminary. Take with dietary fat. Inhibits CYP enzymes; theoretical interactions with anticoagulants and CYP-metabolized drugs. Long-term safety at senolytic doses unknown.

Sources: Yousefzadeh et al. 2018, EBioMedicine (mouse lifespan, senotherapeutic); AFFIRM protocol, Mayo Clinic (frailty RCT, pulsed 20 mg/kg); PMC12341784 2025 (intermittent fisetin, muscle senescence, preclinical); COVID-FIS pilots NCT04537299 / NCT04771611

2 Metformin (drug - noted) Amazon ✗ Website ✗

Evidence: weak/mostly null (medium confidence) | Effect: longevity / healthspan | Prescription-only drug, not sold as a dietary supplement.

THE NOISE Promoted in biohacker circles as an off-label anti-aging 'longevity drug' that slows aging in healthy people.

THE SCIENCE Effect: Konopka et al. 2019, Aging Cell (n=53, 62y, 12 wk aerobic training): metformin abrogated exercise-induced gains in mitochondrial respiration and cardiorespiratory fitness/insulin sensitivity. MASTERS 2019 (resistance training): metformin blunted muscle hypertrophy vs placebo. Observational Bannister 2014: diabetics on metformin had lower mortality than matched non-diabetics - hypothesis-generating, confounded. Clears MCID: no benefit demonstrated in healthy aging; the measurable effect is a negative interaction with exercise. Largest independent trial: TAME (Barzilai/AFAR) - designed but stalled/unfunded through 2026; no results. Population: Type-2 diabetics (observational); older non-diabetic exercisers (mechanistic RCTs). No completed healthy-aging outcome RCT.

THE VERDICT Not a supplement, and not evidence-backed for slowing aging in healthy people. The pivotal TAME trial remains unlaunched/unfunded, the longevity case rests on observational diabetic data, and RCTs show metformin BLUNTS exercise-induced fitness and muscle gains in older adults - a real downside for a healthy user.

BEST FORM Extended-release (metformin ER/XR, e.g. Glucophage XR) - substantially better GI tolerability than immediate-release at equivalent glucose control; prescription-only, not a supplement (Extended-release (ER/XR) - fewer GI side effects, once-daily > immediate-release (IR) - more diarrhea/nausea, multiple daily doses (cheaper))

WATCH Prescription drug, not OTC - longevity use is off-label and debated. Depletes vitamin B12 over time - monitor and supplement B12. GI upset (take with meals, titrate up slowly). Rare lactic acidosis, especially with renal impairment, dehydration, or IV contrast - dose to kidney function. Evidence it can BLUNT exercise-induced fitness/muscle gains, a real trade-off for strength/longevity users. Avoid heavy alcohol.

Sources: Konopka et al. 2019, Aging Cell 18(1):e12880 (blunts exercise mitochondrial adaptation); MASTERS trial 2019, PMC6826125 (blunts hypertrophy); Bannister et al. 2014, Diabetes Obes Metab (observational mortality); TAME rationale, AFAR / Barzilai (unlaunched as of 2026)

2 Resveratrol Amazon ✓ Website ✓

Evidence: weak/mostly null (high confidence) | Effect: longevity / healthspan

THE NOISE Sold as the 'red wine' longevity molecule that activates sirtuins, mimics calorie restriction and extends healthy lifespan.

THE SCIENCE Effect: Lipids: meta-analyses show small reductions in total-C/TG/LDL in some pools, null in others; HDL unchanged. Vascular/endothelial outcomes inconsistent and dose-unrelated. Anthropometric/metabolic outcomes in obesity: no significant clinical improvement. CRP/TNF-alpha modestly reduced in CVD patients; IL-6 unchanged. Clears MCID: no - where significant, effect sizes are small and clinically marginal; heterogeneity high. Largest independent trial: Multiple; e.g., lipid dose-response meta-analysis PMC9506025 - mixed/modest. No landmark independent trial shows a meaningful healthspan benefit. Population: Broad: T2D, obesity/overweight, CVD, cognition, healthy adults.

THE VERDICT **Do not buy. Two decades and 200+ human studies later, the picture is dominated by poor oral bioavailability and inconsistent, often null clinical outcomes. Modest lipid/inflammatory shifts in type-2 diabetics do not generalise, and resveratrol has been shown to blunt exercise-training adaptations.**

BEST FORM Trans-resveratrol (the active isomer), micronized or formulated with fat/piperine to offset its dismal bioavailability - taken with a fatty meal (Trans-resveratrol, micronized/formulated > plain trans-resveratrol > 'resveratrol' blends not specifying trans- content (may be mostly inactive cis- or low-grade polygonum extract))

WATCH Notoriously poor oral bioavailability (rapid glucuronidation) - take with fat; piperine can raise levels. GI cramping/diarrhea common above ~1 g. Inhibits several CYP enzymes and is antiplatelet - real interaction risk with anticoagulants and CYP3A4/2C9/2D6 drugs. Mildly estrogenic. One notable trial found it blunted exercise-training adaptations.

Sources: MDPI IJMS 2024, 25(2):747 (systematic review of resveratrol trials, gaps); Lipid dose-response meta-analysis PMC9506025; Obesity metabolic meta-analysis, ScienceDirect S2667368124000433 (no significant clinical improvement); Gliemann et al. 2013 J Physiol (blunts exercise adaptation)

2 Spermidine Amazon ✓ Website ✓

Evidence: weak/mostly null (high confidence) | Effect: longevity / healthspan

THE NOISE Sold as an autophagy-activating longevity compound that sharpens memory, protects the brain and slows aging.

THE SCIENCE Effect: SmartAge (Schwarz et al. 2022, JAMA Netw Open, n=100, 12 mo): no significant effect on memory (primary) or neuroimaging/inflammatory biomarkers vs placebo. Exploratory hints on verbal memory/inflammation only. Clears MCID: no - primary endpoint null. Largest independent trial: SmartAge, Schwarz et al. 2022 (academic, Germany) - null primary. Population: Older adults (60-90y) with subjective cognitive decline (SmartAge).

THE VERDICT **Do not buy for cognition/longevity. The best RCT (SmartAge) was flatly null on its primary memory and biomarker endpoints. The pro-longevity case rests on observational dietary-intake epidemiology, which cannot establish causation, and trial doses (~0.9 mg/d) may be far below any active dose.**

BEST FORM A standardized wheat-germ extract stating spermidine content per serving (e.g. spermidineLIFE-type extracts) rather than generic 'wheat germ' with unlisted spermidine (Standardized wheat-germ spermidine extract (quantified mg spermidine) > generic wheat germ powder (variable, low spermidine); synthetic spermidine trihydrochloride exists but food-derived extracts dominate the human data)

WATCH Human trial doses are small (~1-6 mg spermidine/day) - check actual spermidine mg, not extract mg. Wheat-germ source contains gluten (celiac caution). Generally well tolerated; mild GI possible. Longevity/cognition benefits are associational/early-trial, not established. Take with food.

Sources: Schwarz et al. 2022, JAMA Netw Open, PMID 35616942 (SmartAge RCT, null primary); SmartAge protocol, Alzheimers Res Ther 2019 PMC6492385

2 Sulforaphane (broccoli sprout extract) Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: longevity / healthspan

THE NOISE Sold as an NRF2-activating detox and anti-aging antioxidant that protects cells and supports longevity.

THE SCIENCE Effect: Singh et al. 2014, PNAS (autism, n=44): ABC improved ~34%, SRS ~17% vs minimal placebo change over 18 wks - but subsequent/larger trials were less impressive or failed. Chemoprevention: modulates carcinogen-detoxification biomarkers; no hard clinical-outcome endpoints. Clears MCID: uncertain - autism signal not robustly replicated; no aging outcome to assess. Largest independent trial: Academic autism RCTs; results mixed and not longevity-relevant. Population: Autism spectrum disorder (young males); cancer-chemoprevention biomarker cohorts; small metabolic studies. No aging cohort.

THE VERDICT **Do not buy for longevity. An NRF2 activator with interesting but inconsistent autism and chemoprevention biomarker data - and zero human healthspan/longevity outcome trials. Early positive autism results have not held up in larger trials.**

BEST FORM Either a preformed/stabilized sulforaphane product (e.g. Prostaphane) OR a glucoraphanin extract that includes ACTIVE myrosinase (e.g. Avmacol) - a glucoraphanin-only extract without myrosinase converts poorly (Preformed sulforaphane / glucoraphanin+active myrosinase > glucoraphanin-only extract (needs gut bacteria to convert, low/variable yield) >> cooked broccoli (heat kills myrosinase))

WATCH Sulforaphane isn't pre-made in most supplements - it forms when glucoraphanin meets the enzyme myrosinase, which heat/processing destroys. If your product is glucoraphanin-only, take it with a pinch of raw mustard-seed powder (a myrosinase source) to rescue conversion. Take with food. Mild GI/gas. Standardize by mg sulforaphane yield, not extract weight.

Sources: Singh et al. 2014, PNAS 111(43):15550 (autism RCT, n=44); NCT02909959 / NCT01474993 (autism RCTs, mixed results)

2 Taurine Amazon ✓ Website ✓

Evidence: weak/mostly null (medium confidence) | Effect: longevity / healthspan

THE NOISE Sold, off the back of the 2023 Science paper, as an anti-aging deficiency to correct that extends healthspan and lifespan.

THE SCIENCE Effect: Singh et al. 2023, Science: taurine supplementation increased median lifespan ~10-12% in mice and improved healthspan markers in monkeys; human data limited to cross-sectional correlations of lower taurine with age-related disease. Cui et al. 2025 (PMC12507425) present experimental evidence AGAINST a consistent age-related taurine decline in humans. Clears MCID: na - no human healthspan outcome measured. Largest independent trial: No human longevity RCT exists; the field is awaiting one. Population: Mice, monkeys (interventional); humans (observational correlations only for aging).

THE VERDICT **Do not buy for longevity. The 2023 Science paper is animal lifespan data plus human correlations - no human RCT shows taurine extends healthspan or lifespan. A 2025 analysis directly disputes that taurine even reliably declines with human age. Wait for the interventional trials.**

BEST FORM n/a - plain taurine (single amino-acid form); cheap bulk powder or capsules are all equivalent, no premium form needed (All retail taurine is the same L-taurine molecule - powder, capsule, or in energy drinks; no meaningful form differences worth paying up for)

WATCH Very well tolerated; typical doses 1-3 g/day (up to ~6 g studied). Water-soluble, take any time with or without food. Longevity data is mostly animal/associational (the cited 2023 aging paper was in mice). Mild theoretical additive effect with blood-pressure-lowering agents. Bulk powder is the value buy.

Sources: Singh et al. 2023, Science 380(6649), doi:10.1126/science.abn9257 (animal lifespan, human correlations); Editorial, Science 2024, doi:10.1126/science.adl2116 (is taurine a biomarker?); PMC12507425 2025 (evidence against taurine deficiency driving human aging)

1.5 Calcium alpha-ketoglutarate (CaAKG / Rejuvant) Amazon ✓ Website ✓

Evidence: none / marketing (high confidence) | Effect: longevity / healthspan · **literature compromised (funding/outlier)**

THE NOISE Promoted as reversing biological age by roughly 8 years and slowing aging at the epigenetic level.

THE SCIENCE Effect: Demidenko et al. 2021, Aging (US): mean biological-age reduction 8.44 y (men) / 6.98 y (women) on TruAge/TruMe saliva methylation - but no placebo arm, retrospective, self-selected, and the clock is not a validated mortality-predictive clock. Effect is uninterpretable. Clears MCID: na - no controlled outcome; the metric itself is not clinically validated. Largest independent trial: None. The only human study is by the product's affiliated group. Population: 42 self-selected Rejuvant users (28 M, 14 F), retrospective, no control group.

THE VERDICT **Do not buy. The sole human 'longevity' result is an uncontrolled, retrospective, manufacturer study using an unvalidated saliva epigenetic clock, reporting a biologically implausible '8-year' age reversal with no placebo group. This is a marketing artifact, not evidence.**

BEST FORM Calcium alpha-ketoglutarate (Ca-AKG, the form used in Rejuvant and the mouse/human aging studies) - distinct from arginine-AKG sold to bodybuilders (Calcium AKG (studied in longevity/epigenetic-age work) > arginine-AKG (AAKG, a pump/NO product, different counter-ion and rationale); ornithine-AKG is yet another use (clinical nutrition))

WATCH Delivers a calcium load - count it toward total daily calcium (excess calcium has cardiovascular/kidney-stone concerns). Human aging evidence is one small uncontrolled biological-age study plus animal data - not proven. Typical dose ~1-2 g Ca-AKG/day. Generally well tolerated; take with food.

Sources: Demidenko et al. 2021, Aging (US), PMC8660611 (retrospective, uncontrolled, TruAge); Asadi Shahmirzadi et al. 2020, Cell Metabolism (mouse healthspan); NCT07114536 (registered CaAKG RCT, pending)

1.5 Pterostilbene Amazon ✓ Website ✓

Evidence: none / marketing (medium confidence) | Effect: longevity / healthspan

THE NOISE Marketed as a better-absorbed resveratrol upgrade that activates sirtuins for anti-aging and metabolic health.

THE SCIENCE Effect: Riche et al. 2014 (n=80, hypercholesterolaemic): 250 mg/day reduced SBP ~7.8 mmHg and DBP ~7.3 mmHg vs placebo - BUT also raised LDL cholesterol ~+24.9 mg/dL (attenuated by statin or added grape extract). Small BMI reduction in non-medicated subgroup. Later NR+pterostilbene RCTs (n=120, 60-80y) showed increased total cholesterol, more pronounced at higher BMI. Clears MCID: BP change would be meaningful if isolated, but the concurrent LDL rise is a clinically adverse trade-off; net benefit unproven. Largest independent trial: Riche et al. 2014 (academic, U. Mississippi) - the LDL-raising finding is independent. Population: Adults with hypercholesterolaemia (Riche 2014); healthy older adults in NR+pterostilbene combos.

THE VERDICT **Do not buy. A better-absorbed resveratrol analog with almost no outcome data and a genuine dose-response warning sign: the one substantive RCT showed pterostilbene RAISED LDL cholesterol. No longevity evidence.**

BEST FORM n/a - plain pterostilbene is already the optimized form: the dimethylated analog of resveratrol with much better oral bioavailability and half-life, so no special delivery system is needed (Pterostilbene (dimethylated, ~4x bioavailability and longer half-life than resveratrol) > trans-resveratrol for absorption; often paired with NR (Basis) - pairing is strategy, not a 'form')

WATCH Better absorbed than resveratrol, but higher doses (a 125 mg twice-daily human trial) RAISED LDL/total cholesterol - monitor lipids. Take with food. Inhibits CYP enzymes - interaction potential with anticoagulants and CYP-metabolized drugs. Long-term human safety data limited; typical doses 50-125 mg/day.

Sources: Riche et al. 2014, PMID 25057276 (metabolic RCT, n=80; BP down, LDL up); Riche et al. safety analysis PMC3575612; NR+pterostilbene RCT (n=120, cholesterol increase)

1.5 Quercetin (senolytic) Amazon ✓ Website ✓

Evidence: none / marketing (high confidence) | Effect: longevity / healthspan

THE NOISE Marketed as a standalone senolytic that flushes senescent cells and, more broadly, supports immunity and anti-aging.

THE SCIENCE Effect: D+Q (Hickson 2019, EBioMedicine, DKD, n=9): reduced adipose p16INK4a+/p21+ cells and circulating SASP (IL-1a, IL-6, MMP-9/-12) at day 11 - but this is the COMBINATION. BP meta-analysis (10 trials, n=841): significant SBP reduction in mixed/normotensive subgroups, DBP in prehypertensives - modest. Clears MCID: no for senescence (no monotherapy data); BP effect small and subgroup-dependent. Largest independent trial: Hickson et al. 2019 (Mayo, D+Q, DKD) - academic, but combination therapy. Population: Diabetic kidney disease (n=9, open-label D+Q); IPF (D+Q). Never quercetin-alone senolytic trials.

THE VERDICT **Do not buy as a senolytic. Every human senolytic result attributed to quercetin used it WITH the prescription leukaemia drug dasatinib (D+Q). Quercetin monotherapy has never been shown to clear senescent cells in humans. As a standalone longevity supplement it is marketing.**

BEST FORM Quercetin phytosome (e.g. Quercetin Phytosome/Quercefit) or enzymatically-modified isoquercetin (EMIQ) - far higher absorption than plain quercetin dihydrate (Quercetin phytosome / EMIQ (up to ~20x absorption) > quercetin + bromelain/vitamin C > plain quercetin dihydrate (poorly absorbed) >> quercetin aglycone alone)

WATCH Take with food (fat helps). Inhibits CYP3A4/P-gp - can raise levels of many drugs (some statins, cyclosporine, certain chemo, anticoagulants); check interactions. Very high chronic doses raise a theoretical kidney concern. In senolytic protocols it's paired with dasatinib (prescription - clinician only). Well tolerated at typical 500-1000 mg doses.

Sources: Hickson et al. 2019, EBioMedicine PMC6796530 (D+Q, DKD, senescent-cell reduction); Justice et al. 2019, EBioMedicine (D+Q, IPF, physical function); Quercetin BP meta-analysis, ScienceDirect S014628062200247X (10 trials, n=841)

Part II — By supplement type (mushrooms, herbs, nootropics, and the rest)

Functional mushrooms (8)

2.5 Turkey tail (Trametes/Coriolus versicolor; PSK 'Krestin' / PSP) Amazon ✓ Website ✓

Evidence: moderate for pharmaceutical PSK as chemo adjunct in Asian GI-cancer cohorts; none for retail powder in healthy adults | Effect: adjuvant overall-survival signal in resected GI cancer (with chemo); none demonstrated in healthy adults | Retail turkey tail extract/powder is freely sold on Amazon and vendor sites; pharmaceutical PSK (Krestin) is a prescription drug in Japan, not the same product.

THE NOISE Sold to healthy adults as an 'immune tonic', gut/microbiome and 'anti-cancer' mushroom. Influencers cite Japanese cancer approval to imply the retail powder fights or prevents cancer and 'boosts immunity'.

THE SCIENCE The genuinely good evidence is for PSK (Krestin), a purified pharmaceutical polysaccharopeptide, given ORALLY at ~3 g/day as an ADJUNCT to chemotherapy in resected colorectal and gastric cancer patients in Japan. Pooled adjunct meta-analysis (23 RCTs, 4,246 pts, Coriolus+Ganoderma) shows overall-survival HR 0.82 (95% CI 0.72-0.94) - real - but relapse-free survival was NULL (HR 1.19, 95% CI 0.91-1.55) and control rate null (RR 1.05, 0.96-1.14). Nearly all trials are Asian, only 4 double-blind, 5 placebo-controlled. PSP RCTs move immune markers (CD4/CD8) but show NO antitumour or survival benefit. There is ZERO credible outcome data in healthy adults for immunity, colds, or gut health.

THE VERDICT Skip as a consumer immune supplement - the survival data belong to a prescription-grade drug (PSK) used with chemo in cancer patients, not the retail powder, and even there disease-free survival is null. If you have resected GI cancer, discuss adjuvant PSK with an oncologist; do not self-treat.

BEST FORM Fruiting-body extract with a stated beta-glucan percentage (aim >20-30%), or a product carrying real PSK (Kureha, CM-101 strain) / PSP (COV-1 strain). Insist on a beta-glucan assay, not just a 'polysaccharides' number. (Fruiting-body dual extract standardised to beta-glucan % (or genuine PSK/PSP protein-bound polysaccharide) vs cheap mycelium-grown-on-grain powder)

WATCH Avoid 'mycelium on grain / full-spectrum' products - residual starch inflates the polysaccharide figure while beta-glucan is low; demand a beta-glucan (not total polysaccharide) %. PSK proper is a prescription adjuvant in Japan. Immunostimulant - use caution alongside immunosuppressants; mild GI/darkened stool. Generally well tolerated.

Sources: PMC6616310 (Coriolus+Ganoderma adjunct MA, 23 RCTs, HR 0.82); Japanese colorectal PSK cooperative-group RCTs (n~1,094); Eliza WLY et al. PSP RCTs - immune markers only, no antitumour outcome

2 Reishi (Ganoderma lucidum) Amazon ✓ Website ✓

Evidence: weak/low-quality: small RCTs, low/very-low GRADE (Cochrane) | Effect: immune-marker modulation (surrogate); weak adjuvant tumour-response signal; none clinically meaningful in healthy adults · **literature compromised (funding/outlier)** | Widely sold as capsules/extract/powder on Amazon and vendor sites; no restriction.

THE NOISE Marketed as the 'mushroom of immortality': immune tonic, adaptogen for stress/sleep, anti-cancer, longevity, liver support and blood-pressure/cholesterol aid.

THE SCIENCE The 2016 Cochrane review (Jin X et al., CD007731) found 5 RCTs, n=373, in cancer patients: G. lucidum was NOT effective as first-line treatment; a weak signal as an add-on to chemo/radiotherapy for tumour response (RR ~1.5) and some immune-marker/QoL shifts, but evidence was rated LOW to VERY LOW quality. Beta-glucan immune-marker RCTs in healthy adults (e.g. CD4/CD8, NK cytotoxicity) show lab changes but NO clinical outcome (fewer infections, better longevity). Cardiometabolic RCTs are small and inconsistent. No convincing healthy-adult benefit.

THE VERDICT Skip. After decades of study the best synthesis (Cochrane) rates the human evidence low/very-low and finds no first-line efficacy; healthy-adult benefits are immune-marker theatre, not outcomes. Hepatotoxicity has been reported with powdered reishi.

BEST FORM Fruiting-body dual extract - hot water pulls the immune beta-glucans, ethanol pulls the lipophilic triterpenes (ganoderic acids, the bitter fraction). Look for stated beta-glucan % and ideally triterpene %; a genuinely bitter taste signals real triterpene content. (Fruiting-body DUAL extract (hot-water + ethanol) vs single hot-water extract or mycelium-on-grain)

WATCH Can potentiate anticoagulants/antiplatelets (bleeding risk) and may add to antihypertensives/hypoglycaemics. Occasional mild GI upset, dry mouth; rare hepatotoxicity with powdered raw product. Avoid mycelium-on-grain (low triterpenes, starch-diluted).

Sources: Cochrane CD007731.pub3 (Jin X, 2016; 5 RCTs, n=373, low/very-low quality); Ganoderma beta-glucan immune-marker RCT PMC9914031

2 Shiitake / lentinan / AHCC (*Lentinula edodes*) Amazon ✓ Website ✓

Evidence: moderate for injectable lentinan in gastric cancer; weak/preliminary single-RCT for AHCC/HPV; none for retail shiitake | Effect: adjuvant survival in gastric cancer (injectable lentinan); persistent-HPV clearance (AHCC, preliminary); none for retail shiitake in healthy adults · **literature compromised (funding/outlier)** | Shiitake extract and branded AHCC are sold on Amazon and vendor sites; lentinan itself is an injectable prescription drug in Japan/China, NOT a supplement.

THE NOISE Culinary shiitake and its extracts (lentinan, AHCC) are marketed for immunity, 'anti-cancer' support, HPV clearance and general wellness - conflating an injectable cancer drug and a proprietary extract with grocery-store mushrooms.

THE SCIENCE Two distinct real signals, neither being retail shiitake powder: (1) LENTINAN - an INJECTABLE pharmaceutical beta-glucan - added to chemo in advanced/gastric cancer improves overall survival (individual-patient MA, 5 RCTs, 650 pts, HR 0.80, 95% CI 0.68-0.95, no heterogeneity; 1-yr survival RR 1.46). (2) AHCC (proprietary Amino-Up extract) - a small phase II RCT in persistent high-risk HPV (NCT02405533, ~50 women, 3 g/day 6 months) reported ~63% clearance vs ~12% placebo, promising but single-centre, industry-sponsored, needs replication (larger trials ongoing). Retail shiitake capsules have no comparable outcome data.

THE VERDICT Skip retail shiitake extract for general 'immunity'. Injectable lentinan is an oncology adjunct (clinician-directed, not a supplement); AHCC for persistent HPV is a reasonable-but-unproven conditional try at 3 g/day pending larger trials - not a healthy-adult tonic.

BEST FORM Match to purpose: for general use a fruiting-body extract with stated beta-glucan %; AHCC only as the specific branded (Amino Up) standardised material - generic 'shiitake mycelium' is not AHCC. Lentinan is hospital IV only, not a supplement. (Three DIFFERENT things: culinary/extract shiitake (beta-glucan) vs lentinan (purified beta-glucan, an injectable drug) vs AHCC (a patented cultured-mycelia extract that is ALPHA-glucan-rich, not a beta-glucan product))

WATCH Eating raw or under-cooked/partly-dried shiitake can cause shiitake (flagellate) dermatitis - a whip-mark rash. AHCC can induce/affect CYP450 and may alter drug levels (e.g. tacrolimus) - check with oncologist/pharmacist. Immunomodulatory - caution with immunosuppressants.

Sources: Individual-patient MA lentinan gastric cancer (PubMed 19596954; ar.iiarjournals 29:2739) HR 0.80; AHCC HPV phase II RCT NCT02405533 / Smith et al.; AHCC review PubMed 35775905

1.5 Maitake (*Grifola frondosa*; D/MD-fraction) Amazon ✓ Website ✓

Evidence: weak/preliminary: single small phase I/II surrogate trial plus uncontrolled series | Effect: immune-marker (NK/cytokine) modulation, direction inconsistent; no clinical outcome in humans · **literature compromised (funding/outlier)** | Sold as capsules/D-fraction liquid on Amazon and vendor sites.

THE NOISE Sold as an immune-activating and 'anti-tumour' mushroom, and separately for blood sugar/blood pressure and weight, citing NK-cell activation.

THE SCIENCE Human data are limited to small early-phase, surrogate-endpoint work. The main study is a Phase I/II dose-escalation in 34 post-menopausal disease-free breast-cancer patients (Deng et al., J Cancer Res Clin Oncol 2009): oral maitake extract produced dose-dependent immunological changes (both immune-stimulatory and -suppressive) with NO clinical outcome measured. Older uncontrolled case-series claims (58-68% 'regression') are not controlled trials. No RCT shows a clinical benefit; glycaemic/BP claims rest on tiny/preclinical data.

THE VERDICT Skip. Only small early-phase trials measuring immune markers, no outcome data, and even those showed mixed immune effects. Nothing here justifies purchase for immunity, cancer, or metabolic health.

BEST FORM Fruiting-body extract standardised to the D-fraction or MD-fraction (the branded proteoglucan), or at least a stated beta-glucan %. (Standardised D-fraction / MD-fraction (beta-1,6/1,3-glucan) fruiting-body extract vs generic maitake powder)

WATCH Notably lowers blood glucose - can be additive with insulin/sulfonylureas (hypoglycaemia); may enhance anticoagulants. Monitor blood sugar if diabetic. Avoid mycelium-on-grain fillers.

Sources: Deng G et al. J Cancer Res Clin Oncol 2009 (PubMed 19253021) phase I/II, n=34; Kodama N et al. NK-activation series PubMed 14977447

1.5 Poria cocos (Wolfiporia extensa / fu ling) Amazon ✓ Website ✓

Evidence: weak/preliminary: 1 uncontrolled study + 1 confounded combination RCT | Effect: modest sleep-duration signal (uncontrolled / combination only); none isolated to Poria · **literature compromised (funding/outlier)** | Sold as powder/capsules on Amazon and vendor/TCM sites.

THE NOISE Marketed for sleep/calm, 'water balance'/diuretic and mild anxiety - a TCM staple repackaged as a natural sleep and de-bloat aid.

THE SCIENCE Human evidence is thin and confounded. An open-label study in 21 poor sleepers (800 mg/night) showed a statistically significant but modest increase in total sleep time (~327 to ~357 min, +30 min) with NO placebo control. The stronger-looking RCT combined Poria with Ziziphus and GABA - so any effect cannot be attributed to Poria. Diuretic and anxiolytic claims rest on rodent/mechanistic (GABAergic, pachymic acid) data. No adequately-powered placebo-controlled monotherapy trial.

THE VERDICT Skip / thin. The only monotherapy human data are an uncontrolled 21-person study; the positive RCT is a GABA-containing combination that doesn't isolate Poria. If you want an evidenced natural sleep-onset aid, the interventions with better data (sleep hygiene, CBT-I, or melatonin for circadian delay) are elsewhere.

BEST FORM A quality sclerotium extract or traditional decoction-grade dried fu ling from a reputable TCM supplier; there is no meaningful 'superior salt/form' to chase. Prioritise clean sourcing and beta-glucan/polysaccharide content if listed. (Single-species sclerotium; standardisation of a defined active is not well established, so form is a minor issue vs sourcing/quality)

WATCH Mild diuretic - can add to prescription diuretics and affect fluid/electrolyte and potassium balance; theoretical additive effect with hypoglycaemics. Usually well tolerated; long traditional use in formulas.

Sources: PMC10574255 / Nutrients 15:4242 (Poria sleep study, n=21, uncontrolled); PMC11167191 (Poria+Ziziphus+GABA combination RCT - confounded)

1.5 Tremella fuciformis Amazon ✓ Website ✓

Evidence: weak/preliminary: 1-2 small RCTs, none for the marketed skin claim | Effect: possible small glycaemic effect (HbA1c/waist) in prediabetics; skin benefit not demonstrated orally in humans · **literature compromised (funding/outlier)** | Sold as powder/capsules/beverages on Amazon and vendor sites.

THE NOISE The 'beauty mushroom' - sold as a natural, plant-based hyaluronic-acid alternative for skin hydration/plumping and anti-ageing, plus cognition and immunity.

THE SCIENCE The oral-skin-benefit claim has NO human trial - the hyaluronic-acid comparison is a chemistry/in-vitro observation (tremella polysaccharide is water-binding), not a clinical skin outcome. Human RCT data are sparse and off-label to the marketing: one manufacturer-linked RCT in overweight/obese prediabetics found a tremella beverage improved HbA1c and waist circumference; a small preliminary cognition RCT reported memory improvement. No oral tremella RCT has measured skin hydration, wrinkles or elasticity in people.

THE VERDICT Skip for skin - the flagship 'natural hyaluronic acid' claim has zero oral human skin trials. The only human RCTs are a small manufacturer glycaemic study and a preliminary cognition study; neither supports the beauty marketing.

BEST FORM Fruiting-body (dual or hot-water) extract with a stated polysaccharide/beta-glucan %; marketed mainly for skin hydration (holds water like hyaluronic acid). (Fruiting-body extract standardised to polysaccharide/beta-glucan vs mycelium-on-grain)

WATCH Generally very well tolerated. Mild theoretical blood-sugar lowering and antiplatelet effect at high doses. As with all these mushrooms, avoid grain-diluted mycelium products - check for a real beta-glucan assay.

Sources: PMC10913275 (Tremella beverage prediabetes RCT - HbA1c/waist); MDPI Foods 14/8/1362 (chemistry: tremella HA-like polysaccharide, in-vitro)

1 Agaricus blazei Murill (A. subrufescens) Amazon ✓ Website ✓

Evidence: weak/preliminary for benefit (surrogate); human case reports of serious hepatotoxicity | Effect: immune-marker modulation only; none clinically demonstrated - net safety concern | Sold as capsules/extract on Amazon and vendor sites despite hepatotoxicity reports.

THE NOISE Marketed as a powerful immune-stimulant and anti-cancer mushroom (Brazilian 'sun/God's mushroom'), often to cancer patients seeking adjuncts.

THE SCIENCE Human data are surrogate-only and safety-negative. A phase I safety study (n=78 cancer patients in remission) and small immunomodulatory pilot RCTs (e.g. NSCLC post-resection multivitamin+ABM) show immune-marker shifts but no proven clinical outcome. Critically, ABM extract is linked to SEVERE hepatotoxicity: published cases of drug-induced liver injury resembling autoimmune hepatitis, including fulminant hepatitis with deaths in cancer patients, and rechallenge-positive liver deterioration.

THE VERDICT Skip - and cancer patients especially should avoid. No demonstrated clinical benefit, immune-marker changes only, and documented fatal fulminant-hepatitis case reports. Risk clearly exceeds any evidenced benefit.

BEST FORM Fruiting-body extract with a stated beta-glucan % from a reputable brand with contaminant testing. (Fruiting-body extract standardised to beta-glucan vs mycelium powder)

WATCH MAJOR: reports of HEPATOTOXICITY (liver injury), particularly in cancer patients using extracts - monitor liver function and avoid with liver disease. Can cause allergic/hypersensitivity reactions, nausea, itching. Strongly immunostimulant - caution with autoimmune disease/immunosuppressants. Can accumulate heavy metals (agrochemical) - buy tested product.

Sources: Ohno S et al. Evid Based Complement Alternat Med 2011 (PMC3092499) phase I; PubMed 17105737 - severe hepatic dysfunction/fulminant hepatitis case series

1 Chaga (Inonotus obliquus) Amazon ✓ Website ✓

Evidence: animal/in-vitro only for benefit; human case reports for renal harm | Effect: none demonstrated in humans (antioxidant claims are in-vitro only) | Sold widely as powder/tea/extract on Amazon and vendor sites; no efficacy trials underlie the marketing.

THE NOISE Sold as the 'king of mushrooms': potent antioxidant, immune booster, anti-cancer, anti-inflammatory and blood-sugar aid, on the strength of its ORAC score and test-tube data.

THE SCIENCE There are essentially NO controlled human efficacy trials. All 'evidence' is in-vitro and rodent (antioxidant, betulinic acid cytotoxicity). What DOES exist in humans is harm: documented case reports of chaga-induced oxalate nephropathy / acute kidney injury (chaga is very high in oxalate), including a nephrotic-syndrome presentation requiring steroids and dialysis, and a 2026 rat model confirming dose-dependent kidney injury.

THE VERDICT Skip - and if you have any kidney risk, avoid. Zero human outcome data and a real oxalate-nephropathy safety signal; the only thing chaga has reliably done in people is damage kidneys at high intake.

BEST FORM Wild-harvested sclerotium, dual (water + alcohol) extract, with a beta-glucan % on the label. Cultured mycelium lacks the birch-derived triterpenes. (Wild sclerotium dual extract standardised to beta-glucan (and betulin/betulinic acid) vs raw ground chaga or mycelium)

WATCH MAJOR: chaga is extremely high in OXALATE - documented oxalate nephropathy/kidney failure with regular use; avoid entirely if you have kidney disease, kidney stones, or on high doses, and keep well hydrated. Can lower blood glucose (additive with antidiabetics) and has antiplatelet activity (bleeding/warfarin risk).

Sources: PMC8913114 (chaga-induced oxalate nephropathy, nephrotic syndrome case); PubMed 35451393; J Korean Med Sci 2026 rat kidney-injury model

Adaptogen & general herbs (10)

3.5 Ginger (*Zingiber officinale*) Amazon ✓ Website ✓

Evidence: moderate-strong for nausea (multiple independent RCTs & metas); weak/heterogeneous for glucose and OA | Effect: anti-nausea (pregnancy/chemo), dysmenorrhea pain; weak/uncertain glycemic effect | Ubiquitous as capsules, powder, and food; no restrictions. Retail doses commonly meet the ~1-1.5 g trial dose.

THE NOISE Sold as a digestive and anti-nausea remedy, an anti-inflammatory 'joint' and blood-sugar herb, and a general immune/metabolic tonic.

THE SCIENCE The one herb here with genuinely robust human data, but only for specific indications. Pregnancy nausea: pooled OR ~4.89 (95% CI 1.88-12.73) for symptom improvement; a 12-RCT/1278-woman analysis found MD 1.20 (95% CI 0.56-1.84) on nausea scales, best at <1500 mg/day. Primary dysmenorrhea: consistent pain reduction. Knee osteoarthritis: pooled SMD only -0.30 (barely above the ~-0.37 relevance threshold) - real but marginal. T2D: some metas show HbA1c ~-1.0% (95% CI -1.56 to -0.44) but with high heterogeneity and small industry-tinged trials, so the glycemic claim is weaker than the anti-emetic one. Anti-inflammatory CRP/TNF-α reductions are biomarker-level.

THE VERDICT Buy for nausea (pregnancy morning sickness, chemo-adjunct, motion) and period pain at 1-1.5 g/day of powdered ginger - one of the few evidence-backed herbal buys. Skip it as a diabetes or arthritis 'cure'; the metabolic signal is inconsistent and OA effect is trivial.

BEST FORM For consistent dosing, an extract standardised to gingerols (and shogaols); plain culinary powder also works and is proven for nausea. Take WITH food. (Whole-root powder vs extract standardised to gingerols/shogaols - single active family, so a good standardised extract is a convenience/potency choice rather than an absorption necessity)

WATCH Antiplatelet at higher doses - bleeding risk with warfarin/aspirin/NSAIDs and around surgery. Can cause heartburn/reflux and mild blood-sugar and blood-pressure lowering. Well tolerated at typical antiemetic doses (~1-1.5 g/day); very high doses may worsen reflux.

Sources: Viljoen 2014 Nutr J meta-analysis pregnancy nausea; Daily 2016 dysmenorrhea meta-analysis; Bartels 2015 Osteoarthritis Cartilage SMD -0.30; Front Pharmacol 2025 PMC12343617 umbrella of metas

2.5 Holy basil / Tulsi (*Ocimum sanctum / tenuiflorum*) Amazon ✓ Website ✓

Evidence: weak-moderate; small RCTs, some manufacturer-linked, variable standardization | Effect: fasting glucose reduction (in metabolic disease); subjective stress/anxiety scores | Sold as capsules, extract, and tea; no restriction. Retail extract doses sometimes below the 300 mg+ standardized trial dose.

THE NOISE Sold as the 'sacred adaptogen' for stress, anxiety, mood and sleep, plus a blood-sugar regulator and immune tonic.

THE SCIENCE Two separate, thin signals. Glycemic: a meta-analysis of RCTs in metabolic-disease adults found tulsi lowered fasting glucose (a crossover T2D trial reported ~-21 mg/dL, ~17.6%) and improved lipids - but in diabetics/metabolic patients, not healthy adults, with small samples and variable extract standardization. Stress: the Holixer™ RCT (PMC9524226) and the 2012 Bhattacharyya trial (300 mg/day) reported reduced stress/anxiety scores over 6 weeks, but samples are small and several trials are extract-manufacturer linked. Effect on a typical healthy adult is unproven; the glucose effect is real but gated to metabolic disease.

THE VERDICT Conditional: reasonable low-risk trial for stress at ~300-600 mg standardized extract, or for glycemic support in metabolic disease alongside standard care - but evidence is small-trial grade and population-gated, so temper expectations. Skip as an 'immune tonic'.

BEST FORM A standardised leaf extract for reliable dosing (tea is fine for gentle daily use). Prefer brands with contaminant testing. (Leaf extract standardised to actives (e.g. ursolic acid / eugenol, or a branded standardised extract) vs plain tea/loose leaf)

WATCH Lowers blood glucose - additive with antidiabetics (hypoglycaemia). Antiplatelet effect - caution with anticoagulants and before surgery. Animal data suggest anti-fertility/anti-spermatogenic effects - avoid if trying to conceive and in pregnancy. May add to thyroid/sedative effects.

Sources: Jamshidi & Cohen 2017 systematic review of tulsi in humans; Holy basil metabolic meta-analysis (ScienceDirect S1756464618301166); Holixer RCT PMC9524226; Bhattacharyya 2008 stress RCT 300 mg

2.5 Maca (*Lepidium meyenii*) [Amazon](#) ✓ [Website](#) ✓

Evidence: weak-moderate; several small RCTs, none large or long | Effect: subjective sexual desire/libido; possible mild ED and semen-parameter effects | Widely sold as powder and capsules; no restriction. Many capsule products deliver well under the 1.5-3 g trial dose.

THE NOISE Marketed as a libido and 'natural Viagra' aphrodisiac, fertility/sperm booster, energy and stamina enhancer, and hormone/menopause balancer.

THE SCIENCE The most defensible claim is a subjective boost in sexual desire. A Cochrane-style systematic review (Shin 2010, 4 RCTs) found 2 trials showing improved sexual desire in healthy men and menopausal women, but a null in cyclists, and judged evidence 'limited' with small, short trials. ED meta-analysis suggests mild benefit in mild ED only. Semen-quality meta-analysis shows some parameter improvement but from small/heterogeneous trials. Crucially, effects are NOT mediated by estrogen or testosterone (Brooks 2008) - so the 'hormone balancing' story is mechanistically wrong. Menopause symptom evidence is mixed and underpowered.

THE VERDICT **Conditional: worth a short trial (1.5-3 g/day dried root) if the goal is subjective libido and you accept placebo-adjacent evidence; do not buy it as a hormone, fertility cure, or energy supplement. Effects are small and self-reported.**

BEST FORM Gelatinised maca - the traditional preparation, better digested and lower in raw goitrogenic glucosinolates than raw powder. Choose colour by goal (black studied for energy/male fertility/cognition, red for prostate/bone in research). (GELATINISED maca (cooked/heat-processed to remove raw starch) vs raw maca powder; also colour-typed (yellow vs red vs black) which affects the effect profile)

WATCH Raw maca contains goitrogenic glucosinolates - not ideal in quantity for thyroid conditions; gelatinising/cooking reduces this. Raw powder can cause bloating/gas. Generally safe; theoretical caution in hormone-sensitive conditions given its traditional hormonal use.

Sources: Shin 2010 BMC Complement Altern Med systematic review (PMID 20691074); Brooks 2008 Menopause (PMID 18784609); Semen-quality meta-analysis PMC9468664; Maca ED meta-analysis JOMH 2023

2.5 Milk thistle (*silymarin*) [Amazon](#) ✓ [Website](#) ✓

Evidence: moderate quantity of RCTs but weak clinical significance; high-quality trials null on outcomes | Effect: small transaminase (ALT/AST) reduction, sub-MCID; possible DILI prophylaxis | Widely sold; caution: many 'milk thistle' capsules list total herb, not standardized silymarin, so actual silymarin can fall well below the 420 mg trial dose.

THE NOISE Sold as a liver 'detox', hepatoprotector and regenerator, hangover/alcohol shield, and fatty-liver treatment.

THE SCIENCE Large evidence base, small clinical payoff. NAFLD/NASH metas (26 RCTs, ~2375 patients) show statistically significant transaminase drops - ALT MD ~-6.4 to -9.2 IU/L, AST MD ~-7 IU/L - but authors explicitly note this is 'without clinical relevance' (below any MCID; it doesn't change fibrosis or hard outcomes). The 2007 Cochrane review of 18 RCTs/1088 patients in alcoholic/viral liver disease found NO significant mortality or histology benefit in the high-quality trials. Best signal is preventing anti-TB drug-induced liver injury (5 RCTs, ~1198 patients, reduced week-4 DILI) - a specific medical use, not general 'detox'.

THE VERDICT **Skip for general liver health, hangovers, or 'detox' - it nudges liver enzymes a few points with no proven clinical benefit. Only conditional value as an adjunct in specific hepatotoxic-drug settings, under medical guidance, at standardized doses (~420 mg silymarin/day).**

BEST FORM A silybin phytosome (silybin-phosphatidylcholine) for meaningfully higher bioavailability; otherwise an extract standardised to 70-80% silymarin (and ideally silibinin content). Take with a fat-containing meal. (Plain seed extract standardised to 70-80% silymarin (poorly water-soluble, low bioavailability) vs silybin PHYTOSOME (e.g. Siliphos/silybin-phosphatidylcholine) which is markedly better absorbed)

WATCH Mild laxative/GI effect. Allergy risk if sensitive to ragweed/daisy/Asteraceae family. Modest CYP/UGT and P-gp effects - space from and review with narrow-window drugs, though clinically usually mild. Can mildly lower blood glucose.

Sources: Rambaldi 2007 Cochrane (alcoholic/viral liver disease); Zhong 2017 Medicine NAFLD meta (PMID 29245314); Administration of silymarin in NAFLD/NASH meta PMID 38579127; Anti-TB DILI meta PMC6348824

2 Astragalus membranaceus (Huang Qi) Amazon ✓ Website ✓

Evidence: weak/compromised - many low-quality, industry-linked trials | Effect: adjunctive renal biomarkers in diabetic kidney disease (not healthy adults) · **literature compromised (funding/outlier)** | Root/extract widely sold; note many efficacy trials used IV astragalus injection, not the oral retail product, so retail doses are not comparable.

THE NOISE Sold as an immune 'tonic', anti-aging/longevity and telomerase activator (TA-65 marketing), and kidney/heart protector.

THE SCIENCE Nearly all positive human data are Chinese-language RCTs of astragalus (often as injection or decoction) as ADD-ON therapy in diabetic kidney disease - metas span 25-66 studies and thousands of patients reporting reduced proteinuria, creatinine and BUN. But reviewers uniformly flag low methodological quality, near-universal industry/publication bias, and absence of large rigorous independent trials. This is a textbook 'positive results concentrated in low-quality studies' pattern. The longevity/telomerase claims rest on the isolated compound cycloastragenol (TA-65) with no credible clinical longevity outcomes. Nothing supports benefit in a healthy adult.

THE VERDICT Skip for healthy adults - the evidence is an adjunctive-nephrology story built on low-quality, bias-prone trials, and the anti-aging pitch is marketing. Any renal use belongs with a nephrologist, not a supplement shelf.

BEST FORM A root extract from a reputable supplier, ideally standardised to astragaloside IV and/or polysaccharides; traditional decoction of sliced root is also legitimate. Ensure it is genuine A. membranaceus root (adulteration/species substitution occurs). (Root extract (sometimes standardised to astragalosides/polysaccharides) vs traditional decoction - single well-defined botanical, so quality/root-vs-adulterant sourcing matters more than 'form')

WATCH Immunostimulant - caution (often avoided) with immunosuppressant therapy (e.g. post-transplant, cyclophosphamide) and active autoimmune disease. May add to antihypertensives/hypoglycaemics/diuretics and to anticoagulants. Generally well tolerated long-term in TCM use.

Sources: Li 2011 J Ethnopharmacol meta DKD (PMID 20951192); Zhang 2019 updated DKD meta (PMID 31034954); Astragalus+RAAS meta PMC11155431; EFSA 2020 AstraGin Novel Food opinion

2 Cinnamon (Cinnamomum cassia / verum) Amazon ✓ Website ✓

Evidence: weak/mixed; many small short RCTs, umbrella review downgrades most outcomes | Effect: modest fasting-glucose reduction; HbA1c effect null; small triglyceride effect | Ubiquitous; supplement capsules (often 500-1000 mg) fall below the >1.5 g dose where any glucose effect concentrates, and higher cassia intake raises coumarin exposure.

THE NOISE Sold as 'natural metformin' - a blood-sugar and HbA1c lowering, insulin-sensitizing metabolic and weight aid.

THE SCIENCE The umbrella review of RCT metas (Front Nutr 2025) is unkind to the headline claim: cinnamon produces a modest fasting-glucose drop (SMD -0.61, 95% CI -0.70 to -0.52; stronger >1.5 g/day) but does NOT meaningfully move HbA1c - the outcome that defines glycemic control - with SMD collapsing to ~-0.09 ('non-significant') after reanalysis. LDL effect became non-significant after publication-bias adjustment; only triglycerides held up ('suggestive'). So it may nudge a fasting reading without changing long-term control. Trials are short (many <=2 months), heterogeneous, and in diabetic/metabolic populations, not healthy adults.

THE VERDICT Skip as a diabetes treatment - it does not reliably lower HbA1c and is not a metformin substitute. If used at all, keep it low-dose and prefer Ceylon (verum) over cassia because of coumarin liver risk. No reason for a healthy adult to supplement.

BEST FORM Ceylon (C. verum) for any regular/supplemental intake - it is very low in coumarin. Reserve cassia (and cassia-based blood-sugar supplements) for occasional culinary amounts. If a product does not specify the species, assume it is cassia. (Ceylon 'true' cinnamon (C. verum, very LOW coumarin) vs Cassia (C. cassia/aromaticum, HIGH coumarin) - the two are pharmacologically different on safety)

WATCH Cassia's COUMARIN is hepatotoxic and anticoagulant - regular high cassia intake risks liver injury and bleeding interactions (warfarin); EFSA sets a coumarin TDI of 0.1 mg/kg/day, easily exceeded with cassia supplements. Both lower blood glucose (additive with antidiabetics). Choose Ceylon to sidestep the coumarin problem.

Sources: Front Nutr 2025 umbrella review PMC12620228 / 10.3389/fnut.2025.1683477; Dose-response meta PMC10346687; Cassia glucose-tolerance RCT PMC4180052; EFSA coumarin TDI; ANSES coumarin advisory

1.5 Eleuthero / Siberian ginseng (*Eleutherococcus senticosus*) Amazon ✓ Website ✓

Evidence: weak; best RCT null, historical studies low quality | Effect: none demonstrated in rigorous human trials · **literature compromised (funding/outlier)** | Root/extract widely sold; no restriction. Retail capsules (300-1200 mg) below the 2 g/day trial dose.

THE NOISE Sold as an adaptogen for energy, stamina, athletic performance, stress resilience and immune/cold defense.

THE SCIENCE Despite decades of use, there is no good modern human evidence for the core claims. The best-designed fatigue RCT (Hartz 2004, ~96 adults, 2 g/day for 2 months) found NO overall difference vs placebo, with only a post-hoc subgroup hint. The much-cited immune/influenza data are 6 Soviet-era studies (1977-1986) of poor quality. WebMD/Drugs.com and Examine converge on 'insufficient evidence' for performance, memory, colds and diabetes. Any Kan Jang COVID signal is from a multi-herb combo (mostly andrographis), not eleuthero alone.

THE VERDICT **Skip - the modern controlled trial is null and the supportive data are old and low-quality. No credible reason for a healthy adult to buy it for energy or immunity.**

BEST FORM Root (not root+rhizome fillers) standardised to eleutherosides B and E (typically $\geq 0.8\%$); reputable extracts state the marker %. Avoid unstandardised bulk powder and beware adulteration/substitution with cheaper *Periploca sepium* (silk vine), a known contaminant. (Root extract standardised to eleutherosides B & E vs raw powder or unlabelled 'root' with no marker)

WATCH Mildly stimulating, take earlier in day to avoid interfering with sleep. Can raise blood pressure in sensitive people, caution if hypertensive. May potentiate/interfere with digoxin assays and affect anticoagulants and hypoglycaemics. Not a true *Panax ginseng*, don't cross-substitute doses.

Sources: Hartz 2004 Psychol Med chronic fatigue RCT; Cizmarova 2020 systematic review of *E. senticosus*; WebMD/Drugs.com eleuthero monographs (insufficient evidence); Examine.com eleuthero research

1.5 Gotu kola (*Centella asiatica*) Amazon ✓ Website ✓

Evidence: weak; cognition meta-analysis null, wound data mostly topical | Effect: none demonstrated for cognition; transient acute mood; topical wound healing | Widely sold as oral capsules and topical creams; no restriction.

THE NOISE Sold as a brain/cognition and memory herb, anxiety reliever, and collagen/skin and wound-healing 'longevity' botanical.

THE SCIENCE The flagship cognition claim fails: the systematic review + meta-analysis (Puttarak 2017, Sci Reports) found NO significant benefit across cognitive domains vs placebo, with only a transient acute mood effect (increased alertness, reduced anger ~1 h post-dose). Wound-healing evidence is real but largely topical/formulation-based (asiaticoside stimulating collagen), not oral cognition. A Phase-1 PK study in MCI established dosing but not efficacy. Anxiety data are thin and preliminary.

THE VERDICT **Skip for memory, focus or anxiety - the meta-analysis is null on cognition. Topical centella has some dermatologic wound/collagen support, but that's a cream, not a nootropic capsule.**

BEST FORM An extract standardised to the total triterpenic (triterpene) fraction - that is where the vascular/wound-healing/cognitive activity sits. Crude powder is far more variable. (Extract standardised to the total triterpenic fraction (asiaticoside, madecassoside, asiatic/madecassic acids - e.g. TTFCA/TECA-type or Centellicum) vs crude leaf powder/tea)

WATCH Reports of HEPATOTOXICITY with prolonged use - limit continuous use (traditional courses ~6 weeks with breaks), monitor if using long-term, and avoid with liver disease. Can be sedating (additive with CNS depressants) and may add to antidiabetic/lipid drugs. Avoid in pregnancy. Possible contact allergy topically.

Sources: Puttarak 2017 Sci Rep meta-analysis cognition/mood (PMC5587720); Wound-healing systematic review PMC8956065; MCI Phase-1 PK RCT PMC8868383; Examine.com gotu kola research summary

1.5 Moringa oleifera Amazon ✓ Website ✓

Evidence: weak / very-low certainty (GRADE) | Effect: none robustly demonstrated (glycemic/lipid effects low-certainty/null) | Powder, leaves and capsules widely sold; no restriction.

THE NOISE Sold as a nutrient-dense 'superfood' and adaptogen for blood sugar, cholesterol, blood pressure, energy and inflammation.

THE SCIENCE The 2025 cardiometabolic meta-analysis with GRADE is the key document: pooled RCTs show NO significant effect on HbA1c or fasting glucose in T2D/prediabetes, and all outcomes were rated low to very-low certainty due to small samples, short durations and heterogeneity. A 2023-25 T2D RCT (20/40/60 g leaves for 14 days) found no significant fasting-glucose difference between doses. Most positive results come from tiny, brief, methodologically weak trials. As a whole-food source of vitamins/minerals it has nutritional value, but the therapeutic supplement claims are not supported.

THE VERDICT Skip as a blood-sugar, cholesterol or blood-pressure supplement - the GRADE-rated evidence is low-certainty and largely null. Fine as a leafy-green food; not worth buying as a medicinal capsule.

BEST FORM Dried LEAF powder or leaf-only capsules (shade-dried to preserve vitamins/isothiocyanates). Choose leaf, not seed or root/bark. Third-party tested for heavy metals as it bioaccumulates from soil. (Dried leaf powder/capsules vs seed or root products; whole-food leaf is what evidence supports)

WATCH Essentially a nutrient-dense whole food, no meaningful 'active-form' issue, dose is about grams of leaf. Avoid root/bark (contains spirochin, potentially toxic and abortifacient). High vitamin K content can antagonise warfarin. May lower blood sugar/BP (additive with medication). Take with food.

Sources: Moringa cardiometabolic meta-analysis w/ GRADE 2025 (MDPI Nutrients 17:3501 / PMC12655524); Afiaenyi 2025 T2D RCT (PMID 37229639); Moringa hyperglycemia/hypertension meta-regression (S2589936826000101)

1.5 Schisandra chinensis (Wu Wei Zi) Amazon ✓ Website ✓

Evidence: weak/preliminary; few small RCTs, liver data preclinical | Effect: possible small muscle-strength gain with exercise; liver effects animal-only · **literature compromised (funding/outlier)** | Berry/extract widely sold; no restriction. Retail capsules often below the ~1 g/day extract used in trials.

THE NOISE Sold as an adaptogen for stress and endurance, a liver-protective 'detox' herb, and a focus/energy tonic.

THE SCIENCE Human evidence is sparse and mostly small Korean trials with likely commercial ties. Two RCTs (54 older adults on 1 g/day for 12 wk; 45 postmenopausal women on 1000 mg/day for 12 wk) report modest gains in quadriceps/skeletal-muscle strength with exercise - but not muscle mass, and not replicated independently. Liver-protection claims rest on preclinical/animal meta-analysis only (no human liver-outcome RCTs). A small menopausal-symptom RCT exists. Nothing supports the core adaptogen/stress or 'detox' marketing in humans.

THE VERDICT Skip - the only human signal is a small, unreplicated muscle-strength effect from industry-adjacent trials, and the liver/adaptogen claims are animal-level. Not worth buying for a healthy adult.

BEST FORM A berry extract standardised to schisandra lignans (schisandrin/schisandrin B/gomisins). Ensure genuine S. chinensis berry. (Berry extract standardised to schisandrins/total lignans vs unstandardised powder or tea)

WATCH IMPORTANT drug interactions: schisandra lignans modulate CYP3A4 and P-gp - it can RAISE levels of co-administered drugs (documented large increases in tacrolimus, and effects on midazolam/statins/other CYP3A4 substrates). Review with a pharmacist before combining with prescription meds. Can cause heartburn/GI upset, appetite changes; avoid in pregnancy and active peptic ulcer/reflux.

Sources: Park 2020 AJCN muscle strength RCT (S0002-9165(22)... older adults); Quadriceps strength RCT PMC7177795; Menopausal symptoms RCT (researchgate 309342787); Liver-injury preclinical meta PMC12358260

Serotonergic / mood (5-HTP class) (5)

3 St John's Wort (*Hypericum perforatum*) Amazon ✓ Website ✓

Evidence: strong for mild-moderate: large multi-trial Cochrane base; weak/minimal for major depression | Effect: antidepressant effect in mild-to-moderate depression | OTC dietary supplement in the US (Amazon + vendor sites); prescription-licensed medicine in some countries (e.g. Germany). The real issue is not availability but interactions.

THE NOISE Marketed as a 'natural Prozac' / natural antidepressant and mood lifter, sold as a gentle, side-effect-free herbal fix for the blues, stress, and 'seasonal' low mood.

THE SCIENCE The strongest evidence in this whole category. Cochrane review (Linde 2008, 29 RCTs, ~5,489 patients) found standardized *Hypericum* extracts beat placebo and matched SSRIs/older antidepressants for MILD-TO-MODERATE depression, with fewer dropouts for side effects (response RR vs placebo roughly 1.3-2.0 depending on trials pooled). Effect clears MCID in mild-moderate disease. Two important caveats: (1) trials from German-speaking countries showed markedly larger effects than trials elsewhere (a country-of-origin/outlier signal suggesting optimism/publication bias), and (2) in the subset of large trials restricted to MAJOR depression the benefit over placebo was minimal. So it is genuinely effective, but for a narrower/milder population than the 'natural antidepressant' label implies.

THE VERDICT **Conditional: a real, evidence-backed option for MILD-to-moderate low mood using a standardized extract (e.g. 300 mg 3x/day, WS 5570/LI 160), but it is a pharmacologically active drug, NOT a benign herb - do not treat it as harmless. Skip if you take ANY other medication without a pharmacist/doctor sign-off, and skip for major depression.**

BEST FORM A standardized extract used in trials - standardized to 0.3% hypericin and a stated hyperforin level, 300 mg 3x/day. If drug interactions are a concern, a low-hyperforin extract (Ze 117) causes markedly less CYP induction. (Clinically-trialed standardized extracts (WS 5570, LI 160, Ze 117) standardized to ~0.3% hypericin and/or 3-6% hyperforin, vs unstandardized raw herb/tea.)

WATCH MAJOR interaction herb: potent inducer of CYP3A4 (also 2C9, 1A2) and P-glycoprotein - it lowers blood levels and can cause failure of oral contraceptives, warfarin, ciclosporin/tacrolimus, antiretrovirals, many statins, chemo agents, immunosuppressants and more. Serotonin-syndrome risk if combined with SSRIs/SNRIs/triptans/tramadol. Causes photosensitivity. Do not start or stop without reviewing every medication; taper rather than stop abruptly.

Sources: Linde K et al. Cochrane Database Syst Rev 2008 CD000448 (St John's wort for major depression); *Hypericum* Depression Trial Study Group. JAMA 2002;287:1807-14 (PMID 11939866); MHRA Drug Safety Update: St John's wort and hormonal contraceptives (2014); StatPearls: St. John's Wort (NCBI Bookshelf NBK557465)

2.5 Lemon balm (*Melissa officinalis*) Amazon ✓ Website ✓

Evidence: weak/preliminary: multiple small short RCTs, moderate pooled SMD but low individual quality and some industry-linked extracts | Effect: mild acute anxiolytic / calming and short-term mood effect | Widely available as tea, capsules, tincture and standardized extracts on Amazon and vendor sites.

THE NOISE Sold as a calming, anti-anxiety, stress-and-mood herb and sleep aid; branded standardized extracts (e.g. Bluenesse) are marketed for acute 'calm alertness,' reduced stress and better mood under pressure.

THE SCIENCE A 2021 Phytotherapy Research meta-analysis and later pooled analyses report statistically significant reductions in anxiety and depression scores (one pooling ~14 RCTs, >800 participants, SMD ~0.65 for anxiety, 95% CI ~0.45-0.85 - a moderate effect). BUT the trials are small, short, heterogeneous in preparation and dose, frequently in specific stressed populations (post-cardiac-surgery, hemodialysis, palpitations, burnout), and several acute-mood studies use industry-linked standardized extracts. The honest read: a plausible mild, mostly ACUTE calming/mood effect, resting on low-to-moderate-quality trials rather than large independent RCTs in general depression.

THE VERDICT **Skip-to-conditional: reasonable low-risk option if you want a mild acute calming aid (standardized extract ~300-600 mg, or ~1.5-3 g dried herb as tea), but do not expect a treatment-grade antidepressant effect - the evidence base is thin and small-trial.**

BEST FORM A standardized aerial-part extract such as Cyracos (standardized for rosmarinic acid), which is what anxiety/sleep trials used, rather than generic leaf powder. Standardization to rosmarinic acid drives the consistency of effect. (Standardized extract (e.g. Cyracos, standardized to rosmarinic acid + hydroxycinnamic acids) vs plain dried leaf/tea.)

WATCH Sedating - can cause drowsiness and may add to alcohol, benzodiazepines, barbiturates and other CNS depressants; take in the evening. Very high doses may affect thyroid signaling (theoretical TSH/thyroid-hormone effects) - caution in thyroid disease and with thyroid meds. Generally well tolerated; avoid combining with strong sedatives before driving.

Sources: Ghazizadeh J et al. Effects of lemon balm (*Melissa officinalis* L.) on depression and anxiety: systematic review and meta-analysis. *Phytother Res* 2021;35:6690 (DOI 10.1002/ptr.7252); 2023 meta-analysis of ~14 RCTs (>800 participants) reporting anxiety SMD ~0.65; Kennedy DO et al. acute *Melissa officinalis* mood/cognition RCTs (standardized extract)

2.5 SAM-e (S-adenosyl-L-methionine) Amazon ✓ Website ✓

Evidence: weak/inconsistent: Cochrane-null monotherapy; conflicting 2024 adjuvant meta-analyses with high heterogeneity | Effect: possible adjunctive antidepressant effect at high dose; monotherapy effect not established | OTC dietary supplement in the US (Amazon + vendor sites); a prescription/licensed drug for depression and liver/joint indications in parts of Europe.

THE NOISE Sold as a fast-acting natural antidepressant plus a joint/liver two-for-one; influencers push it as a mood, methylation and 'brain energy' booster that 'works like an antidepressant without a prescription.'

THE SCIENCE The picture is genuinely mixed and the good headlines don't survive the best analyses. The 2016 Cochrane review (Galizia et al.) found NO significant benefit of SAME over placebo OR over antidepressants as monotherapy. Two competing 2024 meta-analyses disagree with each other: one reports SMD -0.58 (95% CI -0.93 to -0.23) as adjuvant but with high heterogeneity ($I^2=68\%$) driven by dose ranges; another (14 trials, 1,522 subjects) found NO significant difference for SAME added to antidepressants. Signal, if real, is as an ADJUNCT to an SSRI in mild-moderate MDD at high doses (1,600-3,200 mg/day) - not as a standalone. Oral bioavailability is poor and dose-dependent, which is exactly why retail capsules (often 200-400 mg) undershoot the trials badly.

THE VERDICT Skip for general use; conditional only as a doctor-supervised SSRI add-on for mild-moderate depression at 1,600-3,200 mg/day of enteric-coated product. Typical retail doses (200-400 mg) are below the studied active range, and the monotherapy evidence is null.

BEST FORM Enteric-coated tablets of a stabilized salt (butanedisulfonate/disulfonate tosylate), sold in individual blister packs. SAM-e is highly hygroscopic and degrades fast, so the salt, enteric coating and blister packaging are the whole game. (Enteric-coated, stable salt (1,4-butanedisulfonate or tosylate/disulfonate), blister-packed, vs non-enteric or bulk-bottle SAM-e that oxidizes.)

WATCH Take on an empty stomach, 30-60 min before food; start low and titrate. Can trigger mania/hypomania in bipolar disorder - avoid unless supervised. Serotonin-syndrome risk with SSRIs/MAOIs. Requires adequate B12/folate/B6 for methylation (deficiency can raise homocysteine). Can cause insomnia, anxiety, GI upset - dose earlier in the day.

Sources: Galizia I et al. S-adenosyl methionine (SAME) for depression in adults. Cochrane Database Syst Rev 2016 CD011286; SAME as adjuvant therapy for depression: updated systematic review and meta-analysis. J Psychiatr Res / (PMID 38199136), 2024; Efficacy and acceptability of SAME for depressed patients: meta-analysis of RCTs, 2024 (PMC11863058); Sarris J et al. SAME monotherapy/adjunctive RCTs (dose-increase trial, J Affect Disord 2019)

2 Myo-inositol Amazon ✓ Website ✓

Evidence: weak/null for mood: small meta-analysis, no significant effect; real evidence is in metabolic/PCOS, not mood | Effect: none demonstrated for depression/anxiety in humans (weak non-significant PMDD trend only) | Widely sold as a dietary supplement (myo-inositol powder/capsules, often with D-chiro-inositol) on Amazon and vendor sites.

THE NOISE Marketed as 'vitamin B8' for mood, anxiety, panic, OCD and PMS/PMDD, and (via the PCOS crowd) as a natural SSRI-like mood-and-hormone balancer.

THE SCIENCE For mood in a typical adult, the human outcome data are essentially null. The Mukai 2014 meta-analysis (7 depression RCTs, $n=242$; plus 4 anxiety-disorder RCTs, $n=70$) found NO statistically significant effect of inositol on depressive, anxiety or obsessive-compulsive symptoms versus placebo. The only hint of a signal is a non-significant trend in PMDD, and a dedicated PMDD trial concluded myo-inositol had no beneficial effect. Panic disorder rests on a single tiny trial. Inositol's genuinely supported roles are metabolic (PCOS insulin sensitivity / ovulation), NOT general depression or anxiety.

THE VERDICT Skip for depression/anxiety - the pooled RCT evidence is null. If your actual target is PCOS metabolic/ovulatory function that is a different (and better-supported) use; for mood specifically, the effective interventions are psychotherapy and standard antidepressants, not inositol.

BEST FORM Plain myo-inositol powder is a single, freely water-soluble, well-absorbed form - salt/form does not matter. What matters for PCOS/metabolic use is the ratio: use a physiological 40:1 myo-inositol to D-chiro-inositol blend, not high-dose D-chiro alone (excess D-chiro can worsen oocyte quality). (Myo-inositol powder (single, well-absorbed form) - but note the myo-inositol : D-chiro-inositol RATIO in combination products.)

WATCH High doses (typically 2-4 g twice daily) are needed; split the dose. GI effects (loose stool, nausea, flatulence) appear above ~12 g/day. Very safe otherwise. Powder dissolves in water - cheaper and easier to dose than capsules.

Sources: Mukai T et al. A meta-analysis of inositol for depression and anxiety disorders. Hum Psychopharmacol 2014;29:55-63 (DOI 10.1002/hup.2369, PMID 24424706); Nemets B et al. Myo-inositol has no beneficial effect on premenstrual dysphoric disorder (PMDD trial)

2 Saffron (*Crocus sativus* / *affron*) Amazon ✓ Website ✓

Evidence: moderate-by-count but compromised: many small industry-linked RCTs, GRADE 'moderate' on self-report, publication bias flagged | Effect: small improvement in self-rated (subjective) depression/anxiety scores; not confirmed on clinician-rated scales · **literature compromised (funding/outlier)** | Widely sold as a dietary supplement (affron, Safr'Inside and generic) on Amazon and vendor sites.

THE NOISE Sold as a clinically-proven, 'as effective as antidepressants' natural mood and anti-anxiety supplement; branded extracts (affron, Safr'Inside) are marketed with trial citations for low mood, stress, PMS mood and even teen moodiness.

THE SCIENCE Lots of positive trials, but the quality tells a different story. A 2025 GRADE meta-analysis (34 RCTs, 1,769 participants) found saffron cut self-rated Beck Depression Inventory (WMD -4.39, 95% CI -6.64 to -2.15) and Beck Anxiety Inventory scores - but showed NO significant effect on clinician-rated scales (HDRS, HARS) or POMS. Movement on subjective self-report while blinded clinician-rated scales stay flat is a classic expectancy/unblinding signature. Most trials are small, short, conducted in Iran (world's largest saffron producer), and a 2019 review documented publication bias. Branded-extract trials (affron 28 mg) are frequently manufacturer-run: the 2025 J Nutrition low-mood trial was funded by Pharmactive Biotech with two company employees as authors.

THE VERDICT Skip as a serious antidepressant. At best a mild, mostly-subjective mood nudge in low-mood adults, riding on industry-funded, small, self-report-driven trials with weak clinician-rated backing. If you still want to try it, 28-30 mg/day standardized extract is the studied dose - but treat the marketing as marketing.

BEST FORM A standardized stigma extract with defined crocin/safranal content - e.g. affron (standardized 3.5% Lepticrosalides) or Satiereal, at the trial dose of ~28-30 mg/day. Standardization matters because loose saffron powder varies hugely in active crocins and is a frequent adulteration target. (Standardized stigma extracts (affron standardized to Lepticrosalides/crocins, or Satiereal) vs unstandardized saffron powder.)

WATCH Well tolerated at 30 mg/day. Doses above ~1.5 g/day are toxic and >5 g can be abortifacient - avoid high doses and avoid in pregnancy. Mild additive effect with antidepressants (used as adjunct in trials); can mildly lower mood-drug thresholds. Confirm real *Crocus sativus* stigma (dyed/adulterated saffron is common).

Sources: Effect of saffron on depression, anxiety and mood disorder: GRADE meta-analysis of 34 RCTs. Nutritional Neuroscience 2025;29(7) (DOI 10.1080/1028415X.2025.2602153); Lopresti et al. Affron on mood/wellbeing in low-mood adults, J Nutrition 2025 (PMID 40414301) - Pharmactive-funded; Toth B et al. saffron & depression meta-analysis noting publication bias (2019); Marion Nestle, Food Politics, 'Industry-funded study of the week: Saffron and Mood' (2025)

Racetams & synthetic nootropics (10)

4 Modafinil (prescription - included for context) Amazon ✗ Website ✗

Evidence: strong for wakefulness; moderate/modest for healthy-adult cognition | Effect: wakefulness (strong); modest executive-function benefit on complex tasks, larger under sleep deprivation | Prescription-only (Schedule IV in the US). Not a dietary supplement; not sold on Amazon or as a supplement. Online 'pharmacies' selling it without Rx are illegal/gray-market.

THE NOISE Sold in the biohacker world as the ultimate 'limitless' smart drug for all-day focus, wakefulness and productivity.

THE SCIENCE This is the one genuinely effective agent in the category - and it is a prescription drug, not a supplement. For its APPROVED uses (narcolepsy, shift-work disorder, OSA residual sleepiness) the wakefulness evidence is strong. For healthy-adult neuroenhancement the honest picture is modest and task-dependent: meta-analyses (e.g. Battleday & Brem 2015; Repantis; Marraccini) find real but small benefits concentrated in complex, longer-duration executive tasks and in sleep-deprived subjects, with little effect on simple attention and inconsistent gains in rested people - plus headache, insomnia, anxiety and (rarely) serious rash/SJS. Effect is real but oversold as 'limitless'.

THE VERDICT Conditional and by prescription only. Clearly effective for diagnosed sleep-disorder wakefulness; for healthy adults the cognitive lift is modest, task-specific and best under medical supervision - it is the effective NON-supplement in this category, not a nootropic to self-source.

BEST FORM Prescription-only; not a retail supplement. Where a choice exists, armodafinil (R-modafinil) gives smoother, longer-lasting wakefulness at lower mg than racemic modafinil, while racemic modafinil has a faster initial peak - the right pick depends on the clinical goal and prescriber. (Racemic modafinil vs armodafinil (the R-enantiomer) - armodafinil has a longer effective half-life and flatter, more sustained levels.)

WATCH Prescription drug - use only under medical supervision. Induces CYP3A4 and so REDUCES effectiveness of hormonal contraceptives (use a backup/non-hormonal method during and for a month after) and can alter levels of other CYP-metabolized drugs; also inhibits CYP2C19. Rare but serious risk of severe rash/Steven's-Johnson syndrome - stop and seek care for any rash. Common: insomnia (dose in the morning), headache, anxiety, raised BP/heart rate. Interacts with warfarin and some antidepressants.

Sources: Battleday & Brem 2015, Eur Neuropsychopharmacol (modafinil neuroenhancement systematic review); Repantis et al. 2010, Pharmacol Res; Marraccini/Ilieva meta-analyses of stimulant neuroenhancement

2 Aniracetam Amazon ✗ Website ✓

Evidence: weak: small older DB trials in dementia only | Effect: possible slowing of decline in Alzheimer dementia; none demonstrated in healthy adults · **literature compromised (funding/outlier)** | Prescription drug in some countries (e.g. formerly Europe/Japan; withdrawn in some markets); not a lawful US dietary ingredient and not on US Amazon. Gray-market from nootropic vendors.

THE NOISE Sold as a 'mood-lifting' racetam that sharpens memory, verbal fluency, focus and reduces anxiety in healthy users.

THE SCIENCE Human evidence is limited to older double-blind trials in senile dementia of the Alzheimer type - e.g. 6-month multicentre DB trials of aniracetam 1500 mg/day vs placebo and vs piracetam, reporting slowed cognitive decline in the treated group. These are small, dated, largely manufacturer-era studies in a demented population. No credible modern RCT demonstrates memory or mood benefit in healthy adults. Anxiolytic/'mood' claims rest on animal work and open-label impressions.

THE VERDICT Skip for healthy adults - benefit was only ever shown in dementia patients on old trials, and it is not a legal US supplement. Not worth gray-market sourcing on current evidence.

BEST FORM Single defined molecule - form/salt irrelevant; buy purity-tested product. Because it is fat-soluble and short-acting, take it with a meal containing dietary fat and dose 2-3x/day. (Single molecule; the only real variable is that it is fat-soluble, so co-ingested fat matters more than the 'form'.)

WATCH Fat-soluble and short half-life - take with fat, split doses. Pair with a choline source to blunt headaches. Mildly anxiolytic. Metabolized largely to anisic acid; grey-market/unapproved supplement status in many countries.

Sources: Clinical Experience with Aniracetam in SDAT, Clin Drug Investig (Springer BF03258430); Fat-soluble racetam dementia DB trials, 1990s

2 Piracetam Amazon ✗ Website ✓

Evidence: weak: old industry-linked RCTs in disease populations, soft endpoints | Effect: small/uncertain global-rating change in dementia; none demonstrated in healthy adults · **literature compromised (funding/outlier)** | Not FDA-approved and not a lawful DSHEA dietary ingredient; not sold on US Amazon. Prescription/OTC drug in parts of Europe; US buyers get it gray-market from specialist nootropic vendors.

THE NOISE Marketed as the original 'smart drug' - sold to healthy students and professionals as a memory, focus and 'brain energy' enhancer that boosts learning and mental clarity.

THE SCIENCE The positive data is old, industry-linked, and entirely in DISEASE populations. The most-cited pro meta-analysis (Waegemans 2002, 19 double-blind trials) reported a Clinical Global Impression odds ratio of 3.20 (NNT ~3.9) - but CGI is a coarse global-rating scale, not a validated cognitive outcome, and effects on specific memory tests were weak. The 2001 Cochrane review (Flicker & Grimley Evans, 24 RCTs, ~12,000 subjects) concluded evidence was insufficient to support use in dementia/cognitive impairment. A 2024 systematic review (18 studies, 886 patients) found no clinically meaningful memory benefit vs control. There is essentially NO credible RCT evidence in healthy adults - the studied population is elderly dementia/post-stroke/post-CABG patients. Benefit in the deficient brain does not transfer to a healthy one.

THE VERDICT Skip for healthy-adult cognitive enhancement - the human data is in dementia patients on soft global scales, not memory in healthy people, and it is not a legal US supplement. If a clinician prescribes it for a specific cerebrovascular indication that is a different conversation.

BEST FORM Single well-absorbed molecule - form does not matter; buy tested, high-purity powder or capsules with a certificate of analysis. Doses are large (1.6-4.8 g/day) so powder is often practical. (Single defined molecule; only powder vs capsule, no meaningful salt/bioavailability difference.)

WATCH Frequently causes headache unless you co-supplement a choline source (e.g. alpha-GPC/CDP-choline). Cleared renally almost unchanged - caution/lower dose in kidney impairment. Not an approved dietary supplement in the US (grey-market); legality varies. Generally very low toxicity.

Sources: Waegemans et al. 2002, Dement Geriatr Cogn Disord 13(4):217-224; Flicker & Grimley Evans, Cochrane Database Syst Rev 2001;CD001011; 2024 systematic review, ScienceDirect S0303846724002452

2 Sulbutiamine Amazon ✓ Website ✓

Evidence: weak: small trials, largest RCT null on primary outcome | Effect: possible fatigue reduction in specific clinical asthenia; null in largest trial; none in healthy adults | Sold as a dietary supplement (variably available on US Amazon and specialist vendors); prescription (Arcalion) in some countries. Regulatory status is inconsistent.

THE NOISE Sold as a synthetic B1 that crosses the blood-brain barrier to fight fatigue, boost energy, motivation, mood and memory in healthy people.

THE SCIENCE The best test contradicts the hype: a 326-patient double-blind RCT in post-infectious asthenia (sulbutiamine 400 vs 600 mg/day vs placebo x 28 days) found NO significant difference on the primary fatigue measure (MFI); a transient day-7 signal in one female subgroup disappeared by day 28. A small 2017 RCT in multiple sclerosis fatigue (400 mg/day x 2 months) reported reduced fatigue. Net: mostly small trials in clinical fatigue populations, with the largest/most rigorous trial null. No healthy-adult cognition data. Benefit, if any, is a thiamine-derivative effect in specific fatigue states, not general nootropic action.

THE VERDICT Skip for healthy-adult 'energy/focus' - the largest DB RCT was null on its primary endpoint. Meaningful only, if at all, for specific clinical asthenia under supervision; correct genuine thiamine deficiency with thiamine instead.

BEST FORM Single defined molecule (a fat-soluble thiamine derivative) - form does not matter; buy purity-tested product. Because it is lipophilic it crosses into the brain better than plain thiamine; take with food. (Single lipophilic thiamine-disulfide molecule; no meaningful salt/form variants.)

WATCH Take with a fat-containing meal (fat-soluble). Tolerance/diminishing effect builds with continuous daily use - cycle it. Can be activating; caution in bipolar disorder (reports of mood elevation). Dose earlier in day to avoid insomnia.

Sources: DB RCT post-infectious asthenia 400-600 mg vs placebo, ScienceDirect S024886630080096X; Sulbutiamine in MS fatigue RCT 2017, MSARD S2211-0348(17)30119-0

2 Vinpocetine Amazon ✓ Website ✓

Evidence: weak/inconclusive (Cochrane); disease-population only | Effect: cerebral vasodilation; no reliable cognitive benefit demonstrated in healthy adults - **literature compromised (funding/outlier)** | FDA: not a lawful dietary ingredient and carries a pregnancy hazard warning (2019); still widely sold on US Amazon and vendor sites despite FDA's position. Rx (Cavinton) in parts of Europe/Asia.

THE NOISE Marketed as a 'brain blood flow' and memory booster for focus, cognition and even weight loss/energy.

THE SCIENCE Derived from Vinca minor; a cerebral vasodilator/PDE1 inhibitor. The Cochrane review (Szatmari & Whitehouse) concluded evidence is inconclusive/insufficient to support use in cognitive impairment or dementia - trials are old, short and mostly manufacturer-linked. No convincing healthy-adult cognitive benefit. Meanwhile the FDA flagged it as a reproductive hazard (2019) after NTP data showed miscarriage and fetal cardiac/skeletal abnormalities in rats, and has determined vinpocetine does not meet the definition of a dietary ingredient.

THE VERDICT Skip. Efficacy is inconclusive even in dementia and absent in healthy adults, and the FDA warns it can cause miscarriage/fetal harm - women who are or could become pregnant must not take it.

BEST FORM Single defined molecule - form/salt does not matter; choose a COA-verified product and take WITH food, which substantially increases absorption. Typical 5-10 mg up to 3x/day with meals. (Single semi-synthetic molecule; only powder vs capsule.)

WATCH FDA has warned it should NOT be taken by women who are or may become pregnant (associated with reduced fetal weight/miscarriage in animal data). Has antiplatelet/blood-thinning activity - avoid with anticoagulants/antiplatelets and stop before surgery. Can lower blood pressure. Take with food for absorption.

Sources: Szatmari & Whitehouse, Cochrane Database Syst Rev (vinpocetine for cognitive impairment); FDA 2019 vinpocetine pregnancy safety communication; NTP report; FDA 'Vinpocetine in Dietary Supplements' Federal Register determination

1.5 Oxiracetam Amazon ✗ Website ✓

Evidence: weak-moderate in disease; industry-funded; null independent post-stroke RCT | Effect: cognitive score change in vascular/TBI/dementia patients (sponsor trials); none in healthy adults · **literature compromised (funding/outlier)** | Prescription drug in China/Italy and elsewhere; not a lawful US dietary ingredient, not on US Amazon. Gray-market via nootropic vendors.

THE NOISE Promoted as a 'stimulant-type' racetam for logical thinking, focus, memory and attention in healthy users.

THE SCIENCE Better trial machinery than most racetams but still disease-only and industry-funded. Older DB placebo-controlled trials (oxiracetam 800 mg BID x 90 days) showed cognitive gains in mild-moderate Alzheimer/multi-infarct dementia. A recent multicentre randomized DB phase IV trial in vascular cognitive impairment and a phase 3 L-oxiracetam TBI trial (China) report positive cognitive outcomes - but these are sponsor-run in patient populations. Countering that, an independent post-stroke cognitive-impairment RCT found NO significant difference vs placebo on MMSE or CDR. No healthy-adult efficacy data.

THE VERDICT **Skip for healthy adults. The genuinely randomized modern trials are manufacturer-sponsored and in stroke/TBI/dementia patients, with an independent post-stroke trial null; nothing supports use for everyday cognition, and it is not a legal US supplement.**

BEST FORM Single molecule, water-soluble and well absorbed - form does not matter; choose a purity/COA-verified product. Typical 750-1500 mg 1-2x/day. (Single, water-soluble molecule; no meaningful form difference.)

WATCH Mildly stimulating - dose earlier in the day to avoid insomnia. Co-supplement choline to reduce headache. Renally excreted (caution in kidney impairment). Unapproved supplement status in many jurisdictions.

Sources: Oxiracetam DB dementia trial, PubMed 1603291; Phase IV VCI RCT, ScienceDirect S1551714423000319; L-oxiracetam TBI phase 3, PMC12698704

1 DMAE (dimethylaminoethanol / deanol) Amazon ✓ Website ✓

Evidence: weak/negative human data; mostly preclinical rationale | Effect: none demonstrated in humans (null/negative in dementia); adverse: drowsiness/confusion | Sold as a dietary supplement (capsules and topical skincare) on US Amazon and vendor sites.

THE NOISE Sold as a purported acetylcholine precursor for memory, focus, mood and 'anti-aging' brain and skin benefits.

THE SCIENCE The acetylcholine-precursor rationale is biochemically shaky and human outcomes are null-to-negative. A controlled trial in Alzheimer patients found DMAE no better than placebo, with nearly half discontinuing for drowsiness/confusion. Older deanol trials in elderly cognition (Cochrane-indexed) show no reliable benefit. The one 'positive' human signal is a DMAE-pyroglutamate scopolamine-challenge study in young subjects - a lab paradigm, not real-world cognition. This is mechanism-led marketing without a credible human outcome.

THE VERDICT **Skip. Human trials are null-to-negative with real side effects (drowsiness, confusion); there is no credible evidence DMAE improves memory or focus in healthy adults.**

BEST FORM The retail form is essentially always DMAE bitartrate, a stable well-absorbed salt - form is not a meaningful decision point. Buy a purity-tested bitartrate product. (Almost always sold as DMAE bitartrate (a stable salt); occasional L-tartrate - functionally equivalent.)

WATCH Avoid in pregnancy/planning pregnancy (structural analogs raise developmental-toxicity/neural-tube concerns) and while breastfeeding. Can paradoxically worsen mood in some people and may aggravate bipolar depression/agitation - caution in mood disorders. Contraindicated/caution in epilepsy and with cholinergic drugs. Evidence for cognitive benefit is weak; can cause headache, insomnia, muscle tension.

Sources: DMAE vs placebo in Alzheimer's (no benefit, high dropout); Deanol elderly cognition trial, Cochrane CENTRAL CN-00259659; DMAE p-Glu scopolamine study, Psychopharmacology 10.1007/s00213-009-1648-7

1 Noopept Amazon ✗ Website ✓

Evidence: weak/preliminary: single-source, small, open-label human data | Effect: possible cognition/anxiolytic effect in brain-injured patients (open-label); none demonstrated in healthy adults · **literature compromised (funding/outlier)** | Prescription drug in Russia; not a lawful US dietary ingredient and not on US Amazon. Sold gray-market as a 'research chemical'/nootropic online.

THE NOISE Marketed as a piracetam-style dipeptide ~1000x more potent, sold for memory, focus, 'neuroprotection' and NGF/BDNF-driven brain growth.

THE SCIENCE Human data comes essentially from one originating Russian group (Zakusov/Gudasheva program). The key study (Neznamov & Teleshova) was a ~53-patient open-label comparison vs piracetam in mild cognitive disorders from vascular damage or TBI (20 mg/day x 56 days), reporting comparable-or-better cognition plus anxiolytic effect - but it is small, open-label, disease-population, and single-source. NGF/BDNF upregulation is animal/in-vitro. There are no large Western placebo-controlled RCTs and no healthy-adult efficacy data.

THE VERDICT Skip for healthy adults. The evidence is one small open-label Russian trial in brain-injured patients plus rodent mechanism; that is not proof it does anything for a healthy person's memory, and it is not a legal US supplement.

BEST FORM Single defined peptide - form does not matter; buy accurately dosed, COA-verified product. It is roughly 1000x more potent than piracetam, so doses are tiny (10-30 mg/day) - accurate weighing or pre-dosed capsules are essential. (Single dipeptide molecule; sublingual vs oral is a delivery choice, not a different 'form'.)

WATCH Very potent - easy to over-dose from powder; use a milligram scale or capsules. Can cause irritability, headache or overstimulation; a choline source may help. Short-acting, taken 2-3x/day. Grey-market/unapproved supplement status in most countries.

Sources: Neznamov & Teleshova, Zh Nevrol Psikhiatr (Noopept vs piracetam, MCD); Gudasheva et al. preclinical NGF/BDNF studies

1 Phenibut Amazon ✗ Website ✓

Evidence: none for cognition; strong evidence of harm/dependence | Effect: sedation/anxiolysis (GABA-B agonism); dependence and withdrawal; no cognitive enhancement demonstrated · **literature compromised (funding/outlier)** | FDA (2019/2023): phenibut is NOT a lawful dietary ingredient - warning letters issued; banned/removed from US Amazon. Restricted or banned in Australia, UK, several EU states; sold gray-market online.

THE NOISE Sold as a 'calm focus', anxiety-relief, sleep and social-confidence nootropic; marketed as a gentle GABA supplement.

THE SCIENCE Phenibut is a GABA-B agonist anxiolytic/sedative - not a cognitive enhancer, and it has essentially no credible controlled human data supporting nootropic benefit. What IS well documented is harm: >1,300 US poison-center exposure calls (2009-2019) with lethargy, agitation, tachycardia, confusion; >80 associated cases of coma/death; and a severe, benzodiazepine-like dependence and withdrawal syndrome (anxiety, insomnia, psychosis, seizures) sometimes requiring hospitalization. A 2023-2024 systematic review catalogs the withdrawal burden. It is a genuine drug of abuse mis-sold as a supplement.

THE VERDICT Do not buy. This is a dependence-forming GABA-B sedative with documented comas, deaths and severe withdrawal - not a nootropic; the honest recommendation is avoidance, and seek medical help for any existing dependence.

BEST FORM HCl is the standard, reliable retail form; FAA (free amino acid) is less stomach-irritating and more concentrated but must be dosed by weight, not by volume, since it is more potent per gram. Honestly, the bigger message is not which form but that this drug is high-risk. (Phenibut HCl (standard, acidic, water-soluble) vs phenibut FAA (free amino acid - less acidic, more concentrated per gram, better for sublingual).)

WATCH HIGH-RISK: a GABA-B agonist that causes rapid tolerance, physical dependence and a severe, protracted, benzodiazepine-like withdrawal (anxiety, insomnia, psychosis, seizures) - do NOT use daily or long-term. Dangerous additive CNS depression with alcohol, benzodiazepines and opioids (respiratory depression). Delayed onset (2-4 h) leads to accidental re-dosing/overdose. Treat as an intermittent-use drug at most, or avoid.

Sources: FDA warning letters on phenibut (2019-2023); Systematic review of phenibut withdrawal, PubMed 38112312; Cohen et al., quantity of phenibut in supplements before/after FDA warnings

1 Phenylpiracetam Amazon ✗ Website ✓

Evidence: weak: small Russian clinical studies, no registered/Western RCTs | Effect: acute psychostimulation/endurance (WADA-relevant); no demonstrated memory benefit in healthy adults · **literature compromised (funding/outlier)** | Unapproved drug/research chemical; banned by WADA in competition; not a lawful US dietary ingredient and not on US Amazon. Gray-market only.

THE NOISE Hyped as the strongest racetam - a 'clean stimulant' that boosts focus, motivation, physical endurance and cold tolerance; popular pre-exam and pre-workout.

THE SCIENCE Almost the entire human evidence base is small, old, Russian-language clinical literature (Phenotropil/carphedon) in post-stroke, encephalopathy and asthenia patients - not blinded to Western standards and not independently replicated. It does not appear as an intervention in any registered ClinicalTrials.gov study. Stimulant/endurance effects are real enough that WADA bans it (S6 stimulants), which tells you it is pharmacologically active, not that it is safe or memory-enhancing. Zero credible RCT evidence for cognitive enhancement in healthy adults.

THE VERDICT **Skip. Genuinely stimulant (hence WADA-banned) but the cognition evidence is thin Russian clinical lore with no independent RCTs; it is an unapproved research chemical, and athletes will fail drug tests.**

BEST FORM Single, orally well-absorbed molecule - form irrelevant; buy tested product. Low dose (100-200 mg) due to high potency. (Single molecule (phenyl-substituted piracetam); no salt/bioavailability variants of note.)

WATCH Distinctly stimulant; tolerance develops rapidly with daily use, so it is best used occasionally, in the morning. BANNED by WADA - do not use if drug-tested in sport. Can cause overstimulation/insomnia. Choline may help headaches.

Sources: Russian Phenotropil/carphedon clinical literature (Savchenko, Zavalishin et al.); WADA Prohibited List (S6 Stimulants)

Cholinergic & cognition nutrients (6)

2 Acetyl-L-carnitine (ALCAR) Amazon ✓ Website ✓

Evidence: weak-moderate: multiple RCTs but small, short, several manufacturer-sponsored, effect sub-MCID and not replicated in dementia | Effect: small slowing of decline in MCI/mild AD (below MCID); none demonstrated in healthy adults · **literature compromised (funding/outlier)**

THE NOISE Sold as a memory, focus and 'mitochondrial energy' booster for brain fog, anti-aging and studying; influencers claim it sharpens a healthy brain and reverses age-related mental decline.

THE SCIENCE In MCI and mild Alzheimer's, pooled RCTs show a small benefit: standardised effect ~0.20 (95% CI 0.11-0.30) on cognitive scales and ~0.32 (0.18-0.47) on Clinical Global Impression over 3-12 months at 1.5-3 g/day. That is statistically real but below what a clinician would notice as clinically meaningful, and later/larger trials and systematic reviews failed to reproduce it in dementia. There is NO credible RCT showing acute or chronic cognitive gain in healthy adults. A 2024 Mendelian-randomisation study found no causal link between acylcarnitine levels and cognition.

THE VERDICT **Skip for a healthy adult wanting a nootropic - the only signal is a sub-MCID effect in disease populations. Conditional at best: 1.5-3 g/day is worth a clinician-supervised trial in diagnosed MCI/mild AD or diabetic neuropathy, not for cognitive enhancement.**

BEST FORM Acetyl-L-carnitine — the acetylated form crosses the blood-brain barrier and donates acetyl groups for acetylcholine, so it is the correct choice for cognition (plain L-carnitine and L-carnitine-L-tartrate are for muscle/exercise, not brain). Buy on USP/branded purity; free-form powder or capsules are fine as it is highly water-soluble. (Acetyl-L-carnitine (ALCAR) vs plain L-carnitine vs L-carnitine-L-tartrate vs propionyl-L-carnitine)

WATCH Take on an empty stomach for best uptake and dose earlier in the day — it can be mildly stimulating and disrupt sleep. Typical 500-2000 mg/day in divided doses. All carnitines are converted by gut bacteria to TMAO, a marker linked to cardiovascular risk with chronic use. Can potentiate thyroid effects and, theoretically, warfarin. Occasional fishy body odor and GI upset at higher doses.

Sources: Montgomery SA et al. Int Clin Psychopharmacol 2003;18(2):61-71 (meta-analysis, PMID 12598816); Thal LJ et al. Neurology 2000 (1-yr AD RCT, negative); Pennisi M et al. Nutrients 2020;12(5):1389 (critical update review); Two-sample MR study, PMC12224038 (2024, null causal)

2 Centrophenoxine (meclofenoxate) Amazon ✗ Website ✓

Evidence: weak: old, small, methodologically poor RCTs in dementia; unapproved-drug status | Effect: possible modest memory effect in elderly dementia (low-quality, with a worsening subgroup); none in healthy adults · **literature compromised (funding/outlier)** | Unapproved synthetic drug; not a lawful DSHEA dietary ingredient and not compliantly sold on US Amazon. Available gray-market from nootropic vendors; products frequently mislabelled.

THE NOISE Sold by nootropic vendors as an anti-aging 'brain cleaner' that clears lipofuscin, raises acetylcholine and DMAE, and boosts memory in healthy people.

THE SCIENCE The human evidence is old (1960s-1980s), small and low-quality, and confined to elderly dementia/organic-brain-syndrome patients - not healthy adults. The most-cited RCT (50 dementia patients, 2 g/day, 8 wk) reported 48% vs 28% 'memory improvement' but the statistics were not done rigorously and, critically, MORE patients WORSENE on centrophenoxine (5) than placebo (1) - an adverse dose-response signal. A 2022 analysis (Cohen/Clinical Toxicology) found it is an unapproved synthetic drug, not a lawful US dietary ingredient, and that marketed products were badly mislabelled (79-251 mg actual; only 1 of 7 within +/-10% of label).

THE VERDICT Skip. This is an unapproved drug with weak half-century-old dementia data, a worsening signal in the key trial, and severe product-quality/mislabelling problems - not a supplement for a healthy adult.

BEST FORM One defined molecule, so no superior 'form' to select — buy on third-party-verified purity, since it is an unregulated nootropic, not a food supplement. Standard dosing is 250-1000 mg/day. (A single defined molecule (meclofenoxate, a DMAE + pCPB ester) — vendor purity is the variable)

WATCH Cholinergic and mildly stimulating — dose in the morning/early afternoon to avoid insomnia; it liberates DMAE, which is contraindicated in pregnancy and cautioned in bipolar/seizure disorders. Take with food to limit GI upset. Overdosing causes headache, jaw tension, agitation. Not FDA-approved; thin long-term human safety data.

Sources: Cohen PA et al. Clin Toxicol 2022;60(10):1156 (unapproved-drug analysis, PMID 35959800); Marcer & Hopkins 1977 dementia RCT (n=50); ADDF Cognitive Vitality Centrophenoxine researcher report

2 PQQ (pyrroloquinoline quinone) Amazon ✓ Website ✓

Evidence: weak: small RCTs, industry-funded, short duration, no independent replication | Effect: modest attention/working-memory and BDNF changes in small industry trials; clinical significance unestablished · **literature compromised (funding/outlier)**

THE NOISE Sold as a 'mitochondrial biogenesis' and BDNF-boosting nootropic that sharpens memory, attention and energy and 'grows new mitochondria', often paired with CoQ10.

THE SCIENCE Small human RCTs (mostly ~20-40 subjects, 8-24 weeks, 20 mg/day) report modest gains in attention/working memory and a rise in serum BDNF - but nearly all are funded by Mitsubishi Gas Chemical (BioPQQ/MGCPQQ maker) and use manufacturer analyses. Effects are small, inconsistent across domains, and short. The 2024 six-week RCT in MCI (n=34) found raised serum BDNF (P=0.01) but did not establish a clinically meaningful cognitive improvement. No large independent trial exists; the evidence base is industry-concentrated - a classic mechanism-outruns-outcome case.

THE VERDICT Skip / low priority. The efficacy signal is small, short and almost entirely manufacturer-funded. Not worth buying for cognition until an independent, adequately powered trial replicates it.

BEST FORM PQQ disodium salt is the one bioavailable, stable retail form; branded BioPQQ/MGCPQQ material is best-documented. There is no absorption-superior alternative to choose between, so buy on purity/third-party testing rather than 'form.' (PQQ disodium salt (e.g. BioPQQ / MGCPQQ) — the standard and only meaningful consumer form)

WATCH Take with food, often paired with CoQ10/ubiquinol (synergistic for mitochondrial biogenesis). Effective doses are small (10-20 mg/day) — more is not better. Generally very well tolerated but human safety data are sparse, so avoid in pregnancy. Can be mildly activating, so morning dosing is preferable.

Sources: Nakano M et al. 2009/2012 (Japanese cognition RCTs, industry); Shiojima Y et al. Food Funct 2023 D2FO01515C (age-stratified RCT); 2024 dihydrogen-PQQ MCI RCT, PMID 38908296 / PMC12878601; Koikeda et al. 2011

1.5 Choline bitartrate Amazon ✓ Website ✓

Evidence: moderate evidence of NO acute cognitive effect (Bayesian-supported null RCT); nutritional adequacy is a separate matter | Effect: none demonstrated on memory in healthy adults (positive evidence of no effect)

THE NOISE Cheap ubiquitous nootropic sold as a memory and focus booster and 'acetylcholine precursor', and as the fix for racetam headaches.

THE SCIENCE A well-designed double-blind placebo-controlled crossover in healthy young adults (2-2.5 g, tested 60-120 min post-dose across three memory tasks) found NO acute effect on declarative or working memory - and Bayesian analysis positively supported the null, not just failed to reach significance. Choline bitartrate is also a poor brain-delivery form (much is oxidised to trimethylamine peripherally, causing fishy body odour at higher doses). Adequate dietary choline matters for health, but supplementing bitartrate does not sharpen a healthy adult's cognition. Citicoline/alpha-GPC have marginally better (still modest) human data - bitartrate is the weakest form.

THE VERDICT Skip as a nootropic. Meet choline needs from diet (eggs, liver); if you want a choline supplement for a specific reason, citicoline or alpha-GPC have better human data than bitartrate. Bitartrate's only proven job is avoiding dietary deficiency.

BEST FORM For cognition, skip bitartrate and choose Alpha-GPC or CDP-choline (citicoline) — both cross the blood-brain barrier and raise brain acetylcholine efficiently. Bitartrate is cheap but low-yield (~41% choline) and does not meaningfully boost brain choline; it is mainly a basic liver/general choline source. (Choline bitartrate vs Alpha-GPC vs CDP-choline (citicoline))

WATCH All choline sources are converted by gut flora to TMAO (cardiovascular marker). Excess (bitartrate especially) causes fishy body odor, GI upset, sweating, low blood pressure and, at high doses, a depressive/headache effect from cholinergic overshoot. Alpha-GPC has emerging observational (unproven) stroke-risk signals at high chronic doses. Adult UL for choline is 3.5 g/day.

Sources: Lippelt DP et al. PLOS One 2016;11(6):e0157714 (null RCT, PMC4920398); EFSA choline intake opinions; Contrast: McGlade et al. citicoline RCTs (different, better-delivered form)

1.5 Uridine monophosphate (UMP) Amazon ✓ Website ✓

Evidence: animal/mechanistic only for standalone; multinutrient (Souvenaid) evidence is disease-specific and mixed | Effect: none demonstrated for standalone UMP in humans · **literature compromised (funding/outlier)**

THE NOISE Marketed by nootropic vendors as a standalone 'synapse builder' and dopamine/mood enhancer that boosts memory, focus and neurogenesis, often stacked with DHA and choline ('Mr Happy stack').

THE SCIENCE There are essentially NO randomised human trials of standalone UMP for cognition in healthy adults - the human evidence is entirely mechanistic/animal (gerbil, rat) plus its role inside the proprietary Fortasyn Connect / Souvenaid multinutrient. And Souvenaid's own record is mixed-to-null: the largest independent-style RCT, S-Connect (n=527, mild-to-moderate AD, 24 wk), showed NO cognitive benefit vs control; positive signals (Souvenir I, LipiDiDiet) are confined to early/prodromal AD and cannot be attributed to uridine specifically within a 12-ingredient blend. Standalone UMP cognitive benefit in a healthy person is unproven.

THE VERDICT Skip. No standalone human cognition data; the multinutrient it rides in failed in established AD and its prodromal signal is a whole-formula, disease-population result. Do not buy UMP alone as a nootropic.

BEST FORM Retail UMP is effectively a single practical form: uridine-5'-monophosphate disodium salt. Sublingual dosing is marketed as superior but oral UMP is absorbed and raises plasma uridine adequately — do not overpay for sublingual claims. Triacetyluridine is more bioavailable but is not a mainstream consumer product. (UMP disodium salt (retail) vs triacetyluridine/TAU (better absorbed but research-only))

WATCH Best taken with food, and classically stacked with DHA and choline (the Souvenaid/membrane-synthesis rationale). Very well tolerated; high doses can cause mild headache or GI upset. Uridine raises plasma uric acid — caution with gout. Limited long-term human data.

Sources: Shah RC et al. (S-Connect) Alzheimers Res Ther 2013 (n=527, null; PMC3978853); Scheltens P et al. Souvenir I/II trials; Soininen H et al. LipiDiDiet, Lancet Neurol 2017 (prodromal AD); Wang L et al. gerbil study, PMID 18606862 (animal)

1 7,8-Dihydroxyflavone (7,8-DHF) Amazon ✗ Website ✓

Evidence: animal/in-vitro only; no human trials | Effect: none demonstrated in humans (animal/in-vitro only) | Not a DSHEA-compliant dietary ingredient; not sold on US Amazon. Available gray-market from research-chemical and specialist nootropic vendors, sold 'for research use'.

THE NOISE Hyped by biohackers as an oral 'BDNF mimic' / TrkB agonist that grows neurons, boosts memory and treats depression and Alzheimer's - marketed off the back of dramatic mouse data.

THE SCIENCE There are NO human clinical trials of 7,8-DHF for cognition (or anything else) - the entire evidence base is rodent and in-vitro. Worse, its headline mechanism is contested: recent work shows it is NOT a clean TrkB agonist and is instead a direct inhibitor of human pyridoxal phosphatase (a vitamin-B6-handling enzyme), raising an off-target-toxicity question. Oral bioavailability in humans is unknown. This is pure preclinical marketing.

THE VERDICT Do not buy. Zero human data, a disputed mechanism, and an identified off-target enzyme-inhibition liability. It is a research chemical, not a supplement.

BEST FORM Sold as a research chemical, not a regulated supplement — there is no 'better form,' only a better vendor. Insist on a certificate of analysis (HPLC purity, heavy-metal/solvent screen) from a reputable lab, because identity and purity are the real risks. (A single research-chemical compound; the variable is vendor purity, not chemical form)

WATCH Poor, erratic oral bioavailability with a short half-life and rapid glucuronidation — much of the animal TrkB-agonist promise does not translate to oral human dosing. Essentially no controlled human trials or established safe dose, no manufacturing oversight, unknown long-term effects. Treat as experimental; not for pregnancy or alongside other neuroactive drugs.

Sources: Liu C et al. Transl Neurodegener 2016 review, PMC4702337; PLP-phosphatase inhibition paper, PMC11164532 (2024); ADDF Cognitive Vitality 7,8-DHF researcher report (no human studies)

Men's hormonal / libido / prostate (10)

3 Beta-sitosterol Amazon ✓ Website ✓

Evidence: moderate but dated: several small short independent RCTs, consistent symptomatic benefit | Effect: BPH/LUTS urinary symptom-score and flow improvement (symptomatic, not prostate-shrinking) | Widely sold as a dietary supplement (plant sterol) on US Amazon and vendors.

THE NOISE Sold as a prostate/urinary-support supplement for BPH (often blended with saw palmetto and pygeum) and as a cholesterol-lowering plant sterol.

THE SCIENCE Better BPH evidence than saw palmetto. A Cochrane review (4 RCTs, 519 men, 4-26 weeks) found beta-sitosterol improved urinary symptom scores (IPSS weighted mean difference ~-4.9 points) and peak flow versus placebo. The anchor RCT (Berges, Lancet 1995, N=200, 20 mg three times daily) showed IPSS falling -7.4 vs -2.1 for placebo (net ~-5.3, clearing the ~3-point MCID) and improved flow/residual volume. Caveat: trials are small, short and old, and prostate volume did not shrink - it relieves symptoms without treating the enlargement.

THE VERDICT Conditional: a defensible symptomatic option for mild-moderate BPH LUTS at ~60 mg/day beta-sitosterol if you want a supplement route, but the trials are dated/short and it won't reduce prostate size; alpha-blockers/5-ARIs have stronger, longer evidence.

BEST FORM A defined-dose beta-sitosterol (or mixed phytosterol blend stating beta-sitosterol content), the studied BPH dose is ~60-130 mg/day of beta-sitosterol. Form isn't the variable, purity and stated milligrams are, choose a product listing actual beta-sitosterol mg rather than vague 'phytosterol complex'. (Beta-sitosterol is a single defined phytosterol; products differ mainly in dose and purity, not chemical form)

WATCH Take with meals (it works partly by competing with cholesterol absorption). Reduces absorption of dietary cholesterol AND fat-soluble vitamins/beta-carotene over time. Can lower absorption of ezetimibe-class effects overlap. Rare: contraindicated in sitosterolemia (a genetic phytosterol-storage disorder). Mild GI effects. Doesn't shrink prostate but may improve urinary symptoms/flow.

Sources: Cochrane CD001043 (Wilt 1999); Lancet 1995 S0140673695910859 (Berges RCT)

2.5 Tongkat ali (*Eurycoma longifolia*) Amazon ✓ Website ✓

Evidence: moderate but narrow: several small RCTs, one dominant industry extract, benefit gated to deficiency | Effect: total testosterone increase in hypogonadal/aging men; no demonstrated effect in eugonadal healthy men · **literature compromised (funding/outlier)** | Widely sold as a dietary supplement on US Amazon and vendor sites; quality varies enormously.

THE NOISE Sold as a natural testosterone booster, libido/erection enhancer, stress-and-cortisol reducer and muscle builder; influencer marketing pitches it as a legal, side-effect-free replacement for TRT for any man who wants 'higher T'.

THE SCIENCE 2022 meta-analysis (5 RCTs, N=267) found a pooled SMD of 1.352 for total testosterone (95% CI 0.565-2.138), but with severe heterogeneity ($I^2=87\%$) and detected publication bias (Begg $p=0.024$). Crucially the benefit was driven entirely by the hypogonadal subgroup (SMD 1.861, $p=0.002$); in healthy men there was NO significant effect (SMD 0.760, $p=0.249$). Seven of nine trials used one industry extract (Physta). Some sexual-wellbeing and cortisol/stress signals exist but are small and short.

THE VERDICT **Conditional: reasonable to trial only for aging/hypogonadal men with symptoms, 200-400 mg/day of a Physta-grade eurycomanone-standardized extract; for a healthy young man with normal T the RCT evidence shows essentially nothing. Buy a third-party heavy-metal-tested product only.**

BEST FORM Standardised water extract with a stated eurycomanone content (~2%+) or branded Physta/LJ100 standardised to eurycomanone, glycosaponins and polysaccharides. Ignore vague '100:1/200:1' ratios with no marker. Heavy-metal tested (mercury contamination has been reported in cheap root). (Water-extract standardised to eurycomanone / glyco-saponins (e.g. Physta, LJ100) vs unstandardised crude root '200:1' powders)

WATCH Take earlier in the day, can be mildly stimulating/sleep-disrupting for some. Typical 200-400 mg standardised extract daily. May raise testosterone/DHT modestly, caution with hormone-sensitive conditions. Long-term human safety data limited, cycle it. Possible insomnia, restlessness at high doses.

Sources: PMID 36013514 / PMC9415500 (meta-analysis); ScienceDirect S0378512220304497 (6-mo ADAM RCT); PMC3518798 (Physta QoL RCT); ScienceDirect S0278691506000251 (mercury contamination)

2.5 Yohimbine / yohimbe Amazon ✓ Website ✓

Evidence: moderate: multiple RCTs, modest effect, offset by safety and supplement-quality problems | Effect: modest erectile-function improvement (alpha-2 antagonist) · **literature compromised (funding/outlier)** | Yohimbe bark supplements are sold on US Amazon and vendors but are a restricted/gray ingredient; pharmaceutical yohimbine HCl was Rx (brand discontinued); banned or restricted in several countries.

THE NOISE Sold as a natural ED cure, libido enhancer, fat-burner and 'stimulant pre-workout'; yohimbe bark marketed as a testosterone/vitality tonic.

THE SCIENCE Among these compounds yohimbine has the best ED efficacy signal: a 2022 systematic review/meta-analysis (8 studies) found yohimbine monotherapy improved erectile function vs placebo (OR ~2.08, 95% CI 1.30-3.32; older pooling OR ~3.85), i.e. real but only modest. The catch is the supplement form: yohimbe bark extract is wildly mislabeled (an analysis found only ~4% of products accurately stated content, some with pharmaceutical-grade doses), and yohimbine carries genuine cardiovascular risk - hypertension, tachycardia, arrhythmia, seizures, and case reports of cardiac collapse and death.

THE VERDICT **Conditional and cautious: pharmaceutical yohimbine HCl (~5 mg three times daily) has modest RCT support for ED under medical supervision, but avoid OTC yohimbe bark supplements - dose is unreliable and the cardiovascular/seizure risk is real. PDE5 inhibitors remain first-line.**

BEST FORM Standardised yohimbine HCl so you get an exact milligram dose (research fat-loss/ED dosing ~0.2 mg/kg). Avoid crude 'yohimbe bark' extracts, FDA analyses found actual yohimbine content ranging from near-zero to far above label. (Standardised/pharmaceutical yohimbine HCl (precise mg dose) vs crude yohimbe bark extract (wildly variable, often mislabelled alkaloid content))

WATCH Potent alpha-2 antagonist with real cardiovascular risk: raises blood pressure and heart rate, can cause anxiety/panic, palpitations, arrhythmia. Contraindicated with hypertension, heart disease, anxiety disorders, and with MAOIs, stimulants and many antidepressants (hypertensive crisis risk). Take fasted (food blunts effect but also lowers risk). Not for evening use. Start with a low test dose.

Sources: PMID 35118966 (2022 ED meta-analysis); PMID 9123711 (crossover RCT); PMID 26391406 (pharmaceutical quantities in supplements); NCCIH yohimbe safety

2 DHEA (dehydroepiandrosterone) Amazon ✓ Website ✓

Evidence: moderate quantity, weak/inconsistent effect; guideline bodies advise against | Effect: inconsistent/modest ED and androgen effect, mainly in deficiency; largely null in eugonadal men | Sold OTC as a supplement in the US (grandfathered under DSHEA) but is prescription-only or banned in the UK, EU, Australia and Canada; prohibited in sport by WADA.

THE NOISE Sold as an anti-aging 'master hormone', testosterone/androgen precursor, libido and ED aid and muscle preserver for aging men.

THE SCIENCE DHEA is a genuine androgen precursor, so it is not inert - but outcome trials in men are inconsistent and largely disappointing. An early small RCT (Reiter 1999) suggested ED benefit, but the ANDRO 2002-type trials comparing DHEA vs testosterone vs placebo in men with sexual dysfunction found DHEA no better than placebo for erectile function, and it reliably raises testosterone mainly in women and the deficient, not eugonadal men. Endocrine guidance does not recommend it for age-related decline.

THE VERDICT Skip for a typical healthy man; benefit is confined, at best, to documented adrenal/androgen deficiency and even there it is weak. Not a substitute for evaluating and treating true hypogonadism.

BEST FORM Micronised, USP-grade DHEA from a brand with third-party assay (label accuracy is notoriously poor). Know that 7-keto-DHEA is a distinct non-androgenic metabolite, don't confuse the two. Start low (often 25 mg or less); women generally need lower doses. (Pharmaceutical/micronised USP DHEA vs unregulated over-labelled products; also true DHEA vs '7-keto-DHEA' (different molecule))

WATCH It is an actual androgen precursor, banned by WADA (athletes beware). Converts to testosterone/oestrogen, caution with hormone-sensitive cancers (prostate/breast). Can cause acne, hair loss, oily skin, and in women hirsutism/voice changes. Can suppress HDL. Interacts with hormonal therapies and can skew endocrine lab results. Take in the morning to mimic diurnal rhythm.

Sources: Tandfonline 13685530903294388 (T vs DHEA vs placebo RCT); ClinicalTrials NCT00202163 (ANDRO 2002); ScienceDirect S0090429598005718 (Reiter DHEA ED RCT); ScienceDirect S0531556520304587 (dose-response meta-analysis)

2 Fenugreek (Trigonella foenum-graecum) Amazon ✓ Website ✓

Evidence: weak-moderate but industry-concentrated; independent replication weakens the T claim | Effect: self-reported libido/sexual function; equivocal/near-null serum testosterone in independent trials · **literature compromised (funding/outlier)** | Common culinary herb and dietary supplement; freely sold on US Amazon and vendors.

THE NOISE Marketed (mainly as branded Testofen) as a testosterone booster, libido and 'free testosterone' enhancer and strength aid for aging men.

THE SCIENCE Meta-analysis of 4 trials found a statistically significant rise in total testosterone, but the positive body of work is dominated by the proprietary Testofen extract with manufacturer involvement. A 2024 independent dose-ranging RCT (95 men, 40-80y, 600/1200/1800 mg) is telling: plasma total and free testosterone did NOT rise versus placebo - only salivary testosterone did. Self-reported libido/sexual-function improvements are the most consistent finding but are subjective.

THE VERDICT Skip if you want a real serum-testosterone increase; the independent data don't support it. A modest, subjective libido effect exists mostly in industry Testofen trials at 600 mg/day if you want to experiment.

BEST FORM Seed extract standardised to the relevant saponin/glycoside fraction, e.g. Testofen (standardised furostanolic saponins/fenusides) or Furosap (protodioscin-standardised), which is what libido/testosterone trials used. Plain culinary seed powder is fine for blood-sugar/lactation but underdosed for hormonal claims. (Standardised branded seed extracts (Testofen/Fenuside ~50% fenusides, Furosap ~20% protodioscin) vs plain ground seed)

WATCH Strong maple-syrup body odour is common. Lowers blood glucose (additive with diabetes meds) and has anticoagulant activity (caution with warfarin/antiplatelets, and around surgery). High soluble fibre can reduce absorption of other oral drugs, separate timing. Avoid in pregnancy (uterine stimulant). Legume, cross-reacts with peanut/chickpea allergy.

Sources: PMID 32048383 (testosterone meta-analysis); PMC11407615 (2024 independent dose-ranging RCT); Thieme a-2048-5925 (anabolic effect review)

2 Saw palmetto (*Serenoa repens*) Amazon ✓ Website ✓

Evidence: strong evidence of NO effect: large independent RCTs and Cochrane meta-analysis null | Effect: none demonstrated - no improvement in LUTS/urinary flow vs placebo in high-quality trials · **literature compromised (funding/outlier)** | Ubiquitous dietary supplement on US Amazon and vendors.

THE NOISE Sold as a prostate-shrinking, urinary-flow-improving remedy for BPH/enlarged prostate and as a natural DHT-blocker for hair loss.

THE SCIENCE This is a well-documented null. The updated Cochrane review (27 studies, 4,656 men) concluded saw palmetto, alone or in combination, does not improve urologic symptoms or flow versus placebo in the short or long term. The high-quality independent trials (STEP, CAMUS) were negative even at double and triple doses, with clinical responder rates essentially identical to placebo. Older positive signals came from smaller, lower-quality or industry studies.

THE VERDICT **Skip for BPH/LUTS - the best independent evidence says it works no better than placebo. If you have bothersome urinary symptoms, evidence-based options are alpha-blockers (e.g. tamsulosin) and 5-alpha-reductase inhibitors (finasteride/dutasteride).**

BEST FORM Liposterolic extract standardised to ~85-95% free fatty acids and sterols (the Permixon-type CO2/hexane extract used in trials), 320 mg/day. Dried berry powder and cheap low-strength extracts are underdosed in the active lipids. (Liposterolic (hexane/CO2) extract standardised to 85-95% fatty acids/sterols (e.g. Permixon) vs dried berry powder or low-fat-content extract)

WATCH Fat-soluble, take WITH food/fat for absorption. Generally well tolerated, mild GI upset possible. Mild antiplatelet/anticoagulant effect, caution with warfarin/antiplatelets and around surgery. Lowers total but usually not free/serum PSA modestly less than finasteride, tell your doctor if getting a PSA test. Evidence for BPH symptom relief is mixed.

Sources: Cochrane CD001423 (Tacklind 2012, updated wjmh.230222); CAMUS trial JAMA 2011; STEP trial NEJM 2006; AAFP 2024 Cochrane summary

2 Shilajit (purified, PrimaVie) Amazon ✓ Website ✓

Evidence: weak/preliminary: single small manufacturer-funded RCT | Effect: total/free testosterone increase in one small industry RCT (aging men) · **literature compromised (funding/outlier)** | Sold as a dietary supplement (resin and capsules) on US Amazon and vendors; purity highly variable.

THE NOISE Marketed as a Himalayan mineral 'testosterone and energy' tar, boosting T, sperm, mitochondria and vitality; heavy influencer/traditional-medicine hype.

THE SCIENCE The testosterone claim rests essentially on one small industry RCT (Pandit 2016, purified PrimaVie shilajit, 250 mg twice daily, healthy men 45-55y, 90 days) reporting ~20% rise in total testosterone vs ~2% placebo, plus free T and DHEAS increases. It is a single manufacturer-funded study with a small sample; independent replication is essentially absent. Contamination risk (heavy metals) is real for unpurified products.

THE VERDICT **Skip on current evidence - a single small industry trial is not enough to bank on. If experimenting, only use a purified, heavy-metal-tested extract (e.g. PrimaVie) at 250 mg twice daily and treat the T claim as unproven.**

BEST FORM Purified, clinically-studied PrimaVie shilajit standardised to fulvic acid (and dibenzo-alpha-pyrones) with documented heavy-metal and microbial testing. Raw unpurified shilajit is the main risk here. (Purified, standardised shilajit (PrimaVie: standardised fulvic acid + DBPs, heavy-metal tested) vs raw/unpurified 'resin' of unknown origin)

WATCH Unpurified shilajit can be heavily contaminated with lead, mercury, arsenic and mycotoxins, purification/third-party testing is non-negotiable. Fulvic acid chelates minerals and can increase iron absorption, avoid if you have haemochromatosis/iron overload. Separate from other mineral supplements/medications. Typical dose 250-500 mg/day.

Sources: PMID 26395129 / Andrologia 10.1111/and.12482 (Pandit 2016 testosterone RCT)

2 Tribulus terrestris Amazon ✓ Website ✓

Evidence: weak: T effect null in good trials; ED signal from small/low-quality studies | Effect: possible modest self-reported libido/erectile improvement; no reliable testosterone effect - **literature compromised (funding/outlier)** | Freely sold as a dietary supplement on US Amazon and vendor sites.

THE NOISE Sold as a testosterone booster and libido/performance enhancer for athletes and men with low drive; label claims of 'clinically proven' T elevation.

THE SCIENCE The testosterone claim is essentially disproven: controlled trials in eugonadal men and athletes repeatedly show no increase in total testosterone versus placebo. There is a weak, low-quality signal for sexual function/erectile scores - a 2025 systematic review/meta-analysis of ED trials reported IIEF improvements, but the pooled trials are small, short, heterogeneous and often industry-linked or unblinded. Effect on hard hormonal endpoints is null.

THE VERDICT Skip for testosterone (it doesn't raise it). At best a marginal, poorly-evidenced libido product; not worth buying over addressing sleep, weight and underlying cause.

BEST FORM Fruit extract standardised to steroidal saponins as protodioscin (commonly labelled 40-60% 'saponins', but protodioscin is the meaningful fraction). Bulgarian/Tribulus fruit sources are traditional. Verify species identity (adulteration is common). (Fruit/aerial extract standardised to saponins (protodioscin) vs unstandardised or root-based powder)

WATCH Does NOT reliably raise testosterone in men in trials, evidence is mainly libido, set expectations accordingly. Can interfere with some hormone/steroid assays. Reports of hepatotoxicity/nephrotoxicity with certain products (possibly adulteration). Caution with lithium (diuretic effect) and in hormone-sensitive conditions.

Sources: Nature s41443-025-01086-7 (2025 ED meta-analysis); MDPI Nutrients 17/7/1275 / PMC11990417 (systematic review); ScienceDirect S0378512216302973 (Kamenov RCT); PMC9736311 (CrossFit RCT, null)

1 Fadogia agrestis Amazon ✓ Website ✓

Evidence: animal/in-vitro only; no human trials | Effect: none demonstrated in humans (rodent-only testosterone/behavior signal) | Sold as a dietary supplement on US Amazon and specialist vendors despite absence of human safety data.

THE NOISE Marketed (largely off the back of a Huberman podcast mention) as a potent natural testosterone and LH booster, often stacked with tongkat ali; sold as a near-TRT libido and 'drive' enhancer.

THE SCIENCE There are ZERO published human clinical trials of any kind as of 2025-2026. All efficacy claims trace to a single genre of rat studies (aqueous stem extract raising serum testosterone and mounting behavior dose-dependently). The same rodent work documented dose-dependent testicular toxicity (distorted seminiferous tubules) plus liver/kidney signals at higher doses. No human dose has ever been shown safe or effective.

THE VERDICT Skip. This is an animal-only supplement with an unresolved testicular-toxicity signal and no human data at all; there is no responsible dose to recommend.

BEST FORM No reliable standardised retail form exists, there is no agreed active-marker assay, so brand-to-brand potency of '600 mg' products differs enormously. If used at all, pick a brand with third-party identity/heavy-metal testing and lower doses, but form standardisation is not a solved problem here. (No validated standardisation marker exists, products vary wildly by extraction and ratio (e.g. '10:1'))

WATCH Real safety signal: rodent studies show dose-dependent TESTICULAR toxicity plus liver and kidney toxicity at a human-equivalent dose (~600 mg) close to what many products sell. No human safety or efficacy trials exist. Cycle conservatively or avoid, monitor liver/kidney function, do not stack with other hepatotoxic supplements/alcohol.

Sources: Yakubu et al. rodent studies (aqueous stem extract, testosterone & testicular toxicity); Examine.com research summary confirming absence of human trials

1 Horny goat weed (Epimedium / icariin) Amazon ✓ Website ✓

Evidence: animal/in-vitro only for the isolated herb; human data confounded by multi-herb blends | Effect: none demonstrated in humans as monotherapy (in-vitro PDE5 inhibition only) · **literature compromised (funding/outlier)** | Freely sold as a dietary supplement on US Amazon and vendors.

THE NOISE Sold as a natural Viagra/PDE5 alternative for erectile dysfunction and libido, and as a testosterone and bone tonic.

THE SCIENCE There is no adequate standalone human RCT showing that oral Epimedium or icariin treats ED. Mechanistically icariin does inhibit PDE5 in vitro, but it is roughly 80x less potent than sildenafil, and a human pharmacokinetic study found oral icariin blood levels very low or undetectable (poor bioavailability). The only 'positive' human data come from multi-herb formulas containing 12+ ingredients, which cannot be attributed to Epimedium.

THE VERDICT **Skip. The bioavailability and potency numbers make an oral effect implausible, and there is no credible monotherapy human trial. If you have ED, a PDE5 inhibitor (sildenafil/tadalafil) is the evidence-based option.**

BEST FORM Aerial-part extract with a stated icariin content (commonly 10-20%, some 40-60%). The active is icariin (a PDE5-inhibitor-like flavonoid), so the icariin % is what matters, unstandardised powder is near-useless. Verify Epimedium species and third-party testing. (Extract standardised to icariin % vs unstandardised 'horny goat weed' bulk powder)

WATCH Acts on the same PDE5 pathway as ED drugs, do NOT combine with sildenafil/tadalafil or nitrates (blood-pressure drop). Caution with antihypertensives and anticoagulants. Can cause tachycardia, dizziness. Adulteration with synthetic PDE5 inhibitors has been found in some sexual-enhancement products, buy reputable brands only.

Sources: Sage 1934578X19856789 (human icariin PK/safety RCT); ScienceDirect S1743609515329830 (icariin in vitro/in vivo, potency); Examine.com horny goat weed research

Women's / hormonal / urinary (8)

4 Cranberry (proanthocyanidins) - UTI prevention Amazon ✓ Website ✓

Evidence: moderate-strong: large 2023 Cochrane review with independent RCTs | Effect: recurrent UTI prevention (women with prior recurrent UTI, children, peri-procedural) | Widely sold as juice, capsules and PAC-standardized supplements; no restriction. Watch for products that don't disclose PAC content.

THE NOISE Sold as a natural way to 'flush out' and prevent urinary tract infections - pills, juice, and PAC-standardized capsules marketed to anyone prone to UTIs, and often pitched to elderly and pregnant women too.

THE SCIENCE The 2023 Cochrane review (Williams et al., 50 RCTs, 8,857 participants) found cranberry products cut the risk of recurrent UTI in otherwise-healthy women with a history of recurrent UTI (RR approx 0.74, 95% CI ~0.55-0.99, moderate certainty), and in children (RR approx 0.46), and in people made susceptible by an intervention (e.g. bladder procedures). It did NOT help elderly institutionalised adults, pregnant women, or those with neuropathic bladder/catheters. A 2024 dose-response meta-analysis found benefit only when standardized PAC intake was ≥ 36 mg/day (approx 18% risk reduction). This is a genuine, prevention-only effect that is tightly gated by population and by an adequate PAC dose - it does not treat an established infection.

THE VERDICT **Conditional BUY for women with recurrent UTIs or peri-procedural risk, using a product delivering ≥ 36 mg standardized proanthocyanidins/day; SKIP if you are elderly-institutionalised, pregnant, catheterised, or hoping to treat an active UTI (that needs antibiotics).**

BEST FORM Concentrated extract delivering ≥ 36 mg soluble A-type proanthocyanidins/day, quantified by the BL-DMAC method (the industry-standard assay). Sugary juice cocktails and 'cranberry 500 mg' with no PAC figure are unreliable, the A-type PAC dose is the whole point. (PAC-standardised extract (BL-DMAC assay, A-type PACs) vs cranberry 'juice cocktail' or unquantified whole-fruit powder)

WATCH Needs the ≥ 36 mg PAC/day threshold and 12+ weeks of consistent use to show benefit, sub-threshold doses show no effect. It's prevention, not treatment of active infection. High doses may raise oxalate (kidney-stone caution). Can mildly potentiate warfarin (monitor INR). Sugar load if using juice, prefer capsules for diabetics.

Sources: Cochrane CD001321.pub7 (2023), PMID 37068952; Front Nutr 2024 dose-response meta-analysis, PMID 39668896 / PMC11635990

3.5 Soy isoflavones (genistein/daidzein) Amazon ✓ Website ✓

Evidence: moderate: multiple independent RCT meta-analyses, modest and dose-dependent effect | Effect: vasomotor symptoms (hot-flash frequency and severity) | Sold freely as soy isoflavone / genistein supplements; no restriction.

THE NOISE Marketed as a natural, hormone-free estrogen alternative that melts away hot flashes and night sweats and protects bones - the plant-based answer to menopause.

THE SCIENCE Independent meta-analyses of RCTs show a real but modest effect: pooled reduction in hot-flash frequency of roughly 20% (approx 95% CI -28% to -13%) and severity approx 26% versus placebo, with onset over weeks-to-months (median dose ~54 mg aglycone equivalents, 6 weeks-12 months). Effect is dose- and composition-dependent: supplements delivering >18.8 mg genistein were about twice as effective, and benefit is larger in equol-producers. A 2024 meta-analysis found a small effect on overall menopausal symptoms (Hedges g approx -0.25). Real signal, smaller than hormone therapy, and it hinges on adequate genistein and slow onset.

THE VERDICT **Conditional BUY for menopausal women wanting a modest, non-hormonal option - use a genistein-rich extract (>=18-20 mg genistein, ~50 mg total isoflavones/day) and give it 12 weeks; SKIP if you need rapid or large relief (menopausal hormone therapy is far more effective) or have estrogen-sensitive cancer without oncology sign-off.**

BEST FORM Extract standardised to total isoflavones with genistein and daidzein content stated, ideally aglycone (unconjugated) forms which are better absorbed than glycosides. Some benefit depends on gut conversion of daidzein to equol, non-equol-producers respond less; S-equol supplements exist for them. (Standardised isoflavone extract (genistein+daidzein aglycones stated) vs glycoside forms; and whether you're an equol-producer)

WATCH Take with food. Modest phytoestrogen activity, discuss if you have oestrogen-sensitive breast cancer (data reassuring for dietary levels but caution with high-dose supplements). Can interfere with thyroid hormone absorption, separate from levothyroxine by several hours, and ensure adequate iodine. May reduce non-heme iron absorption. Effects on hot flushes are modest and slow (weeks).

Sources: Taku et al. Menopause 2012, PMID 22433977; Adv Nutr 2024 estrogenicity meta-analysis, PMC11784794; PeerJ 2024/2025 perimenopause meta-analysis, PMC12296567

3 Chasteberry (Vitex agnus-castus) - PMS Amazon ✓ Website ✓

Evidence: moderate but bias-compromised: multiple RCTs, high risk of bias and publication bias, manufacturer-linked | Effect: premenstrual syndrome symptom relief · **literature compromised (funding/outlier)** | Sold freely as chasteberry/Vitex supplements; no restriction.

THE NOISE Sold as a hormone-balancing 'women's herb' that fixes PMS, PMDD, irregular cycles, low progesterone, acne and even fertility - the go-to influencer supplement for anything cyclical.

THE SCIENCE For PMS specifically the signal is the strongest in this category: a meta-analysis of 14 double-blind RCTs found Vitex more effective than placebo, with most individual trials positive. But the authors flagged high risk of bias, high heterogeneity and likely publication bias, meaning the pooled effect is 'explorative' and probably overstates the truth. Much of the best data uses one standardized extract (Ze 440) from a single manufacturer, so the positive evidence is concentrated in industry-linked trials. Evidence for cycle regulation, fertility and menopause is weaker still.

THE VERDICT **Conditional/cautious BUY for PMS or cyclical breast tenderness using a standardized extract (e.g. Ze 440, ~20-40 mg/day) for 3 cycles; treat the effect size skeptically and SKIP for fertility, 'hormone balancing', or menopause claims where evidence is thin. Avoid with dopaminergic drugs and in pregnancy.**

BEST FORM Standardised fruit (berry) extract as used in trials, e.g. Ze 440 or BNO 1095, standardised to agnuside/casticin, ~20-40 mg/day. Unstandardised berry powder is inconsistent. Take once daily in the morning. (Standardised fruit extract (branded Ze 440 / BNO 1095, standardised to agnusides/casticin) vs unstandardised berry powder)

WATCH Acts on dopamine (D2) receptors and lowers prolactin, so it interacts with dopamine agonists/antagonists (antipsychotics, metoclopramide) and can affect fertility hormones. Avoid with hormonal contraceptives/HRT and in pregnancy. Can reduce effectiveness of/interact with dopaminergic Parkinson's meds. Effects build over 2-3 cycles. Generally well tolerated (mild GI/headache/acne).

Sources: Verkaik et al. Am J Obstet Gynecol 2017, PMID S0002-9378(17)30319-8; Csupor et al. Complement Ther Med 2019 meta-analysis (S0965229919306053); NCCIH Chasteberry monograph

2 Black cohosh (*Cimicifuga racemosa*) Amazon ✓ Website ✓

Evidence: weak/conflicting: null in independent pooled RCTs, positive mainly in manufacturer-linked extract trials | Effect: vasomotor symptoms - inconsistent / not demonstrated in independent pooled data · **literature compromised (funding/outlier)** | Sold freely; some jurisdictions require a liver-warning label on the product.

THE NOISE Marketed as the leading herbal fix for menopausal hot flashes, night sweats and mood - a natural, estrogen-free alternative to hormone therapy.

THE SCIENCE Evidence is genuinely mixed. A Cochrane/Leach systematic review of 16 RCTs (2,027 women) found no significant difference in hot-flash frequency versus placebo. The positive pooled analyses cluster around one proprietary isopropanolic extract (iCR / Remifemin, Schaper & Brummer) and are largely manufacturer-linked, so the 'it works' data is funding-concentrated while the independent data are null-to-weak. Mechanism (it is not meaningfully estrogenic) outruns consistent clinical outcome.

THE VERDICT **SKIP for most people - independent trials show no reliable hot-flash benefit and there is a liver-safety advisory. At best a cautious, time-limited trial of a standardized 40 mg extract if hormone therapy is off the table, stopping at any sign of liver trouble.**

BEST FORM Standardised rhizome/root extract as used in menopause trials, e.g. Remifemin, standardised to triterpene glycosides (often ~2.5%/dosed to ~1 mg per tablet as 27-deoxyactein). Adulteration with cheaper Asian *Actaea* species is common, buy DNA/identity-tested brands. (Standardised rhizome extract (branded Remifemin, standardised to triterpene glycosides) vs unstandardised or species-adulterated products)

WATCH Rare but documented reports of hepatotoxicity, stop and get liver tests if jaundice/dark urine/abdominal pain appear, avoid with existing liver disease and other hepatotoxic agents/alcohol. Not a phytoestrogen in the classic sense but avoid in oestrogen-sensitive cancers pending advice. Give it several weeks. Possible mild GI upset. Interactions with some chemotherapy/statins reported.

Sources: Leach & Moore, Cochrane CD007244 (2012); Newton et al. HALT trial Ann Intern Med 2006; EMA HMPC public statement on *Cimicifuga racemosa* rhizoma; LiverTox NBK547990

2 Red clover isoflavones (*Trifolium pratense*) Amazon ✓ Website ✓

Evidence: weak: Cochrane-level null, inconsistent smaller meta-analyses | Effect: vasomotor symptoms - not reliably demonstrated | Sold freely as red clover / Promensil isoflavone supplements; no restriction.

THE NOISE Sold (often as Promensil) as a potent phytoestrogen that relieves hot flashes and night sweats and supports bone and heart health in menopause.

THE SCIENCE The Cochrane phytoestrogen review (43 RCTs, 4,364 women) found no significant reduction in hot-flash frequency for Promensil versus placebo (MD -0.93/day, 95% CI -1.95 to +0.10). Smaller meta-analyses are inconsistent - some show a modest reduction, several show non-significant effects (e.g. MD -1.99, p=0.067). Net: the best controlled evidence does not support a reliable benefit; any effect is small and inconsistent.

THE VERDICT **SKIP - the Cochrane-level evidence shows no reliable hot-flash benefit. If you want a phytoestrogen, genistein-rich soy has better (still modest) data.**

BEST FORM Extract standardised to total isoflavones with the four isoflavones quantified (e.g. Promensil ~40-80 mg isoflavones/day), the studied form. Whole-herb/tea powders don't deliver a reliable isoflavone dose. (Standardised isoflavone extract (branded Promensil, stated biochanin A/formononetin/genistein/daidzein mg) vs unquantified whole-herb powder)

WATCH Phytoestrogenic, avoid in oestrogen-sensitive cancers and with tamoxifen/HRT without medical advice, and in pregnancy. Contains coumarin-type compounds with mild anticoagulant potential, caution with warfarin/antiplatelets and around surgery. May interact with CYP-metabolised drugs. Take with food. Benefits for hot flushes are modest and take weeks.

Sources: Lethaby et al. Cochrane CD001395 (phytoestrogens); Ghazanfarpour et al. meta-analysis PMID 26693407; Kanadys et al. 2021 clinical-meaningfulness meta-analysis, PMC8069620

1.5 Evening primrose oil (GLA) Amazon ✓ Website ✓

Evidence: weak-to-null: multiple negative RCTs and Cochrane reviews | Effect: none reliably demonstrated in humans for the marketed indications | Sold freely as EPO / GLA capsules; former UK eczema/mastalgia prescription licenses were withdrawn for lack of efficacy.

THE NOISE Promoted for hot flashes, PMS, breast pain (mastalgia), eczema and 'hormone balance' - a GLA-rich essential-fatty-acid cure-all for women.

THE SCIENCE Controlled trials are consistently null for the headline claims. For menopausal hot flashes a 6-month double-blind RCT (56 women, 2,000 mg twice daily) found no significant difference versus placebo. Cochrane concluded evening primrose oil is not effective for eczema (oral GLA no better than placebo, so a former license was withdrawn) and the PMS evidence is weak/negative. Mastalgia data are weak and largely superseded by better options. Mechanism (GLA to prostaglandins) has never translated into a reliable clinical outcome.

THE VERDICT **SKIP - the best controlled evidence shows no benefit for hot flashes, PMS or eczema. For cyclical breast pain, first-line measures and a proper clinical assessment beat EPO.**

BEST FORM Softgels stating actual GLA content (evening primrose oil is ~8-10% GLA; borage oil is a more concentrated GLA alternative at ~20%). Choose cold-pressed, antioxidant-protected (vitamin E), opaque packaging, fish/plant oils oxidise easily. Judge by GLA mg, not total oil mg. (GLA-standardised oil (state mg GLA, ~8-10% of oil) in dark/nitrogen-flushed softgels vs generic 'EPO 1000 mg' with no GLA figure or oxidised oil)

WATCH Take with food. Prone to rancidity, buy fresh, store cool/dark, discard if it smells off. Mild anticoagulant/antiplatelet effect, caution with warfarin/antiplatelets and stop before surgery. May lower seizure threshold, caution with epilepsy and with phenothiazine antipsychotics. Mild GI upset/headache. Evidence for eczema/PMS/mastalgia is mixed, GLA can take 6-12 weeks.

Sources: Bamford et al. Cochrane CD004416 (evening primrose/borage oil for eczema); Farzaneh et al. menopausal hot-flash RCT (Arch Gynecol Obstet); MHRA license-withdrawal record for Epogam/Efamast

1 DIM (diindolylmethane) Amazon ✓ Website ✓

Evidence: weak/preliminary: small RCTs on biomarkers, no clinical endpoints | Effect: urinary estrogen-metabolite ratio (surrogate only) - no demonstrated clinical outcome | Sold freely as DIM supplements; no restriction.

THE NOISE Sold as an 'estrogen detox' / hormone-balancer for acne, PMS, weight, menopause, estrogen dominance and cancer prevention - the influencer poster-child for 'metabolising bad estrogen'.

THE SCIENCE Human data stop at surrogate biomarkers. The best RCT (breast-cancer patients on tamoxifen, 150 mg DIM twice daily, >=12 months) shifted urinary estrogen metabolites toward a higher 2-OH:16-OH ratio and raised SHBG - but showed NO change in breast density and actually reduced tamoxifen active-metabolite (endoxifen) levels, a potential harm. No trial demonstrates a clinical outcome (acne cleared, symptoms improved, cancer prevented). It moves a lab ratio; it has never been shown to move how anyone feels or their disease.

THE VERDICT **SKIP - there is no human outcome evidence that DIM improves acne, PMS, menopause or cancer risk, and it can lower the active metabolite of tamoxifen. Anyone on tamoxifen or other hormone-sensitive therapy should specifically avoid it.**

BEST FORM A bioavailability-enhanced (microencapsulated/absorption-formulated) DIM such as BioResponse-DIM. Crystalline DIM is very poorly absorbed; I3C converts to DIM erratically in the stomach and also produces other metabolites, so standardized enhanced DIM is preferable to I3C. (Bioavailability-enhanced/microencapsulated DIM (e.g. BioResponse-DIM, DIM emulsified with vitamin E TPGS/phosphatidylcholine) vs plain crystalline DIM vs the precursor indole-3-carbinol (I3C).)

WATCH Take with food/fat to aid absorption. Commonly turns urine amber/orange (harmless dye effect); higher doses cause headache, nausea and occasionally a temporary drop in estrogen symptoms. It modulates estrogen metabolism and induces CYP enzymes, so avoid in pregnancy/breastfeeding and use caution with hormone therapy, tamoxifen or other CYP-metabolized drugs. Typical dose 100-200 mg/day of enhanced DIM; do not megadose.

Sources: Thomson et al. Cancer Epidemiol Biomarkers Prev 2017;26(3):435 (DIM + tamoxifen RCT); Dalessandri et al. pilot study, PMID 15623462

1 Dong quai (*Angelica sinensis*) Amazon ✓ Website ✓

Evidence: null: negative RCTs | Effect: none demonstrated in humans | Sold freely as dong quai supplements and in TCM blends; no restriction, but bleeding/photosensitivity cautions apply.

THE NOISE Branded 'female ginseng' - marketed for hot flashes, menstrual cramps, irregular cycles and overall 'women's hormone balance'.

THE SCIENCE Controlled evidence is negative. A double-blind placebo-controlled RCT for hot flashes found dong quai no better than placebo (in the androgen-deprivation population the dong quai arm even reported slightly more hot flashes). The earlier well-known menopause RCT (Hirata et al.) also found no benefit on hot flashes or vaginal cytology versus placebo. There is no credible human outcome evidence for any of the marketed indications, plus real safety concerns (photosensitising furocoumarins, anticoagulant potentiation).

THE VERDICT **SKIP - no controlled evidence of benefit for hot flashes or menstrual symptoms, and it interacts dangerously with warfarin/anticoagulants and can cause photosensitivity.**

BEST FORM Form/salt is not the issue - there is no single well-validated active or standardization marker. Buy a reputable extract from a brand that species-verifies the root (adulteration/substitution is common) and ideally states ferulic acid or ligustilide content; source quality matters more than 'form'. (Whole-root powder vs hydroalcoholic extracts loosely standardized to ferulic acid or ligustilide (no consensus marker).)

WATCH Contains coumarins and furocoumarins: real bleeding-risk interaction with warfarin/anticoagulants/antiplatelets, and photosensitivity (avoid strong sun/UV). Phytoestrogenic - avoid in estrogen-sensitive cancers (breast, uterine). Uterine-stimulant/emmenagogue: avoid in pregnancy. May potentiate other photosensitizing drugs.

Sources: Hirata et al. Fertil Steril 1997 (dong quai menopause RCT); ADT hot-flash RCT, Can Urol Assoc J, PMID 20165579; Memorial Sloan Kettering / NCCIH dong quai monographs

Metabolic / blood sugar / lipids / cardio (9)

4 Red yeast rice (monacolin K) Amazon ✓ Website ✓

Evidence: strong for the active (lovastatin), but retail product quality/dose is unreliable | Effect: LDL-cholesterol lowering (statin effect) · **literature compromised (funding/outlier)** | Sold on Amazon and vendor sites, but US products cannot legally disclose monacolin K content and many are deliberately near-zero to stay legal - so retail monacolin often falls BELOW the effective dose. EU capped at <3 mg/day. Citrinin contamination risk.

THE NOISE Sold as a 'natural,' gentler, statin-free way to lower cholesterol without a prescription.

THE SCIENCE The active ingredient monacolin K is chemically IDENTICAL to lovastatin, a prescription statin. It genuinely lowers LDL (roughly 15-25% at meaningful monacolin doses) - the efficacy is real precisely because it is a statin in disguise. That is also the problem: it carries statin risks (myopathy, rhabdomyolysis, hepatotoxicity) plus supplement-specific hazards - wildly variable monacolin content between brands (from ~0 to ~10+ mg), and possible citrinin (nephrotoxic mold toxin) contamination. The EU (2022) capped monacolins at <3 mg/day and mandated warnings, which pushes products toward sub-therapeutic dosing; in the US, FDA treats products with lovastatin-equivalent levels as unapproved drugs, so labels legally cannot state the monacolin dose - you don't know what you're taking.

THE VERDICT **Skip in favor of a prescribed statin - you get the same drug with a known dose, quality control and monitoring. Buying red yeast rice is self-prescribing an unregulated, variably-dosed, possibly contaminated statin. is_compromised: it is effectively a drug sold as a supplement.**

BEST FORM Buy a product that discloses/standardizes monacolin K content AND certifies it is citrinin-free (citrinin is a nephrotoxic fermentation contaminant). Potency varies wildly between brands, and US products often can't legally state monacolin content — a third-party COA is essential. Note the EU now caps monacolin K at <3 mg/serving. (Standardized, citrinin-tested monacolin K product vs unstandardized red yeast rice of unknown potency)

WATCH Monacolin K is chemically identical to lovastatin — same risks: myopathy/rhabdomyolysis, raised liver enzymes and CoQ10 depletion (consider co-supplementing CoQ10). Do NOT combine with a prescription statin or with fibrates. Avoid grapefruit and other CYP3A4 inhibitors. Contraindicated in pregnancy and liver disease. Take in the evening.

Sources: EFSA Journal 2018;16(8):5368 & 2025;23 (monacolin safety); EU Commission Reg 2022/860; China Coronary Secondary Prevention Study (Lancet/Xuezhikang); EU Reg 2023/915 (citrinin)

3.5 Plant sterols / phytosterols (and stanols) Amazon ✓ Website ✓

Evidence: strong for LDL surrogate (large consistent dose-response); none for hard CVD outcomes | Effect: LDL-cholesterol lowering (surrogate) | Sold as supplements on Amazon/vendor sites and in fortified foods (Benecol, Flora ProActiv). Retail servings (~2 g/day) match the effective dose; capsules sometimes under-dose vs food matrix.

THE NOISE Sold in supplements and fortified spreads/yogurt-drinks (Benecol, ProActiv) as clinically proven to lower cholesterol and 'support heart health.'

THE SCIENCE This is one of the best-evidenced items in the category for its SURROGATE endpoint: a large, consistent, dose-responsive body of RCTs shows ~2 g/day lowers LDL-C by roughly 6-12% (and adds ~8-9% on top of a statin), reliably clearing the LDL MCID. The honest caveats: (1) there are still NO randomized trials showing plant sterols reduce heart attacks, strokes or mortality - the outcome data simply do not exist; and (2) Mendelian/genetic studies link higher circulating phytosterol levels to atherosclerosis (the sitosterolemia signal), so some bodies (e.g., German Cardiac Society) withhold a blanket recommendation pending outcome trials. Strong on LDL, unproven on events.

THE VERDICT **Conditional buy - a reasonable, well-tolerated LDL-lowering adjunct at ~2 g/day (spreads/drinks or capsules with meals) for people with mild hypercholesterolemia or on top of a statin. Just don't assume LDL-lowering equals proven event reduction; it hasn't been tested. Avoid if you have sitosterolemia.**

BEST FORM Sterols and stanols lower LDL comparably, so isomer choice barely matters — what matters is delivery: they work best esterified and taken within a fat-containing meal (fortified spreads, yogurt drinks, or capsules taken with food), spread across meals. ~2 g/day is the effective plateau; more doesn't help. (Sterols vs stanols (near-equivalent LDL lowering); the real variable is free vs esterified and food-matrix vs pill delivery)

WATCH Must be taken WITH meals containing fat to work at all. They lower absorption of fat-soluble vitamins and especially carotenoids (beta-carotene) — eat plenty of colorful vegetables. They add to statin LDL-lowering. Contraindicated in sitosterolemia (rare genetic disorder). Modestly reduce absorption of fat-soluble drugs.

Sources: Ras et al. Br J Nutr 2014 dose-response MA (PMC4071994); Efficacy of phytosterols umbrella review 2026 (S0261561426001251); EFSA plant sterol health claim opinion

3 Berberine Amazon ✓ Website ✓

Evidence: moderate: many RCTs but small, largely single-region, quality-limited; consistent direction | Effect: fasting glucose / HbA1c / LDL & triglycerides (in dysglycemic populations) | Widely sold as a dietary supplement on Amazon and vendor sites; retail 500 mg capsules match the trial dose. Dihydroberberine sold as a higher-bioavailability variant.

THE NOISE Sold as 'Nature's Ozempic' - a natural GLP-1 / metabolic miracle for blood sugar, weight loss, insulin resistance and cholesterol. TikTok/influencer weight-loss framing dominates.

THE SCIENCE Genuine, reproducible drug-like glycemic effect in people WITH type 2 diabetes/prediabetes: pooled HbA1c ~-0.73% (95% CI -0.97 to -0.51), FPG ~-0.86 mmol/L (-1.10 to -0.62), plus real LDL/TG reductions. Clears the MCID for glycemia. BUT the evidence base is dominated by small, older, mostly Chinese trials with quality/publication-bias concerns; the 'Ozempic' weight-loss claim is not supported (weight change is minor and not GLP-1-mediated - it acts via AMPK). No large independent Western RCT and no cardiovascular-outcome data.

THE VERDICT **Conditional buy for people with prediabetes/type 2 diabetes or metabolic syndrome who cannot/will not use first-line drugs, at 500 mg 2-3x/day - not a weight-loss aid and not for a healthy adult. Skip the 'Nature's Ozempic' hype entirely.**

BEST FORM Dihydroberberine (roughly 5x the absorption, allowing lower doses) or a berberine phytosome is markedly better absorbed than ordinary berberine HCl, whose bioavailability is under 1%. If using plain HCl, split into 2-3 daily doses to compensate. (Berberine HCl (standard, <1% oral bioavailability) vs dihydroberberine vs berberine phytosome (BervoElite))

WATCH A potent inhibitor of CYP3A4, CYP2D6 and P-glycoprotein — major interaction potential with statins, cyclosporine and many drugs, and it stacks with metformin/other hypoglycemics to risk low blood sugar. Take with meals; GI effects (cramping, diarrhea, constipation) are common. Avoid in pregnancy/breastfeeding and in neonates (displaces bilirubin — kernicterus risk).

Sources: PMID 38016844 (Clin Ther 2023 umbrella MA); PMID 34956436 (metabolic profile MA); Cardiology Discovery 2024 double-blind MA (doi:10.1097/CD9.000000000000087)

2.5 Alpha-lipoic acid (ALA) Amazon ✓ Website ✓

Evidence: weak/preliminary for oral (Cochrane 2024: little/no effect at 6 mo); moderate short-term for IV | Effect: diabetic neuropathy symptom relief (modest); minimal glucose effect | Sold freely as a supplement (typically 300-600 mg) on Amazon and vendor sites; retail dose matches oral trial dose.

THE NOISE Marketed as a universal antioxidant for blood-sugar control, insulin sensitivity, weight loss, 'nerve health' and anti-aging.

THE SCIENCE The only reasonably supported use is symptomatic diabetic peripheral neuropathy, and even there the signal is soft: IV ALA (600 mg) trials (ALADIN, SYDNEY) showed short-term symptom relief, but the 2024 Cochrane review concluded ORAL ALA probably has little or no effect on neuropathy symptoms or impairment at 6 months. Effects on glucose/HbA1c and weight are small and inconsistent. There is essentially no benefit signal for a healthy adult - benefit is gated to diabetic-neuropathy patients.

THE VERDICT Skip for general 'metabolic support' or weight loss. Only conditional - as an adjunct for diabetic neuropathy symptoms at 600 mg/day, and even then oral evidence is weak and IV is where the effect lives.

BEST FORM R-ALA is the biologically active isomer; a stabilized R-lipoate salt (sodium R-lipoate) is best absorbed and most potent per mg. Plain R-ALA free acid is heat/stomach-unstable and can polymerize, so stabilized salt forms are preferable; racemic ALA works but delivers half inactive S-isomer. (Racemic ALA (R+S, ~50/50) vs stabilized R-ALA (sodium R-lipoate / Na-RALA))

WATCH Take on an empty stomach (30-60 min before food) — food substantially cuts absorption. Can lower blood glucose (additive with diabetes drugs — hypoglycemia risk) and chelates metals/minerals, so separate from iron, calcium, magnesium. May deplete biotin with chronic use. Rare autoimmune insulin syndrome (hypoglycemia) reported, mainly in HLA-predisposed (often East Asian) individuals.

Sources: Cochrane CD012967.pub2 (Baicus 2024, PMID 38205823); Nutrients 2023;15:3634 neuropathy MA; Diabetes Care 2011;34:e146 (IAS case); PMID 30086435 (ALA-IAS case series)

2.5 Chromium picolinate Amazon ✓ Website ✓

Evidence: weak: low/very-low certainty, quality- and deficiency-dependent | Effect: fasting glucose (small, low-certainty, in T2D only) - **literature compromised (funding/outlier)** | Ubiquitous cheap supplement (typically 200-1000 mcg) on Amazon and everywhere; retail doses meet or exceed trial doses.

THE NOISE Sold as a blood-sugar stabilizer, insulin-sensitizer, carb-craving crusher and weight-loss/fat-burner aid.

THE SCIENCE Meta-analyses show at best small, low-certainty reductions in fasting glucose and HbA1c in type 2 diabetics, and the effect is inconsistent and likely concentrated in lower-quality studies and/or people who are chromium-deficient (rare). Weight-loss effect is trivial (~0.5-1 kg, not clinically meaningful). The American Diabetes Association (2024) explicitly states there is insufficient evidence to recommend chromium for glycemia. No credible benefit in a healthy, replete adult.

THE VERDICT Skip. Weak, low-certainty glycemic signal that guidelines reject; the craving/weight-loss claims are essentially marketing. Not worth buying.

BEST FORM Chromium picolinate (or nicotinate) is the well-absorbed retail form used in most trials; avoid inorganic chromium chloride, which is very poorly absorbed. Typical doses 200-1000 mcg/day. (Chromium picolinate / chromium nicotinate (well absorbed) vs chromium chloride (poorly absorbed))

WATCH Real-world benefits on glucose/weight are modest to marginal. Take with food. Can add to hypoglycemic drugs; separate from calcium/antacids and high-dose vitamin C (which alter absorption). Rare renal/hepatic case reports at very high chronic doses. Supplements are trivalent chromium — not the toxic hexavalent form.

Sources: ADA Standards of Care 2024 (micronutrients); Asbaghi et al. Pharmacol Res 2020 (S1043661820314067); Linus Pauling Institute chromium review

2 Bergamot (Citrus bergamia) Amazon ✓ Website ✓

Evidence: weak/moderate: multiple small RCTs, industry-concentrated, no outcome data | Effect: LDL / total cholesterol / triglyceride lowering (short-term, small trials) · **literature compromised (funding/outlier)** | Sold as citrus bergamot capsules on Amazon and vendor sites; retail standardized extracts roughly match trial doses.

THE NOISE Marketed as a natural statin alternative that lowers LDL and triglycerides, raises HDL and 'supports healthy cholesterol' via polyphenols.

THE SCIENCE Small RCTs report LDL reductions of roughly 8-30% and total-cholesterol/triglyceride improvements over 30-120 days - superficially impressive. But the evidence base is dominated by small, short, mostly Italian trials, several tied to the producers of specific standardized extracts (BPF / Bergamet / branded phytocomplexes), with heterogeneous formulations and doses. Independent, large, long-duration replication is thin, and there are no cardiovascular-outcome trials. Effect size is real in these trials but the industry concentration and small n keep confidence low.

THE VERDICT **Skip as a primary lipid therapy; at most a conditional experiment (500-1000 mg standardized extract or ~150 mg flavonoids) for someone with mild hypercholesterolemia who declines a statin - understanding the evidence is small, short and largely industry-linked.**

BEST FORM Choose a standardized bergamot polyphenolic fraction (BPF) branded extract (Bergavit 40, Bergamonte, BergaMet) standardized to defined flavonoids (brutieridin/melitidin, naringin, neohesperidin). Generic 'bergamot' or bergamot essential oil is not the studied cardiometabolic material. (Standardized bergamot polyphenolic fraction (BPF) — Bergavit, Bergamonte, BergaMet — vs generic non-standardized bergamot or essential oil)

WATCH Take with meals; typical studied dose ~500-1000 mg BPF/day. Because it acts on the HMG-CoA pathway it can be additive with statins (both benefit and myopathy potential) — coordinate with a clinician. Bergamot polyphenols/furanocoumarins can inhibit CYP3A4 (grapefruit-like drug interactions). Mild GI effects and heartburn possible.

Sources: Foods 2024;13(23):3883 (PMC11641072); Arch Med Sci bergamot phytocomplex 3-arm RCT; Nutrients artichoke+bergamot phytosome RCT (PMC8746931)

2 Nattokinase Amazon ✓ Website ✓

Evidence: weak/preliminary: few small RCTs, no outcome data | Effect: small blood-pressure reduction; fibrinolysis unproven in vivo | Sold as capsules (measured in FU - fibrinolytic units, e.g., 2000 FU) on Amazon and vendor sites; retail doses roughly match trial doses.

THE NOISE Sold as a natural blood-thinner/'clot-buster' and fibrinolytic that cleans arteries, lowers blood pressure and prevents heart attack and stroke.

THE SCIENCE The fibrinolytic 'clot-dissolving' story is almost entirely in-vitro/mechanistic; oral enzyme survival and systemic activity are questionable. The only reproducible human signal is a modest blood-pressure reduction: a 2023 meta-analysis of 6 small RCTs (n=546, some Asian/industry-linked) found SBP -3.45 mmHg and DBP -2.32 mmHg - real but small, below the impact of any standard antihypertensive. Crucially, no trial has shown reduced heart attack, stroke, clot events or mortality, and one signal suggested unfavorable effects on cholesterol/HDL at low doses. Thin overall.

THE VERDICT **Skip. The evidence is a small blood-pressure nudge from a handful of small trials and zero hard-outcome data; the 'natural clot-buster' claim is not supported. If blood pressure is the goal, proven lifestyle/drug options dominate.**

BEST FORM One enzyme, so buy on standardized activity (aim ~2000 FU/dose) from a reputable brand with good stability/handling, rather than by weight. Check whether the product still contains vitamin K2 (some are K2-depleted) if that interaction matters to you. (Activity graded in fibrinolytic units (FU), not mg; that plus vitamin K2 content is the real variable, not molecular form)

WATCH Has fibrinolytic/antiplatelet activity — real bleeding-risk interaction with warfarin, DOACs, aspirin and other anticoagulants/antiplatelets; stop before surgery. As a protein enzyme it is best taken on an empty stomach. If it contains vitamin K2 it can antagonize warfarin. Human efficacy data are limited; avoid in pregnancy and bleeding disorders.

Sources: Rev Cardiovasc Med 2023;24(8):234 (PMID 39076715, PMC11266782)

2 Niacin (immediate-release, for lipids) Amazon ✓ Website ✓

Evidence: strong outcome evidence of NO benefit + harm (two large RCTs) | Effect: lipid biomarkers (HDL up, TG/Lp(a) down) - but null/harmful on CV outcomes · **literature compromised (funding/outlier)** | IR/'flush' niacin sold freely as a supplement on Amazon and vendor sites, but typical supplement doses are far below the ~1.5-3 g/day lipid dose - and self-dosing to therapeutic levels is effectively unsupervised drug use.

THE NOISE Sold as a natural, cheap cholesterol fixer that raises 'good' HDL, lowers triglycerides and Lp(a) - the old-school heart vitamin.

THE SCIENCE Niacin does dramatically move lipid BIOMARKERS (raises HDL, lowers TG and Lp(a)) - but the hard-outcome evidence is the cautionary tale of cardiology. On top of a statin, AIM-HIGH (n~3,400) and HPS2-THRIVE (n~25,673) found NO reduction in cardiovascular events, and HPS2-THRIVE found significant HARMS: increased diabetes, infections, bleeding, GI and musculoskeletal events. Add flushing and hepatotoxicity (worse with sustained-release). A textbook case of a significant surrogate effect that is clinically null-to-harmful.

THE VERDICT Skip for self-directed cholesterol management. High-dose niacin is a drug with a proven no-benefit/net-harm outcome record on top of statins; if Lp(a) or specific lipid issues are in play, that's a physician decision, not a supplement-aisle one.

BEST FORM For lipids use immediate-release nicotinic acid (niacin). Do NOT use 'no-flush' inositol hexanicotinate (releases little free niacin, no lipid effect) and avoid OTC sustained/timed-release niacin (greater hepatotoxicity). Nicotinamide/niacinamide does nothing for lipids. (Immediate-release nicotinic acid vs sustained/extended-release niacin vs 'no-flush' inositol hexanicotinate vs nicotinamide)

WATCH Expect cutaneous flushing — dose with food, start low and titrate, and 325 mg aspirin ~30 min before blunts it. Hepatotoxicity (worse with SR forms), raised blood glucose and uric acid (caution in diabetes/gout), and it potentiates statin myopathy. Note: large outcome trials (AIM-HIGH, HPS2-THRIVE) showed no cardiovascular benefit added to statins, so lipid niacin has fallen out of favor — use only with clinician oversight.

Sources: AIM-HIGH, NEJM 2011;365:2255; HPS2-THRIVE, NEJM 2014;371:203 (NEJMoa1300955); HPS2-THRIVE design, Eur Heart J 2013;34:1279

1 Policosanol Amazon ✓ Website ✓

Evidence: none (independent trials null; positive data are industry-only) | Effect: none demonstrated in independent human trials · **literature compromised (funding/outlier)** | Sold cheaply as capsules (typically 10-20 mg) on Amazon and vendor sites; retail dose matches (irrelevant, since even higher doses were null independently).

THE NOISE Sold as a natural cholesterol-lowering agent (sugarcane wax alcohols) claimed to rival statins for LDL reduction without side effects.

THE SCIENCE The poster child for funding-source bias. Roughly 80+ Cuban trials from a single patent-holding group reported large LDL reductions (~20-30%) rivaling statins - and virtually every one was sponsored by the Cuban commercial enterprise that markets it. Independent, non-Cuban RCTs uniformly FAILED to replicate any cholesterol benefit: a rigorous German RCT (Berthold, JAMA 2006) tested up to 80 mg/day and found policosanol no better than placebo. Even studies using the authentic Cuban product at up to 8x dose saw LDL drops matched by placebo. No credible independent evidence of any lipid effect.

THE VERDICT Skip - do not buy. This is marketing built on non-replicable, single-source industry trials; independent science shows it does nothing to cholesterol.

BEST FORM Only the original Cuban sugarcane-wax material showed lipid effects, and even that has repeatedly failed to replicate in independent non-Cuban trials — so no form is reliably effective. If tried, use the sugarcane-derived product but keep expectations low. (Original Cuban sugarcane-derived policosanol (Cuban trials) vs beeswax/other-source policosanol)

WATCH Independent replication is largely negative for LDL lowering, so it is a weak/likely-ineffective lipid agent. It does have mild antiplatelet activity — additive bleeding risk with aspirin/anticoagulants. Generally well tolerated (occasional insomnia, GI upset). Do not rely on it in place of proven lipid therapy.

Sources: Berthold et al. JAMA 2006;295:2262; Marinangeli/independent replication reviews; AJCN 2006 'lack of efficacy of Cuban policosanol' (S0002916523291431)

2.5 Glucomannan (konjac fibre) Amazon ✓ Website ✓

Evidence: moderate for LDL; weak/mixed for weight | Effect: Satiety / appetite; modest LDL reduction. Weight effect marginal

THE NOISE Sold as a natural appetite suppressant that 'expands in your stomach' to make you eat less; label claims dramatic weight loss and blood-sugar/cholesterol benefits.

THE SCIENCE A viscous soluble fibre with a plausible satiety mechanism, but weight-loss trials are underwhelming when done well. Onakpoya 2014 meta-analysis (JACN) concluded glucomannan does NOT produce statistically significant weight loss vs placebo in adults. A 2020 meta-analysis similarly found only small/non-robust effects. EFSA does grant a qualified claim that 3 g/day (in 1 g doses before meals with water) contributes to weight loss within an energy-restricted diet - the strongest formal endorsement, but the independent RCT weight effect is marginal. It does reliably lower LDL cholesterol (AJCN 2023 meta-analysis), which is its better-supported benefit.

THE VERDICT Conditional/skip for weight loss. If used, take 3 g/day split before meals with a large glass of water as a satiety aid alongside a calorie deficit - but don't expect the scale to move much. Its cleaner win is modest LDL lowering.

BEST FORM High-viscosity pure konjac glucomannan (the EFSA weight-management claim requires 3 g/day in 3 doses of 1 g). Powder stirred into water works; if using capsules ensure adequate dose. (High-purity, high-molecular-weight/high-viscosity konjac glucomannan powder or capsules vs low-viscosity or blended 'fibre complex' products; powder vs pressed tablets.)

WATCH MUST be taken with a large glass of water (250 ml) 15-30 min before meals - it swells massively. NEVER swallow the dry tablet/powder without ample fluid: documented choking and oesophageal/GI obstruction, especially pressed tablets. Can reduce absorption of oral medications and other nutrients - separate dosing by 1-2 h. Causes bloating/flatulence. Avoid if swallowing difficulty or GI strictures.

Sources: Onakpoya et al. J Am Coll Nutr 2014;33(1):70 (PMID 24533610); Sood et al. Am J Clin Nutr 2008 (KGM meta-analysis); AJCN 2023 konjac glucomannan LDL meta-analysis (S0002-9165(22)04891-2); EFSA NDA Panel opinion glucomannan 2010

2.5 White kidney bean extract (Phaseolus vulgaris, alpha-amylase inhibitor / 'carb blocker') Amazon ✓

Website ✓

Evidence: moderate but product-specific; industry-linked | Effect: Reduced starch digestion / modest fat and weight reduction (product-specific)

THE NOISE Sold as a 'carb blocker' / 'starch neutralizer' that stops you absorbing the carbs you eat by inhibiting alpha-amylase, so you can eat pasta and still lose weight.

THE SCIENCE The mechanism (alpha-amylase inhibition delaying starch digestion) is real and there IS a modest human signal, but it hinges on one specific standardized product. A meta-analysis (11 studies, 573 subjects) found -1.08 kg body weight and -3.26 kg body fat - almost entirely from the proprietary aqueous extract Phase2 standardized to ≥ 3000 amylase-inhibiting units/g. Generic or under-standardized extracts show little. Effect on body weight is below the MCID; the fat-mass figure is more interesting but rests on a manufacturer-standardized product with industry involvement. A newer 2026 review (8 RCTs, 543 participants) confirms modest weight/adiposity reduction with acceptable safety.

THE VERDICT Conditional. If you want to try it, only the standardized Phase2-type extract (~1000-1500 mg with starchy meals) has data, and even then expect a small effect that requires eating a carb-containing diet. Mild GI side effects. Not a substitute for a calorie deficit.

BEST FORM A standardised Phase 2 (Phaseolamin) extract taken immediately before starch-containing meals; typical studied dose ~445 mg-1 g before meals. (Standardised, activity-assured extract (e.g. Phase 2 / Phaseolamin, dosed by amylase-inhibiting units) vs generic 'white kidney bean' powder with unverified inhibitory activity.)

WATCH Only affects starch (not sugar or fat) and only works when taken right before a starchy meal - useless taken alone. Undigested starch reaching the colon causes gas, bloating, cramping and diarrhoea. Heat/moisture degrades activity. Weight-loss effect is small. Raw bean products must be properly processed (raw phytohaemagglutinin is toxic).

Sources: Nolan et al. Nutrients 2020 - Phaseolus meta-analysis (PMC5920428); 2024 12-wk RCT (PMC11148185); 2026 systematic review (S0271531726000400)

2 Apple cider vinegar (ACV) Amazon ✓ Website ✓

Evidence: weak/mixed: small RCTs, key trial retracted | Effect: Modest post-meal glycemic blunting / satiety; weight effect small and uncertain - **literature compromised (funding/outlier)**

THE NOISE The wellness-influencer staple: a 'tonic' or gummy claimed to melt belly fat, curb appetite, detox, and fix blood sugar - taken as a shot or gummy before meals.

THE SCIENCE A plausible glycemic/satiety mechanism (acetic acid slows gastric emptying, blunts post-meal glucose) with genuinely modest human data - but the eye-catching numbers are unreliable. A 2025 meta-analysis (10 RCTs, 789 participants) reported large weight loss (~7.5 kg / -2.0 kg/m²), BUT this was heavily leveraged by a single 2024 Lebanese trial (Abou-Khalil) that was **RETRACTED** in Sept 2025 for implausible data and poor analysis. Sensitivity analyses excluding low-quality trials still show a small benefit, but no large high-quality RCT supports dramatic weight loss. The reproducible effect is modest post-meal glucose blunting, not fat loss.

THE VERDICT **Skip for weight loss (conditional at best). ~1-2 tbsp (15-30 mL) diluted before a carb meal may slightly blunt glucose and appetite, but the big weight-loss claims rest on a retracted study. Cheap and low-risk if you like it - just don't expect the scale results the internet promises. Dilute to protect tooth enamel.**

BEST FORM Liquid ACV (~1-2 tablespoons = ~750-1500 mg acetic acid) diluted in water, taken with or just before a carb-containing meal. Prefer liquid over gummies for dose certainty. (Liquid ACV containing acetic acid (the active) 'with the mother' vs ACV capsules/gummies, which often contain little real acetic acid and are unreliably dosed.)

WATCH Always dilute - undiluted vinegar erodes tooth enamel and can burn the oesophagus; rinse mouth after. Can slow gastric emptying (worsens gastroparesis) and may lower potassium/interact with diuretics, insulin and glucose-lowering drugs. Capsules risk oesophageal injury if lodged. Effects on weight are modest at best.

Sources: 2025 ACV body-composition meta-analysis (PMC12472926); Abou-Khalil et al. BMJ Nutr Prev Health 2024 - **RETRACTED** 2025 (Healio/Retraction coverage); Khezri et al. J Funct Foods 2018; T2DM glycemic meta-analysis PMC11821484

2 Capsaicin / Capsimax (capsaicinoids, capsinoids) Amazon ✓ Website ✓

Evidence: weak-moderate: RCTs mostly null for weight, small subgroup/fat signal | Effect: Acute thermogenesis / fat oxidation and mild appetite suppression; small % body-fat effect

THE NOISE Sold as a thermogenic 'metabolism booster' (Capsimax caps) that burns extra calories, raises fat oxidation and suppresses appetite via TRPV1 activation.

THE SCIENCE Capsaicinoids do measurably raise energy expenditure and fat oxidation acutely and can modestly blunt appetite - the mechanism is real. But translation to weight loss is weak. A dose-response meta-analysis of capsinoids found **NO** significant effect on body weight, BMI, waist, fat mass or % body fat overall; only a subgroup of overweight subjects in trials ≥12 weeks showed a small weight reduction. A separate capsaicinoid meta-analysis found small decreases in % body fat/fat mass after covariate adjustment. It also modestly improves lipids (lower triglycerides/total cholesterol). Real thermogenic effect, marginal scale effect.

THE VERDICT **Skip-to-conditional. As a standalone weight-loss pill the effect is small and inconsistent; a modest appetite/thermogenic nudge at best in overweight users over 12+ weeks. Dietary chili/capsaicin is fine; don't expect a supplement to move the needle. GI irritation at higher doses.**

BEST FORM Capsimax controlled-release capsaicinoids (~2 mg capsaicinoids per typical serving) - the encapsulation is the whole point: same thermogenic exposure without gastric burning. (Beadlet/controlled-release capsaicinoid (Capsimax) that releases in the intestine vs raw cayenne powder or uncoated capsaicin that irritates the stomach.)

WATCH Uncoated/high-dose capsaicin causes heartburn, gastric pain and GI upset; the coated form avoids this so don't chew/open the capsule. Take with food. May increase absorption/effect of some drugs and can add to bleeding risk at high doses. Thermogenic/appetite effects are real but small.

Sources: Capsinoids dose-response meta-analysis, Clin Nutr ESPEN 2022 (S2405457722004843); Whiting et al. capsaicin weight-loss meta-analysis, Br J Nutr; Capsaicinoids body-fat post-hoc analysis (PMC6088424); Capsaicin lipid meta-analysis metabolic syndrome (PMC8923259)

2 Conjugated linoleic acid (CLA) Amazon ✓ Website ✓

Evidence: moderate: multiple RCTs, effect real but below MCID | Effect: Small fat-mass reduction (not body-weight)

THE NOISE Sold as a 'body recomposition' and belly-fat burner that shifts fat to lean mass by modulating lipid metabolism and PPAR pathways; popular in gym/bodybuilding marketing.

THE SCIENCE Multiple meta-analyses show a real but tiny effect that sits below clinical relevance. Whigham 2007 (humans): ~0.09 kg/week fat loss, i.e. modest, at 3.2 g/day. Pooled body-weight effect ~-0.7 kg; body-fat ~-0.7 to -1.2 kg over 6-16 weeks. A 2023 review found CLA + exercise lowers body fat and insulin resistance but does NOT reduce body weight. Effects are largest in women/postmenopausal subgroups. All below the MCID and offset by GI side effects and signals of raised inflammatory markers / insulin resistance at higher intakes.

THE VERDICT Skip for weight loss. The fat-mass change is real but too small to notice on the scale, and higher doses carry metabolic downsides. Diet, resistance training and protein do the 'recomposition' job better.

BEST FORM A standardised ~80% CLA softgel (Tonalin or Clarinol) delivering ~3-3.4 g/day of the two active isomers, taken with meals containing fat. (Bioassay-grade 80% CLA free-fatty-acid/triglyceride softgels standardised to the active c9,t11 + t10,c12 isomers (e.g. Tonalin/Clarinol) vs cheap 'CLA' or generic safflower oil with unspecified isomer content.)

WATCH Fat-soluble, so take with food for absorption. Effect on body fat is small and slow (months). The t10,c12 isomer can worsen insulin sensitivity and lower HDL and has been linked to fatty-liver/inflammation signals in some trials - not for people with metabolic syndrome or hepatic issues. Common GI upset (loose stools, nausea). Doses above ~3.4 g/day give no added benefit.

Sources: Whigham et al. Am J Clin Nutr 2007 (fat-mass meta-analysis); Gaullier et al. Am J Clin Nutr 2004 (PMID 15159244); Nutrition Reviews 2023 CLA+exercise meta-analysis (81:397); BJN dose-response meta-analysis

2 Green coffee bean extract (chlorogenic acid, GCA) Amazon ✓ Website ✓

Evidence: weak/preliminary; foundational evidence retracted | Effect: Possible small body-weight change; not reliably demonstrated · **literature compromised (funding/outlier)**

THE NOISE Marketed as a 'miracle' fat-burner that blocks glucose absorption and burns fat via chlorogenic acid; Dr Oz called it 'magic' and 'unprecedented', driving a global diet craze.

THE SCIENCE The evidence base is contaminated at the root. The foundational 2012 Vinson/Applied Food Sciences trial (the '18 lb in 22 weeks' study) was RETRACTED in 2014, and the FTC fined the sponsor \$3.5M for fabricated data. Recent independent meta-analyses are tiny and fragile: one 2023 analysis found -1.30 kg from just 3 RCTs / 103 people at >=500 mg/day CGA. Earlier reviews noted high risk of bias, extreme heterogeneity and industry funding of most positive trials. Any real effect is small and unproven at scale.

THE VERDICT Skip. This is the textbook case of hype built on retracted, industry-funded data; independent trials are too small and short to trust. Not worth buying.

BEST FORM Pick an extract standardized to chlorogenic acid content (Svetol/GCA type, ~45-50% CGA) with a stated (ideally low) caffeine level, rather than a vague 'green coffee' powder. Even then, treat efficacy claims skeptically. (Standardized chlorogenic acid extract (e.g. Svetol/GCA, ~45-50% CGA, low caffeine) vs generic low-CGA extract)

WATCH The headline weight-loss trial (Vinson) was retracted and the evidence base is weak. Watch caffeine content (insomnia, jitters, raised heart rate/BP). Chlorogenic acid can modestly lower blood pressure and glucose (additive with medications) and, like coffee, may raise homocysteine at high intake. Take with food; caution with anxiety, arrhythmia or pregnancy.

Sources: Kanchanasurakit et al. Syst Rev 2023;12:163 (PMID 37710316); Vinson et al. 2012 - RETRACTED 2014; FTC press release Sept 2014; Onakpoya green coffee meta-analysis Gastroenterol Res Pract 2011

1.5 Forskolin (*Coleus forskohlii*) Amazon ✓ Website ✓

Evidence: weak/preliminary: small RCTs, mostly null | Effect: None reliably demonstrated for weight; possible minor body-composition shift in men (unreplicated)

THE NOISE Marketed as 'nature's lipolysis switch' that raises cAMP to melt fat and preserve lean mass; another Dr Oz-hyped 'belly blaster'.

THE SCIENCE Human evidence is thin and mostly null for weight loss. The most-cited trial (Godard 2005, n=30 men) showed changes in body composition/testosterone but no significant body-weight loss. A 2015 double-blind RCT (n=30, 250 mg 2x/day, 12 wk) found waist/hip and lipid changes in BOTH the forskolin and placebo groups - i.e. the diet, not the supplement, drove results. A small open-label trial showed weight and fat drop but also lost lean mass (opposite of the marketed mechanism) and wasn't blinded. No adequately powered independent RCT supports weight loss.

THE VERDICT Skip. Tiny, mostly null or unblinded trials; the one decent RCT showed placebo did just as well. Not worth buying.

BEST FORM A root extract standardised to 10% forskolin (ForsLean), ~250 mg twice daily is the studied regimen - though weight-loss evidence is weak. (Root extract standardised to 10% forskolin (e.g. ForsLean) vs unstandardised whole-herb powder or 20%+ products of dubious labelling.)

WATCH Forskolin is a vasodilator and lowers blood pressure - avoid with antihypertensives, and it can potentiate bleeding (avoid with anticoagulants/antiplatelets and before surgery). Contraindicated with nitrates and with PDE5 inhibitors. Human fat-loss data are minimal; body-composition claims are overstated.

Sources: Godard et al. *Obes Res* 2005;13:1335 (PMID 16129715); Loftus et al. 2015 RCT metabolic syndrome (PMC4663611); Henderson open-label body-composition trial (S2186495013000059)

1.5 Garcinia cambogia (hydroxycitric acid, HCA) Amazon ✓ Website ✓

Evidence: weak: multiple small short RCTs, effect below MCID | Effect: Marginal, sub-clinical body-weight change; effectively none demonstrated

THE NOISE Sold as a 'fat blocker' and 'appetite suppressant' that stops your body making fat by inhibiting ATP-citrate lyase; label and influencer claims promise effortless weight loss without diet change, popularised by Dr Oz as a 'revolutionary fat buster'.

THE SCIENCE Pooled RCTs show a trivial, borderline-null effect: the classic Onakpoya meta-analysis (9 RCTs) gave a mean difference of -0.88 kg (95% CI -1.75 to -0.00) vs placebo over 2-12 weeks. That upper CI bound literally touches zero, and the point estimate is far below the ~2.3 kg / 5% body-weight MCID. Later 2020 dose-response meta-analyses (8 trials, ~530 subjects) found only small changes in obesity indices. In practice indistinguishable from placebo.

THE VERDICT Skip. The effect is below the minimum clinically important difference, is statistically fragile, and there are documented idiosyncratic hepatotoxicity case reports (and the Hydroxycut recall). No healthy adult should buy this for weight loss.

BEST FORM If used at all, choose an extract standardized to ~50-60% HCA as a calcium-POTASSIUM (or magnesium-potassium) salt (Super CitriMax type) — calcium-only salts are poorly soluble and absorbed. Evidence for meaningful weight loss is weak regardless. (Calcium-potassium HCA salt (e.g. Super CitriMax, ~60% HCA) vs plain calcium-HCA salt (poorly soluble/absorbed))

WATCH Documented cases of hepatotoxicity/acute liver failure (Garcinia/HCA is a recurring signal in supplement-induced liver injury, including some Hydroxycut formulations) — stop immediately for any jaundice/dark urine. Can cause serotonergic effects (caution with SSRIs). Take before meals on an empty stomach. May lower blood sugar. Avoid in pregnancy and liver disease.

Sources: Onakpoya et al. *J Obes* 2011 (PMID 21197150); Heymsfield et al. *JAMA* 1998 (PMID 9820262); Golzarand 2020 dose-response meta-analysis (PMID 32951714); NCCIH Garcinia cambogia review

1.5 Synephrine (bitter orange / Citrus aurantium, p-synephrine) Amazon ✓ Website ✓

Evidence: moderate evidence of NO weight benefit; safety signal present | Effect: Acute thermogenesis / mild BP+HR rise; no demonstrated weight loss

THE NOISE Sold as the 'ephedra replacement' thermogenic that boosts metabolism, energy and fat-burning, especially stacked with caffeine in pre-workout and fat-burner blends.

THE SCIENCE A 2022 systematic review/meta-analysis (18 RCTs) found weight loss with synephrine was NON-SIGNIFICANT after prolonged treatment - no evidence it facilitates weight loss. What it does do is raise blood pressure with prolonged use (systolic +6.4 mmHg, diastolic +4.3 mmHg) and increase heart rate, especially combined with caffeine. Acute thermogenesis is measurable but does not translate into meaningful fat loss.

THE VERDICT Skip. It fails on efficacy and carries a cardiovascular risk signal, magnified by the caffeine it's almost always stacked with. Avoid.

BEST FORM A single-ingredient extract standardised to p-synephrine content, isomer specified, at a modest dose (~10-20 mg). Avoid stimulant stacks. (Standardised p-synephrine extract with the isomer stated (natural bitter orange is p-synephrine, NOT the more cardioactive m-synephrine/ephedrine analogues) vs unspecified 'bitter orange' blends stacked with caffeine.)

WATCH Sympathomimetic - can raise blood pressure and heart rate, especially combined with caffeine (which most fat-burners do) or other stimulants; cardiovascular events reported. Avoid with hypertension, arrhythmia, or MAOI/stimulant medication. Inhibits CYP3A4/CYP2D6 - drug interactions. Banned/restricted in some sport contexts; check anti-doping status.

Sources: Nutrients 2022;14(19):4019 - Citrus aurantium/p-synephrine meta-analysis (PMC9572433); Stohs et al. Int J Med Sci 2011; Nasir et al. variant angina case, Mayo Clin Proc 2006 (PMID 16610576); NCCIH Bitter Orange; ANSES 2014 opinion

1 Raspberry ketones Amazon ✓ Website ✓

Evidence: animal/in-vitro only | Effect: None demonstrated in humans

THE NOISE Heavily marketed (again a Dr Oz favourite, 'miracle fat-burner in a bottle') as boosting adiponectin and fat oxidation for rapid weight loss.

THE SCIENCE There are essentially no human efficacy trials. Claims rest entirely on rodent and in-vitro studies using doses vastly higher (per kg) than any human supplement delivers. No RCT has shown raspberry ketones cause weight loss in people, and safety in humans is uncharacterised. This is pure marketing extrapolated from a petri dish.

THE VERDICT Skip. Zero credible human outcome data. Do not buy.

BEST FORM Form is irrelevant because there is no credible human efficacy - do not buy for weight loss. If bought at all, a plainly-labelled single-ingredient product beats proprietary 'fat-burner' blends. (Synthetic vs 'natural' raspberry ketone - chemically identical single molecule; no meaningful retail form distinction.)

WATCH No human weight-loss evidence; effects seen only in rodents at doses ~100x typical human intake. Frequently sold in stimulant blends (caffeine, synephrine, bitter orange) - the real (and risky) effects come from those, not the ketone. Money better spent elsewhere.

Sources: Morimoto et al. Life Sci 2005 (rodent, PMID 15862604); IFT / OPSS raspberry ketone reviews (no human efficacy data); Medical News Today / NCCIH summaries

General vitamins & minerals (uncovered) (8)

4 Folate / folic acid Amazon ✓ Website ✓

Evidence: strong: large independent RCT (MRC) + population fortification data for NTD; null-to-weak for CVD/cancer | Effect: prevention of neural tube defects (periconception); homocysteine lowering with minimal hard-outcome payoff otherwise

THE NOISE Sold broadly as a heart-health (homocysteine-lowering), energy, mood, hair and 'methylation' supplement for everyone, alongside its legitimate prenatal use.

THE SCIENCE This is the one genuine winner in the category - but only for a specific job. Periconceptional folic acid (400-800 mcg/day) prevents neural tube defects: the MRC Vitamin Study RCT (n=1,817) showed a 72% reduction in NTD recurrence (RR 0.28), and population fortification data confirm a large real-world drop in spina bifida/anencephaly. That benefit is huge and clinically decisive for women who could become pregnant. OUTSIDE pregnancy the marketing collapses: folic acid lowers homocysteine but the large HOPE-2/VITATOPS/SEARCH RCTs show little-to-no reduction in heart attacks or overall mortality (a modest ~10% stroke signal at best). The 2013 Lancet meta-analysis (50,000 individuals) found no overall change in cancer incidence (RR 1.05, 0.99-1.11). High folic acid can also mask B12-deficiency anemia while neuropathy progresses.

THE VERDICT **Buy - but it is a targeted drug, not a wellness pill. Every woman who could become pregnant should take 400 mcg/day (higher if prior NTD or on certain medications), ideally starting before conception. For everyone else it is not an energy/heart/mood supplement and offers little; watch B12 status if taking it long-term.**

BEST FORM L-5-methylfolate (5-MTHF) at ~400 mcg DFE for general use / preconception. Folic acid is still proven for neural-tube-defect prevention, but 5-MTHF is preferable for MTHFR variants and avoids unmetabolised folic acid. (L-methylfolate (5-MTHF, e.g. Quatrefolic/Metafolin) or folinic acid - the bioactive forms that bypass MTHFR conversion - vs synthetic folic acid, which requires enzymatic reduction and can leave unmetabolised folic acid in blood.)

WATCH Critical: high folate can mask B12-deficiency anaemia while nerve damage progresses - always ensure adequate B12 alongside, especially in older adults/vegans. Preconception folate must start before conception. Folic acid interacts with methotrexate, some anticonvulsants (phenytoin) and sulfasalazine. Upper limit for folic acid is 1000 mcg/day (applies to synthetic, not food folate).

Sources: MRC Vitamin Study, Lancet 1991; Vollset et al. (folic acid & cancer meta), Lancet 2013, 50,000 individuals; USPSTF NTD recommendation 2023

3 Potassium / electrolyte blends Amazon ✓ Website ✓

Evidence: strong (large independent RCT + dose-response meta) for salt-substitute/dietary potassium; weak for OTC pills and 'wellness' hydration | Effect: blood-pressure lowering and stroke reduction from dietary/salt-substitute potassium; genuine rehydration during large fluid/electrolyte losses

THE NOISE Electrolyte powders and potassium pills sold for hydration, muscle cramps, energy, 'adrenal support' and blood-pressure/heart health; premium hydration brands imply broad performance and wellness gains.

THE SCIENCE The blood-pressure and stroke evidence for potassium is real - but it points to salt substitution and diet, not to the 99 mg pill or a hydration sachet. SSaSS (n=20,995, rural China): replacing regular salt with 75% NaCl/25% KCl cut stroke 14% (RR 0.86), major CV events 13%, and death 12% over ~5 years, via a ~3.3 mmHg systolic drop. Dose-response meta-analyses confirm higher potassium intake lowers BP, mainly in people with high sodium intake or hypertension. The catch: those effects come from grams of potassium (from food or salt substitute), whereas US OTC potassium supplements are capped at 99 mg (FDA, due to small-bowel ulceration from concentrated KCl) - about 3% of a day's target, i.e. clinically trivial. 'Hydration' electrolyte powders provide modest potassium and mainly benefit real fluid/sodium losses (heavy sweating, illness), not everyday wellness. Cramps are not reliably fixed by electrolytes.

THE VERDICT **For blood pressure, the winning move is a potassium-enriched salt substitute plus potassium-rich foods (fruit, vegetables, legumes) - not 99 mg pills, which are too small to matter. Salt substitutes are contraindicated in kidney disease or with potassium-sparing/ACE/ARB drugs (hyperkalemia risk). Electrolyte blends are worth it only around genuine heavy sweating or illness; as a daily wellness or cramp product, skip.**

BEST FORM For most, get potassium from food; for exercise/electrolyte use choose a blend with meaningful sodium (the main sweat loss) plus modest potassium and magnesium, low added sugar. OTC potassium is limited to ~99 mg/dose by design. (Well-tolerated potassium salts (citrate, chloride, gluconate) in modest per-dose amounts vs single large boluses. OTC supplements are capped low (~99 mg) for safety; food and prescription KCl deliver more. Choose blends by sodium:potassium:magnesium ratio and sugar content.)

WATCH Potassium is the dangerous electrolyte - excess causes hyperkalaemia and cardiac arrhythmia. HIGH RISK with kidney disease, ACE inhibitors/ARBs, potassium-sparing diuretics (spironolactone), NSAIDs - do not supplement without medical advice. Potassium chloride tablets can cause GI ulceration; take with food/fluid. Don't confuse electrolyte needs (mostly sodium) with a potassium requirement.

Sources: SSaSS: Neal et al., NEJM 2021; SSaSS mechanism: J Hum Hypertens 2024 (PMC11001572); NIH ODS Potassium Fact Sheet (99 mg OTC limit)

2.5 Multivitamin/multimineral (daily) [Amazon ✓](#) [Website ✓](#)

Evidence: strong evidence of NO mortality/CVD benefit; weak-moderate for a small cognition effect in the elderly | Effect: small episodic-memory/global-cognition signal in adults 60+; none for mortality, CVD, energy, or immunity · **literature compromised (funding/outlier)**

THE NOISE Sold as daily 'insurance', an energy and immune booster, and lately a brain/memory pill ('clinically shown to support memory in older adults'). Marketed as something every healthy adult should take to 'fill nutritional gaps'.

THE SCIENCE Large independent RCTs in generally-nourished adults show NO all-cause mortality, cardiovascular or energy benefit. COSMOS (n=21,442, adults 60+): CVD HR 0.98 (0.86-1.12), all-cause mortality HR 0.93 (0.81-1.08), invasive cancer HR 0.97 (0.86-1.09) - all null. Physicians' Health Study II (n=14,641 male physicians) found a small total-cancer reduction (HR 0.92, 0.86-0.998) but no effect on CVD or mortality. The only positive signal is cognition: COSMOS meta-analysis of its 3 sub-studies found global cognition and episodic memory improved by roughly 0.07 SD (~equivalent to slowing cognitive aging by a couple of years) - real but small, in adults 60+, and the MVM (Centrum Silver) was supplied by the manufacturer. No demonstrated benefit for a healthy adult under 60 with a normal diet.

THE VERDICT Skip as routine 'insurance' for a well-fed healthy adult - it does not extend life or lower disease risk. Reasonable ONLY for people with genuinely poor intake, older adults concerned about cognition, or defined deficiency states; if you take one, a plain low-dose Centrum-type product is fine and cheap.

BEST FORM A reputable multi in active/chelated forms at roughly RDA levels (not mega-doses), third-party tested (USP/NSF). Choose iron-free unless you are menstruating or iron-deficient. (Products using better-absorbed/active forms - methylfolate or folic acid over folic acid, methylcobalamin/adenosylcobalamin, magnesium glycinate/citrate over oxide, chelated (bisglycinate) zinc/iron, K2 as MK-7 - vs cheap oxide/carbonate/folic-acid-only formulas. Also iron-free 'men's/50+' versions.)

WATCH Avoid mega-dose formulas: excess vitamin A (retinol), iron and selenium are the main toxicity risks; smokers should avoid high beta-carotene. Take fat-soluble vitamins (A, D, E, K) with a fatty meal. Separate from thyroid meds/antibiotics (minerals chelate them). A multi does not replace treating a diagnosed deficiency. More is not better - stay near 100% DV.

Sources: COSMOS CVD/cancer: Sesso et al., Am J Clin Nutr 2022, PMC9170475; COSMOS cognition: Vyas et al., Am J Clin Nutr 2023, S0002-9165(23)66342-7; PHS-II cancer: Gaziano et al., JAMA 2012

2 Iodine / kelp [Amazon ✓](#) [Website ✓](#)

Evidence: strong for deficiency correction; kelp harm from case reports; no benefit for the marketed uses | Effect: corrects deficiency-related goiter and protects fetal neurodevelopment; no metabolism/weight/energy effect in replete adults

THE NOISE Kelp sold as a 'natural' thyroid booster, metabolism/energy and weight aid, and iodine source for 'thyroid support' and detox.

THE SCIENCE Iodine is an essential nutrient and correcting deficiency genuinely matters - it prevents goiter, and in pregnancy adequate iodine protects fetal neurodevelopment (deficiency lowers child IQ). BUT most Western adults are iodine-sufficient (iodized salt, dairy), and MORE iodine is not better: excess causes BOTH hyper- and hypothyroidism. Kelp specifically is a bad delivery vehicle: iodine content is wildly variable (up to ~1500 mcg/g dry), and multiple case reports document kelp-induced thyrotoxicosis and thyroid dysfunction in previously normal people. Kelp products are also documented sources of arsenic - 8 of 9 products in one FDA-cited survey exceeded the 2 ppm tolerance (up to 65 ppm).

THE VERDICT Skip kelp - uncontrolled iodine dose plus arsenic risk, and no metabolism/weight benefit. Iodine itself matters only for deficiency and pregnancy: pregnant/planning women should get ~150-220 mcg from a standardized prenatal (potassium iodide), not kelp. Well-nourished adults using iodized salt need nothing extra.

BEST FORM A precisely dosed potassium iodide supplement at ~150 mcg/day (RDA; 220 mcg pregnancy, 290 mcg lactation). Avoid kelp for routine dosing because of unpredictable, often huge iodine loads. (Defined-dose potassium iodide/potassium iodate (precise microgram dosing) vs kelp/seaweed products, whose iodine content is wildly variable and can be excessive or contaminated with heavy metals (arsenic).)

WATCH Both too little and too much cause thyroid dysfunction - excess iodine can trigger hypo- OR hyperthyroidism and thyroiditis, especially with underlying Hashimoto's/Graves' or nodular goitre. Adult UL ~1100 mcg/day; high-dose 'thyroid support' and kelp products routinely exceed this. Kelp may carry inorganic arsenic. Interacts with anti-thyroid drugs, lithium, amiodarone.

Sources: Arsenic in kelp: Amster et al., Environ Health Perspect 2007 (PMC1852683); Kelp thyrotoxicosis: Di Matola 2014 (PMC4216876); Med (Baltimore) 2019 case; Pregnancy iodine: Gowachirapant et al., Lancet Diabetes Endocrinol 2017

2 Selenium Amazon ✓ Website ✓

Evidence: moderate-strong: two independent RCTs show no cancer benefit and a diabetes signal | Effect: none demonstrated in replete adults; adverse signal for type 2 diabetes

THE NOISE Sold as a thyroid supporter, immune and 'antioxidant' mineral, fertility aid, and cancer preventive (often as selenomethionine or 'from selenium yeast', 100-200 mcg).

THE SCIENCE In selenium-replete populations (most of the US) there is no benefit and a diabetes signal. SELECT (n=35,533, 200 mcg/day): no prostate-cancer or mortality benefit; a non-significant excess of type 2 diabetes. NPC trial (n=1,312): the earlier cancer 'benefit' did not replicate and long-term follow-up showed type 2 diabetes INCREASED (HR ~1.55) in the top tertile of baseline selenium. Thyroid-antibody reductions in Hashimoto's are small, surrogate, and don't reliably change clinical outcomes. Benefit is plausible only in genuinely selenium-deficient regions (parts of China, Europe), not the replete US adult.

THE VERDICT Skip for a US/replete healthy adult - no proven benefit and a real type 2 diabetes risk signal above adequate intake. Only consider under supervision for diagnosed deficiency; do not stack multiple selenium-containing products.

BEST FORM L-selenomethionine (or selenium-enriched yeast), ~55-100 mcg/day. Do not stack multiple selenium-containing products. (Selenomethionine (well-absorbed, builds stores) or high-selenium yeast vs sodium selenite/selenate (less bioavailable). Selenomethionine is the studied form for most outcomes.)

WATCH Narrow safety window - UL is 400 mcg/day; chronic excess causes selenosis (hair/nail loss and brittleness, garlic breath, GI upset, neuropathy) and is linked to raised type-2 diabetes risk (NPC/SELECT). Brazil nuts are very high - easy to double up. Most replete people gain nothing from supplementing. Don't exceed ~200 mcg without cause.

Sources: SELECT: Lippman et al., JAMA 2009; NPC diabetes: Stranges et al., Ann Intern Med 2007; Vinceti Cochrane review 2018

1.5 Vitamin A / beta-carotene Amazon ✓ Website ✓

Evidence: strong: large independent RCTs show harm (beta-carotene) and benefit only in deficiency (vit A) | Effect: none beneficial in replete adults; beta-carotene increases lung cancer/mortality in smokers; vitamin A benefits only true deficiency

THE NOISE Beta-carotene sold as an antioxidant that protects lungs, eyes and skin and 'prevents cancer'; retinol/vitamin A sold for immunity, vision and skin. Framed as a safe 'provitamin'.

THE SCIENCE In well-nourished adults, supplemental beta-carotene is useless-to-harmful. CARET (n=18,314 smokers/asbestos workers, 30 mg beta-carotene + 25,000 IU retinyl palmitate): lung cancer +28% (RR 1.28) and total mortality +17%; trial stopped early. ATBC (n=29,133 male smokers, 20 mg beta-carotene): lung cancer +18%. Genuine vitamin A benefit exists ONLY in deficiency: high-dose vitamin A cuts child mortality and measles complications in deficient populations and treats night blindness/xerophthalmia - irrelevant to a well-fed Western adult. Preformed vitamin A above ~3000 mcg/day (10,000 IU) raises fracture and, in pregnancy, teratogenic risk.

THE VERDICT Skip beta-carotene supplements entirely (harm in smokers, no benefit otherwise) and avoid standalone high-dose retinol unless treating diagnosed deficiency. Eat carotenoids from food. Preformed vitamin A is dangerous in pregnancy above RDA.

BEST FORM For most people, get vitamin A largely as beta-carotene/mixed carotenoids from food or low-dose supplement; use preformed retinol only to correct a diagnosed deficiency and keep well under the upper limit. (Preformed retinol/retinyl palmitate (potent, storable, teratogenic) vs beta-carotene (provitamin, converted on demand). Beta-carotene is safer for general use but hazardous at high dose in smokers.)

WATCH High-dose beta-carotene supplements INCREASE lung cancer risk in smokers and asbestos-exposed people (ATBC/CARET trials) - they must avoid it. Preformed vitamin A is teratogenic - do not exceed ~3000 mcg (10,000 IU)/day in pregnancy or while trying to conceive; chronic excess causes liver damage and bone loss (UL 3000 mcg retinol). Fat-soluble - take with fat. Note retinol activity equivalents, not just IU.

Sources: CARET: Omenn et al., NEJM 1996; ATBC: NEJM 1994; Cochrane vitamin A child mortality (Imdad 2022)

1.5 Vitamin B6 (pyridoxine) Amazon ✓ Website ✓

Evidence: weak/none for benefit; strong regulatory + case evidence for dose-related neuropathy | Effect: none demonstrated for the marketed uses in replete adults; net harm (sensory neuropathy) at chronic/high dose

THE NOISE Sold for energy, mood/PMS, nausea, nerve health, homocysteine and immune support - often hidden inside magnesium, 'stress', sleep and B-complex blends at 25-100 mg.

THE SCIENCE For a healthy, non-deficient adult there is no robust outcome benefit, and B6 carries an active regulatory harm signal. Excess pyridoxine causes a dose- and duration-dependent sensory peripheral neuropathy - and crucially it can occur BELOW 50 mg/day, especially with chronic use or stacked products. Australia's TGA, after 32+ neuropathy reports, mandated warnings above 10 mg and on 25 Nov 2025 finalized rescheduling: >50-200 mg/day becomes pharmacist-only and >200 mg/day prescription (from June 2027); New Zealand and Health Canada issued parallel reviews. The narrow genuine uses (some morning-sickness regimens, treating diagnosed deficiency, certain drug-induced deficiency) are clinical, not wellness. Homocysteine-lowering B-vitamin trials did not deliver cardiovascular benefit.

THE VERDICT Skip standalone/high-dose B6 and audit your other supplements - people most often overdose unknowingly via magnesium or 'stress' blends. Keep total B6 low (well under 50 mg/day; ideally <=10-25 mg) unless a clinician prescribes it; it causes nerve damage, it doesn't protect nerves.

BEST FORM Pyridoxal-5'-phosphate (P5P) at a low dose - needs are only ~1.3-2 mg/day; keep total well under the neuropathy threshold. (Pyridoxal-5'-phosphate (P5P, the active coenzyme form) vs pyridoxine hydrochloride. P5P may be preferable for people with impaired hepatic activation and is less implicated (though not exempt) in the neuropathy seen with pyridoxine.)

WATCH Chronic high-dose B6 causes a dose-dependent SENSORY PERIPHERAL NEUROPATHY (numbness, tingling, unsteady gait) - reported even below the old 100 mg 'safe' level with long use; several regulators have cut recommended limits. Avoid mega-dose B-complex/energy products that stack hundreds of mg. Reduces efficacy of levodopa (without carbidopa) and interacts with some anticonvulsants. Usually reversible if caught early.

Sources: TGA final decision, 25 Nov 2025 (pyridoxine scheduling); TGA safety review of B6 peripheral neuropathy 2022; Medsafe NZ B6 neuropathy article, Jun 2025

1.5 Vitamin E (alpha-tocopherol, high-dose) Amazon ✓ Website ✓

Evidence: strong: large independent RCTs and meta-analyses show harm | Effect: none beneficial; net harm (increased prostate cancer, increased mortality at high dose)

THE NOISE Sold as a heart-protective and anti-cancer antioxidant, an anti-aging 'cellular protection' pill, and a skin/immune supplement, typically at 200-1000 IU.

THE SCIENCE Independent mega-trials show no benefit and net HARM at supplement doses. SELECT (n=35,533 men, 400 IU/day): prostate cancer INCREASED by 17% (HR 1.17, 1.004-1.36) versus placebo - 76 vs 65 cancers per 1000 men. Miller meta-analysis (19 trials, n=135,967): doses >400 IU/day associated with a ~4-6% rise in all-cause mortality. HOPE, GISSI and others found no CVD benefit. Any deficiency is essentially nonexistent in adults eating a normal diet (nuts, seeds, oils provide it).

THE VERDICT Do not buy high-dose vitamin E. It provides no proven benefit and at 400 IU/day and above raises prostate-cancer and all-cause mortality risk. There is no healthy-adult use case.

BEST FORM Natural d-alpha (RRR-alpha-tocopherol), ideally as mixed tocopherols, at a modest dose (~15 mg / ~22 IU meets needs). Avoid high-dose supplementation unless directed. (Natural d-alpha-tocopherol (RRR-) - roughly twice as bioavailable as synthetic dl-alpha (all-rac); and mixed tocopherols/tocotrienols (including gamma-tocopherol) vs alpha-tocopherol alone.)

WATCH High-dose vitamin E is harmful, not protective: >=400 IU/day is associated with increased all-cause mortality and (in the SELECT trial) higher prostate cancer risk; it also raises haemorrhagic stroke and bleeding risk - stop before surgery and avoid with anticoagulants/antiplatelets. Fat-soluble, take with food. Upper limit ~1000 mg/day. High-dose alpha alone depletes gamma-tocopherol.

Sources: SELECT: Klein et al., JAMA 2011; Miller et al., Ann Intern Med 2005 (meta, 135,967); ATBC Study Group, NEJM 1994

4 Beetroot / dietary nitrate Amazon ✓ Website ✓

Evidence: moderate-strong: many independent RCTs and meta-analyses, consistent direction, small effect | Effect: endurance time-trial performance (<40 min); modest resting blood-pressure reduction

THE NOISE Sold as a natural 'blood-flow' and endurance booster: bigger pumps, more oxygen to muscles, faster times, lower blood pressure - the flagship 'clean' ergogenic aid.

THE SCIENCE The best-supported item here. ~15-20 meta-analyses agree on a small but real endurance benefit for time-trials under ~40 min: pooled improvements on the order of 1-3% and effect sizes generally SMD ~0.2-0.4. Nitrate → nitrite → NO lowers the O₂ cost of exercise. The catch is population transfer: the signal is strongest in recreationally active / untrained people and shrinks toward null in elite endurance athletes (who already have high NO bioavailability). Also lowers resting SBP ~3-5 mmHg. Needs 8.4-16.4 mmol nitrate (~400-800 mg), taken 2-3 h pre-exercise or loaded 3+ days.

THE VERDICT **Buy - conditional. Worthwhile for recreational/sub-elite endurance and short TTs; take a concentrated 'shot' delivering ≥6.4 mmol (400 mg) nitrate 2-3 h before. Elite athletes should expect little. Cheaper as actual beetroot juice than pills.**

BEST FORM A concentrated beetroot juice/shot standardised to ~6.4-12.8 mmol (~400-800 mg) nitrate, taken ~2-3 hours before exercise. Dose by mmol nitrate, not by mg of powder. (Products dosed by actual NITRATE content in mmol (concentrated beetroot 'shots' or standardised juice) vs beetroot powder/capsules that don't state nitrate mmol and are often too low. Betaine/'beet root' capsules for BP are usually under-dosed.)

WATCH Do NOT use antibacterial mouthwash or chewing gum around dosing - it kills the oral bacteria that convert nitrate to nitrite and abolishes the effect. Causes harmless red/pink urine and stool (beeturia). Can lower blood pressure - caution with antihypertensives/PDE5 inhibitors. Load 2-3 h pre-event, or several days for a bigger effect. Oxalate content is relevant for stone-formers.

Sources: Umbrella review PMC12106159 (2024); Cycling sprint MA PMC11357493; Single/repeated high-intensity MA MDPI Antioxidants 12(6):1194; IOC consensus statement on dietary supplements (BJSM 2018)

4 Sodium bicarbonate Amazon ✓ Website ✓

Evidence: moderate: multiple RCTs and an ISSN position stand; quality varies by task | Effect: high-intensity anaerobic capacity / repeated sprint performance (~45 s-8 min)

THE NOISE 'Baking soda' buffer - sold as a legal blood-pH buffer that lets you push harder and delay the burn in sprints, rowing, HIIT and lifting.

THE SCIENCE A genuine, decades-old ergogenic with an ISSN position stand behind it (PMID 34503527). Raises extracellular bicarbonate, buffering H⁺ efflux, so it helps high-intensity efforts lasting ~45 s to ~8 min: Wingate peak/mean power and Yo-Yo test are moderate-quality evidence; ~2000 m rowing and muscle endurance low-quality. Pooled effect sizes are small-to-moderate (SMD ~0.3-0.4). Two honest caveats: (1) GI distress (bloating, cramping, vomiting) is common and can wipe out the benefit if you don't dose-titrate or use enteric/serial dosing; (2) women may respond less. No benefit for continuous distance running in a 2025 MA.

THE VERDICT **Buy - conditional, for trained athletes doing repeated high-intensity efforts (45 s-8 min). 0.2-0.3 g/kg (~15-20 g for a 70 kg adult) 60-180 min before, ideally split-dosed with food to cut GI upset; trial it in training first. Skip for steady-state endurance.**

BEST FORM Dose by body weight: ~0.2-0.3 g/kg taken 60-180 min before exercise, ideally as enteric-coated capsules or split doses with a carb-rich meal and plenty of water to blunt GI upset. (Enteric-coated/delayed-release capsules or divided dosing (and the emerging hydrogel formulations) vs a single large dose of plain baking soda, which reliably causes GI distress.)

WATCH The limiting factor is GI distress - nausea, cramping, diarrhoea - worst with a single bolus on an empty stomach; enteric coating, split dosing and taking with carbohydrate/water reduce it, and individuals should trial it in training first. High sodium load - caution with hypertension, heart failure or renal disease. Separate from medications whose absorption is pH-sensitive. Don't confuse with sodium citrate (gentler alternative).

Sources: ISSN position stand PMID 34503527; Umbrella review (ResearchGate 356193110); Female athlete MA (EJSS 2022, doi:10.1080/17461391.2021.1880649); Running MA 2025 (ResearchGate 394039535)

3 Chlorella [Amazon](#) ✓ [Website](#) ✓

Evidence: weak-moderate: several RCTs, conflicting BP results, GRADE ratings low | Effect: modest total-cholesterol/LDL and fasting-glucose reductions; inconsistent blood-pressure effect

THE NOISE Green algae 'detox' that supposedly binds heavy metals, boosts immunity and energy, and cleanses the body.

THE SCIENCE Similar profile to spirulina: modest cardiometabolic signal, in cardiometabolic populations. A 19-RCT MA (n=797) reported SBP -4.51 mmHg, DBP -1.64 mmHg, total cholesterol -9.1 mg/dL, LDL -8.3 mg/dL, fasting glucose -4.2 mg/dL - but a more recent GRADE-assessed analysis found a neutral BP effect, so consistency is shaky. The famous 'heavy-metal detox' claim rests on animal/in-vitro work, not controlled human outcomes. Small salivary-IgA immune signal is preliminary.

THE VERDICT **Conditional / low-priority. Minor lipid/glucose benefit possible in people with cardiometabolic risk; the 'detox' marketing is unsupported in humans. Not needed by a healthy adult.**

BEST FORM Buy 'broken cell wall' or 'cracked cell wall' chlorella. Chlorella's tough cellulose wall is largely indigestible, so cracking it markedly improves absorption of its nutrients and pigments. Species is usually Chlorella vulgaris or pyrenoidosa; either is fine if the wall is broken and the product is purity-tested. (Broken/cracked cell-wall chlorella vs intact-cell-wall chlorella)

WATCH As with spirulina, heavy-metal and contaminant testing is essential (chlorella is often marketed as a metal 'binder', which cuts both ways). Can cause GI upset, gas and green stools initially - start low. Photosensitivity has been reported. High vitamin K content (warfarin interaction). Possible immune activation - caution in autoimmune disease/immunosuppression. Mould-allergy cross-reactivity in sensitive people.

Sources: Cardiovascular MA PMID 29037431; Chlorella GRADE MA (ScienceDirect S2211926425003923); Chlorella vulgaris lipid dose-response MA (S0965229922000243)

3 Krill oil [Amazon](#) ✓ [Website](#) ✓

Evidence: moderate for omega-3 status; bioavailability edge over fish oil is small and industry-tinged | Effect: raises omega-3 index / EPA-DHA status; downstream triglyceride-lowering (modest, shared with fish oil) - **literature compromised (funding/outlier)**

THE NOISE Sold as a 'superior', better-absorbed omega-3 (phospholipid-bound EPA/DHA plus astaxanthin) for heart, joints, brain and inflammation - premium-priced over fish oil.

THE SCIENCE Two-part answer. (1) The delivery claim has some support: phospholipid-bound EPA/DHA in krill shows equal-to-modestly-higher plasma incorporation than fish-oil triglycerides/ethyl esters (early study AUC 80% vs 60% vs 48%, but wide SDs, often NS; a 2024 network MA leans pro-krill). (2) The outcome that matters is the omega-3 effect itself, which is real but modest (triglyceride lowering, minor CV signal) and identical in kind to cheap fish oil. Krill capsules carry low EPA/DHA per softgel, so per-gram of omega-3 it's expensive, and much of the bioavailability literature is Aker BioMarine-linked.

THE VERDICT **Conditional. If you specifically want omega-3s, plain fish oil (or algal oil) delivers the same EPA/DHA far cheaper; krill's absorption edge is small and doesn't justify the premium for most people. Buy krill only if fish-oil burp/tolerance is a real problem for you.**

BEST FORM Krill delivers its omega-3s bound to phospholipids, which are absorbed efficiently and integrate into cell membranes well, plus it carries astaxanthin and choline. Per-gram absorption can beat cheap ethyl-ester fish oil. However, each krill capsule contains far LESS total EPA/DHA than a fish-oil capsule, so it is a premium low-dose option; for high-dose omega-3 needs a re-esterified triglyceride (rTG) fish oil is more cost-effective. Check the mg of actual EPA+DHA, not just 'krill oil mg'. (Krill oil (phospholipid-bound EPA/DHA) vs standard triglyceride/ethyl-ester fish oil)

WATCH Shellfish allergen - avoid if allergic to crustaceans. Compare on EPA+DHA content, not oil weight, as krill is dilute and expensive per mg omega-3. Mild fishy burps/GI upset. Antiplatelet effect at high doses - caution with anticoagulants/before surgery. Choose sustainably sourced, low-oxidation (low TOTOX) product.

Sources: Omega-3 bioavailability network MA S2590157524007685 (2024); FO vs krill PK study PMID 21854650 / PMC3168413; Krill phospholipid crossover PK study (S2213434424000434)

3 L-carnitine (L-tartrate) Amazon ✓ Website ✓

Evidence: weak-moderate: small RCTs, soreness/biomarker endpoints, thin performance data | Effect: reduced muscle soreness / lowered exercise-induced muscle-damage markers

THE NOISE Marketed for fat-burning, endurance, faster recovery and less muscle soreness; a staple of pre-workout and 'recovery' stacks.

THE SCIENCE Fat-burning claim in healthy adults is essentially null (muscle carnitine is hard to raise without co-ingested carbs/insulin). The recovery/soreness claim is the real one: a 2020 MA (7 RCTs, PMID 32154768) found significantly lower muscle soreness at 0-96 h post-exercise and reduced CK/myoglobin/malondialdehyde. But performance markers of recovery did not improve with 2 g/d for 3 weeks. So it blunts soreness and damage markers without clearly making you faster/stronger. Small trials, some overlap with supplement-industry authors.

THE VERDICT **Conditional / low-priority. If you chase reduced DOMS from eccentric/resistance work, 2 g/day L-carnitine L-tartrate has modest support; don't expect fat loss or performance gains. Most people can skip it.**

BEST FORM For exercise/recovery buy L-carnitine L-tartrate (LCLT), the form used in most sports trials (typically 1-2 g/day, standardised so ~68% is carnitine). Choose acetyl-L-carnitine (ALCAR) instead only if the goal is cognitive/neurological, since it crosses the blood-brain barrier; propionyl-L-carnitine is the niche choice for peripheral vascular/blood-flow use. Plain unbranded carnitine base offers no advantage. (L-carnitine L-tartrate (LCLT) vs plain L-carnitine base vs acetyl-L-carnitine (ALCAR) vs propionyl-L-carnitine)

WATCH Oral bioavailability is low (~15-18%) and falls further at higher single doses, so split into 2 doses; taking with a carbohydrate-containing meal raises insulin and improves muscle uptake/retention. Main real concern is that gut bacteria convert carnitine to TMAO (a marker linked to cardiovascular risk) - a reason for heavy meat-eaters and cardiac patients to be cautious. GI upset and a fishy body odour occur at multi-gram doses. Can modestly interact with warfarin (raised INR reported).

Sources: Damage MA PMID 32154768 (2020); 2025 EIMD MA (ScienceDirect S2950273X25000256); 5-week recovery RCT PMC8541253

3 Spirulina Amazon ✓ Website ✓

Evidence: moderate for lipids/BP; weak/preliminary for exercise | Effect: modest reductions in blood pressure and LDL/total cholesterol in cardiometabolic populations

THE NOISE 'Superfood' algae sold as an immune tonic, detoxifier, energy/endurance booster and protein source with anti-inflammatory 'phycocyanin' magic.

THE SCIENCE Real but condition-gated signal. Meta-analyses show modest cardiometabolic effects: SBP ~-4.6 mmHg and DBP down (larger in hypertensives), improved lipid profile, and reduced body fat - mostly in overweight/obese or hypertensive populations, not healthy athletes. Exercise/antioxidant and muscle-damage data are small and preliminary; anti-inflammatory effects are inconclusive. Population transfer is the key caveat: benefits shown in metabolically impaired people don't clearly carry to a healthy adult. Contamination risk (microcystins, heavy metals) if not from a reputable source.

THE VERDICT **Conditional. A reasonable nutrient-dense adjunct if you're overweight/hypertensive and want a small lipid/BP nudge (1-8 g/day); not a proven ergogenic aid for healthy athletes. Buy third-party-tested product only.**

BEST FORM Form does not matter - spirulina is a whole dried cyanobacteria (Arthrospira), sold as powder or tablet with no meaningful bioavailability difference. What matters is sourcing: buy from a producer with third-party testing for microcystins and heavy metals (controlled tank-grown, e.g. certified USP/clean-label brands) rather than cheap open-pond product. (Whole-food cyanobacterium; the variable is source purity/testing, not chemical form)

WATCH Contamination is the key risk: some spirulina is adulterated or co-cultured with microcystin-producing blue-green algae (hepatotoxins) and can carry heavy metals (lead, arsenic) - insist on tested product. Contains phenylalanine (avoid in phenylketonuria). Theoretical immune-stimulation concern in autoimmune disease and with immunosuppressants. High in vitamin K (relevant to warfarin) and can carry an iodine load.

Sources: BP MA PMC8468496; Lipid dose-response MA (ScienceDirect S1043661823001585); Spirulina+exercise cardiometabolic MA PMID 40655486; Umbrella review S2211926426000561

2 CBD (cannabidiol) Amazon ✗ Website ✓

Evidence: strong for Rx epilepsy; weak/preliminary for OTC wellness claims | Effect: proven only for specific epilepsies (Rx); anxiety/sleep/pain in healthy adults unproven at retail doses | US Amazon prohibits CBD/hemp-extract supplements; sold via specialty/vendor websites and dispensaries. FDA holds it is not a lawful dietary supplement.

THE NOISE Sold everywhere for anxiety, sleep, pain, inflammation, 'recovery' and general calm - the wellness cure-all of the 2020s.

THE SCIENCE Only one thing is proven to a drug standard: prescription CBD (Epidiolex) for specific pediatric epilepsies. For the mass-market anxiety/sleep/pain/recovery claims in healthy adults, controlled evidence is thin, doses in retail products are far below therapeutic-trial doses (trials often use hundreds of mg/day; gummies contain 10-25 mg), and product-label accuracy is notoriously poor. Regulators are explicit: the FDA has repeatedly stated CBD is NOT a lawful dietary supplement, is not GRAS, and flags a real dose-dependent liver-injury signal (elevated transaminases) plus drug interactions (CYP inhibition).

THE VERDICT Skip as a supplement. The general-wellness claims are largely unproven at retail doses, quality control is poor, and there's a genuine liver-enzyme safety signal. If you have a diagnosed seizure disorder, that's a prescription (Epidiolex) conversation, not a gummy.

BEST FORM Oral CBD has poor and variable bioavailability with a strong food effect - a high-fat meal can increase absorption up to ~4-fold, so take oil-based CBD with fatty food, or choose a nano-emulsified/liposomal product for more consistent uptake. On spectrum: full-spectrum (with trace THC and other cannabinoids) may give an 'entourage' effect but carries THC; broad-spectrum removes THC; isolate is pure CBD - pick based on drug-testing risk. Above all, buy a product with a batch-specific Certificate of Analysis, as CBD is notoriously mislabelled. (Full-spectrum vs broad-spectrum vs isolate; plus delivery (oil with fat / nano-emulsion / liposomal vs plain))

WATCH CBD inhibits CYP3A4/CYP2C19/2C9 and can raise levels of many drugs - clinically important with warfarin (raised INR/bleeding), clobazam, and other narrow-margin medicines; it also carries the grapefruit-warning interaction profile. High doses can elevate liver enzymes (seen with prescription CBD). Full-spectrum products can contain enough THC to trigger a positive drug test. Product quality/dose accuracy varies widely - insist on third-party COA for potency and contaminants.

Sources: FDA 'Regulation of Cannabis/CBD' consumer update; FDA warning letters (2019, 2022); EFSA novel-food CBD safety statement (PMC12884199); Epidiolex FDA approval basis

2 Cod liver oil Amazon ✓ Website ✓

Evidence: strong evidence of NO infection benefit; nutrient value depends on baseline deficiency | Effect: none for infection prevention; supplies vitamin D / omega-3 (nutrient replacement only)

THE NOISE Traditional winter 'tonic' sold to prevent colds/flu, support immunity, joints, heart and bones via omega-3 plus vitamins A and D.

THE SCIENCE The infection-prevention claim was tested at scale and failed: a Norwegian RCT of 34,601 adults (BMJ 2022, NCT04609423) found daily cod liver oil (5 mL: 10 ug vitamin D, 250 ug vitamin A, 1.2 g omega-3) did NOT reduce COVID-19, serious COVID-19, or other acute respiratory infections vs placebo. It remains a legitimate source of vitamin D and omega-3, but if you're vitamin-D deficient the effective, precise intervention is a vitamin D supplement - not a fixed oil that also loads vitamin A (preformed retinol carries a cumulative toxicity/bone-fracture ceiling). Omega-3 dose per teaspoon is modest.

THE VERDICT Skip as an immune/wellness tonic - the large trial is null. If your goal is vitamin D, dose vitamin D directly; if omega-3, use fish/algal oil. Don't rely on cod liver oil to megadose either, because the vitamin A rides along.

BEST FORM Form is not the issue - it is a whole fish-liver oil supplying EPA/DHA PLUS vitamins A and D. Choose a fresh, low-oxidation product from a reputable brand and, importantly, one that DECLARES its vitamin A and D content per dose so you can control intake. 'Fermented' cod liver oil is a marketing niche with quality-control concerns and unpredictable vitamin levels - standard, molecularly-tested oil is safer. (Traditional cod liver oil; variable is vitamin A/D content and freshness, not chemical form)

WATCH The dominant risk is vitamin A: cod liver oil is a concentrated retinol source, so stacking it with a multivitamin or taking large doses risks hypervitaminosis A (liver toxicity, bone effects) - especially dangerous in pregnancy (teratogenic), where high preformed vitamin A should be avoided. Don't simply scale up the dose to chase omega-3s, or you overdose vitamin A/D. Antiplatelet effect at high omega-3 intake. Take with a meal; check for oxidation.

Sources: Cod liver oil RCT BMJ 2022, NCT04609423 (n=34,601); Companion vitamin D/ARI trials (CNN/Healthgrades coverage of BMJ 2022)

2 Exogenous ketones (BHB salts/esters) [Amazon ✓](#) [Website ✓](#)

Evidence: moderate evidence of NO ergogenic effect: pooled RCTs cross zero | Effect: none demonstrated for endurance performance (raises BHB, lowers glucose only)

THE NOISE Pitched as a fourth fuel that spares glycogen, boosts endurance, sharpens focus and speeds recovery - the premium 'biohacker' fuel.

THE SCIENCE The performance claim is a clean null. A 2022 MA of 8 RCTs / 80 participants found no endurance benefit: Hedges $g = 0.136$ (95% CI -0.195 to 0.467, $p = .42$) across time-trials and time-to-exhaustion. They do exactly what they say metabolically - raise blood BHB, lower glucose - but that doesn't move performance, and esters often cause GI distress that can hurt it. Cognitive/therapeutic uses are a separate, still-preliminary story and not what most retail buyers are chasing.

THE VERDICT Skip for athletic performance. Expensive, GI-unfriendly, and the endurance benefit is statistically zero. Real fuel for endurance is carbohydrate.

BEST FORM Ketone monoesters raise blood BHB highest and fastest and are the form used in the strongest performance research, but they are expensive and taste unpleasant. BHB salts are cheaper and better-tasting but deliver a much smaller ketone rise and come bound to a large mineral load. Critically, buy a product specifying the D- (R-) enantiomer: many cheap salts are racemic D/L, and only the D isomer is efficiently used, so half the labelled BHB is largely inert. R-1,3-butanediol is a good-value ester precursor. (Ketone monoester (R-1,3-butanediol / R-BHB ester) vs BHB mineral salts, and D (R) vs racemic D/L enantiomers)

WATCH GI distress (nausea, cramping, diarrhoea) is common, especially with esters and on an empty stomach. Mineral salts add substantial sodium/potassium/calcium/magnesium - check the per-serving sodium if you are watching intake or have hypertension. Exogenous ketones do not replicate the metabolic state of true dietary ketosis and can transiently blunt fat oxidation. Modest performance benefit at best in trials.

Sources: Endurance MA IJSNEM 32(3):214 (2022); Blood-glucose MA PMID 35380602 (2022); GSSI review on ketones as ergogenic aids

2 Glutathione (oral / liposomal) [Amazon ✓](#) [Website ✓](#)

Evidence: weak: mostly biomarker/bioavailability trials, often industry-run | Effect: raises measured blood glutathione stores; no demonstrated clinical outcome in healthy adults · **literature compromised (funding/outlier)**

THE NOISE 'Master antioxidant' - sold for detox, immune support, anti-aging, liver health and (especially in Asian markets) skin-whitening.

THE SCIENCE Almost all human data is bioavailability, not outcomes. Plain oral glutathione is largely degraded in the gut; liposomal/micellar forms do raise measured blood/PBMC GSH stores (e.g. ~40% whole-blood, ~100% PBMC increase at 500-1000 mg/day; PMID 28853742). But raising a lab marker is not a clinical benefit: there is little independent, controlled evidence that oral glutathione improves any hard endurance, immune, liver, or longevity outcome in healthy adults, and the bioavailability studies are frequently run/funded by formulation makers (e.g. proprietary LipoMicel, LipoDuo). Skin-whitening evidence is weak and mostly IV-based.

THE VERDICT Skip for general wellness. You're paying for a bigger antioxidant number with no proven downstream benefit. Cheaper, evidenced ways to raise glutathione status: adequate protein and NAC/cysteine precursors; even those lack strong healthy-adult outcome data.

BEST FORM Plain oral glutathione is largely broken down in the gut, though some studies show reduced (GSH) forms modestly raise body stores over time. Prefer a liposomal or the branded Setria glutathione, both of which have human data for raising glutathione status. An efficient and cheaper route to raising endogenous glutathione is the precursor NAC (N-acetylcysteine), often with glycine. Avoid oxidised (GSSG) or unspecified forms. (Liposomal / Setria clinically-studied reduced glutathione vs plain oral reduced glutathione (GSH) vs precursor NAC)

WATCH Buy the REDUCED (GSH) form, not oxidised. Sublingual/liposomal delivery matters because of gut degradation. Generally well tolerated; occasional bloating. NAC can cause GI upset and interacts with nitroglycerin (hypotension) and can affect clotting. Sulphur-sensitive individuals may react. Evidence for skin-whitening claims is weak.

Sources: Sinha et al. PMID 28853742; LipoMicel crossover MDPI Antioxidants 15(3):354; gamma-glutamylcysteine pilot PMC5284489

2 Greens / superfood powders Amazon ✓ Website ✓

Evidence: weak: company-funded biomarker/safety trials; no independent outcome RCTs | Effect: raises circulating micronutrient levels; no demonstrated hard health outcome · **literature compromised (funding/outlier)**

THE NOISE Marketed as a daily nutritional insurance policy - 'foundational nutrition', gut health, immunity, energy, replacing your vegetables in one scoop (e.g. AG1/Athletic Greens).

THE SCIENCE The evidence gap is the story. As of 2025 there are essentially no independent, health-outcome RCTs on branded greens powders; company-linked trials mostly show a single dose raises blood micronutrient levels and doesn't harm safety biomarkers - i.e. absorption and 'no harm', not benefit. Any BP/antioxidant hints come from small studies of generic greens, not the products themselves. Whole vegetables and fruit have vastly stronger outcome evidence. Proprietary blends also obscure whether any ingredient is present in an effective dose.

THE VERDICT **Skip for most people. You're paying a premium for a multivitamin-plus-fiber with 'raises nutrient levels' data and no proven health outcome. Eat vegetables and, if needed, take a cheap multivitamin. Consider only if your actual diet is poor and convenience genuinely changes adherence.**

BEST FORM Form does not meaningfully matter - these are dried, blended plant/algae powders. What matters is buying one with transparent per-ingredient dosing (avoid 'proprietary blends' that hide token 'fairy-dusted' amounts) and third-party/heavy-metal testing. Don't pay for long ingredient lists; a shorter, honestly-dosed product is better. Treat as an insurance top-up, not a vegetable replacement. (Blended whole-food/vegetable powders; the variable is formulation transparency and testing, not a chemical form)

WATCH Heavy-metal contamination is a recurring problem in this category (plant concentrates accumulate lead/cadmium/arsenic) - look for tested products. Many are high in vitamin K and/or contain green tea, licorice, or added botanicals that interact with medications (warfarin, blood pressure). They do NOT replace whole vegetables/fibre. Watch added iodine (kelp) and stimulant botanicals. Marketing claims usually outrun the evidence.

Sources: AG1 safety-biomarker RCT PMC12481523; AG1 micronutrient-absorption study (NutraIngredients 2026-07-23); ConsumerLab AG1 assessment

2 L-arginine Amazon ✓ Website ✓

Evidence: weak/conflicting: mixed RCTs, poor oral bioavailability | Effect: inconsistent; small aerobic/anaerobic signal at high dose, often no NO change in trained subjects

THE NOISE Sold as a 'NO booster' for pumps, vasodilation, endurance and even growth-hormone release; core of many pre-workouts.

THE SCIENCE Mechanism outruns outcome. Oral arginine has poor bioavailability (heavy first-pass gut/liver metabolism by arginase) and several trials show it fails to raise plasma NO or performance in trained athletes (PMID 18708287). Some meta-analyses report small VO2max/performance gains at high doses (~0.15 g/kg, 10-11 g, 60-90 min pre), but results are inconsistent and skewed to lower-quality or acute studies. L-citrulline is the better-absorbed NO precursor and the more defensible buy.

THE VERDICT **Skip. Poorly absorbed and unreliable; if you want the NO-precursor route, buy L-citrulline (or citrulline malate) instead. High doses of arginine mostly buy you GI upset.**

BEST FORM Counter-intuitively, L-citrulline (3-6 g, or citrulline malate 6-8 g) raises steady-state plasma arginine and nitric-oxide precursor levels MORE reliably than L-arginine itself, because oral arginine is largely destroyed by intestinal and hepatic arginase (first-pass) and needs very large doses. If buying arginine directly, plain L-arginine base/HCl is fine - salt form is not the issue; the dose and first-pass loss are. For most consumers seeking pump/blood-flow effects, citrulline is the better buy. (L-arginine (free base/HCl) vs L-citrulline / citrulline malate as the arginine-raising alternative)

WATCH Take on an empty stomach; large arginine doses (>5-9 g) commonly cause diarrhoea, nausea and cramping. Arginine can reactivate herpes simplex (cold sores) in susceptible people - lysine is the antagonist. Vasodilatory: use caution with blood-pressure meds, nitrates and PDE5 inhibitors (additive hypotension). Avoid soon after a heart attack. Citrulline is far better tolerated at the gut than arginine.

Sources: VO2max MA doi:10.14814/phy2.14739 (2021); Arginine energy-metabolism MA doi:10.3390/nu12051300 (2020); Null trial PMID 18708287; Food-source NO MA PMC8715640

2 MCT oil Amazon ✓ Website ✓

Evidence: moderate (small effect) for modest weight loss; evidence of no ergogenic effect | Effect: small weight/body-composition reduction vs long-chain fat; no endurance/ergogenic benefit

THE NOISE Keto/biohacker staple sold for fast clean energy, endurance fuel, appetite control, 'brain fuel' and effortless fat loss (bulletproof coffee).

THE SCIENCE Endurance/ergogenic claim is basically null - a systematic review found little-to-no performance or substrate-utilization benefit, and high pre-exercise doses mainly cause GI distress. The weight claim is real but trivial: a classic MA found ~0.5 kg greater weight loss over ~10 weeks vs long-chain fats, with small reductions in waist/fat; more recent MAs echo small effects with mixed independence. It's a modestly thermogenic fat, not a metabolic hack, and it displaces other calories rather than creating a deficit by magic.

THE VERDICT Skip for performance; low-value for weight. If you tolerate it and like it in a ketogenic context, fine - but expect at most a fraction of a kilogram, not transformation, and titrate to avoid the GI hit.

BEST FORM Buy C8 (caprylic acid) or a labelled C8/C10 blend. C8 is the most rapidly converted to ketones and is the reason people take MCT for energy/ketogenic support. Avoid cheap 'MCT oils' that are largely C12 (lauric acid) - lauric acid metabolises more like a regular long-chain fat and is much less ketogenic, so a coconut-derived product heavy in C12 is inferior for the intended effect. Check the C8/C10/C12 breakdown on the label. (C8 (caprylic) or C8/C10 blends vs cheap 'MCT' oils padded with C12 lauric acid)

WATCH GI distress is the big one: MCTs commonly cause cramping, nausea and urgent diarrhoea, especially C8 and especially on an empty stomach or above ~1 tablespoon - start with a teaspoon and titrate up with food. Do not use for high-heat cooking (low smoke point). It is calorie-dense. Caution in liver disease (some conditions impair MCT handling). No true bioavailability problem, just tolerance and the C8-vs-C12 quality issue.

Sources: Weight-loss MA PMID 25636220; MCT blood-lipids MA (J Nutr S0022-3166(22)00366-2); Endurance/substrate systematic review PMID 36096496

1 Sea moss (Irish moss / Chondrus crispus) Amazon ✓ Website ✓

Evidence: animal/in-vitro only; no human RCTs on Chondrus crispus | Effect: none demonstrated in humans; delivers uncontrolled iodine (a risk, not a benefit)

THE NOISE Viral 'billion-dollar' superfood claiming 92 minerals, thyroid support, immunity, gut/digestion, skin, energy, libido and detox - the influencer cure-all.

THE SCIENCE This is marketing. There are essentially no human RCTs on Chondrus crispus for any of these claims; the cited support is seaweed-in-general, test-tube or animal work. Its one real, measurable property - high, variable iodine - is a liability, not a feature: uncontrolled iodine can trigger thyroid dysfunction in both directions. It also concentrates heavy metals and contains carrageenan-type polysaccharides. The '92 minerals' line is nutritionally meaningless at the gram doses consumed.

THE VERDICT Skip / recommend against. No credible human outcome evidence, plus a genuine uncontrolled-iodine and heavy-metal risk. If you want the actual nutrients, eat a normal varied diet; if you need iodine, use a defined-dose supplement.

BEST FORM Form does not matter - gel, powder and capsule are the same seaweed; there is no bioavailability upgrade to buy. The decisive variable is IODINE content, which is high and highly variable batch-to-batch, plus heavy-metal load. Choose a product that actually tests and states iodine and heavy-metal levels; ocean-harvested and pool-grown differ. Be sceptical of the '92 minerals' marketing claim. (Whole seaweed as gel/powder/capsule; the variable is iodine content and contamination, not chemical form)

WATCH Iodine is the main hazard: sea moss can deliver excessive iodine, causing thyroid dysfunction (both hyper- and hypothyroid) - risky for anyone with thyroid disease or on thyroid medication, and easy to overdo daily. Heavy-metal accumulation (lead, arsenic, mercury) is a real concern with wild seaweed. Carrageenan is derived from this species; degraded carrageenan raises gut-inflammation concerns. Can affect anticoagulation (vitamin K/blood-thinning effects). Use small, tested doses.

Sources: OPSS (DoD Operation Supplement Safety) sea moss review; CHOP Health/Poison-control sea moss statement; National Geographic/Banner Health evidence reviews

Part III — Teas, coffee & other beverages

Beverages — teas (8)

3 Black tea (brewed) Amazon ✓ Website ✓

Evidence: moderate: large cohorts + meta-analysed RCTs, effects small/sub-MCID, confounded | Effect: CV/all-cause mortality association (cohort); sub-MCID BP reduction (RCT); acute alertness | No liver warning. Tannins can modestly reduce non-heme iron absorption — relevant to iron-deficient drinkers taking tea with meals.

THE NOISE Heart-healthy, gut-friendly, and life-extending; theaflavins lower cholesterol and blood pressure.

THE SCIENCE Black tea dominates the UK Biobank cohort where ≥ 2 cups/day tracked with all-cause mortality HR $\sim 0.87\text{--}0.91$ (observational, healthy-user/reverse-causation confounded). RCTs: black-tea BP meta-analysis (13 trials, 22 arms) found SBP -1.04 mmHg and DBP -0.59 mmHg — smaller than green tea and well below the ~ 5 mmHg MCID. Lipid effects are inconsistent and small. The cohort mortality signal is the strongest data and cannot prove causation.

THE VERDICT Drink it — safe, hydrating, mildly protective in cohorts and with a trivially small real BP effect; treat the mortality benefit as an association, not a guarantee, and don't rely on it to manage hypertension or cholesterol.

BEST FORM n/a - the beverage is the form. Full oxidation converts catechins into theaflavins/thearubigins; brew at full boil ($\sim 95\text{--}100^\circ\text{C}$) for 3-5 min. No milk needed, but note below.

WATCH Adding milk binds some polyphenols (casein) and blunts the flavanol effect on vascular function - drink it plain if that is your goal. Higher caffeine than green tea ($\sim 40\text{--}70$ mg/cup). Still chelates non-heme iron - separate from iron sources. Theaflavins are the distinctive black-tea polyphenol, not 'weaker green tea.'

Sources: Black tea & BP dose-response meta-analysis of RCTs (PMID 33237083); UK Biobank tea & mortality, Ann Intern Med 2022 (PMID 36037472); Tea BP meta-analysis, PLOS One 2014 (PONE 0103247)

3 Green tea (brewed) Amazon ✓ Website ✓

Evidence: moderate: large prospective cohorts + many small RCTs, but effects sub-MCID and confounded | Effect: acute alertness (caffeine/L-theanine); small CV/all-cause mortality association (cohort); sub-MCID BP/LDL changes (RCT) | Brewed green tea has no liver warning; the hepatotoxicity signal is specific to concentrated EXTRACT capsules, not the beverage.

THE NOISE Marketed as a fat-burning, cancer-preventing, life-extending 'superfood' — melts belly fat, detoxes, and adds years, all from catechins/EGCG.

THE SCIENCE Best data are OBSERVATIONAL: in UK Biobank ($n=498,043$, median 11.2y) ≥ 2 cups/day tracked with all-cause mortality HR $\sim 0.87\text{--}0.91$ — small and confounded by healthy-user bias/reverse causation. RCTs of green-tea catechins show only biomarker nudges: SBP ~ -2.1 mmHg, DBP ~ -1.7 mmHg, LDL ~ -0.16 mmol/L (~ 6 mg/dL) — all below any minimum clinically important difference (MCID ~ 5 mmHg BP). No RCT shows brewed green tea prevents cancer or produces meaningful weight loss. Real but tiny; a pleasant habit, not a therapy.

THE VERDICT Drink it freely — safe, near-zero-calorie, mildly stimulating, and the cohort signal is reassuring, but do not expect measurable fat loss, cancer protection, or a clinically real drop in BP/LDL from the cup.

BEST FORM n/a - the beverage is the form. To maximize catechins, steep loose leaf or quality bags at $\sim 70\text{--}80^\circ\text{C}$ (not boiling) for 2-3 min; a squeeze of lemon (vitamin C/citrate) stabilizes EGCG and boosts catechin bioavailability.

WATCH Tannins/catechins chelate non-heme iron - separate green tea from iron-rich meals or iron supplements by 1-2 hours, important for anyone iron-deficient. Catechins absorb best on an empty stomach but that can cause nausea in sensitive people. Caffeine ($\sim 25\text{--}45$ mg/cup) is real; L-theanine softens the jitter. Brewed tea itself has no hepatotoxicity concern (unlike concentrated extract).

Sources: UK Biobank tea & mortality — Inoue-Choi et al., Ann Intern Med 2022;175:1201 (PMID 36037472; DOI 10.7326/M22-0041); Green tea catechins & BP meta-analysis (PMID 24861099); Green tea & blood lipids meta-analysis of RCTs (PMC7240975); Umbrella review of green tea on BP/glycemic/lipid outcomes, Nutr Metab Cardiovasc Dis 2022

3 Matcha (powdered whole green tea leaf) Amazon ✓ Website ✓

Evidence: weak-moderate: a few small short RCTs; most claims extrapolated from green-tea catechin data | Effect: acute attention/stress-buffering (small RCT); acute alertness (caffeine/L-theanine) | Ceremonial/culinary powder; no liver warning. Higher caffeine per serving than steeped green tea — relevant for sensitive/pregnant drinkers.

THE NOISE 'Cleaner, calmer' caffeine; 137x the antioxidants of regular green tea; boosts focus, metabolism, and mood via L-theanine + EGCG.

THE SCIENCE Because you ingest the whole leaf, a 2 g serving delivers more catechins/caffeine/L-theanine than an equal weight of steeped leaf (~170 mg catechins, ~70 mg caffeine, ~50 mg L-theanine per 2 g). Small placebo-controlled RCTs show modest, real acute effects: better attention/reduced stress reactivity under mild stress. A 12-month RCT in older adults with cognitive decline (n=99, 2 g/day) improved a facial-emotion 'social acuity' score (p=0.028) but was largely null on core cognition. No matcha-specific metabolism/weight-loss RCT. The '137x' antioxidant claim is a lab ORAC talking point with no outcome meaning.

THE VERDICT Fine to drink and a legitimately higher-catechin/caffeine way to take green tea; expect a mild focus/calm effect from caffeine+L-theanine, not measurable fat loss or cognitive rescue.

BEST FORM n/a - the powder whisked into water IS the form, and because you ingest the whole ground leaf you get more catechins, L-theanine and caffeine per serving than steeped tea. Choose ceremonial-grade, vivid green, from a brand that publishes lead/cadmium testing.

WATCH Whole-leaf means higher caffeine (~60-70 mg/serving) and higher intake of whatever the leaf accumulated - lead and other heavy metals concentrate in tea leaves, so third-party testing matters more than for steeped tea (you swallow the leaf). Same iron-chelation caveat with meals. Shading raises L-theanine, giving the calmer 'matcha' feel.

Sources: Matcha & attention under acute stress RCT, *Physiol Behav* 2020 (S0271531720306084); Matcha 12-month cognition/sleep RCT, *PLOS One* 2024 (PMC11364242); Therapeutic potential of matcha — critical review, 2022 (PMC9792400)

3 Sencha / Gyokuro (premium Japanese green teas) Amazon ✓ Website ✓

Evidence: thin (variety-specific): no dedicated outcome RCTs; borrows green-tea evidence | Effect: acute alertness/calm (caffeine + elevated L-theanine); inherits green-tea cohort/BP data | Loose-leaf specialty teas; no liver warning. Gyokuro delivers more caffeine/L-theanine and fewer catechins per cup than sencha.

THE NOISE Shade-grown gyokuro and top-grade sencha are sold as elevated-catechin/L-theanine 'longevity' teas with superior health benefits over ordinary green tea.

THE SCIENCE These are green teas with a different chemistry: shading (gyokuro) raises L-theanine and caffeine and lowers catechins, giving a smoother, more umami, more stimulating cup; sencha is a standard high-catechin green tea. There are essentially NO variety-specific outcome trials isolating sencha or gyokuro — their evidence is inherited wholesale from green-tea catechin/cohort data (same sub-MCID BP/LDL effects, same confounded mortality association). No head-to-head trial shows gyokuro or premium sencha outperforms cheap green tea on any clinical endpoint.

THE VERDICT Drink for flavor and the same modest green-tea profile; the premium grade buys taste and a higher L-theanine/caffeine ratio (gyokuro), not proven extra health benefit.

BEST FORM n/a - premium brewed green teas; the leaf-and-water is the form. Gyokuro is shade-grown, giving markedly higher L-theanine (smoother, umami, calmer caffeine) - brew it cool (~50-60C); sencha brews ~70C.

WATCH Gyokuro's shading also raises caffeine and can raise nitrate/theanine - it is a stronger cup than it tastes. Cool, short steeps favor theanine/sweetness; hot, long steeps pull more bitter catechins and caffeine. Same iron-with-meals caveat as other green teas.

Sources: (Uses green-tea evidence base) UK Biobank 2022 (DOI 10.7326/M22-0041); Green tea catechin BP meta-analysis (PMID 24861099)

2 Green tea EXTRACT / EGCG capsules (concentrated supplement) Amazon ✓ Website ✓

Evidence: moderate evidence of small/absent benefit + moderate evidence of idiosyncratic harm | Effect: clinically trivial weight change; NULL cancer-biomarker effect; hepatotoxicity signal (harm) · **literature compromised (funding/outlier)** | Green tea EXTRACT capsules carry a documented hepatotoxicity signal and USP labeling caution; brewed green tea does not. Retail doses can equal/exceed the trial/safety EGCG threshold, so retail is NOT safely below trial exposure. Highest-risk pattern is high-dose EGCG taken as a fasting bolus.

THE NOISE Concentrated 'fat-burner' and 'cancer-preventer' — get the benefits of 10+ cups in one capsule for weight loss and prevention.

THE SCIENCE This is where a modest brewed-tea profile becomes a net-negative supplement. Efficacy: weight-loss meta-analyses show only small, clinically trivial changes; the largest independent prevention RCT — the Minnesota Green Tea Trial (n=1,075 postmenopausal women, 843 mg EGCG/day, 1 year) — was NULL on breast-cancer biomarkers. Harm: the SAME trial found ALT elevation in 6.7% vs 0.7% on placebo (P<0.001; ALT +5.4 U/L, 95% CI 3.6–7.1), 1.3% with serious hepatic events, with a COMT-genotype susceptibility signal. USP's 2020 review documents idiosyncratic hepatotoxicity; EFSA found ≥800 mg EGCG/day linked to liver injury and could not set a fully safe bolus dose (case reports from ~140 mg upward). Bolus dosing on an empty stomach spikes EGCG bioavailability and risk.

THE VERDICT Skip it — the concentrated capsule offers no clinically meaningful weight or cancer benefit while carrying a real, idiosyncratic risk of liver injury; if you want green tea's mild upside, drink the beverage, which does not carry this signal.

BEST FORM If you use it at all, choose a modest-dose, standardized extract and TAKE IT WITH FOOD, not fasted high-dose. Prefer products staying under ~300-400 mg EGCG/day; avoid mega-dose fat-burner 'fasted' formulas. (With-food, moderate-dose standardized extract SAFEST; high-dose EGCG taken on an empty stomach (common in weight-loss/fat-burner products) is the risky form - fasting sharply increases EGCG absorption and hepatotoxicity risk.)

WATCH Concentrated EGCG can cause idiosyncratic liver injury; EFSA flagged ≥800 mg EGCG/day from supplements as a hepatotoxicity signal and taking it fasted raises plasma levels several-fold. This is the one green-tea form where safety, not just absorption, is the issue. Also chelates iron and can interact with some drugs. Brewed tea is far safer than pills - most people do not need the capsule.

Sources: Minnesota Green Tea Trial safety, Cancer Prev Res 2017 (PMID 26051348); USP comprehensive review of green tea extract hepatotoxicity, Toxicol Rep 2020 (PMID 32140423); NIH LiverTox: Green Tea (NBK547925); EFSA scientific opinion on green tea catechins/EGCG safety (2018)

2 Oolong tea Amazon ✓ Website ✓

Evidence: weak: small short crossover RCTs; surrogate endpoints only | Effect: acute fat-oxidation marker (small RCT); no demonstrated weight loss | No liver warning for brewed oolong. Some concentrated 'oolong slimming' extracts/blends carry the same caffeine-load and unverified-additive concerns as other slimming products.

THE NOISE The 'weight-loss tea' — boosts metabolism, burns fat overnight, and shrinks belly fat between green and black tea.

THE SCIENCE The metabolic story rests on small, short crossover RCTs: 2 weeks of oolong/caffeine raised fat oxidation by ~20% and lowered sleeping respiratory quotient, but 24-hour energy expenditure was UNCHANGED and no meaningful weight loss was demonstrated. Older work shows a small acute rise in metabolic rate driven largely by caffeine. Higher fat oxidation without higher total energy expenditure or a calorie deficit does not equal weight loss. Longer-term body-composition benefits are modest, plateau by ~8–12 weeks, and lean on small/insulin-resistant subgroups.

THE VERDICT A tasty tea, but skip it as a weight-loss tool — the 'fat-burning' effect is a short-term metabolic marker (mostly caffeine), not demonstrated fat loss; drink for enjoyment, not your waistline.

BEST FORM n/a - the beverage is the form; partial oxidation gives a catechin/theaflavin mix. Gongfu-style short multiple infusions extract more polyphenols across steeps.

WATCH Caffeine is intermediate between green and black and varies widely by oxidation/roast. Same iron-with-meals caveat. Marketing claims of dramatic 'fat burning' are overstated - modest thermogenic effect at best.

Sources: Subacute caffeine & oolong tea, fat oxidation & energy expenditure crossover RCT, Nutrients 2020;12:3671 (PMC7760339); Oolong tea metabolic rate/fat oxidation study (S0022316622144937)

2 Pu-erh tea (fermented) Amazon ✓ Website ✓

Evidence: weak: small, partly industry/region-concentrated extract RCTs; brewed-tea data near-absent | Effect: lipid lowering in dyslipidemic subjects (small extract RCTs); benefit in disease, not proven in healthy adults · **literature compromised (funding/outlier)** | Sold as aged cakes/loose leaf and as extract capsules. Brewed pu-erh has no liver warning; the studied lipid benefit used concentrated extract, not the cup.

THE NOISE Fermented 'gut-microbiome' and 'cholesterol-melting' tea that lowers lipids, aids weight loss, and detoxifies.

THE SCIENCE Lipid claims come from a few small RCTs of pu-erh EXTRACT (not casual brewed tea): e.g., 3 g/day extract for 20 weeks in 59 hyperlipidemic/overweight subjects improved lipid profile and reduced body fat; a 90-patient metabolic-syndrome RCT lowered TC/TG/LDL and raised HDL. These are small, mostly in dyslipidemic (not healthy) populations, several China/industry-adjacent, and use standardized extract doses a casual drinker won't reach. Brewed pu-erh has essentially no rigorous outcome data. Naturally fermented pu-erh can also carry mycotoxin/quality concerns if poorly produced.

THE VERDICT **Enjoy brewed pu-erh for its flavor; if you specifically want the lipid effect you'd need standardized EXTRACT at trial doses in a dyslipidemic context, and the evidence is too small and non-independent to recommend it as a cholesterol treatment.**

BEST FORM n/a - the beverage is the form; microbial fermentation (ripe/shou or aged raw/sheng) changes the polyphenol profile and produces statin-like compounds in some batches. Rinse the first short steep and re-steep.

WATCH Fermented/aged product: quality and storage vary, and poorly stored cakes can harbor mold/mycotoxins - buy from reputable sources. Caffeine is still present (often substantial). The initial quick rinse also removes dust from aged compressed cakes.

Sources: Pu-erh extract & body fat/lipids in hyperlipidemia RCT (PMID 27069360); Pu'er tea extract & metabolic syndrome RCT (PMID 21725873)

1 White tea Amazon ✓ Website ✓

Evidence: none/very thin: no human outcome RCTs; lab and cosmetic data only | Effect: none demonstrated in humans (in-vitro/animal antioxidant claims only) | No liver warning. Marketed on lab antioxidant numbers with no human outcome backing.

THE NOISE The 'least processed, most antioxidant-rich, anti-aging' tea — highest polyphenols, anti-cancer, skin-protecting, and best for you of all teas.

THE SCIENCE This is essentially a marketing category. Claims rest on in-vitro antioxidant assays, cell/animal work, and cosmetic-extract studies — not human outcome trials. There are effectively NO clinical outcome RCTs on brewed white tea for mortality, cancer, weight, or cardiovascular endpoints. Its chemistry overlaps green tea (catechins, caffeine, L-theanine), so any real effect would plausibly mirror green tea's small ones, but this is inference, not evidence. The 'more antioxidants than green tea' line is not consistently true and has no demonstrated clinical meaning.

THE VERDICT **Drink it because you like the delicate flavor — a fine, low-caffeine tea, but every specific health claim is unproven in humans; buy it for taste, not benefits.**

BEST FORM n/a - the beverage is the form; minimally processed, so it retains high catechins similar to green tea. Brew gently (~75-85C).

WATCH Despite a mild flavor it is not automatically low-caffeine - buds (silver needle) can be quite caffeinated. Same iron chelation caveat. 'Highest antioxidant tea' claims are inconsistent and depend on the specific tea and brew.

Sources: (No human outcome RCTs) claims derive from in-vitro antioxidant and cosmetic-extract literature; nearest human data are green-tea catechin trials (PMID 24861099)

4 Coffee (caffeinated, paper-filtered / drip) Amazon ✓ Website ✓

Evidence: strong prospective cohort; RCT-null for hard outcomes; caffeine alertness = strong RCT | Effect: CV & all-cause mortality (cohort association, J-curve); acute alertness (RCT) | Ubiquitous as beans/ground/pods; brewed coffee carries no supplement warnings. Caffeine pills are a separate, higher-risk product.

THE NOISE Marketed as a longevity 'superfood' — drink coffee to add years, torch fat, prevent diabetes, dementia and heart disease. Wellness culture treats a large cohort correlation as if it were a proven drug.

THE SCIENCE Large, consistent PROSPECTIVE COHORT data (no RCT on hard outcomes exists). Pooled dose-response: all-cause mortality RR ~0.86 (95% CI 0.82–0.89) and CVD mortality RR ~0.85 (0.77–0.93) at ~3–4 cups/day, a J-curve that plateaus — more is not better. Umbrella review (Poole, BMJ 2017) finds the largest relative risk reductions at 3 cups/day across ~200 meta-analyses. Signal survives adjustment and appears in non-smoker strata (RR ~0.94 per cup). CAVEAT: observational only — residual confounding, reverse causation (sick people quit coffee) and healthy-user bias are not fully excluded; benefit in a cohort ≠ a treatment effect. The one RCT-grade fact is caffeine's ACUTE alertness/vigilance effect, which is robust.

THE VERDICT Drink it if you enjoy it — 2–4 cups/day of filtered coffee is associated with the best outcomes and is safe for most adults; it is not a reason to START drinking coffee for health. Cap caffeine at ~400 mg/day.

BEST FORM Paper-filtered drip/pour-over is the best-tolerated cardiovascular form - the paper traps the cholesterol-raising diterpenes cafestol and kahweol. (Paper-filtered (drip, pour-over, AeroPress with paper) GOOD - lowest LDL impact; unfiltered (French press, boiled, Turkish, espresso to a lesser degree) raises LDL because diterpenes pass through.)

WATCH The filter, not decaf-vs-regular, is what controls the LDL/diterpene effect. Caffeine ~95 mg/cup - mind afternoon timing for sleep and interactions with anxiety/arrhythmia. Coffee also inhibits non-heme iron absorption, so separate from iron supplements/meals. Adding sugar/cream is usually the real health variable.

Sources: Poole et al., BMJ 2017;359:j5024 (umbrella review) doi:10.1136/bmj.j5024; Grosso et al. dose-response meta-analysis, all-cause/CVD/cancer mortality PMID:27699514; EFSA Scientific Opinion on caffeine safety, EFSA Journal 2015;13(5):4102

3 Cocoa flavanols / dark chocolate (COSMOS trial standardized extract) Amazon ✓ Website ✓

Evidence: moderate: large RCT null on primary, positive secondary + Cochrane BP | Effect: Blood pressure (small, RCT); CV death (secondary endpoint only); primary CV endpoint null | Cocoa-flavanol capsules (e.g., CocoaVia) sold on Amazon and direct; unlike green-tea EXTRACT, cocoa extract carries no hepatotoxicity warning. Chocolate bars deliver far less standardized flavanol.

THE NOISE Dark chocolate marketed as heart-healthy, blood-pressure-lowering, brain-boosting 'flavanol' medicine. Ads imply a chocolate bar delivers the trial dose.

THE SCIENCE The rigorous test is COSMOS (NIH/NHLBI-led, Brigham/Harvard; Mars supplied the standardized 500 mg flavanol/80 mg epicatechin extract), n=21,442, ~3.6 yr. PRIMARY endpoint — total CV events — was NULL: HR 0.90 (95% CI 0.78–1.02, p=0.11). A PRE-SPECIFIED SECONDARY endpoint, CV death, was lower: HR 0.73 (0.54–0.98) — hypothesis-generating, not confirmatory, and all-cause mortality was non-significant (0.89, 0.77–1.03). Separately, Cochrane (Ried 2017, 40+ comparisons) shows cocoa flavanols lower BP by ~1.8 mmHg overall, more in hypertensives (~4 mmHg SBP) and at ≥900 mg flavanols/day — a small, short-term effect. NOTE: this evidence is for a STANDARDIZED EXTRACT; a normal chocolate bar is NOT this.

THE VERDICT The standardized 500 mg-flavanol extract has a genuine but modest, mostly-secondary-endpoint signal (CV death, small BP drop) — reasonable for those wanting a low-risk adjunct, not a proven therapy. A dark chocolate bar is a treat, not the trial: eat it for pleasure, keep portions small for the sugar/calories.

BEST FORM For the evidence-based cardiovascular benefit, use a standardized cocoa-flavanol supplement delivering ~500 mg cocoa flavanols/day (the COSMOS trial dose) - not ordinary chocolate bars. (Standardized flavanol extract (COSMOS-type, ~500 mg/day) BEST and dose-verified; dark chocolate bars are inconsistent and 'Dutched'/alkalized cocoa is WORST (alkalizing destroys most flavanols); milk chocolate is negligible.)

WATCH You cannot reliably get the COSMOS dose from chocolate without a lot of sugar and fat, and flavanol content is usually unlabeled and degraded by processing. The COSMOS trial showed cardiovascular-death benefit for the extract, not for candy. Bars also carry cadmium/lead concerns (see cacao). Caffeine/theobromine are mild but present.

Sources: Sesso et al., COSMOS trial, Am J Clin Nutr 2022;115(6):1490 doi:10.1093/ajcn/nqac055 (PMC9170467); Ried et al., 'Effect of cocoa on blood pressure', Cochrane Database Syst Rev 2017 CD008893.pub3

3 Decaffeinated coffee Amazon ✓ Website ✓

Evidence: moderate/weak: observational-only | Effect: Type 2 diabetes risk & mortality (cohort association); no acute alertness | Widely sold; 'Swiss Water'/CO₂-decaffeinated options market solvent-free processing.

THE NOISE Framed either as 'all the benefits, none of the jitters' or dismissed as 'chemical-laden and pointless'. Both are overstatements.

THE SCIENCE COHORT data suggest decaf carries much of coffee's metabolic/mortality association independent of caffeine: in type 2 diabetics, ≥ 2 cups/day decaf associated with RR ~ 0.76 all-cause and ~ 0.96 CVD mortality; decaf lowers type 2 diabetes risk in dose-response meta-analysis ($\sim 6\%$ per cup) and is linked to lower HbA1c. This argues the benefit is partly from non-caffeine coffee compounds (chlorogenic acids). All OBSERVATIONAL; no hard-outcome RCT. Removes caffeine's acute alertness benefit and its downsides (anxiety, sleep disruption, pregnancy caution).

THE VERDICT **A legitimate choice for caffeine-sensitive people, the evening, or pregnancy — it appears to retain most of coffee's metabolic association without the caffeine. Choose water-processed (e.g., Swiss Water) if solvent residue concerns you.**

BEST FORM Choose paper-filtered decaf, and if solvent exposure concerns you, look for Swiss Water or CO₂-processed decaf (no methylene chloride). (Filtered decaf keeps the LDL-friendly profile; the decaffeination method (Swiss Water/CO₂ vs solvent) is the other axis - Swiss Water avoids residual solvents.)

WATCH Decaf still contains chlorogenic acids and trace caffeine, and unfiltered decaf still raises LDL via diterpenes - decaffeination does not remove cafestol. Good option for reflux-prone or caffeine-sensitive people; still mildly inhibits non-heme iron absorption.

Sources: Ding et al., 'Caffeinated and decaffeinated coffee and risk of type 2 diabetes: dose-response meta-analysis', Diabetes Care 2014;37(2):569 doi:10.2337/dc13-1203; Coffee & CVD/mortality in type 2 diabetes meta-analysis, Nutr Metab Cardiovasc Dis 2021, S0939-4753(21)00237-4

3 Espresso Amazon ✓ Website ✓

Evidence: moderate: large cross-sectional + mechanistic RCT on diterpenes | Effect: Small rise in total/LDL cholesterol (diterpenes); shares coffee's cohort mortality signal | Beans/pods widely sold; no warning label. Cholesterol effect is brew-method-driven, not a product defect.

THE NOISE Sold as the 'purest', most antioxidant-dense coffee — a concentrated health shot. Espresso-forward wellness content ignores that it is only lightly filtered.

THE SCIENCE Espresso shares coffee's cohort mortality association BUT is metal/paper-lightly-filtered, so it retains cholesterol-raising diterpenes (cafestol/kahweol). Tromsø Study (n=21,083, cross-sectional): 3–5 espressos/day associated with higher total cholesterol — $+0.16$ mmol/L (~ 6 mg/dL) in men, $+0.09$ mmol/L in women. Effect is real but SMALL and below the LDL rise seen with boiled/French-press coffee; likely below most clinicians' MCID for a single lifestyle factor, though additive at high intake.

THE VERDICT **Fine in moderation; if you have high LDL or familial hypercholesterolemia, prefer paper-filtered coffee and watch total espresso volume. A modest, dose-dependent cholesterol nudge, not a health elixir.**

BEST FORM Fine as a small serving; note it is semi-filtered (metal/paper puck) so it carries some diterpenes - moderate cups are low-volume so total cafestol stays modest. (Espresso sits between fully paper-filtered (lowest LDL) and fully unfiltered French press/Turkish (highest) - intermediate diterpene load per volume, but small servings limit the dose.)

WATCH Caffeine per fluid ounce is high but a single shot (~ 63 mg) is less than a mug of drip. If drinking many shots daily and watching LDL, prefer paper-filtered brews. Same iron-with-meals caveat.

Sources: Svaton et al., 'Association between espresso coffee and serum total cholesterol: the Tromsø Study 2015–2016', Open Heart 2022, PMID:35537850; Jee et al., coffee & serum cholesterol meta-analysis of RCTs, Am J Epidemiol 2001

3 Instant coffee Amazon ✓ Website ✓

Evidence: moderate (inferred from general coffee cohorts); instant-specific data thin | Effect: Shares coffee's cohort mortality/diabetes association; negligible lipid effect | Ubiquitous; 'mushroom/functional' instant coffees add unproven ingredients at a premium.

THE NOISE Dismissed as nutritionally inferior 'fake' coffee, or conversely marketed with added 'functional' mushrooms/collagen as an upgrade.

THE SCIENCE Spray/freeze-dried soluble coffee is essentially filtered (near-zero diterpenes, so no meaningful LDL effect) and appears in the same cohorts underlying coffee's mortality/diabetes associations, contributing comparable benefit per cup. It has somewhat higher acrylamide than brewed, but at intakes far below toxicological concern for the association. No instant-specific hard-outcome RCT. 'Functional' add-ins (lion's mane, collagen, ashwagandha) are separate ingredients with their own — mostly thin — evidence and don't inherit coffee's data.

THE VERDICT A fine, low-diterpene everyday choice with the same broad coffee association and less cholesterol risk than French press. Ignore 'functional coffee' health halos — judge each added ingredient on its own (usually thin) evidence.

BEST FORM n/a - the beverage is the form. Convenient and, because it is filtered during manufacture, it is low in the LDL-raising diterpenes (a plus vs French press).

WATCH Instant tends to have somewhat higher acrylamide than fresh-brewed and generally lower caffeine per cup; chlorogenic-acid/antioxidant content is a bit lower than fresh but still meaningful. Same non-heme iron inhibition. Quality/flavor aside, it is cardiovascularly comparable to filtered coffee.

Sources: UK Biobank coffee-type & mortality analyses (e.g., Eur J Prev Cardiol 2022); Poole et al., BMJ 2017;359:j5024 (umbrella review, coffee overall)

3 Unfiltered coffee (French press, boiled/Scandinavian, Turkish, cezve) Amazon ✓ Website ✓

Evidence: strong RCT (for the adverse lipid effect) | Effect: Raises LDL/total cholesterol (causal, RCT, dose-dependent) | The beans are ordinary coffee — risk is the BREW METHOD, not the product; a paper filter neutralizes most of it.

THE NOISE Sold as artisanal, 'full-bodied', more antioxidant/oil-rich and therefore healthier than 'stripped' filtered coffee. The oils are exactly the problem.

THE SCIENCE This is the one place with strong CAUSAL RCT evidence — of HARM to lipids. Cafestol and kahweol (diterpenes retained without a paper filter) reduce hepatic LDL clearance and raise LDL. RCTs: ~63 mg/day cafestol for 4 weeks raised cholesterol ~17% and triglycerides sharply; boiled coffee raised LDL ~0.4 mmol/L (~16 mg/dL); RCT meta-analysis average LDL rise ~5.4 mg/dL, concentrated in unfiltered arms and dose-dependent. Diterpene load: boiled coffee ~900+ mg/L cafestol vs paper-filtered ~12 mg/L; French press intermediate (~90 mg/L). Dose-response direction is unambiguous and adverse at high intake.

THE VERDICT Skip or minimize if you have elevated LDL or CVD risk — or run it through a paper filter, which removes ~80% of the diterpenes. Occasional French press is fine for low-risk people; daily 4+ cups of boiled/press coffee measurably raises LDL.

BEST FORM If you love it, keep intake modest and monitor LDL; switching to a paper filter (or a paper insert) is the single change that removes most of the cholesterol-raising diterpenes. (Unfiltered is the highest-diterpene form - boiled/Scandinavian and French press deliver the most cafestol/kahweol and measurably raise LDL vs paper-filtered.)

WATCH This is the coffee form with a documented adverse cardiovascular signal - cafestol is one of the most potent dietary cholesterol-raising compounds known. People with high LDL or CVD risk should favor filtered. Otherwise same caffeine and iron-chelation notes as regular coffee.

Sources: Urgert & Katan, 'The cholesterol-raising factor from coffee beans', Annu Rev Nutr 1997; Jee et al., RCT meta-analysis of coffee and serum lipids, Am J Epidemiol 2001; Cafestol/kahweol brew-concentration analysis, Nutr Metab Cardiovasc Dis 2025, PMID:40089392

2 Cacao (raw cacao powder / nibs / 'superfood' marketing) Amazon ✓ Website ✓

Evidence: weak: mechanism/flavanol-content only; no cacao-specific outcome RCT | Effect: None demonstrated for 'cacao' as such; any effect is the cocoa-flavanol BP/endothelial signal — dose-dependent and often not met | Powder/nibs sold everywhere with 'superfood' claims; flavanol content is rarely disclosed or standardized and dutching silently strips it — 'processed with alkali' on the label signals low flavanols.

THE NOISE Sold as a 'raw superfood' — highest antioxidants, mood enhancer, magnesium powerhouse, ancient heart medicine. Positioned as strictly superior to processed cocoa.

THE SCIENCE The health-relevant compound is the SAME cocoa flavanol tested in COSMOS/Cochrane — there is no separate outcome literature for 'raw cacao'. Flavanol content is highly variable and destroyed by processing: natural cocoa ~34.6 mg flavanols/g, but alkalization ('dutching') cuts it ~60% (light) to ~90% (heavy); many retail 'raw' products are un-standardized and much of the marketed antioxidant/mood claims have no human outcome data. To reach the COSMOS 500 mg standardized dose from powder requires a specific, verified high-flavanol product — most bars and dutched powders fall well short. 'Raw' status does not itself confer proven benefit.

THE VERDICT Skip the superfood premium. If you want the flavanol effect, use a standardized cocoa-flavanol supplement (the COSMOS/CocoaVia route) or a verified natural (non-dutched) cocoa — 'raw cacao' branding buys marketing, not a documented outcome. Enjoy it as food, not medicine.

BEST FORM Choose natural (non-alkalized/non-Dutched) cocoa or raw cacao from a brand publishing heavy-metal testing; 'raw superfood' branding alone guarantees nothing. (Natural non-alkalized cocoa/raw cacao retains more flavanols than Dutch-processed (alkalized) cocoa, which is largely stripped - but 'raw' does NOT reliably mean higher usable flavanols or a standardized dose.)

WATCH Cacao is a notable dietary source of cadmium and lead (Consumer Reports/As You Sow flagged many dark-chocolate/cacao products above levels of concern) - heavy-metal testing matters more here than the 'raw' label. Theobromine and some caffeine can disturb sleep in large servings. Not a validated substitute for the standardized COSMOS flavanol dose.

Sources: Miller et al., 'Impact of Alkalization on Antioxidant and Flavanol Content of Commercial Cocoa Powders', J Agric Food Chem 2008 doi:10.1021/jf801670p; Sesso et al., COSMOS, Am J Clin Nutr 2022 (the only hard-outcome cocoa-flavanol RCT)

Beverages — other drinks (10)

3.5 Hibiscus tea (Hibiscus sabdariffa) Amazon ✓ Website ✓

Evidence: moderate: multiple RCTs and meta-analyses, high heterogeneity | Effect: blood pressure lowering (RCT) | Brewed tea and dried calyces widely sold; effect requires regular daily intake, not occasional cups.

THE NOISE 'Natural blood pressure medicine' and heart tonic - marketed as a drug-free way to lower BP.

THE SCIENCE The strongest evidence in this herbal family. Meta-analyses of RCTs (up to 26 trials, ~1,797 participants) show dose-dependent BP lowering: pooled systolic ~ -7 mmHg (95% CI roughly -13 to -1) and diastolic ~ -3 mmHg (often not statistically significant), with the largest effect in people with elevated baseline BP and trials ≥4 weeks. The systolic effect clears a plausible ~5 mmHg MCID in hypertensives, though heterogeneity is high (I² ~90%) and some trials are small.

THE VERDICT A legitimate adjunct for prehypertension/mild hypertension: 2-3 cups/day of steeped hibiscus. Real but modest - not a replacement for prescribed antihypertensives, and tell your doctor if you take BP meds.

BEST FORM n/a - the beverage is the form; brewed hibiscus (2-3 cups/day in BP studies) is what carries the modest blood-pressure-lowering effect. Steep dried calyces in hot water.

WATCH Genuine drug interactions: hibiscus can add to antihypertensive and diuretic effects (watch for low BP), may reduce absorption/levels of acetaminophen and possibly some other drugs, and has estrogenic/emmenagogue activity - avoid in pregnancy. Very acidic (dental erosion). Effect on BP is real but mild, not a substitute for prescribed therapy. Caffeine-free.

Sources: Nutr Rev 2022;80(6):1723 PMID 34927694; ScienceDirect 2025 S096522992500010X overview/dose-response

3 Ginger tea (*Zingiber officinale*) Amazon ✓ Website ✓

Evidence: moderate for ~1 g ginger; thin for typical tea dose | Effect: nausea reduction (RCT evidence for ~1 g standardized ginger) | Nausea RCTs use ~1-1.5 g standardized ginger; tea bags/slices are often below this and variable.

THE NOISE Nausea cure-all, potent anti-inflammatory and immune booster.

THE SCIENCE Ginger genuinely reduces nausea, but the trial evidence is for standardized ginger ~1-1.5 g/day (powder/capsules): pregnancy-nausea meta-analysis of 12 RCTs (n=1,278) shows significant symptom improvement vs placebo (MD 1.20, 95% CI 0.56-1.84); chemotherapy nausea data are weaker/mixed. A cup of ginger tea often delivers well under 1 g of actual ginger and is unstandardized, so effect depends heavily on how strong you make it. Anti-inflammatory/immune claims lack solid clinical-outcome data.

THE VERDICT Genuinely useful for mild/pregnancy nausea if brewed strong (aim toward ~1 g ginger); weak tea underdoses. Skip the 'immune booster' framing.

BEST FORM For flavor/warmth, fresh grated ginger simmered ~10 min releases gingerols; brief steeping under-extracts. For reliable ANTI-NAUSEA dosing (pregnancy, motion, chemo), a standardized ginger capsule/extract is better — tea delivers an unknown, variable dose, whereas clinical benefit is seen around 1-1.5 g/day, hard to hit consistently with tea. (Standardized ginger extract capsule (consistent dose, best for nausea) > fresh grated root simmered (good, variable) > dried ginger tea bags (weaker, gingerol degrades) > 'ginger flavored' products (often little real ginger).)

WATCH Antiplatelet effect at higher doses — caution with warfarin/aspirin/anticoagulants and before surgery. Can lower blood glucose and blood pressure (watch with diabetes/BP meds). Large amounts may cause heartburn. Well-supported and considered safe for pregnancy nausea up to ~1 g/day, but discuss higher doses with a clinician.

Sources: Nutr J 2014 ginger pregnancy nausea meta-analysis (PMC3995184); PMC9739555 ginger CINV systematic review

2.5 Chamomile tea (*Matricaria chamomilla*) Amazon ✓ Website ✓

Evidence: moderate for standardized extract; thin for brewed tea | Effect: sleep quality; mild GAD (mostly extract, not tea) | Trials used standardized extract (500-1500 mg/day); brewed tea is a much lower, variable dose.

THE NOISE Bedtime sleep aid and natural anxiety cure - the classic 'calming' tea.

THE SCIENCE Meta-analyses show a real but modest signal: improved sleep quality (SMD ~ -0.73; PSQI WMD ~ -1.88) and a small GAD benefit (HAM-A MD -1.43, 95% CI -2.47 to -0.39), while state anxiety and insomnia are non-significant. The key catch: nearly all positive trials use standardized chamomile EXTRACT capsules (e.g. 500-1500 mg/day), not brewed tea - a cup of tea delivers a far lower, unstandardized dose, so the trial evidence does not transfer cleanly to the teabag.

THE VERDICT A calming bedtime ritual with a modest genuine sleep-quality signal - but most of that evidence is for extract capsules, not tea. Reasonable low-risk habit; do not expect an anxiolytic drug.

BEST FORM The brewed tea IS the form, but preparation matters: use whole dried German chamomile flowers, steep 5-10 min with the cup COVERED to trap the volatile apigenin-rich oils that otherwise evaporate. Covered steeping meaningfully raises active yield versus an open mug. (Whole dried German chamomile flowers, covered steep (best) > standard tea bags (fine, often finer cut) > Roman chamomile (different species, more for flavor/topical). Standardized apigenin capsules exist for sleep studies but the tea is the traditional form.)

WATCH Cross-reactivity: avoid if allergic to ragweed/Asteraceae (daisy, marigold). Contains natural coumarins — can potentiate warfarin and other anticoagulants, so watch bleeding risk. Sedative; can add to other CNS depressants. May inhibit CYP enzymes (e.g., affecting cyclosporine). Best taken in the evening for its mild calming effect.

Sources: Phytother Res 2019 chamomile anxiety/sleep meta-analysis (ptr.6349); Complement Ther Med 2024 chamomile & sleep meta-analysis (PMID 39106912)

2.5 Guarana (Paullinia cupana, drink/added to beverages) Amazon ✓ Website ✓

Evidence: moderate for the caffeine effect; weak for any 'beyond caffeine' claim | Effect: acute alertness/reaction time (caffeine) | Sold as powder, capsules and in energy drinks/sodas; 'natural' framing masks that potency = caffeine content.

THE NOISE 'Natural energy stronger than caffeine,' fat-burner, focus enhancer - marketed as if its effect is something beyond caffeine.

THE SCIENCE Guarana seed is essentially a concentrated caffeine source (up to ~4-5% caffeine by weight, vs ~1-2% in coffee). Acute alertness and reaction-time benefits in small crossover RCTs (e.g. n=25) are consistent with its caffeine content; claims that co-occurring saponins/tannins add a distinct cognitive effect are not established beyond the caffeine dose. As a 'beverage' it usually appears in sugary energy/soda drinks.

THE VERDICT A caffeine delivery vehicle - it works because it is caffeine. No reason to pay a premium over coffee/tea; watch the sugar in guarana sodas and energy drinks.

BEST FORM Prefer a standardized guarana extract with the caffeine content stated on the label so you can count total caffeine; treat it as a caffeine source, not a benign herb. (Standardized, caffeine-labeled extract BEST for dose control; unlabeled guarana in energy drinks/'proprietary blends' WORST - caffeine stacks invisibly on top of any added caffeine.)

WATCH Guarana seeds are among the most caffeine-dense plants (~2-4x coffee beans by weight, ~40-80+ mg caffeine per gram) and the caffeine is often ADDED to products that also list caffeine separately - easy to overshoot. Watch total daily caffeine (~400 mg cap for most adults; less if pregnant, arrhythmia-prone, or on stimulants). Slow-release effect from tannins.

Sources: Br J Nutr 2023 PMID 36146946; Nutrients 2024;16(12):1892 acute guarana vs low-dose caffeine (PMC11206275)

2.5 Yerba mate (Ilex paraguariensis, brewed) Amazon ✓ Website ✓

Evidence: weak/observational for metabolic claims; consistent observational signal for hot-beverage esophageal cancer | Effect: acute alertness (caffeine); esophageal cancer risk when drunk very hot (case-control/cohort) | Widely sold as loose leaf, tea bags and canned drinks; weight-loss capsule extracts use higher standardized doses than a brewed gourd.

THE NOISE Sold as a 'metabolic superfood' and the 'healthiest energy drink' - claims of fat-burning, weight loss, appetite control and antioxidant supremacy over green tea.

THE SCIENCE Contains caffeine (~30-50 mg per 150 mL, ~80 mg per traditional gourd refill) so acute alertness is real and is just the caffeine. Metabolic claims rest on small, short (3g/day capsule) RCTs and animal work - modest, unreplicated lipid/weight signals. The robust human finding is a HARM: case-control/cohort data (IARC Group 2A) link mate drunk very hot ($\geq 65^{\circ}\text{C}$) to esophageal squamous-cell carcinoma; the driver is temperature, not the leaf.

THE VERDICT Fine as a caffeinated tea for alertness; ignore the fat-burning marketing. Let it cool below ~60C before drinking - the cancer signal is about scalding temperature, not the herb.

BEST FORM n/a - the beverage is the form. Brew and drink it WARM, not scalding, and favor loose-leaf brewing over commercial energy-drink versions loaded with sugar.

WATCH Two real cautions: (1) drinking it very hot is associated with esophageal cancer - let it cool below ~60C; (2) traditionally cured mate can contain PAHs (polycyclic aromatic hydrocarbons) - air-dried/uncured styles reduce this. Caffeine ~80 mg/cup plus theobromine - real stimulant load. Same iron-chelation caveat as tea.

Sources: IARC Monograph Vol 116 (2018) very hot beverages; Cancer Epidemiol Biomarkers Prev 2014;23(1):107 mate & ESCC pooled; PMC5831222 hot beverage esophageal risk assessment

2 Chicory 'coffee' (Cichorium intybus root) Amazon ✓ Website ✓

Evidence: weak; inulin fiber evidence is for supplement-scale doses, not a cup | Effect: none clinically meaningful from the beverage; prebiotic/GI effects at gram doses | Sold as roasted chicory grounds/instant and coffee blends; inulin fiber supplements (grams) are where the metabolic data sit, not the drink.

THE NOISE Gut-healthy, blood-sugar-friendly, caffeine-free coffee alternative loaded with prebiotic fiber.

THE SCIENCE Caffeine-free is true and is the main honest selling point. Chicory root's inulin is a genuine prebiotic and can blunt post-meal glucose when it replaces sugar - but that evidence is for gram-scale inulin doses, and a cup of roasted-chicory 'coffee' contains relatively little inulin, so beverage-level clinical benefit is unproven. The most reliable real-world effect of higher intake is GI: gas, bloating and a mild laxative action.

THE VERDICT A fine caffeine-free coffee substitute for taste; the 'gut/blood-sugar' benefits need far more inulin than a cup provides. Go easy if you are FODMAP-sensitive - it can cause bloating.

BEST FORM The roasted-root brew IS the form. Pick 100% roasted chicory root (or a chicory-heavy blend) as a caffeine-free coffee substitute; roasting develops the flavor. Naturally caffeine-free, so no decaffeination-residue concerns. (Pure roasted chicory root brew (best caffeine-free substitute) > chicory/coffee blends (partial caffeine) > instant chicory products (convenient, similar). All deliver inulin regardless of format.)

WATCH Very high in INULIN (prebiotic fiber) — commonly causes gas, bloating and loose stools, so start small. Stimulates bile flow and gallbladder contraction: avoid with gallstones/biliary obstruction. Cross-reacts with ragweed/Asteraceae allergy. Traditionally avoided in pregnancy (possible uterine/emmenagogue effect). May modestly lower blood sugar.

Sources: inulin glycemic/prebiotic reviews; chicory inulin & post-meal glucose studies

2 Peppermint tea (Mentha x piperita) Amazon ✓ Website ✓

Evidence: strong for enteric-coated oil; none for brewed tea | Effect: IBS symptoms (peppermint OIL only); none demonstrated for tea | IBS benefit belongs to enteric-coated peppermint oil capsules; tea delivers menthol to the stomach, not the gut, and can worsen reflux.

THE NOISE Digestive cure and IBS remedy - marketed to relieve bloating, cramps and gut upset.

THE SCIENCE The strong IBS evidence is for enteric-coated peppermint OIL capsules (meta-analysis of ~10 RCTs: significant improvement in global IBS symptoms and abdominal pain, NNT ~3-4), NOT tea. Enteric coating delivers menthol to the intestine where it relaxes gut smooth muscle; a cup of tea releases menthol in the stomach at a far lower dose, so there is no IBS-tea trial evidence. Tea can soothe subjectively and may ease nausea, but by relaxing the lower esophageal sphincter it can worsen reflux/GERD.

THE VERDICT Pleasant and soothing, but for actual IBS use enteric-coated peppermint OIL capsules, not tea. Avoid if you have reflux/GERD, which peppermint can aggravate.

BEST FORM For a beverage, loose dried leaf steeped covered 5-7 min (cover retains menthol volatiles). Key distinction: to treat IBS or cramping, the evidence-based form is ENTERIC-COATED peppermint OIL capsules — the coating delivers menthol to the small intestine and avoids the reflux uncoated oil/tea can trigger. The tea itself is only mildly antispasmodic. (Enteric-coated peppermint oil capsules (therapeutic, IBS-grade) > covered-steep loose leaf tea (mild carminative) > standard tea bags > non-enteric oil (worst — releases in stomach, causes reflux).)

WATCH Peppermint RELAXES the lower esophageal sphincter, so it can worsen GERD, reflux, hiatal hernia and heartburn — avoid if prone to these. Helpful for gas, bloating and IBS-type cramping. Non-enteric peppermint oil can cause heartburn and should not be given to infants (menthol/laryngospasm risk).

Sources: BMC Complement Altern Med 2019;19:21 peppermint oil IBS meta-analysis (PMC6337770)

2 Rooibos (*Aspalathus linearis*) Amazon ✓ Website ✓

Evidence: weak - one small human trial | Effect: lipids/oxidative markers (single small RCT); none robust | Caffeine-free; commonly sold as red (fermented) and green rooibos - lipid trial used fermented, six cups/day.

THE NOISE Antioxidant 'anti-aging' heart tonic; caffeine-free wellness tea.

THE SCIENCE Caffeine-free is genuinely true and useful. Cardiometabolic benefit rests on essentially one small human trial (n=40, six cups/day for 6 weeks) reporting reduced LDL, increased HDL and improved oxidative markers in adults at CV risk - a single, small, unreplicated study. The aspalathin/antioxidant mechanism is well characterized but human outcome data are thin.

THE VERDICT A pleasant caffeine-free tea; drink it if you want an evening cup without caffeine. Do not rely on it to fix cholesterol - the lipid data are one small trial.

BEST FORM Green (unfermented) rooibos over the common red (fermented) type — green retains far more aspalathin and total polyphenols. Loose-leaf steeped 5-10 min in freshly boiled water; unlike black/green tea it doesn't turn bitter or over-extract tannins with long steeping. (Green/unfermented rooibos (highest aspalathin & antioxidants) > red/fermented rooibos (pleasant taste, lower antioxidant load) > flavored/blended bags (lowest, often diluted).)

WATCH Naturally caffeine-free and very low in tannins, so it does NOT block non-heme iron the way black tea does — a good choice with meals. Generally very safe; isolated case reports of hepatotoxicity exist only with heavy chronic intake. Mild effects on CYP enzymes are possible at extreme intakes, but normal tea amounts are not a concern.

Sources: Marnewick et al. J Ethnopharmacol 2011;133(1):46 rooibos & oxidative stress/lipids

1.5 Energy drinks (caffeine + taurine, e.g. Red Bull-type) Amazon ✓ Website ✓

Evidence: strong for acute cardiovascular effects; benefit is caffeine alone | Effect: acute alertness (caffeine); adverse BP and QTc/arrhythmia signals (RCT) · **literature compromised (funding/outlier)** | Ubiquitous; AAP advises against for children/adolescents. High-caffeine variants can hit or exceed 200-300 mg/can.

THE NOISE Boosts energy, focus and physical/mental performance - taurine framed as a special edge ('gives you wings').

THE SCIENCE The alertness is the caffeine (~80-160 mg/can); taurine adds no clearly demonstrated independent benefit and may even slightly shorten QTc. Randomized trials show acute harms: systolic BP rises ~3-5 mmHg, QTc prolongs ~6-10 ms after high-volume intake, and supraventricular extrasystoles increase - risk concentrated in high consumers, adolescents, and people with long-QT/underlying heart disease. Plus a large sugar load. Net: a caffeine-sugar stimulant with a measurable cardiovascular downside and no health benefit.

THE VERDICT Skip as a health product. If used, treat as caffeine only, cap intake, and avoid entirely if pregnant, adolescent, or with any arrhythmia/long-QT history.

BEST FORM Choose a SUGAR-FREE product with a clearly LABELED caffeine amount (avoid 'proprietary energy blends' that hide the dose). Taurine and B-vitamins are not the concern — the caffeine and sugar are. Plain coffee or a labeled low-sugar option gives the same caffeine with fewer additives; if buying an energy drink, pick one stating mg caffeine per can. (Sugar-free, caffeine-labeled single-serve (best) > regular sugared version (adds 25-40 g sugar) > mega-size/high-caffeine cans and undisclosed 'blends' (worst — easy to overdose caffeine).)

WATCH Keep total caffeine under ~400 mg/day (adults); one large can plus coffee can exceed this. Do NOT combine with alcohol (masks intoxication) or use before intense exercise with heart risk — linked to arrhythmias and BP spikes. Avoid in pregnancy, adolescents and children. Cut off 8+ hours before bed. Added niacin can cause flushing; taurine itself is well-tolerated.

Sources: JAHA 2019 high-volume energy drink ECG/BP RCT (doi:10.1161/JAHA.118.011318); Int J Cardiol energy drinks & long-QT crossover study

1.5 Kombucha (fermented tea) Amazon ✓ Website ✓

Evidence: weak/pilot-only | Effect: none clinically demonstrated; microbiome shifts of uncertain benefit | Commercial bottled kombucha is pasteurization-controlled; home brews carry contamination and over-acidification risk.

THE NOISE 'Detox,' gut-healing, immune-boosting, blood-sugar-fixing living tonic - a near cure-all.

THE SCIENCE Human trials are tiny and preliminary. A diabetes pilot (n~12) showed lower fasting glucose (164->116 mg/dL, p=0.035) but is underpowered and unreplicated; controlled microbiome studies show modest bacterial shifts with no demonstrated clinical outcome, and one 8-week trial saw rising fasting insulin/HOMA-IR in the kombucha arm. It carries real caveats: residual sugar, trace alcohol (~0.5-1.5%), and documented case reports of lactic acidosis and hepatotoxicity, plus contamination risk in home brews.

THE VERDICT Enjoy it as a low-sugar fizzy drink if you like the taste; do not treat it as medicine. Skip home-brew if immunocompromised or pregnant. The health claims are pilot-grade at best.

BEST FORM n/a - the beverage is the form. Choose commercial, pasteurized-or-controlled, low-sugar brands with stated alcohol <0.5%; be cautious with home-brew.

WATCH Home or long-fermented kombucha can exceed 0.5% alcohol and, if brewed in ceramic/lead-glazed vessels or contaminated, has caused rare toxicity - buy commercial or brew carefully with tested equipment. It is acidic (dental erosion, reflux) and contains caffeine and residual sugar. Live cultures may not suit immunocompromised or pregnant people. Probiotic/detox health claims are largely unproven.

Sources: Ann Nutr Metab / kombucha diabetes pilot (PMC10426908); Sci Rep 2024 microbiome (PMC11686376); case reports of kombucha hepatotoxicity/lactic acidosis

Method. Four adversarial research workflows (~150 agents, ~5.7M tokens, 0 errors). Every supplement graded 1–5 from primary human trials against seven bias checks (funding-source subgroup, outlier leverage, effect-size vs MCID, trial vs retail dose, population transfer, dose–response direction, regulatory advisories). High-stakes and compromised grades were handed to independent reviewers instructed to refute them. Beverages distinguish the brewed drink from the extract/capsule form. Bioavailability guidance covers best form, form ranking, and what to watch.

Caveats & not medical advice. Grades measure evidence quality, not personal fit; several are conditional on a deficiency or a specific complaint. Availability (Amazon / website) reflects typical US retail status and moves over time; ✗ commonly means prescription-only or restricted from Amazon (racetams, noopept, phenibut, modafinil, rapamycin). Supplement potency and purity are loosely regulated — confirm the active/elemental dose and third-party testing before buying. Talk to a clinician about your labs, conditions and medications. Full dataset: data-full.json.